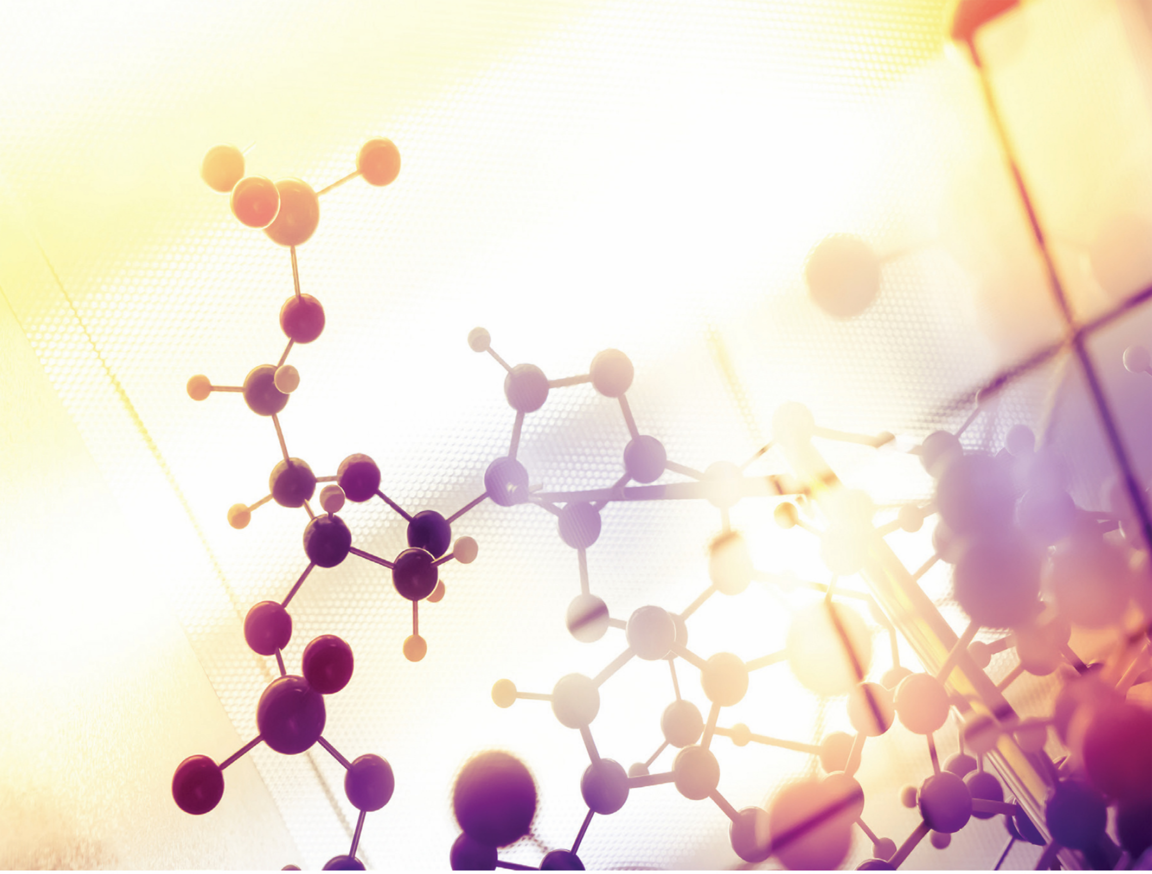


Biochemistry and Molecular Biology: Principles and Applications

Edited by
Fischer Dimitrion

BIBLIOTEX
Digital Library

BIOCHEMISTRY AND MOLECULAR BIOLOGY: PRINCIPLES AND APPLICATIONS

Biochemistry and Molecular Biology: Principles and Applications

Editor:

Fischer Dimitrion

BIBLIOTEX
Digital Library
www.bibliotex.com

Biochemistry and Molecular Biology: Principles and Applications

Editor: Fischer Dimitrion

www.bibliotex.com

email: info@bibliotex.com

e-book Edition 2024

ISBN: 978-1-98469-331-0 (e-book)

In Collaboration with Intelliz Press. Originally Published in printed book format by Intelliz Press with ISBN 978-1-68251-966-0

TABLE OF CONTENTS

PREFACE

Biochemistry, with its wide-ranging applications, has had a profound impact on various industries and aspects of society. While its significance is most evident in the medical and pharmaceutical sectors, biochemistry permeates everyday life, influencing fields such as retail, food, cosmetics, fashion, and healthcare. One of the notable contributions of biochemistry lies in the development of medical products. Through biochemical research, scientists have discovered and improved upon various medications that have revolutionized healthcare. Furthermore, biochemistry has played a crucial role in the advancement of DNA recombinant technology, which enables the production of essential molecules like insulin and food additives. These breakthroughs have enhanced the treatment of diseases and improved the overall quality of life for countless individuals.

Beyond medicine, biochemistry has also had a positive impact on food production and safety. Agrochemicals developed through biochemistry have enhanced crop yield and protected plants from pests and diseases, leading to increased food production. Additionally, biochemistry has facilitated the creation of foods that promote human health, such as pre- and probiotics, which support gut health, and antioxidants, which help combat oxidative stress. These advancements in food science have contributed to a healthier and more sustainable food supply.

This book provides a comprehensive overview of the storage and flow of genetic information, serving as a valuable resource for students and researchers in the field. It emphasizes the integration of cutting-edge research into biochemistry and molecular biology education, highlighting the importance of a supportive academic environment with access to advanced facilities and mentoring.

The applications of molecular biology methods extend beyond fundamental scientific inquiries. These methods are instrumental in disease prevention and treatment, enabling the development of innovative therapies and diagnostic tools. They also facilitate the generation of novel protein products with various applications, from industrial processes to biotechnology. Moreover, molecular biology methods are employed in the manipulation of plants and animals to achieve desired phenotypic traits, enhancing agricultural practices and animal husbandry.

CHAPTER 1

INTRODUCTION TO BIOCHEMISTRY AND MOLECULAR BIOLOGY

INTRODUCTION

Biochemistry and molecular biology are closely related fields that study the chemical processes and molecular mechanisms that occur within living organisms. They provide a foundation for understanding the structure, function, and regulation of biological macromolecules such as proteins, nucleic acids, carbohydrates, and lipids. Let's explore each field individually:

Biochemistry focuses on the chemical processes and transformations that take place within living organisms. It investigates the structure and function of biomolecules and their interactions within cells. Key areas of study in biochemistry include:

- Proteins: Biochemists examine the structure, function, and properties of proteins, which play crucial roles in cell signaling, enzymatic reactions, and structural support.
- Enzymes: Enzymes are proteins that catalyze biochemical reactions, and their study involves understanding their mechanisms, kinetics, and regulation.
- Metabolism: Biochemistry investigates metabolic pathways, which involve the conversion of food into energy and the synthesis or breakdown of molecules in cells.

- Nucleic Acids: This includes studying DNA and RNA, their structures, replication, transcription, translation, and the regulation of gene expression.
- Carbohydrates and Lipids: Biochemists examine the structure, function, and metabolism of carbohydrates and lipids, which are essential for energy storage, cell signaling, and structural integrity.
- Molecular Biology: Molecular biology focuses on the study of biological molecules and their interactions within cells to understand cellular processes and the flow of genetic information. Key areas of study in molecular biology include:
- DNA Replication and Repair: Molecular biology investigates the mechanisms of DNA replication, ensuring the faithful duplication of genetic material, as well as the repair of damaged DNA.
- Gene Expression: This involves the processes of transcription and translation, where the information encoded in DNA is transcribed into RNA and then translated into proteins.
- Regulation of Gene Expression: Molecular biologists study the mechanisms that control gene expression, including transcription factors, epigenetic modifications, and non-coding RNAs.
- Recombinant DNA Technology: Molecular biology utilizes techniques such as genetic engineering to manipulate DNA, creating recombinant molecules for various applications, including gene therapy and biotechnology.
- Molecular Genetics: Molecular biology investigates the structure, function, and regulation of genes, including the study of mutations and their effects on phenotypes.

Biochemistry and molecular biology are interdisciplinary fields that draw from various branches of science, including chemistry, biology, genetics, and biophysics. They are fundamental to understanding the complexities of life at the molecular level and have significant applications in medicine, agriculture, biotechnology, and pharmacology, among other areas.

1.1 BIOCHEMICAL AND MOLECULAR BIOLOGY

Biochemistry and molecular biology are closely related fields that focus on the study of the chemical processes and molecular mechanisms occurring within living organisms. These disciplines provide a foundation for understanding the structure, function, and regulation of biological macromolecules such as proteins, nucleic acids, carbohydrates, and lipids.

Biochemistry investigates the chemical reactions and transformations that take place within living organisms. It delves into the structure and function of biomolecules and their interactions within cells. One crucial area of study in biochemistry is proteins. Biochemists examine the structure, function, and properties of proteins, which play essential roles in cell signaling, enzymatic reactions, and structural support. They investigate the folding and three-dimensional structure of proteins, as well as their involvement in various biological processes.

Enzymes, a type of protein, are of particular interest in biochemistry. Enzymes act as catalysts, speeding up biochemical reactions. Biochemists study enzyme kinetics, mechanisms of action, and regulation. Understanding enzymes is vital for comprehending the metabolic pathways that occur within cells. Metabolism, another key area of biochemistry, involves the conversion of food into energy and the synthesis or breakdown of molecules in cells. Biochemists explore the intricate networks of metabolic pathways and the regulation of these processes.

The study of nucleic acids, including DNA and RNA, is also a significant aspect of biochemistry. Biochemists investigate the structure, replication, transcription, and translation of nucleic acids. They unravel the processes by which genetic information is stored, copied, and utilized within cells. Additionally, biochemistry examines carbohydrates and lipids, which are essential molecules for energy storage, cell signaling, and structural integrity in organisms.

Molecular biology, on the other hand, focuses on the study of biological molecules and their interactions within cells to understand cellular processes and the flow of genetic information. It encompasses several key areas of investigation. DNA replication and repair are fundamental processes studied in molecular biology. Scientists examine the mechanisms of DNA replication, ensuring the faithful duplication of genetic material, as well as the repair mechanisms that safeguard the integrity of DNA.

Gene expression is another critical area within molecular biology. It involves the processes of transcription and translation, where the genetic information encoded in DNA is transcribed into RNA and then translated into proteins. Molecular biologists investigate the regulation of gene expression, including the roles of transcription factors, epigenetic modifications, and non-coding RNAs in controlling gene activity.

Recombinant DNA technology is an essential technique employed in molecular biology. It allows scientists to manipulate DNA, creating recombinant molecules for various applications. This technology has contributed to advancements in gene therapy, biotechnology, and genetic engineering.

Molecular genetics is an integral part of molecular biology, focusing on the structure, function, and regulation of genes. Scientists study the inheritance of traits, the effects of mutations, and the relationship between genotype and phenotype.

Biochemistry and molecular biology are interdisciplinary fields that draw from various branches of science, including chemistry, biology, genetics, and biophysics. Together, they provide a comprehensive understanding of the complexities of life at the molecular level. These fields have significant applications in medicine, agriculture, biotechnology, and pharmacology. They contribute to advancements in drug development, genetic research, personalized medicine, and the improvement of agricultural practices. Overall, biochemistry and molecular biology are crucial for unraveling the mysteries of life and advancing our knowledge of living organisms.

1.1.1 Protein Structure and Function

Proteins are essential macromolecules that perform a wide array of biological functions in living organisms. Understanding protein structure and function is crucial for unraveling their roles in cellular processes, disease mechanisms, and drug discovery. Proteins are composed of long chains of amino acids, linked together by peptide bonds. The sequence of amino acids determines the primary structure of a protein.

Protein structure can be described at several levels. The primary structure refers to the linear sequence of amino acids. The secondary structure involves local folding patterns, such as alpha-helices and beta-sheets, stabilized by hydrogen bonding between backbone atoms. Tertiary structure describes the overall three-dimensional arrangement of the protein, determined by various forces, including hydrophobic interactions, electrostatic interactions, and disulfide bonds. Finally, some proteins have a quaternary structure, which involves the association of multiple polypeptide chains. The structure of a protein is intricately linked to its function. The specific arrangement of amino acids allows proteins to adopt unique shapes, which are essential for carrying out their biological roles. Proteins can have enzymatic functions, acting as catalysts for biochemical reactions. Enzymes bind to specific substrates and facilitate the conversion of reactants into products. The active site of an enzyme, a region within its structure, plays a crucial role in substrate binding and catalysis.

Proteins are also involved in cell signaling and communication. Receptor proteins on the cell surface recognize and bind to signaling molecules, triggering intracellular signaling pathways. These pathways transmit signals within the cell, leading to specific cellular responses. For example, G-protein coupled receptors (GPCRs) are a large family of cell surface receptors that play key roles in various physiological processes, including sensory perception and hormone signaling.

Structural proteins provide support and stability to cells and tissues. For instance, collagen, a fibrous protein, forms a major component of connective tissues, providing strength and flexibility to skin, tendons, and bones. Actin and tubulin, proteins involved in the cytoskeleton, are crucial for maintaining cell shape and facilitating cell movement.

Proteins are also involved in transport and storage. Carrier proteins transport molecules across cell membranes, facilitating their entry or exit from cells.

Hemoglobin, an oxygen-binding protein found in red blood cells, transports oxygen from the lungs to tissues. Storage proteins, such as ferritin, store and release essential ions, like iron, as needed by the body.

Furthermore, proteins have immune functions. Antibodies, also known as immunoglobulins, are proteins produced by immune cells that recognize and neutralize foreign molecules, such as bacteria or viruses. Antibodies are crucial components of the immune response, protecting the body against pathogens.

Protein function can be modulated by various factors, including post-translational modifications. These modifications, such as phosphorylation, acetylation, or glycosylation, can alter protein activity, stability, and interactions with other molecules. Protein-protein interactions are vital for many cellular processes. Proteins can bind to other proteins, forming complexes that perform specific functions, such as DNA replication, cell signaling, or gene regulation.

Studying protein structure and function relies on various experimental techniques, including X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and cryo-electron microscopy (cryo-EM). These techniques allow researchers to determine protein structures at atomic or near-atomic resolution, providing insights into their functional mechanisms.

1.1.2 Metabolism and Metabolic Pathways

Metabolism is the set of chemical reactions that occur within an organism to sustain life. It involves the processes of converting food into energy, synthesizing complex molecules, and breaking down molecules to obtain building blocks for cellular processes. Metabolism is essential for maintaining cellular homeostasis and providing the energy required for various biological activities.

Metabolic pathways are a series of interconnected chemical reactions that occur within cells to transform molecules. These pathways are highly organized and tightly regulated to ensure proper coordination of metabolic processes. They involve a variety of enzymes, cofactors, and intermediates that facilitate the conversion of one molecule into another.

One of the most well-known metabolic pathways is glycolysis, which occurs in the cytoplasm of cells and involves the breakdown of glucose into pyruvate. Glycolysis produces a small amount of ATP (adenosine triphosphate), the primary energy currency of cells, and generates intermediates that feed into other metabolic pathways.

Another essential metabolic pathway is the tricarboxylic acid (TCA) cycle, also known as the Krebs cycle or citric acid cycle. The TCA cycle takes place within the mitochondria and oxidizes acetyl-CoA derived from carbohydrates, fats, and proteins. Through a series of reactions, the TCA cycle generates high-energy electron carriers, including NADH (nicotinamide adenine dinucleotide) and FADH2 (flavin adenine dinucleotide), which play crucial roles in oxidative phosphorylation.

Oxidative phosphorylation is the process by which ATP is generated in the mitochondria. It involves the transfer of electrons from NADH and FADH2 to molecular oxygen, coupled with the pumping of protons across the mitochondrial membrane. The resulting proton gradient drives ATP synthesis through a complex protein structure called ATP synthase. Lipid metabolism is another important aspect of metabolism. It involves the synthesis, breakdown, and modification of lipids, which serve as a major energy source and structural components of cell membranes. Fatty acid synthesis occurs in the cytoplasm and involves the stepwise addition of acetyl-CoA units to form long-chain fatty acids. On the other hand, fatty acid oxidation takes place in the mitochondria and involves the breakdown of fatty acids to generate acetyl-CoA and produce ATP.

Amino acid metabolism plays a crucial role in protein synthesis, as well as the synthesis and degradation of other important molecules. Amino acids can be used for energy production, converted into glucose or fatty acids, or incorporated into proteins. The process of deamination removes the amino group from amino acids, producing ammonia, which is further processed into urea for excretion.

Metabolic pathways are highly regulated at multiple levels to maintain metabolic balance and respond to the changing needs of the organism. Regulatory mechanisms include feedback inhibition, allosteric regulation, hormonal control, and gene expression regulation. These mechanisms ensure that the products of metabolic pathways are produced in appropriate quantities and at the right times.

Understanding metabolism and metabolic pathways is crucial for various fields, including medicine, nutrition, and bioengineering. Dysregulation of metabolic pathways can lead to metabolic disorders, such as diabetes, obesity, and inborn errors of metabolism. Targeting specific metabolic pathways is also a promising approach for the development of therapeutic interventions, particularly in the context of cancer metabolism

1.1.3 Nucleic Acids and Gene Expression

Nucleic acids and gene expression are integral components of biochemistry and molecular biology, playing crucial roles in storing and transmitting genetic information and controlling the functioning of cells. Nucleic acids, specifically DNA and RNA, are the molecules responsible for encoding and expressing the genetic information that determines an organism's characteristics.

DNA (deoxyribonucleic acid) is a double-stranded helical molecule found in the nucleus of cells. It consists of a sequence of nucleotides, each containing a sugar (deoxyribose), a phosphate group, and one of four nitrogenous bases: adenine (A), cytosine (C), guanine (G), and thymine (T). The specific sequence of bases within DNA carries the genetic code.

Gene expression is the process by which the information encoded in DNA is converted into functional gene products, primarily through the synthesis of proteins. It involves two main steps: transcription and translation. Transcription occurs in the nucleus and involves the synthesis of a complementary RNA molecule, called messenger RNA (mRNA), from a DNA template. RNA polymerase enzymes catalyze this process by adding RNA nucleotides to the growing mRNA strand, following the base pairing rules (A-U, C-G).

After transcription, the mRNA molecule undergoes several modifications, such as the removal of introns (non-coding regions) and the addition of a modified guanine nucleotide, forming a 5′ cap, and a string of adenine nucleotides, forming a poly-A tail. These modifications protect the mRNA molecule and facilitate its transport to the cytoplasm.

Translation, the second step of gene expression, takes place in the cytoplasm and involves the synthesis of proteins using the information encoded in mRNA. Ribosomes, composed of ribosomal RNA (rRNA) and proteins, bind to the mRNA molecule and facilitate the decoding of the mRNA sequence into a specific sequence of amino acids, the building blocks of proteins. Transfer RNA (tRNA) molecules carry the appropriate amino acids and match them to the codons (three-nucleotide sequences) on the mRNA, according to the genetic code.

The process of translation continues until a stop codon is reached, at which point the newly synthesized protein is released. Post-translational modifications, such as folding, cleavage, and addition of chemical groups, may further modify the protein to achieve its functional form. The final protein product carries out various cellular functions, such as enzymatic activity, structural support, signaling, and regulation of gene expression.

Gene expression is highly regulated to ensure that the right genes are expressed in the appropriate cells and at the correct times. Transcriptional regulation involves the interaction of specific proteins, called transcription factors, with regulatory regions of DNA, such as promoters and enhancers.

Transcription factors can activate or repress gene expression by influencing the binding of RNA polymerase to DNA or by modulating chromatin structure.

Epigenetic modifications, such as DNA methylation and histone modifications, also play a role in gene expression regulation. These modifications can alter the accessibility of DNA to transcriptional machinery, affecting gene expression patterns and cell differentiation.

Understanding nucleic acids and gene expression is crucial for various fields of research, including genetics, molecular biology, developmental biology, and medicine. Dysregulation of gene expression can lead to diseases, such as cancer, where genes involved in cell growth and division are aberrantly expressed.

Technological advancements, such as DNA sequencing and gene editing techniques like CRISPR-Cas9, have revolutionized the study of nucleic acids and gene expression. These tools enable researchers to investigate the roles of specific genes and their regulatory elements, paving the way for advancements in personalized medicine, genetic engineering, and gene therapy.

1.1.4 Cellular Signaling and Communication

Cellular signaling and communication are vital processes that allow cells to coordinate their activities, respond to their environment, and maintain proper functioning.

Cells utilize complex signaling pathways to transmit information through various signaling molecules, receptors, and intracellular signaling cascades. This communication is essential for development, homeostasis, immune response, and numerous physiological processes.

Signaling pathways can be classified into several types based on the signaling molecules involved:

1. Endocrine Signaling: In this form of signaling, hormones are produced by specialized cells and released into the bloodstream to act on distant target cells. Examples include insulin, estrogen, and adrenaline.
2. Paracrine Signaling: Signaling molecules are released by a cell and act on nearby cells. This type of signaling is crucial for local communication and coordination. Growth factors, such as epidermal growth factor (EGF), are examples of paracrine signaling molecules.
3. Autocrine Signaling: In autocrine signaling, cells release signaling molecules that act on the same cell or nearby cells of the same type. This form of signaling plays a role in cell proliferation and self-regulation.
4. Synaptic Signaling: Nerve cells, or neurons, communicate through synaptic signaling. Neurotransmitters are released from the presynaptic cell and bind to receptors on the postsynaptic cell, transmitting electrical and chemical signals across synapses.

Cellular signaling involves several key components and processes:

1. Signaling Molecules: These can be small molecules (e.g., neurotransmitters), proteins (e.g., growth factors), or gases (e.g., nitric oxide) that serve as messengers to transmit signals between cells.

2. Receptors: Receptors are proteins located on the cell surface or within the cell that bind specific signaling molecules. The binding of a signaling molecule to its receptor initiates a series of intracellular events.
3. Signal Transduction: Signal transduction refers to the process by which extracellular signals are converted into intracellular responses. It involves a cascade of events, including receptor activation, intracellular signaling molecule recruitment, and amplification of the signal.
4. Intracellular Signaling Cascades: These are complex networks of proteins that transmit and amplify signals from the cell surface to specific targets within the cell. Common signaling pathways include the MAPK (mitogen-activated protein kinase) pathway and the PI3K (phosphoinositide 3-kinase)/ Akt pathway.
5. Second Messengers: Second messengers, such as cyclic AMP (cAMP) and calcium ions (Ca2+), play crucial roles in signal transduction by relaying and amplifying signals from the cell membrane to downstream signaling components.
6. Transcription Factors: Signaling pathways often activate or inhibit specific transcription factors, which are proteins that regulate gene expression. Transcription factors can influence the synthesis of new proteins, thereby altering cellular responses and functions.
7. Feedback Regulation: Signaling pathways are regulated by feedback mechanisms that can modulate the intensity and duration of the signal. Feedback mechanisms include positive feedback loops that amplify the signal and negative feedback loops that dampen or terminate the signal.

Disruptions in cellular signaling can lead to various diseases, including cancer, autoimmune disorders, and metabolic disorders. Understanding the mechanisms of cellular signaling allows researchers to develop targeted therapies that modulate specific signaling pathways to restore proper cell function.

Advancements in techniques such as live-cell imaging, proteomics, and computational modeling have greatly enhanced our understanding of cellular signaling and communication. These tools enable the visualization and quantification of signaling events, identification of signaling components, and modeling of signaling networks.

1.1.5 Genetic Engineering and Recombinant DNA Technology

Genetic engineering and recombinant DNA technology are powerful tools that have revolutionized the field of molecular biology. They involve the manipulation and modification of DNA to create new genetic combinations, introduce specific genes into organisms, and produce valuable products for various applications. Recombinant DNA technology involves the combination of DNA molecules from different sources, creating a novel DNA sequence that can be replicated and expressed in host organisms. The key steps involved in this process include:

1. Isolation of DNA: DNA containing the gene of interest is extracted from a donor organism or synthesized using laboratory techniques.
2. DNA Fragmentation: The isolated DNA is then fragmented using restriction enzymes, which recognize specific DNA sequences and cut the DNA at those sites.
3. DNA Ligation: The desired DNA fragments, including the gene of interest, are joined together using DNA ligase, an enzyme that catalyzes the formation of phosphodiester bonds between DNA fragments.
4. Introduction into Host Organisms: The recombinant DNA construct is introduced into host organisms, such as bacteria, yeast, or mammalian cells, using various methods like transformation, transfection, or microinjection.
5. Selection and Expression: Host cells containing the desired recombinant DNA are selected using selectable markers, such as antibiotic resistance genes. The recombinant DNA is then expressed in the host cells, allowing the production of the desired protein or the alteration of a specific cellular function.

Genetic engineering and recombinant DNA technology have numerous applications across different fields:

1. Biomedical Research: Genetic engineering enables the study of gene function, protein expression, and disease mechanisms. It allows the creation of transgenic animal models, knockout or knock-in animals, and the production of recombinant proteins for research purposes.
2. Agriculture: Genetically modified (GM) crops have been developed with traits such as increased yield, improved

nutritional content, pest and disease resistance, and tolerance to environmental stress. Genetic engineering is used to introduce desirable traits into plants, enhancing crop productivity and sustainability.

3. Pharmaceutical Industry: Recombinant DNA technology is widely used in the production of therapeutic proteins, such as insulin, growth hormones, and monoclonal antibodies. These proteins can be produced in large quantities using genetically engineered host cells, ensuring a consistent and reliable supply for medical treatments.
4. Environmental Applications: Genetic engineering plays a role in environmental remediation by developing organisms capable of degrading pollutants or cleaning up contaminated environments. It also contributes to the production of biofuels through the modification of microorganisms for enhanced energy production.
5. Forensic Science: DNA profiling techniques, including polymerase chain reaction (PCR) amplification and DNA sequencing, are employed in forensic analysis for identifying individuals, determining paternity, and solving criminal cases.

Ethical considerations surrounding genetic engineering include potential risks to the environment and human health, as well as the social and economic impact of genetically modified organisms (GMOs). Regulations and strict safety protocols are in place to ensure responsible use and minimize potential harm.

Genetic engineering and recombinant DNA technology have transformed various fields by providing the means to manipulate and harness the power of DNA. These tools continue to advance our understanding of genetics, improve agricultural practices, develop novel therapeutics, and address pressing societal and environmental challenges.

1.1.6 Bioinformatics and Computational Biology

Bioinformatics and computational biology are interdisciplinary fields that involve the application of computational and statistical methods to analyze and interpret biological data. They play a crucial role in modern biological research, enabling scientists to extract valuable insights from large-scale genomic, proteomic, and other omics datasets. Bioinformatics involves the development and application of computational tools, algorithms, and databases for storing, retrieving, and analyzing biological information. It encompasses various areas, including sequence analysis, structural bioinformatics, comparative genomics, functional genomics, and systems biology.

Sequence analysis is a fundamental aspect of bioinformatics, focusing on the analysis of DNA, RNA, and protein sequences. This includes tasks such as sequence alignment, identifying patterns or motifs, predicting protein structure and function, and studying sequence evolution.

Structural bioinformatics involves the prediction and analysis of protein structures using computational techniques. This field aids in understanding protein folding, interactions, and drug design. It includes methods like homology modeling, protein structure prediction, and molecular docking.

Comparative genomics involves the comparison of genomic sequences across different species to identify conserved regions, evolutionary relationships, and functional elements. It helps in studying genome evolution, gene annotation, and identifying disease-related genetic variations.

Functional genomics investigates the functions and interactions of genes and their products on a large scale. It includes transcriptomics (study of gene expression), proteomics (study of proteins), and metabolomics (study of metabolites). Computational methods are used for data analysis, identifying gene networks, and inferring biological functions.

Systems biology integrates computational modeling and experimental data to understand biological systems as a whole. It aims to study the interactions and dynamics of biological components to gain insights into complex biological processes and behaviors.

Computational biology encompasses a broader range of computational approaches applied to various biological problems. It includes mathematical modeling, statistical analysis, data mining, machine learning, and network analysis. These techniques help in deciphering biological networks, predicting gene function, understanding disease mechanisms, and facilitating personalized medicine.

Bioinformatics and computational biology have had a significant impact on many areas of research and applications:

1. Genomics: These fields have revolutionized genomics research by enabling the analysis of large-scale genome sequences, identifying genetic variations, understanding gene regulation, and studying genomic evolution.

2. Drug Discovery and Development: Computational methods are extensively used in drug discovery, including virtual screening, molecular dynamics simulations, and structure-based drug design. These techniques facilitate the identification and optimization of potential drug candidates.
3. Personalized Medicine: Bioinformatics and computational biology play a crucial role in personalized medicine by analyzing individual genomic and clinical data to predict disease risk, select appropriate treatments, and develop targeted therapies.
4. Agriculture and Crop Improvement: Computational approaches assist in improving crop yield, disease resistance, and nutritional quality through genomic analysis, molecular breeding, and marker-assisted selection.
5. Infectious Disease Research: Bioinformatics helps in tracking disease outbreaks, studying pathogen genomics, understanding drug resistance, and developing diagnostic tools for infectious diseases.

The field of bioinformatics and computational biology continues to evolve rapidly with advances in high-throughput technologies, data integration, and machine learning. The interdisciplinary nature of these fields promotes collaborations between biologists, statisticians, computer scientists, and mathematicians to tackle complex biological questions and drive innovation in biological research and applications.

1.1.7 Cellular and Molecular Immunology

Cellular and molecular immunology is the branch of immunology that focuses on understanding the cellular and molecular mechanisms underlying the immune response. It explores how the immune system recognizes and responds to foreign substances, such as pathogens, and how it maintains self-tolerance to prevent autoimmune reactions. The immune system is a complex network of cells, tissues, and molecules that work together to defend the body against infections and diseases. Cellular and molecular immunology delves into the following key aspects:

1. Cells of the Immune System: The immune system comprises various types of cells, including white blood cells such as lymphocytes (B cells and T cells), macrophages, dendritic cells, and natural killer (NK) cells. Each cell type has specific functions and plays a crucial role in the immune response.

2. Antigen Recognition: The immune system is capable of recognizing foreign substances, called antigens, through antigen receptors. B cell receptors (BCRs) on B cells and T cell receptors (TCRs) on T cells recognize specific antigens. Antigen presentation by antigen-presenting cells (APCs), such as macrophages and dendritic cells, is necessary for effective immune activation.
3. Immune Response Activation: Once antigens are recognized, a series of cellular and molecular events occur to initiate an immune response. This involves the activation, proliferation, and differentiation of immune cells. Cytokines, small signaling molecules secreted by immune cells, play a vital role in coordinating immune responses.
4. Antibody Production and Humoral Immunity: B cells play a central role in humoral immunity. When activated, B cells differentiate into plasma cells, which produce and secrete antibodies specific to the antigen encountered. Antibodies bind to antigens, neutralize them, and facilitate their elimination by other immune cells.
5. Cell-Mediated Immunity: T cells are primarily responsible for cell-mediated immunity. They recognize antigens presented by APCs and play critical roles in killing infected cells, activating other immune cells, and regulating immune responses. T cells can be further classified into helper T cells (CD4+) and cytotoxic T cells (CD8+).
6. Major Histocompatibility Complex (MHC): MHC molecules are key components of the immune system that present antigens to T cells. MHC class I molecules present antigens to CD8+ T cells, while MHC class II molecules present antigens to CD4+ T cells. MHC diversity contributes to the immune system's ability to recognize a wide range of antigens.
7. Immune Regulation and Tolerance: The immune system must strike a balance between robust responses to pathogens and tolerance to self-antigens. Regulatory T cells (Tregs) play a crucial role in immune regulation and prevent autoimmune reactions by suppressing excessive immune responses.
8. Immunological Memory: Following an initial encounter with an antigen, the immune system develops immunological memory. This memory allows for a rapid and enhanced response upon re-exposure to the same antigen, resulting in a more efficient clearance of the pathogen.

Understanding cellular and molecular immunology is essential for advancing knowledge of immune-related disorders, developing vaccines, and designing immunotherapies. It provides insights into how dysregulation of the immune system can lead to autoimmune diseases, allergies, immunodeficiencies, and cancer. Recent advancements in cellular and molecular techniques, such as flow cytometry, gene expression analysis, and single-cell sequencing, have revolutionized the study of immunology. These tools enable researchers to characterize immune cell populations, identify novel immune receptors, and dissect intricate immune responses.

1.2 UNIT OF MEASUREMENT IN BIOCHEMICAL AND MOLECULAR BIOLOGY

Biochemical and molecular biology research utilizes various units of measurement to quantify and describe biological phenomena. Here are some commonly used units in these fields:

1. **Mass and Weight:**
   - Gram (g): It is the base unit of mass in the International System of Units (SI). It is commonly used to measure the mass of biological samples or reagents.
2. **Volume:-** Liter (L): It is the base unit of volume in the SI system and is used to measure the volume of solutions, reagents, or biological samples.
   - Microliter (μL): It is a commonly used unit for small volumes, such as the volume of liquid added to experiments or the volume of a cell culture.
3. **Concentration:-**
   - Molar (M): It represents the number of moles of a solute per liter of solution. It is commonly used to express the concentration of a chemical substance in a solution.
   - Millimolar (mM): It is equal to one thousandth of a mole per liter and is used for lower concentrations.
   - Micro molar (μM): It is equal to one millionth of a mole per liter and is used for even lower concentrations.
4. **Time:-** Seconds (s): It is the base unit of time and is commonly used to measure reaction times, incubation periods, or reaction rates.

- Minutes (min) and hours (h): These units are used for longer time intervals, such as incubation times or experimental durations.

5. Temperature:-

- Celsius (°C): It is the commonly used unit for temperature in biochemical and molecular biology experiments. It is used to measure the temperature of incubators, water baths, or sample storage conditions.
- Kelvin (K): It is an absolute temperature scale and is also used in scientific research, particularly in thermodynamic calculations.

6. Enzyme Activity:

- International Unit (IU): It is a unit of measurement for enzyme activity and is defined as the amount of enzyme that catalyzes a specific reaction under defined conditions.
- Enzyme Unit (U): It is another unit of measurement for enzyme activity, representing the amount of enzyme that catalyzes the conversion of one micromole of substrate per minute.

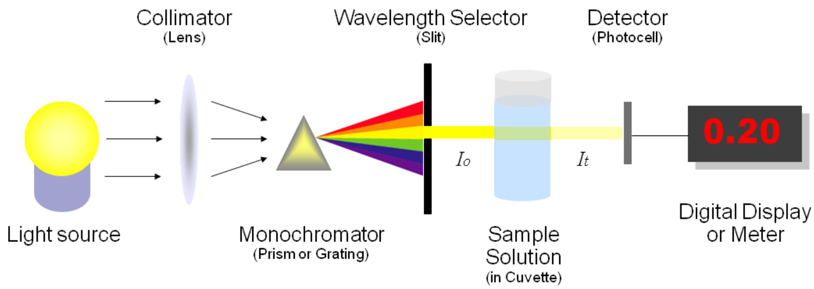
These are just a few examples of the units commonly used in biochemical and molecular biology research. It's important to note that different laboratories or publications may use alternative units or conversion factors depending on their specific requirements or conventions.

1.2.1 Spectroscopy Units

Spectroscopy is a powerful technique used in biochemical and molecular biology research to study the interaction of light with biological molecules. It provides valuable information about the structural, chemical, and functional properties of molecules. Spectroscopic measurements are often expressed in specific units that quantify the observed phenomena. Here, we will discuss spectroscopy units commonly used in these fields.

Absorbance is a fundamental unit in spectroscopy and is typically measured using spectrophotometry.

It represents the amount of light absorbed by a substance at a particular wavelength. Absorbance is dimensionless and is denoted by the symbol "A." It is directly proportional to the concentration of the absorbing species and the path length through which the light passes. The Beer-Lambert law, $A = \varepsilon lc$, relates absorbance (A) to the molar absorptivity (ε), the concentration (c) of the absorbing species, and the path length (l) of the light through the sample. Absorbance values can range from 0 to infinity, with higher values indicating greater absorption.



Fluorescence is another important spectroscopic phenomenon used in biochemical and molecular biology research. It involves the absorption of light by a fluorophore and subsequent emission of light at a longer wavelength. Fluorescence intensity is often expressed in units such as arbitrary fluorescence units (AFU) or relative fluorescence units (RFU). These units provide a relative measure of the fluorescence emission from a fluorophore, taking into account factors such as instrument sensitivity, detection settings, and fluorophore concentration.

In addition to absorbance and fluorescence, other spectroscopy units are used to quantify specific measurements. For instance, in circular dichroism (CD) spectroscopy, which assesses the differential absorption of left- and right-circularly polarized light, the measured quantity is often reported as the ellipticity (θ) in degrees. It represents the extent and direction of the molecular secondary structure, such as alpha-helices or beta-sheets, in proteins or nucleic acids.

Raman spectroscopy measures the inelastic scattering of light, providing information about molecular vibrations and chemical bonds. The Raman shift, expressed in units of wavenumber (cm^{-1}), represents the difference between the incident and scattered light wavelengths. It is used to identify specific chemical groups and characterize molecular structures.

In nuclear magnetic resonance (NMR) spectroscopy, which examines the interaction of atomic nuclei with a strong magnetic field, chemical shifts are employed to describe the resonance frequencies of nuclei relative to a standard compound. Chemical shifts are expressed in parts per million (ppm) and provide information about the chemical environment and molecular structure.

It is important to note that specific spectroscopy units can vary depending on the technique, instrument, and scientific community. Units such as absorbance, fluorescence intensity, ellipticity, Raman shift, and chemical shifts are widely used and standardized within their respective fields. Understanding and correctly utilizing these spectroscopy units is crucial for accurate interpretation and comparison of spectroscopic data in biochemical and molecular biology research.

Spectroscopy plays a vital role in studying biomolecules, elucidating structural information, monitoring biochemical reactions, and understanding molecular interactions. The use of appropriate spectroscopy units ensures consistency and facilitates effective communication of research findings, enabling scientists to make meaningful conclusions and advancements in their fields.

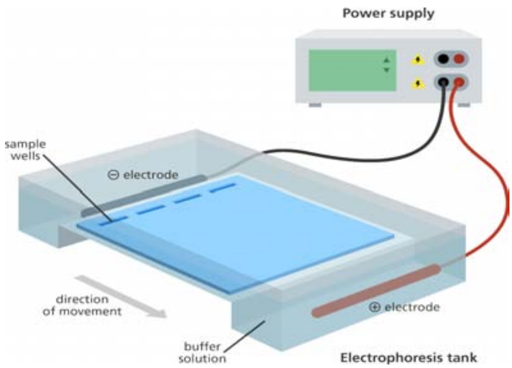
1.2.2 Gel Electrophoresis

Gel electrophoresis is a method for separation and analysis of biomacromolecules (DNA, RNA, proteins, etc.) and their fragments, based on their size and charge. It is used in clinical chemistry to separate proteins by charge or size (IEF agarose, essentially size independent) and in biochemistry and molecular biology to separate a mixed population of DNA and RNA fragments by length, to estimate the size of DNA and RNA fragments or to separate proteins by charge.

The principle of gel electrophoresis relies on the migration of charged molecules in an electric field. The gel matrix, typically made of agarose or polyacrylamide, provides a porous network through which the molecules can move. The gel is cast into a thin, rectangular slab or in the form of a cylindrical tube called a gel or polyacrylamide gel electrophoresis (PAGE).

In nucleic acid gel electrophoresis, the DNA or RNA samples are mixed with a loading buffer containing tracking dyes and a denaturing agent (in the case of DNA) to denature the molecules into single strands. The samples are loaded into wells made in the gel, and an electric field is applied across the gel. The negatively charged nucleic acid molecules migrate towards the positive electrode (anode) through the gel matrix. Smaller molecules move faster and migrate farther through the gel than larger molecules.

Nucleic acid molecules are separated by applying an electric field to move the negatively charged molecules through a matrix of agarose or other substances. Shorter molecules move faster and migrate farther than longer ones because shorter molecules migrate more easily through the pores of the gel. This phenomenon is called sieving. Proteins are separated by the charge in agarose because the pores of the gel are too small to sieve proteins. Gel electrophoresis can also be used for the separation of nanoparticles.



Gel electrophoresis uses a gel as an anticonvective medium or sieving medium during electrophoresis, the movement of a charged particle in an electrical current. Gels suppress the thermal convection caused by the application of the electric field, and can also act as a sieving medium, slowing the passage of molecules; gels can also simply serve to maintain the finished separation so that a post electrophoresis stain can be applied. DNA gel electrophoresis is usually performed for analytical purposes, often after amplification of DNA via polymerase chain reaction (PCR), but may be used as a preparative technique prior to use of other methods such as mass spectrometry, RFLP, PCR, cloning, DNA sequencing, or Southern blotting for further characterization.

Types of gel

The types of gel most typically used are agarose and polyacrylamide gels. Each type of gel is well-suited to different types and sizes of the analyte. Polyacrylamide gels are usually used for proteins and have very high resolving power for small fragments of DNA (5-500 bp). Agarose gels, on the other hand, have lower resolving power for DNA but have a greater range of separation, and are therefore used for DNA fragments of usually 50–20,000 bp in size, but the resolution of over 6 Mb is possible with pulsed field gel electrophoresis (PFGE).[9] Polyacrylamide gels are run in a vertical configuration while agarose gels are typically run horizontally in a submarine mode. They also differ in their casting methodology, as agarose sets thermally, while polyacrylamide forms in a chemical polymerization reaction.

Agarose

Agarose gels are made from the natural polysaccharide polymers extracted from seaweed. Agarose gels are easily cast and handled compared to other matrices because the gel setting is a physical rather than chemical change. Samples are also easily recovered. After the experiment is finished, the resulting gel can be stored in a plastic bag in a refrigerator.

Agarose gels do not have a uniform pore size, but are optimal for electrophoresis of proteins that are larger than 200 kDa. Agarose gel electrophoresis can also be used for the separation of DNA fragments ranging from 50 base pair to several megabases (millions of bases), the largest of which require specialized apparatus. The distance between DNA bands of different lengths is influenced by the percent agarose in the gel, with higher percentages requiring longer run times, sometimes days. Instead high percentage agarose gels should be run with a pulsed field electrophoresis (PFE), or field inversion electrophoresis.

"Most agarose gels are made with between 0.7% (good separation or resolution of large 5–10kb DNA fragments) and 2% (good resolution for small 0.2–1kb fragments) agarose dissolved in electrophoresis buffer. Up to 3% can be used for separating very tiny fragments but a vertical polyacrylamide gel is more appropriate in this case. Low percentage gels are very weak and may break when you try to lift them. High percentage gels are often brittle and do not set evenly. 1% gels are common for many applications."

Polyacrylamide

Polyacrylamide gel electrophoresis (PAGE) is used for separating proteins ranging in size from 5 to 2,000 kDa due to the uniform pore size provided by the polyacrylamide gel. Pore size is controlled by modulating the concentrations of acrylamide and bis-acrylamide powder used in creating a gel. Care must be used when creating this type of gel, as acrylamide is a potent neurotoxin in its liquid and powdered forms.

Traditional DNA sequencing techniques such as Maxam-Gilbert or Sanger methods used polyacrylamide gels to separate DNA fragments differing by a single base-pair in length so the sequence could be read. Most modern DNA separation methods now use agarose gels, except for particularly small DNA fragments.

It is currently most often used in the field of immunology and protein analysis, often used to separate different proteins or isoforms of the same protein into separate bands. These can be transferred onto a nitrocellulose or PVDF membrane to be probed with antibodies and corresponding markers, such as in a western blot.

Typically resolving gels are made in 6%, 8%, 10%, 12% or 15%. Stacking gel (5%) is poured on top of the resolving gel and a gel comb (which forms the wells and defines the lanes where proteins, sample buffer, and ladders will be placed) is inserted. The percentage chosen depends on the size of the protein that one wishes to identify or probe in the sample. The smaller the known weight, the higher the percentage that should be used. Changes in the buffer system of the gel can help to further resolve proteins of very small sizes.

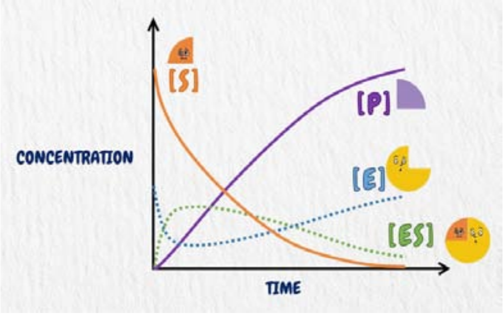
Starch

Partially hydrolysed potato starch makes for another non-toxic medium for protein electrophoresis. The gels are slightly more opaque than acrylamide or agarose. Non-denatured proteins can be separated according to charge and size. They are visualised using Napthal Black or Amido Black staining. Typical starch gel concentrations are 5% to 10%.

1.2.3 Enzyme Kinetics

Enzyme kinetics is a branch of biochemistry that studies the rates of enzyme-catalyzed reactions and the factors that influence those rates. Enzymes are biological catalysts that enhance the rate of chemical reactions in living organisms by lowering the activation energy required for the reactions to occur. Understanding enzyme kinetics is crucial for unraveling the mechanisms of enzyme action and for designing and optimizing enzymatic processes.

The study of enzyme kinetics involves the measurement of reaction rates under varying conditions. One of the key parameters in enzyme kinetics is the reaction velocity, which represents the rate at which the substrate is converted to the product. The reaction velocity is typically determined by measuring changes in concentration, absorbance, fluorescence, or other detectable properties of the reactants or products over time.



The Michaelis-Menten model is a widely used mathematical model that describes the relationship between substrate concentration ([S]) and reaction velocity (V) for many enzymatic reactions. According to this model, the reaction velocity initially increases with increasing substrate concentration until it reaches a maximum, known as Vmax, at high substrate concentrations. The Michaelis constant (Km) is the substrate concentration at which the reaction velocity is half of the maximum velocity. It represents the affinity of the enzyme for the substrate, with lower Km values indicating higher affinity.

Enzyme kinetics experiments often involve determining the initial rates of reactions at different substrate concentrations. This information is used to construct a Lineweaver-Burk plot, also known as a double reciprocal plot, which is a graphical representation of the Michaelis-Menten equation. The Lineweaver-Burk plot allows for the determination of Vmax and Km values by extrapolating the data to the y-intercept and x-intercept, respectively.

In addition to the Michaelis-Menten model, enzyme kinetics can be more complex in some cases. For enzymes that exhibit cooperative behavior or multiple substrate binding sites, alternative models such as the Hill equation or the Briggs-Haldane model may be used.

Enzyme kinetics experiments also provide insights into other important kinetic parameters. The turnover number (kcat) represents the number of substrate molecules converted to product per unit time when the enzyme is fully saturated with substrate. It reflects the catalytic efficiency of the enzyme. The specificity constant (kcat/Km) combines both the rate of catalysis and the affinity of the enzyme for the substrate and provides a measure of the overall efficiency of the enzyme-substrate interaction.

Enzyme kinetics studies help in understanding the mechanism of enzyme action, the effect of inhibitors or activators on enzyme activity, and the optimization of enzymatic reactions for various applications. By determining the kinetic parameters, researchers can gain insights into enzyme properties, design more efficient enzyme variants, and develop strategies to modulate enzyme activity for therapeutic or industrial purposes.

1.2.4 Protein Concentration

Protein concentration refers to the amount of protein present in a given sample or solution. It is an important parameter in biochemical and molecular biology research as it allows researchers to quantify and compare the abundance of proteins, assess protein purity, and determine protein concentrations for various experimental applications.

There are several methods commonly used to measure protein concentration:

1. Absorbance-based methods: The most common method is the absorbance measurement at a specific wavelength using a spectrophotometer. The absorbance is correlated with the concentration of protein in the sample. The most widely used method is the Bradford assay, where the absorbance of a dye (Coomassie Brilliant Blue G-250) in the presence of protein is measured. Other colorimetric assays, such as the Lowry assay and the bicinchoninic acid (BCA) assay, are also frequently employed.
2. Fluorescence-based methods: Fluorescence-based assays use protein-specific dyes or probes that bind to proteins and generate fluorescence signals. The fluorescence intensity is proportional to the protein concentration. Fluorescence-based assays are often more sensitive than absorbance-based methods and can be used for high-throughput protein quantification.
3. Bicinchoninic acid (BCA) assay: This assay is based on the reduction of Cu2+ ions to Cu+ ions by protein in an alkaline medium. The resulting Cu+ ions form a colored complex with bicinchoninic acid, and the absorbance of the complex is measured. The intensity of the color is proportional to the protein concentration.
4. Bradford assay: The Bradford assay uses the binding of Coomassie Brilliant Blue dye to proteins. The binding of the dye to proteins results in a shift in the absorbance spectrum, and the change in absorbance is used to determine protein concentration.
5. UV absorbance at 280 nm: Proteins containing aromatic amino acids (tryptophan, tyrosine, and phenylalanine) can be quantified based on their UV absorbance at 280 nm. This method requires knowledge of the protein's extinction coefficient, which is a measure of how strongly the protein absorbs light at that wavelength.

It is important to note that different protein assays may have different sensitivities, linearity ranges, and interferences. The choice of method depends on factors such as the protein type, sample matrix, desired sensitivity, and the availability of instrumentation.

Accurate protein quantification is crucial for various applications, including protein purification, enzymatic assays, protein-protein interaction studies, and protein expression analysis. Additionally, protein concentration measurements are essential for calculating enzyme activities, protein concentrations in cell lysates, and determining protein loading amounts for gel electrophoresis or other analytical techniques.

In summary, protein concentration measurement methods are diverse and range from colorimetric assays to fluorescence-based techniques. The selection of a specific method depends on the specific requirements of the experiment, the type of protein being analyzed, and the available resources. Accurate protein concentration determination is vital for a wide range of biochemical and molecular biology research applications.

1.2.5 Nucleic Acid Amplification

Nucleic acid amplification is a technique used to exponentially increase the amount of specific DNA or RNA molecules in a sample. It is a crucial tool in molecular biology and biotechnology research, as it allows for the detection, quantification, and analysis of nucleic acids, including genes, genetic mutations, and infectious agents. One of the most widely used nucleic acid amplification methods is the polymerase chain reaction (PCR).

PCR involves a cyclic series of temperature changes that enable the selective amplification of a specific region of DNA. The process requires a DNA template, a pair of specific primers that flank the target region, DNA polymerase enzyme, and nucleotides (dNTPs). The DNA template is denatured by heating to separate the double-stranded DNA into single strands. The temperature is then lowered to allow the primers to anneal to their complementary sequences on the template. The temperature is raised again, and the DNA polymerase enzyme extends the primers by adding new nucleotides, synthesizing a new DNA strand. This cycle of denaturation, annealing, and extension is repeated multiple times, resulting in the exponential amplification of the targeted DNA sequence.

PCR can be used for various applications, including DNA cloning, genetic testing, mutation analysis, forensic analysis, and pathogen detection. It allows researchers to obtain a large amount of specific DNA sequence from a small initial sample. The amplified DNA can be further analyzed using techniques such as DNA sequencing, restriction enzyme digestion, or gel electrophoresis.

Another important nucleic acid amplification technique is reverse transcription polymerase chain reaction (RT-PCR), which is used to amplify RNA molecules. RT-PCR combines reverse transcription, where RNA is converted into complementary DNA (cDNA) using the enzyme reverse transcriptase, followed by PCR amplification of the cDNA. RT-PCR is commonly used in gene expression studies, as it allows researchers to measure the levels of specific RNA transcripts.

Other nucleic acid amplification methods have been developed to suit specific applications. Some examples include quantitative PCR (qPCR), which allows for the quantification of DNA or RNA in real-time, and loop-mediated isothermal amplification (LAMP), which amplifies DNA under isothermal conditions without the need for a thermal cycler.

Nucleic acid amplification techniques have revolutionized molecular biology and have numerous applications in research, clinical diagnostics, forensics, and biotechnology. They have enabled the detection and analysis of genetic diseases, identification of pathogens, monitoring of gene expression, and investigation of genetic variation. The continuous advancements in nucleic acid amplification technologies have led to improved sensitivity, specificity, and speed of analysis, making them indispensable tools in modern biological research and medical diagnostics.

1.3 WEAK ELECTROLYTES

Weak electrolytes are substances that partially dissociate into ions when dissolved in a solvent, such as water. Unlike strong electrolytes that completely dissociate into ions, weak electrolytes only undergo partial ionization, resulting in a mixture of ions and uncharged molecules in the solution. This behavior is due to the equilibrium between the dissociated and undissociated forms of the weak electrolyte.

One of the key characteristics of weak electrolytes is their ability to establish an equilibrium between the dissolved ions and the undissociated molecules. The extent of ionization is influenced by factors such as the concentration of the weak electrolyte, the nature of the solvent, and the temperature. The equilibrium is described by the equilibrium constant, known as the ionization constant (Ka), which is a measure of the extent of ionization at a given concentration.

An important example of a weak electrolyte is acetic acid (CH_3COOH). When acetic acid dissolves in water, it undergoes partial ionization to form acetate ions (CH_3COO^-) and hydronium ions (H_3O^+). The equilibrium between the undissociated acetic acid and the dissociated ions can be represented by the following equation:

$CH_3COOH \rightleftharpoons CH_3COO^- + H_3O^+$

The ionization constant (Ka) for acetic acid is a measure of the extent of ionization and is defined as the ratio of the concentration of the products (CH_3COO^- and H_3O^+) to the concentration of the undissociated acetic acid (CH_3COOH). Ka is temperature-dependent and provides important information about the strength of the weak electrolyte.

In addition to acetic acid, there are many other examples of weak electrolytes. Some common examples include weak acids, such as carbonic acid (H_2CO_3), phosphoric acid (H_3PO_4), and citric acid ($C_6H_8O_7$), as well as weak bases, such as ammonia (NH_3) and pyridine (C_5H_5N). These substances exhibit partial ionization in water and establish an equilibrium between the ions and the undissociated molecules.

The behavior of weak electrolytes has important implications in various areas of chemistry and biology. In aqueous solutions, weak electrolytes play a crucial role in maintaining pH balance and buffering capacity. They act as weak acids or bases and can donate or accept protons to help regulate the acidity or alkalinity of a solution. Additionally, weak electrolytes are involved in many biological processes, including enzyme-catalyzed reactions, protein folding, and cell signaling.

The understanding of weak electrolytes is essential in the design and optimization of chemical reactions, as the extent of ionization can affect reaction rates and equilibria. The pH-dependent behavior of weak electrolytes is also utilized in analytical chemistry for the determination of acid-base properties and the titration of solutions.

In summary, weak electrolytes are substances that only partially dissociate into ions when dissolved in a solvent. They establish an equilibrium between the ions and the undissociated molecules, and the extent of ionization is determined by factors such as concentration, solvent, and temperature. Understanding the behavior of weak electrolytes is crucial in various fields, including chemistry, biology, and analytical sciences, and has broad implications in areas such as pH regulation, chemical reactions, and acid-base titrations.

1.3.1 Characteristics of Weak Electrolytes

Weak electrolytes possess certain characteristics that distinguish them from strong electrolytes. Here are some key characteristics of weak electrolytes:

1. Partial ionization: Unlike strong electrolytes that completely dissociate into ions when dissolved in a solvent, weak electrolytes only undergo partial ionization. This means that only a fraction of the weak electrolyte molecules dissociate into ions, while the remaining molecules remain undissociated.
2. Equilibrium: Weak electrolytes establish an equilibrium between the dissociated ions and the undissociated molecules in the solution. The forward and backward reactions occur simultaneously, leading to a dynamic equilibrium. The equilibrium is governed by the ionization constant (Ka), which represents the ratio of the concentrations of the products (ions) to the concentration of the weak electrolyte.
3. Ionization constant: The ionization constant (Ka) is a measure of the extent of ionization of a weak electrolyte. It indicates the equilibrium position between the ions and the undissociated molecules. A higher Ka value indicates a greater degree of ionization, while a lower Ka value suggests a weaker ionization tendency.
4. pH dependence: The ionization of weak electrolytes is pH dependent. The concentration of ions in a weak electrolyte solution affects the pH of the solution. Weak acids, for example, increase the concentration of hydronium ions (H3O+) in solution, leading to acidity, while weak bases increase the concentration of hydroxide ions (OH-) and result in alkalinity.
5. Conductivity: Due to their partial ionization, weak electrolyte solutions exhibit lower electrical conductivity compared to strong electrolytes. Since only a fraction of the weak electrolyte molecules dissociate into ions, the number of charge carriers is reduced, resulting in lower conductivity.
6. Buffering capacity: Weak electrolytes play a crucial role in buffering systems. Buffer solutions consist of a weak acid and its conjugate base (or a weak base and its conjugate acid). These solutions can resist changes in pH upon the addition of small amounts of acid or base. Weak electrolytes act as both an acid and a base in buffering systems, enabling them to accept or donate protons to maintain a relatively constant pH.
7. Temperature dependency: The degree of ionization of weak electrolytes can be influenced by temperature. Generally, an increase in temperature leads to an increase in the extent of ionization, while a decrease in temperature results in a decrease in ionization.

Understanding the characteristics of weak electrolytes is important in various scientific disciplines. These characteristics impact areas such as acid-base chemistry, pH regulation, buffering systems, and the behavior of solutions containing weak electrolytes. By studying these properties, researchers can gain insights into the behavior of weak electrolytes and utilize them in various applications, ranging from chemical reactions and analytical chemistry to biological and environmental processes.

1.3.2 Ionization Constants and Equilibrium

Ionization constants and equilibrium are fundamental concepts in chemistry that play a crucial role in understanding the behavior of weak acids and bases. These concepts describe the equilibrium between the dissociated ions and the undissociated molecules of a weak electrolyte in a solution.

Ionization Constants: Ionization constants are quantitative measures of the extent of ionization of a weak electrolyte. They are denoted by Ka for weak acids and Kb for weak bases. The ionization constant represents the ratio of the concentrations of the products (ions) to the concentration of the undissociated weak electrolyte.

For a weak acid HA, the ionization constant can be expressed as:

$HA \rightleftharpoons H^+ + A^-$

$Ka = [H^+][A^-] / [HA]$

For a weak base B, the ionization constant can be expressed as:

$B + H_2O \rightleftharpoons BH^+ + OH^-$

$Kb = [BH^+][OH^-] / [B]$

The ionization constant (Ka or Kb) provides information about the strength of the weak acid or base. Higher values of Ka or Kb indicate a stronger acid or base with a higher degree of ionization.

Equilibrium: When a weak electrolyte dissolves in a solvent, such as water, it establishes an equilibrium between the dissociated ions and the undissociated molecules. This equilibrium is represented by a chemical equation and is governed by the principles of dynamic equilibrium.

For example, consider the dissociation of a weak acid HA:

$HA \rightleftharpoons H^+ + A^-$

At equilibrium, the rate of the forward reaction (dissociation of HA) is equal to the rate of the backward reaction (recombination of H+ and A-). The concentration of the ions and the undissociated molecules remains constant over time, giving rise to a dynamic balance between the reactants and products.

The equilibrium position of a weak electrolyte can be quantified by the equilibrium constant, which is the ratio of the concentration of the products to the concentration of the reactants at equilibrium. For a weak acid, the equilibrium constant is the ionization constant Ka, while for a weak base, it is the ionization constant Kb.

Manipulating Equilibrium: The equilibrium of a weak electrolyte can be shifted by changing external factors such as concentration, temperature, or pressure. Le Chatelier's principle can be applied to predict the effect of these changes on the equilibrium position.

For example, increasing the concentration of the weak acid or base will shift the equilibrium towards the production of more ions. Similarly, changing the temperature can affect the equilibrium position. In general, increasing the temperature favors the endothermic direction of the reaction (absorbing heat), leading to a higher degree of ionization.

Equilibrium calculations using ionization constants and other equilibrium principles are vital in understanding the behavior of weak acids and bases, as well as other weak electrolytes.

They provide a quantitative understanding of the extent of ionization, the concentrations of ions and molecules at equilibrium, and the relationship between the equilibrium constants and the strength of the weak electrolytes. These concepts have broad applications in fields such as acid-base chemistry, pharmaceuticals, environmental science, and chemical analysis.

1.3.3 pH and Weak Electrolytes

pH and weak electrolytes are closely related concepts as the ionization of weak acids and bases influences the acidity or alkalinity of a solution. pH is a measure of the concentration of hydronium ions (H_3O_+) in a solution, and weak electrolytes can affect the pH due to their ability to donate or accept protons (H_+). Here's a discussion on the relationship between pH and weak electrolytes:

1. Weak Acids: When a weak acid, such as acetic acid (CH_3COOH), dissolves in water, it undergoes partial ionization, releasing hydronium ions (H_3O^+) and the corresponding conjugate base (CH_3COO^-). The concentration of H_3O^+ ions increases, leading to an acidic solution. The pH of a weak acid solution

is lower than 7, indicating its acidic nature. The extent of ionization and the concentration of H_3O^+ ions depend on the concentration of the weak acid and its ionization constant (Ka).

2. Weak Bases: Weak bases, such as ammonia (NH_3), accept protons from water molecules to form hydroxide ions (OH^-) and the corresponding conjugate acid (NH_4^+). The presence of OH- ions increases the hydroxide ion concentration in the solution, resulting in alkalinity. The pH of a weak base solution is higher than 7, indicating its basic nature. Similar to weak acids, the extent of ionization and the concentration of OH- ions depend on the concentration of the weak base and its ionization constant (Kb).
3. pH Calculation: The pH of a solution containing a weak electrolyte can be calculated using the ionization constant and the concentrations of the weak acid or base and its conjugate species. For weak acids, the pH can be determined using the equation:

$pH = -log[H_3O^+]$,

where $[H_3O^+]$ is the concentration of hydronium ions.

For weak bases, the pOH (negative logarithm of hydroxide ion concentration) is first calculated using a similar equation, and then the pH can be determined using the relation:

$pH = 14 - pOH$.

4. Buffering Capacity: Weak electrolytes are essential components of buffer solutions, which are able to resist changes in pH upon the addition of small amounts of acid or base. Buffer solutions consist of a weak acid and its conjugate base, or a weak base and its conjugate acid. When an acid or base is added to a buffer, the weak electrolyte components can absorb or release protons to maintain the pH relatively stable. This buffering capacity arises from the equilibrium established between the weak acid and its conjugate base, or the weak base and its conjugate acid.

Understanding the relationship between pH and weak electrolytes is crucial in various fields, including biochemistry, pharmaceutical sciences, and environmental science. pH regulation is critical for maintaining optimal conditions for biological reactions and enzyme activity. Moreover, the behavior of weak electrolytes in solution affects solubility, chemical reactions, and the behavior of biological systems. pH measurements and control are important in numerous applications, such as in the development of drugs, analysis of biological samples, and environmental monitoring.

1.3.4 Biological Importance of Weak Electrolytes

Weak electrolytes play a crucial role in maintaining the pH balance and regulating biochemical processes in biological systems. Here are some of the biological importance of weak electrolytes:

1. Acid-Base Balance: Weak electrolytes, particularly weak acids and bases, are vital in maintaining the acid-base balance within living organisms. Biological processes produce acidic or basic byproducts, and weak electrolytes help buffer these changes in pH. They act as components of buffer systems, which consist of a weak acid and its conjugate base, or a weak base and its conjugate acid. Buffer systems prevent drastic pH fluctuations, ensuring the proper functioning of enzymes and maintaining cellular homeostasis.
2. Cellular pH Regulation: Weak electrolytes participate in the regulation of pH within cells. Intracellular pH plays a crucial role in various cellular processes, including enzyme activity, protein structure, and cell signaling. Weak acids and bases in the cytoplasm and organelles act as buffers, helping to maintain the optimal pH range for these cellular processes.
3. Carbon Dioxide Transport and Blood pH Regulation: In the blood, weak electrolytes are involved in the transport and regulation of carbon dioxide (CO2) levels. Carbon dioxide is transported in the form of bicarbonate ions (HCO3-) through the carbonic acid-bicarbonate buffering system. This system involves the equilibrium between carbonic acid (H2CO3) and bicarbonate ions. The presence of weak electrolytes, such as carbonic acid, helps maintain the pH of the blood within a narrow range, facilitating efficient oxygen delivery and waste removal.
4. Acidic and Alkaline Digestive Environments: Weak electrolytes are present in the digestive system, where they contribute to the acidic or alkaline environments necessary for optimal digestion and enzyme activity. For instance, gastric acid in the stomach contains weak electrolytes such as hydrochloric acid (HCl), which aids in the breakdown of

food and activation of digestive enzymes. Additionally, weak electrolytes in the pancreatic secretions help neutralize the acidic stomach contents as they enter the small intestine.

5. Enzyme Activity and Protein Structure: Weak electrolytes are critical for maintaining the optimal pH for enzyme activity. Enzymes often exhibit pH-dependent activity, and deviations from their optimal pH can impact their efficiency and substrate binding. Weak electrolytes, as part of buffer systems, help regulate the pH around enzymes, ensuring their proper functioning. Moreover, weak electrolytes also influence protein structure and stability, as alterations in pH can disrupt hydrogen bonding and electrostatic interactions within proteins.
6. Transport of Molecules across Cell Membranes: Weak electrolytes are involved in the transport of molecules across cell membranes. Ion channels, transporters, and pumps are often pH-dependent, and the presence of weak electrolytes can affect the ionization state of these transport proteins. This, in turn, influences the movement of ions and other molecules across cell membranes, maintaining the electrochemical balance and facilitating cellular processes such as nerve conduction and muscle contraction.
7. Osmotic Balance: Weak electrolytes, such as amino acids and small organic molecules, contribute to osmotic balance within cells and tissues. They help maintain the osmotic pressure, which is crucial for regulating water movement and cellular hydration. The presence of weak electrolytes ensures the proper distribution of ions and solutes across cell membranes, preventing osmotic imbalances that can be detrimental to cellular function.

Understanding the biological importance of weak electrolytes is essential for comprehending the intricate processes that occur within living organisms. Their role in pH regulation, enzyme activity, transport processes, and overall cellular homeostasis highlights their significance in maintaining proper biological function.

1.3.5 Titration and Weak Electrolytes

Titration is a widely used analytical technique that involves the gradual addition of a titrant solution of known concentration to a solution of the analyte (the substance being analyzed) until the reaction between the titrant and the analyte is complete. In the case of weak electrolytes, titration can be specifically employed to determine their concentration or to characterize their acid-base properties. Here's a discussion on titration and its application to weak electrolytes:

1. Acid-Base Titration: Titration is commonly used to determine the concentration of weak acids or bases. In an acid-base titration, a solution of a strong acid or base, known as the titrant, is added to the solution of the weak electrolyte until the equivalence point is reached. The equivalence point corresponds to the complete neutralization of the weak acid or base, resulting in a pH change. The pH of the solution is monitored during the titration using an indicator or a pH meter. The volume of the titrant required to reach the equivalence point allows for the calculation of the concentration of the weak electrolyte.
2. Indicator Selection: Selecting an appropriate indicator is crucial in acid-base titrations involving weak electrolytes. Indicators are substances that change color depending on the pH of the solution. For weak acids, indicators with a transition range that includes the pKa of the acid are preferred. Similarly, for weak bases, indicators with a transition range that includes the pKb of the base are suitable. The color change of the indicator signifies the proximity to the equivalence point and aids in determining the endpoint of the titration.
3. Buffered Titration: Weak electrolytes, especially weak acids or bases, can also serve as components of buffer solutions used in titrations. Buffers resist changes in pH and help maintain a stable environment during the titration process. They can prevent large pH shifts and ensure accurate determination of the equivalence point. By incorporating a weak acid or base into the titration system, the buffer capacity can be increased, allowing for greater control over the pH changes during the titration.
4. Polyprotic Weak Acids: Titration can also be employed to determine the concentration and acid dissociation constants of polyprotic weak acids, which can donate multiple protons. Each proton-donation step in the acid dissociation process corresponds to a separate equivalence point. By performing a stepwise titration, the concentrations and dissociation constants of the successive acidic protons can be determined.

5. Solubility Determination: Titration can be used to determine the solubility of weak electrolytes, particularly sparingly soluble salts. By titrating a solution containing the weak electrolyte with a solution of a strong electrolyte, the point of precipitation or complete reaction can be observed. The volume of the strong electrolyte solution required to reach this point provides information about the solubility of the weak electrolyte.

Titration serves as a valuable tool in analyzing weak electrolytes, allowing for the determination of their concentration, acid-base properties, solubility, and other relevant parameters. By carefully selecting indicators, employing buffer systems, and considering the unique characteristics of weak electrolytes, titration techniques can provide valuable insights into their behavior and chemical properties.

1.3.6 Comparison with Strong electrolytes

When comparing weak electrolytes to strong electrolytes, several key differences arise in terms of their behavior in solution. Here are some of the primary points of comparison between weak and strong electrolytes:

1. Ionization: Strong electrolytes undergo complete ionization when dissolved in water, meaning that they dissociate fully into their constituent ions. In contrast, weak electrolytes undergo partial ionization, resulting in a mixture of both ionized and un-ionized species in solution. This difference in ionization behavior leads to variations in conductivity and other properties.
2. Conductivity: Due to their complete ionization, strong electrolytes exhibit high electrical conductivity in solution. This is because the abundance of ions enables the easy flow of electric current. Weak electrolytes, on the other hand, have lower electrical conductivity due to their partial ionization and the presence of a significant proportion of un-ionized molecules.
3. Dissociation: Strong electrolytes dissociate rapidly and to a large extent in water, resulting in a high concentration of ions. Weak electrolytes, however, dissociate to a lesser degree, leading to a lower concentration of ions in solution.
4. pH Effects: Strong acids and bases have a significant impact on pH when added to a solution due to their complete ionization. The presence of a strong acid leads to a decrease in pH (acidic effect), while a strong base increases the pH (alkaline effect).

Weak acids and bases, although they can affect pH to some extent, have a less pronounced impact due to their partial ionization.

5. Buffering Capacity: Weak electrolytes are essential components of buffer systems due to their ability to resist changes in pH. Buffer solutions typically consist of a weak acid and its conjugate base or a weak base and its conjugate acid. These systems are capable of absorbing or releasing protons to maintain the pH within a relatively narrow range. Strong electrolytes, in contrast, do not exhibit significant buffering capacity.
6. Solubility: Strong electrolytes generally exhibit high solubility in water, as they dissociate into ions that are stabilized by the solvent. Weak electrolytes can have varying solubilities, depending on factors such as the strength of the electrostatic forces between the ions and the nature of the solvent. Weak acids and bases may exhibit limited solubility due to their partial ionization.
7. Reaction Rates: The complete ionization of strong electrolytes allows for rapid reaction rates because all reactant species are in the ionized form. In contrast, the partial ionization of weak electrolytes may lead to slower reaction rates due to the presence of both ionized and un-ionized species, which can limit the frequency of productive collisions.

Understanding the differences between weak and strong electrolytes is crucial for various applications in chemistry and biochemistry. These differences influence the behavior and properties of solutions containing these electrolytes, including their electrical conductivity, pH effects, solubility, and reaction kinetics.

1.3.7 Factors Affecting the Extent of Ionization

The extent of ionization of a weak electrolyte, such as a weak acid or weak base, in a solution depends on several factors. These factors can significantly influence the degree of ionization and, consequently, the behavior and properties of the weak electrolyte. Here are some of the key factors that affect the extent of ionization:

1. Concentration: The concentration of the weak electrolyte in solution has a direct impact on its ionization. As the concentration increases, the probability of collisions between the molecules of the weak electrolyte and the solvent molecules also increases. This leads to a higher likelihood of ionization, resulting in a greater extent of ionization.

2. Nature of the Solvent: The solvent in which the weak electrolyte is dissolved can influence its ionization behavior. Different solvents have varying abilities to solvate and stabilize ions. For example, water is a highly polar solvent that can effectively solvate ions, making it favorable for the ionization of many weak electrolytes. Nonpolar solvents, on the other hand, may not provide sufficient stabilization for ionization, resulting in lower extents of ionization.
3. Temperature: Temperature plays a role in the extent of ionization of weak electrolytes. In general, an increase in temperature enhances the kinetic energy of the molecules, leading to more frequent collisions and increased ionization. However, the effect of temperature on ionization can vary depending on the specific weak electrolyte. Some weak electrolytes may exhibit decreased ionization at higher temperatures due to the disruption of intermolecular forces that stabilize the non-ionized species.
4. Ionic Strength: The presence of other ions in the solution, known as the ionic strength, can affect the extent of ionization. High ionic strength can decrease the extent of ionization by increasing the ionic atmosphere around the weak electrolyte ions, reducing their mobility and hindering further ionization. Conversely, a low ionic strength environment may enhance ionization by minimizing the ion-ion interactions.
5. Presence of Common Ion: The presence of a common ion, which is an ion already present in the solution due to another source, can influence the extent of ionization. If a weak electrolyte and its conjugate ion are both present in the solution, the Le Chatelier's principle predicts that the equilibrium will shift towards the non-ionized form, reducing the extent of ionization.
6. Molecular Structure: The molecular structure of the weak electrolyte can also impact its ionization behavior. Factors such as the strength of the bond holding the ions together, the presence of functional groups, and the stability of the resulting ions can influence the ease of ionization. For example, a weak acid with a more stable conjugate base is likely to exhibit higher ionization.

Understanding the factors that affect the extent of ionization is essential for predicting and manipulating the behavior of weak electrolytes in various chemical and biological systems. By considering these factors, researchers can optimize conditions for desired ionization levels, design effective buffer systems, and better understand the behavior of weak acids and bases in solution.

REFERENCES

1. Alberts, B., Johnson, A., Lewis, J., Raff, M., Roberts, K., Walter, P. Molecular Biology of the Cell. Garland Science, 2014.
2. Berg, J. M., Tymoczko, J. L., Gatto, G. J. Stryer, L. Biochemistry. W.H. Freeman and Company, 2018.
3. Cox, M. M., Nelson, D. L. Lehninger Principles of Biochemistry. W.H. Freeman and Company, 2017.
4. Garrett, R. H., Grisham, C. M. Biochemistry. Cengage Learning, 2016.
5. Lodish, H., Berk, A., Zipursky, S. L., Matsudaira, P., Baltimore, D., Darnell, J. Molecular Cell Biology. W.H. Freeman and Company, 2016.
6. Nelson, D. L., Cox, M. M. Lehninger Principles of Biochemistry. W.H. Freeman and Company, 2017.
7. Voet, D., Voet, J. G., Pratt, C. W. Fundamentals of Biochemistry: Life at the Molecular Level. Wiley, 2016.

CHAPTER

2

CELL CULTURE TECHNIQUES AND CENTRIFUGATION

INTRODUCTION

Cell culture is the process by which cells are grown under controlled conditions, generally outside of their natural environment. The term «tissue culture» was coined by American pathologist Montrose Thomas Burrows. This technique is also called micropropagation. After the cells of interest have been isolated from living tissue, they can subsequently be maintained under carefully controlled conditions. They need to be kept at body temperature (37°C) in an incubator.

These conditions vary for each cell type, but generally consist of a suitable vessel with a substrate or rich medium that supplies the essential nutrients (amino acids, carbohydrates, vitamins, minerals), growth factors, hormones, and gases (CO2, O2), and regulates the physio-chemical environment (pH buffer, osmotic pressure, temperature).

Centrifugation is a method of separating molecules having different densities by spinning them in solution around an axis (in a centrifuge rotor) at high speed. It is one of the most useful and frequently employed techniques in the molecular biology laboratory. Centrifugation is used to collect cells, to precipitate DNA, to purify virus particles, and to distinguish subtle differences in the conformation of molecules.

2.1 CENTRIFUGATION TECHNIQUES-BASIC PRINCIPLES, ANALYTICAL AND PREPARATIVE CENTRIFUGATION, THEIR APPLICATIONS.

Centrifugation is a process used to separate or concentrate materials suspended in a liquid medium. The theoretical basis of this technique is the effect of gravity on particles (including macromolecules) in suspension.

The two particles of different masses will settle in a tube at different rates in response to gravity.

It is a process that involves the use of the centrifugal force for the sedimentation of heterogeneous mixtures with a centrifuge used in industry and in laboratory settings.

This process is used to separate two immiscible liquids. More dense components of the mixture migrate away from the axis of the centrifuge, while less-dense components of the mixture migrate towards the axis.

2.1.1 Principle

It is a device for separating particles from a solution according to their size shape, density, viscosity of the medium and rotor speed.

In a solution particles whose density is higher than that of the solvent sink sediment and the particles that are lighter than it floats to the top. The greater the difference in density in the faster they move.

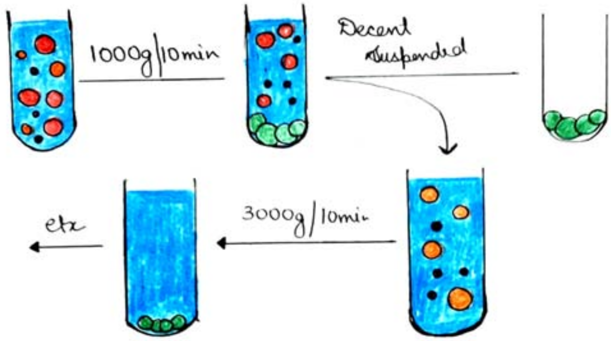
Differential centrifugation

The density of a liquid is uniform. The density of liquid << Density of particles. The viscosity of the liquid is low. Consequence: The rate of particle sedimentation depends mainly on its size and the applied *g*-force.

*S*ize of major cell organelles

• Nucleus	4-12 µm
• Plasma membrane sheets	3-20 µm
• Golgi tubules	1-2 µm
• Mitochondria	0.4-2.5 µm
• Lysosomes/peroxisomes	0.4-0.8 µm
• Microsomal vesicles	0.05-0.3µm

Differential centrifugation of a tissue homogenate



Differential centrifugation of a tissue homogenate

1. Homogenate – 1000g for 10 min
2. Supernatant from 1 – 3000g for 10 min
3. Supernatant from 2 – 15,000g for 15 min
4. Supernatant from 3 – 100,000g for 45 min
   - Pellet 1 – nuclear
   - Pellet 2 – "heavy" mitochondrial
   - Pellet 3 – "light" mitochondrial
   - Pellet 4 – microsomal

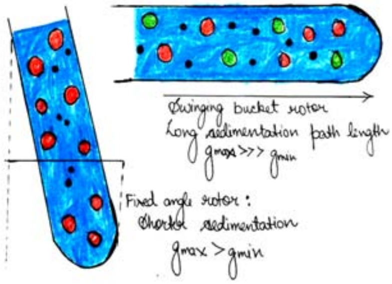
Differential centrifugation (III) expected content of pellets

- 1000g pellet: nuclei, plasma membrane sheets
- 3000g pellet: large mitochondria, Golgi tubules
- 15,000g pellet: small mitochondria, lysosomes, peroxisomes
- 100,000g pellet: microsomes

Differential centrifugation

- Poor resolution and recovery because of particle size heterogeneity.
- Particles starting out at r_{min} have the furthest to travel but initially experience lowest RCF.
- Smaller particles close to r_{max} have only a short distance to travel and experience the highest RCF.

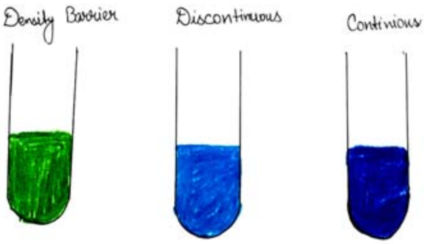
Differential centrifugation



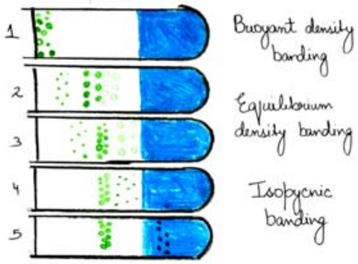
Differential centrifugation

The rate of sedimentation can be modulated by particle density. The nuclei have an unusually rapid sedimentation rate because of their size and high density. Golgi tubules do not sediment at 3000g, in spite of their size: they have an unusually low sedimentation rate because of their very low density: (ρ_p - ρ_l) becomes rate-limiting.

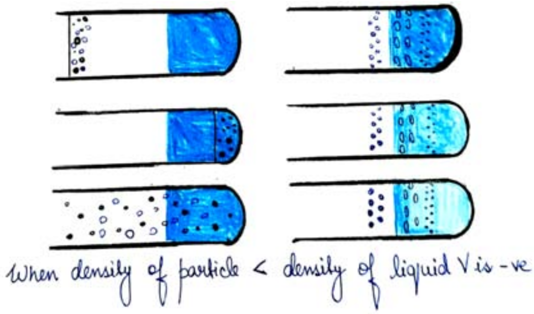
Density gradient centrifugation



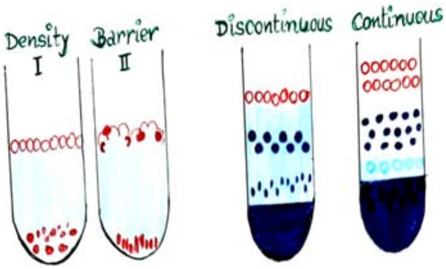
A particle will sediment through a solution if particle density > solution density. If particle density < solution density, the particle will float through the solution. When particle density = solution density the particle stop sedimenting or floating.



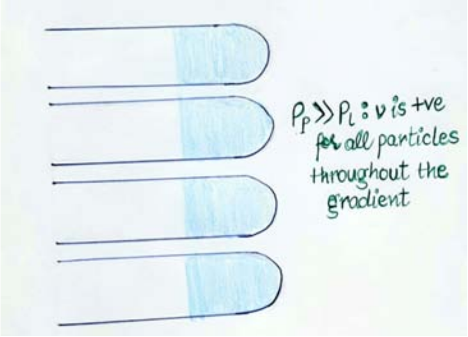
It has formats for separation of particles according to their density.



Resolution of density gradients



Separation of particles according to size



2.1.2 Application of Centrifugation

- Centrifugation can be employed to separate a mixture of two different miscible liquids.
- This technique can also be used in order to study and analyze macromolecules and their hydrodynamic properties.

- Mammalian cells can be purified with the help of a special type of centrifuge.
- Centrifugation is known to have a vital role in the fractionation of many subcellular organelles. Furthermore, centrifugation is also useful in the fractionation of membrane fractions and membranes.
- Centrifugation also has applications in the fractionation of membrane vesicles.
- Chalk can be separated from water with the help of a centrifuge.
- Skimmed milk is a form of milk that has a lower amount of dissolved fats. Skimmed milk can be obtained from regular milk with the help of the process of centrifugation. Here, the centrifuge serves to separate the fat from the milk, leaving the required skimmed milk behind.
- Cyclonic separation is an important process that has vital applications in the separation of particles from air flows.
- Another important application of this technique is in the stabilization and clarification of wine.
- This technique, in combination with other purification techniques, is extremely helpful while separating proteins.
- Other techniques that are used include salting out techniques such as ammonium sulfate precipitation.
- Centrifuges are widely used in the field of forensic chemistry.
- In this field, the technique is employed for the separation of blood components from blood samples.
- Furthermore, the technique is also employed in certain laboratories for the separation of urine components from urine samples.
- Differential centrifugation, a distinct type of centrifugation, is known to have applications in the identification of organelles.

2.1.3 Applications of Cell Culture

Mass culture of animal cell lines is fundamental to the manufacture of viral vaccines and other products of biotechnology. Culture of human stem cells is used to expand the number of cells and differentiate the cells into various somatic cell types for transplantation. Stem cell culture is also used to harvest the molecules and exosomes that the stem cells release for the purposes of therapeutic development. Biological products produced by recombinant DNA (rDNA) technology in animal cell cultures include enzymes, synthetic hormones, immunobiologicals (monoclonal antibodies, interleukins, lymphokines), and anticancer agents.

Although many simpler proteins can be produced using rDNA in bacterial cultures, more complex proteins that are glycosylated (carbohydrate-modified) currently must be made in animal cells. An important example of such a complex protein is the hormone erythropoietin. The cost of growing mammalian cell cultures is high, so research is underway to produce such complex proteins in insect cells or in higher plants, use of single embryonic cell and somatic embryos as a source for direct gene transfer via particle bombardment, transit gene expression and confocal microscopy observation is one of its applications. It also offers to confirm single cell origin of somatic embryos and the asymmetry of the first cell division, which starts the process.

Cell culture is also a key technique for cellular agriculture, which aims to provide both new products and new ways of producing existing agricultural products like milk, (cultured) meat, fragrances, and rhino horn from cells and microorganisms. It is therefore considered one means of achieving animal-free agriculture. It is also a central tool for teaching cell biology.

Cell Culture in two Dimensions

This technique is known as two-dimensional (2D) cell culture, and was first developed by Wilhelm Roux who, in 1885, removed a portion of the medullary plate of an embryonic chicken and maintained it in warm saline for several days on a flat glass plate. From the advance of polymer technology arose today›s standard plastic dish for 2D cell culture, commonly known as the Petri dish. Julius Richard Petri, a German bacteriologist, is generally credited with this invention while working as an assistant to Robert Koch. Various researchers today also utilize culturing laboratory flasks, conicals, and even disposable bags like those used in single-use bioreactors.

Aside from Petri dishes, scientists have long been growing cells within biologically derived matrices such as collagen or fibrin, and more recently, on synthetic hydrogels such as polyacrylamide or PEG. They do this in order to elicit phenotypes that are not expressed on conventionally rigid substrates. There is growing interest in controlling matrix stiffness, a concept that has led to discoveries in fields such as:

- Stem cell self-renewal
- Lineage specification
- Cancer cell phenotype
- Fibrosis
- Hepatocyte function
- Mechanosensing

Cell culture in three dimensions

Cell culture in three dimensions has been touted as «Biology›s New Dimension». At present, the practice of cell culture remains based on varying combinations of single or multiple cell structures in 2D. Currently, there is an increase in use of 3D cell cultures in research areas including drug discovery, cancer biology, regenerative medicine, nanomaterials assessment and basic life science research. 3D cell cultures can be grown using a scaffold or matrix, or in a scaffold-free manner. Scaffold based cultures utilize an acellular 3D matrix or a liquid matrix. Scaffold-free methods are normally generated in suspensions. There are a variety of platforms used to facilitate the growth of three-dimensional cellular structures including scaffold systems such as hydrogel matrices and solid scaffolds, and scaffold-free systems such as low-adhesion plates, nanoparticle facilitated magnetic levitation, hanging drop plates, and rotary cell culture. Culturing cells in 3D leads to wide variation in gene expression signatures and partly mimics tissues in the physiological states. A 3D cell culture model showed cell growth similar to that of in vivo than did a monolayer culture, and all three cultures were capable of sustaining cell growth. As 3D culturing has been developed it turns out to have a great potential to design tumors models and investigate malignant transformation and metastasis, 3D cultures can provide aggerate tool for understanding changes, interactions, and cellular signaling.

3D cell culture in scaffolds

Eric Simon, in a 1988 NIH SBIR grant report, showed that electrospinning could be used to produced nano- and submicron-scale polystyrene and polycarbonate fibrous scaffolds specifically intended for use as *in vitro* **cell substrates. This early use of** electrospun fibrous lattices for cell culture and tissue engineering showed that various cell types including Human Foreskin Fibroblasts (HFF), transformed Human

Carcinoma (HEp-2), and Mink Lung Epithelium (MLE) would adhere to and proliferate upon polycarbonate fibers. It was noted that, as opposed to the flattened morphology typically seen in 2D culture, cells grown on the electrospun fibers exhibited a more histotypic rounded 3-dimensional morphology generally observed *in vivo*.

3D cell culture in hydrogels

As the natural extracellular matrix (ECM) is important in the survival, proliferation, differentiation and migration of cells, different hydrogel culture matrices mimicking natural ECM structure are seen as potential approaches to in vivo–like cell culturing.

Hydrogels are composed of interconnected pores with high water retention, which enables efficient transport of substances such as nutrients and gases. Several different types of hydrogels from natural and synthetic materials are available for 3D cell culture, including animal ECM extract hydrogels, protein hydrogels, peptide hydrogels, polymer hydrogels, and wood-based nanocellulose hydrogel.

3D Cell Culturing by Magnetic Levitation

The 3D Cell Culturing by Magnetic Levitation method (MLM) is the application of growing 3D tissue by inducing cells treated with magnetic nanoparticle assemblies in spatially varying magnetic fields using neodymium magnetic drivers and promoting cell to cell interactions by levitating the cells up to the air/liquid interface of a standard petri dish.

The magnetic nanoparticle assemblies consist of magnetic iron oxide nanoparticles, gold nanoparticles, and the polymer polylysine. 3D cell culturing is scalable, with the capability for culturing 500 cells to millions of cells or from single dish to high-throughput low volume systems.

Tissue culture and engineering

Cell culture is a fundamental component of tissue culture and tissue engineering, as it establishes the basics of growing and maintaining cells in vitro. The major application of human cell culture is in stem cell industry, where mesenchymal stem cells can be cultured and cryopreserved for future use.

Tissue engineering potentially offers dramatic improvements in low cost medical care for hundreds of thousands of patients annually.

Vaccines

Vaccines for polio, measles, mumps, rubella, and chickenpox are currently made in cell cultures. Due to the H5N1 pandemic threat, research into using cell culture for influenza vaccines is being funded by the United States government. Novel ideas in the field include recombinant DNA-based vaccines, such as one made using human adenovirus (a common cold virus) as a vector, and novel adjuvants.

Cell co-culture

The technique of co-culturing is used to study cell crosstalk between two or more types of cells on a plate or in a 3D matrix. The cultivation of different stem cells and the interaction of immune cells can be investigated in an in vitro model similar to biological tissue. Since most tissues contain more than one type of cell, it is important to evaluate their interaction in a 3D culture environment to gain a better understanding of their interaction and to introduce mimetic tissues.

There are two types of co-culturing: direct and indirect. While direct interaction involves cells being in direct contact with each other in the same culture media or matrix, indirect interaction involves different environments, allowing signaling and soluble factors to participate.

Cell differentiation in tissue models during interaction between cells can be studied using the Co-Cultured System to simulate cancer tumors, to assess the effect of drugs on therapeutic trials, and to study the effect of drugs on therapeutic trials.

The co-culture system in 3D models can predict the response to chemotherapy and endocrine therapy if the microenvironment defines biological tissue for the cells.

A co-culture method is used in tissue engineering to generate tissue formation with multiple cells interacting directly.

Cell culture in microfluidic device

Microfluidics technique is developed systems that can perform a process in a flow which are usually in a scale of micron. Microfluidics chip are also known as Lab-on-a-chip and they are able to have continuous procedure and reaction steps with spare amount of reactants and space. Such systems enable the identification and isolation of individual cells and molecules when combined with appropriate biological assays and high-sensitivity detection techniques.

Organ-on-a-chip

OoC systems mimic and control the microenvironment of the cells by growing tissues in microfluidics. Combining tissue engineering, biomaterials fabrication, and cell biology, it offers the possibility of establishing a biomimetic model for studying human diseases in the laboratory.

In recent years, 3D cell culture science has made significant progress, leading to the development of OoC. OoC is considered as a preclinical step that benefits pharmaceutical studies, drug development and disease modeling.

OoC is an important technology that can bridge the gap between animal testing and clinical studies and also by the advances that the science has achieved could be a replace for in vivo studies for drug delivery and pathophysiological studies.

2.1.4 Spectrophotometry: UV-Visible Spectrophotometer

Photometry refers to the science of light measurement in terms of the light's brightness that the human eye can perceive. As a basic principle, photometers examine how light interacts with reflective materials. The array of photometers allow for photometry to apply in different fields of study. Some instruments are used to observe how light absorbs and/or reflects wavelengths. Other instruments measure light by converting light into an electrical current and then measuring the electrical current produced by the light. Photometry is often used in the study of liquids and solutions in chemistry. Photometers can help measure masses of organic or inorganic materials in a solution or liquid.

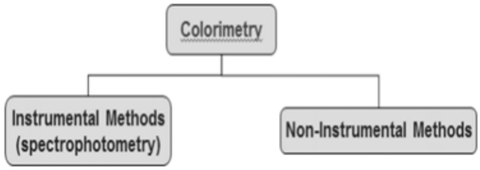
Colorimetry

An analytical technique in which the concentration of an analyte is measured by its ability to produce or change the color of a solution.

- Changes the solution's ability to absorb light.

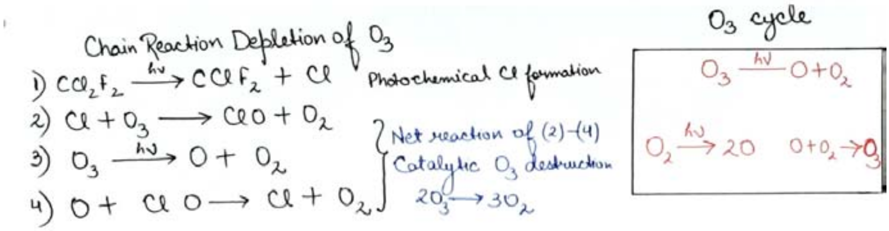
Spectrophotometry

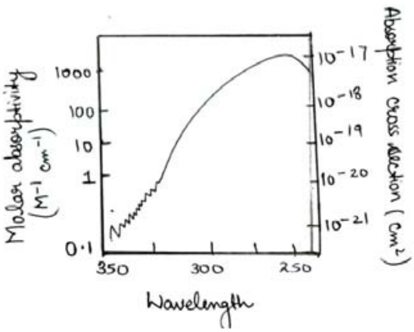
Any technique that uses light to measure chemical concentrations. A colorimetric method where an instrument is used to determine the amount of analyte in a sample by the sample's ability or inability to absorb light at a certain wavelength.



Measurement of Ozone (O_3) Above the South Pole

- O_3 provides protection from ultraviolet radiation
- Seasonal depletion due to chlorofluorocarbons





Properties of Light

Particles and waves

Light waves are consist of perpendicular, oscillating electric and magnetic fields.

- Parameters used to describe light.
- Amplitude (A): height of wave's electric vector.
- Wavelength (l): distance (nm, cm, m) from peak to peak
- Frequency (n): number of complete oscillations that the waves make each second

Hertz (Hz): unit of frequency, second^{-1} (s^{-1})

1 megahertz (MHz) = $10^6 s^{-1}$ = 10^6Hz

Properties of light

Particles and waves

- Parameters used to describe light.
- **Energy (E):** the energy of one particle of light (photon) is proportional to its frequency

E = hv

where: E = photon energy (Joules)

n = frequency (sec^{-1})

h = Planck's constant ($6.626x10^{-34}$J-s)

- As frequency (n) increases, energy (E) of light increases

Relationship between frequency and wavelength

λv = c = V

V = c/λ

where: c = speed of light ($3.0x10^{8}$ m/s in vacuum))

n = frequency (sec^{-1})

l = wavelength (m)

Relationship between energy and wavelength

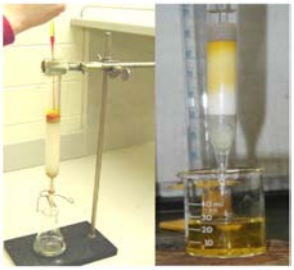
E = hc/λ = hcv

where: = (1/l) = wavenumber

- As frequency (l) decreases, energy (E) of light increases.

2.1.5 Column chromatography – An Example of The Equipment Used in Low-Performance Liquid

The sample is usually applied directly to the top of the column. Detection is by fraction collection with later analysis of each fraction.



Advantages

- Simple system requirements.
- Low cost.
- Popular in sample purification.
- Used in the removal of interferences from samples.
- Used in some analytical applications not common due to low efficiency, long analysis times and poor limits of detection.

2.1.6 High-Performance Liquid Chromatography (HPLC)

LC methods that use small, uniform, rigid support material. Particles < 40 mm in diameter. Usually 3-10 mm in practice. Good system efficiencies. Small plate heights. Such systems have the following Characteristics

- Narrow peaks
- Low limits of detection
- Short separation times
- Columns can only tolerate high operating pressures and faster flow-rates

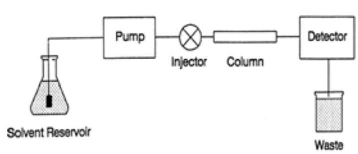
Higher operating pressures need for mobile phase delivery requires special pumps and other system components. The sample applied using a closed system (*i.e.*, injection valve). The detection uses a flowthrough detector.

Advantages

- Fast analysis time.
- Ease of automation.
- Good limits of detection.
- Preferred choice for analytical applications.
- Popular for purification work.

Disadvantages

- Greater expense
- Lower sample capacities



Ion Exchange Chromatography

Ion exchange chromatography (or ion chromatography) is a process that allows the separation of ions and polar molecules based on their affinity to ion exchangers. The principle of separation is thus by a reversible exchange of ions between the target ions present in the sample solution to the ions present on ion exchangers. In this process, two types of exchangers i.e., cationic and anionic exchangers can be used.

1. ***Cationic exchangers*:** They possess negatively charged groups, and these will attract positively charged cations. These exchangers are also called "Acidic ion exchange" materials, because their negative charges result from the ionization of the acidic group.
2. ***Anionic exchangers*:** They have positively charged groups that will attract negatively charged anions. These are also called "Basic ion exchange" materials.

Ion exchange chromatography is most often performed in the form of column chromatography. However, there are also thin-layer chromatographic methods that work basically based on the principle of ion exchange his form of chromatography relies on the attraction between the oppositely charged stationary phase, known as an ion exchanger, and analyte. The ion exchangers basically contain charged groups covalently linked to the surface of an insoluble matrix. The charged groups of the matrix can be positively or negatively charged. When suspended in an aqueous solution, the charged groups of the matrix will be surrounded by ions of the opposite charge. In this "ion cloud", ions can be reversibly exchanged without changing the nature and the properties of the matrix.

On exchange Material

Ion exchange material is available naturally and they also can be synthesis.

A. ***Natural*:** The naturally obtained ion exchangers are different. They are,

1. Cation exchanger like zeolite
2. Anion exchangers like dolomite

B. ***Synthetic*:** The synthetic Ion exchange resins are based on cross-linked polystyrene prepared by co- polystyrene prepared by co-polymersition of styrene and divinyl benzene.

- Other synthetic ion exchange resin used are :-
- Co-polymer of acrylic acid derivative and divinyl benzene
- Cross-linked dextrane.
- Cellulose fibers with introduced charged group.

Gas chromatography

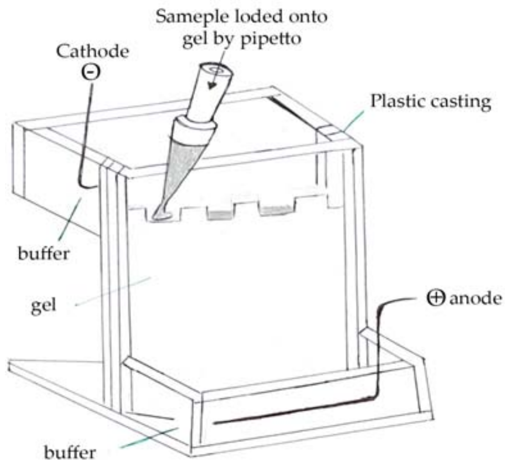
It is an analytical separations technique useful for separating volatile organic compounds consists of : Flowing mobile phase (inert gas - Ar, Ne, N). Injection port (rubber septum - syringe injects sample). kept at a higher temperature than the boiling point. Separation due to differences in partitioning behavior selective retardation. organic compounds separated due to differences in their participating behavior between the mobile gas phase and the stationary phase in the column. in contrast to other types of chromatography, the mobile phase does not interact with molecules of the analyte; its only function is to transport the analyte through the column. The good for volatile samples (up to about 250 ^{o}C). 0.1-1.0 microliter of liquid or 1-10 ml vapor. It can detect <1 ppm with certain detectors. It can be easily automated for injection and data analysis.

Components of a gas chromatography

- Gas Supply: (usually N_2 or He).
- Sample Injector: (syringe / septum).
- Column: 1/8″ or 1/4″ x 6-50′ tubing packed with small uniform size, inert support coated with a thin film of nonvolatile liquid
- Detector: TC - thermal conductivity and FID - flame ionization detector

2.1.7 Electrophoresis: Theory and Different Types – Page, SDS-Page, Capillary Electrophoresis, and IEF

Electrophoresis is a method whereby charged molecules in solution, chiefly proteins and nucleic acids, migrate in response to an electrical field. Their rate of migration through the electrical field depends on the strength of the field, on the net charge, size, and shape of the molecules, and also on the ionic strength, viscosity, and temperature of the medium in which the molecules are moving. As an analytical tool, electrophoresis is simple, rapid and highly sensitive. It can be used analytically to study the properties of a single charged species or mixtures of molecules. It can also be used preparatively as a separating technique. Electrophoresis is usually done with gels formed in tubes, slabs, or on a flat bed. In many electrophoresis units, the gel is mounted between two buffer chambers containing separate electrodes, so that the only electrical connection between the two chambers is through the gel. In most electrophoresis units, the gel is mounted between two buffer chambers containing separate electrodes so that the only electrical connection between the two chambers is through the gel.



The interrelation of resistance, voltage, current and power. Two basic electrical equations are important in electrophoresis.

- The first is Ohm's Law, I = E/R
- The second is P = EI
- This can also be expressed as P = I^2R

In electrophoresis, one electrical parameter, either current, voltage, or power, is always held constant. Under constant current conditions (velocity is directly proportional to current), the velocity of the molecules is maintained, but heat is generated. Under constant voltage conditions, the velocity slows, but no additional heat is generated during the course of the run. Under constant power conditions, the velocity slows but heating is kept constant. The net charge is determined by the pH of the medium. Proteins are amphoteric compounds, that is, they contain both acidic and basic residues. Each protein has its own characteristic charge properties depending on the number and kinds of amino acids carrying amino or carboxyl groups. Nucleic acids, unlike proteins, are not amphoteric. They remain negative at any pH used for electrophoresis.

During Polymerization

- Exothermic reaction
- Gel irregularities
- Pore size

During electrophoresis

- Denaturation of proteins
- Smile effect
- Temperature regulation of buffers

Role of the Solid Support Matrix

It inhibits convection and diffusion, which would otherwise impede the separation of molecules. It allows a permanent record of results through staining after the run. It can provide additional separation through molecular sieving.

Agarose and Polyacrylamide

Although agarose and polyacrylamide differ greatly in their physical and chemical structures, they both make porous gels. A porous gel acts as a sieve by retarding or, in some cases, completely obstructing the movement of macromolecules while allowing smaller molecules to migrate freely. By preparing a gel with restrictive pore size, the operator can take advantage of molecular size differences among proteins. Because the pores of an agarose gel are large, agarose is used to separate macromolecules such as nucleic acids, large proteins and protein complexes. Polyacrylamide, which makes a small pore gel, is used to separate most proteins and small oligonucleotides. Both are relatively electrically neutral.

Agarose Gels

Agarose is a highly purified uncharged polysaccharide derived from agar. Agarose dissolves when added to boiling liquid. It remains in a liquid state until the temperature is lowered to about 40° C at which point it gels. The pore size may be predetermined by adjusting the concentration of agarose in the gel. Agarose gels are fragile, however, they are actually hydrocolloids, and they are held together by the formation of weak hydrogen and hydrophobic bonds.

Polyacrylamide Gels

Polyacrylamide gels are tougher than agarose gels. Acrylamide monomers polymerize into long chains that are covalently linked by a crosslinker. Polyacrylamide is chemically complex, as is the production and use of the gel.

SDS Gel Electrophoresis

In SDS separations, migration is determined not by intrinsic electrical charge of polypeptides but by molecular weight. Sodium dodecyl sulfate (SDS) is an anionic detergent that denatures secondary and non-disulfide–linked tertiary structures by wrapping around the polypeptide backbone. SDS confers a net negative charge to the polypeptide in proportion to its length. When treated with SDS and a reducing agent, the polypeptides become rods of negative charges with equal "charge densities" or charge per unit length.

Continuous and Discontinuous Buffer Systems

A continuous system has only a single separating gel and uses the same buffer in thetanks and the gel. In a discontinuous system a non-restrictive large pore gel, called a stacking gel, is layered on the top of a separating gel. The resolution obtainable in a discontinuous system is much greater than that obtainable in a continuous one. However, the continuous system is a little easier to set up.

Determining molecular weights of proteins by SDS-PAGE. Run a gel with standard proteins of known molecular weights along with the polypeptide to be characterized. A linear relationship exists between the $\log_{10}$ of the molecular weight of a polypeptide and its R_f. Rf = ratio of the distance migrated by the molecule to that migrated by a marker dye-front. The R_f of the polypeptide to be characterized is determined in the same way, and the $\log_{10}$ of its molecular weight is read directly from the standard curve

Blotting

Blotting is used to transfer proteins or nucleic acids from a slab gel to a membrane such as nitrocellulose, nylon, DEAE, or CM paper. The transfer of the sample can be done by capillary or Southern blotting for nucleic acids (Southern, 1975) or by electrophoresis for proteins or nucleic acids.

Isoelectric Point

There is a pH at which there is no net charge on a protein and this is the isoelectric point (pI). Above its isoelectric point, a protein has a net negative charge and migrates toward the anode in an electrical field. Below its isoelectric point, the protein is positive and migrates toward the cathode.

Isoelectric Focusing

Isoelectric focusing is a method in which proteins are separated in a pH gradient according to their isoelectric points. Focusing occurs in two stages; first, the pH gradient is formed. In the second stage, the proteins begin their migrations toward the anode if their net charge is negative or toward the cathode if their net charge is positive. When a protein reaches its isoelectric point (pI) in the pH gradient, it carries a net charge of zero and will stop migrating.

Two-dimensional gel electrophoresis

Two-dimensional gel electrophoresis is widely used to separate complex mixtures of proteins into many more components than is possible in conventional one-dimensional electrophoresis. Each dimension separates proteins according to different properties.

2.2 MOLECULAR BIOLOGY TECHNIQUES-SOUTHERN BLOTTING, NORTHERN BLOTTING, WESTERN BLOTTING, MICROARRAY TECHNOLOGY

DNA, RNA and proteins are separated by blotting techniques. These are described below:

Southern Blotting

A method, developed genes in a DNA restriction fragment is called as Southern blotting technique. Southern blots can easily provide a physical map of restriction sites within a gene located normally and reveal the number of copies of the gene in the genome, and the degree of similarity of the gene when compared with the other complementary genes. The procedure starts with the digestion of the DNA population by one or many restriction enzymes. Consequently, DNA fragments of unequal length are produced. This preparation is passed through agarose gel electrophoresis which results in the separation of DNA molecules based on their size. DNA restriction fragments present in gel are denatured by alkali treatment. Gel is then put on top of the buffer saturated filter paper. The upper surface of the gel is covered with nitrocellulose filter and overlaid with dry filter paper. The dry filter paper draws the buffer through the gel. The buffer contains single-stranded DNA. Nitrocellulose filter binds DNA fragments strongly. After baking at 80°C, DNA fragments are permanent and fixed to the when come in contact of the nitrocellulose filter. Then the filter is placed in a solution containing radio-labeled RNA or denatures DNA probe of known sequences. These are complementary in sequence to the blot-transferred DNA. The radiolabeled nucleic acid probe hybridizes the DNA on the nitrocellulose filter. The filter is thoroughly washed to remove the probe. The hybridized regions are autoradiographically detected by placing the nitrocellulose filter in contact with a photographic film.

Northern Blotting

The southern blotting technique could not be directly applied to the blot transfer of mRNA separated by gel electrophoresis, because RNA was found not to bind with nitrocellulose, devised a technique in which RNA bands are blot transferred from the gel to chemical reactive paper. The hybridized bands are found by autoradiography.Thus, Alwine's method extends that of the Southern method and tor this reason it has been grunt the jargon Northern blotting. "There is not this northern or western like Southern." These blot transfers are reusable because of the firm covalent bonding of RNA to the reactive paper. Effective in binding the DNA as well Small fragments of DNA can most effectively be transferred to the diazotized paper derivative to nitrocellulose. These techniques were more advanced and demonstrated that mRNA bands can also be blotted directly on nitrocellulose paper and appropriate conditions.

Western Blotting

The western blotting technique to find out the newly encoded protein by a transformed cell. Its working principle lies on antigen-antibody reaction: hence is an immuno detection technique. In this method, radiolabeled nucleic acid probes are not used. This technique follows the following steps: (Extraction of protein from transformed cells. (I) Separation of protein by using SDS-PAGE (sodium dodecyl sulfate polyacrylamide electrophoresis) where SDS acts as a solvent for electrophoresis. Transfer of electrophoresed gel in a buffer at low temperatmre (40° C) for half an hour (h) Blotting of proteins on nitrocellulose filter paper of nitrocellulose filter, Whaiman filter and coarse filter in transfer buffer. So a kind of nitrocellulose filters Whatman filter and coarse filter in transfer buffer. Placing of Whatman filter paper on a cathode plate followed by stack of coarse filter Whatman filter, electrophoresed gel, nitrecellulose filter, Whatman filter paper, coarse filter stack, Whatman filter and anode plate transferred. Putting the complete set up in transfer tank containing sufficient transfer buffer. Application of an electric field (30 V overnight for 5 hours) to cause the migration of proteins from the gel to nitrocellulose filter and binding on its surface. The nitrocellulose filter has exact image of pattern of proteins as present in the gel. This type of blotting is called western blotting. Hybridization of proteins by using radiolabeled antibodies (I123- antibodies) of known structure, isolated from the rabbit. Washing of nitrocellulose filter with a wash solution (Tris-buffered saline + tween 20) to facilitate the removal of unhybridised antibodies. Detection of hybridized sequences by autoradiography. The dots of diagram show the presence of desired protein.

Microarray

Allows simultaneous measurement of the level of transcription for every gene in a genome (gene expression). Differential expression, changes over time. Single microarray can test ~10k genes. Data obtained faster than can be processed. Want to find genes that behave similarly. A pattern discovery problem (Green = repressed and Red = induced).

2.2.1 Complementation Techniques; Polymerase Chain Reaction (PCR), Molecular Markers and Plant Diversity

During the last few years, with the improvement of PCR protocols and due to the availability of automatic thermal cyclers commercially the applications of PCR have increased manifold. However, for the application of PCR, we need a pair of primers which should be based on the knowledge of the nucleotide sequence of DNA to be amplified. Therefore, the non-availability of this information about the DNA segment or gene to be amplified becomes a limitation in the applications of PCR, although this difficulty has been overcome for the study of DNA polymorphism. Some of the applications of PCR technology of random DNA primer are described in this section.

Polymerase chain reaction (PCR)

We discussed the techniques of recombinant DNA and gene cloning, which permit us to obtain an unlimited supply of identical copies of a gene sequence or DNA segment, which is cloned in a prokaryotic a eukaryotic cell with the help of a vector.

This technique, discovered around 1975, proved extremely useful in experiments of molecular biology and genetic engineering and thus became an essential tool in all molecular biology laboratories. In 1985, yet another remarkable tool in molecular biology was discovered, which is known as polymerase chain reaction (PCR) nicknamed now as people's choice reaction.

- The methodology is so important that the prestigious journal Science published from the USA considered PCR as the major scientific development of the year 1989, and had chosen Taq DNA polymerase, the enzyme used in PCR, in the year 1989. Kary Mullis who discovered this Scientific American, how he got the basic idea of this reaction, while driving through the Redwood Mountains of California, USA.
- The polymerase chain reaction is such a powerful technique that may replace completely the gene cloning with vectors in due course of time.
- The DNA replication involves polymerization of nucleotides using a template DNA strand with the help of an enzyme DNA polymerase, but this reaction invariably requires a primer strand to which further nucleotides can be added using the DNA polymerase enzyme. If the primer strand is not available, the reaction cannot proceed.
- In the living cells, this primer strand is not a DNA strand but is a small single-stranded RNA molecule synthesized with the help of RNA polymerase enzyme.
- In PCR, a similar reaction takes place in an 'Eppendorf tube', where the primer strand is added from outside in the form of a deoxyoligonucleotide, and DNA polymerase enzyme is added to help in polymerization.
- An unlimited supply of amplified NA is obtained by repeating the reaction, which is made possible by regular denaturation of freshly synthesized double-stranded DNA molecules by heating it to 90-98°C.

- At this high temperature the two strands separate. Once the double-stranded DNA is made single-stranded by heating up to 90-98°C, the mixture with two primers, recognizing the two strands and bordering the sequence to be amplified is cooled to 40-60̊ C.
- This allows the primers (which are in excess) to bind to their complementary strands through renaturation.
- The presence of Taq DNA polymerase enzyme and all the four essential nucleoside triphosphates allow the synthesis of complementary strands in the usual manner.
- In a thermal cycler this process is automatically repeated 20 and 30 times so that in a single afternoon, a billion copies of the sequence flanked by the left and right primers can be produced.
- In order to continue the synthesis, the temperature of the mixture is alternately increased (for denaturation) and decreased (for renaturation) once every 1-3 minutes (as fixed by the computer device).
- Therefore during temperature rise, the enzymatic activity of DNA polymerase should not be destroyed, otherwise one may have to add a fresh aliquot of the enzyme in each cycle of amplification.
- This became possible only by the discovery of the thermostable enzyme Taq DNA polymerase, isolated from *Thermus aquaticus* growing in hot springs.
- This enzyme acts best at 72°C and the need denaturation temperature of 90°C does not destroy its enzymatic activity.
- Later, other thermostable enzymes like *Pflu* DNA polymerase isolated from *Pyrococcus furiosus* and Vent polymerase isolated from *Thermococcus litoralis* were discovered and were found more efficient.
- These enzymes allowed automation of the entire process and automatic PCR thermal cyclers are now available, which can amplify DNA sequences at a fast speed unattended.
- **As pointed** out earlier, PCR may eventually replace the gene cloning technique. This technology is much easier and requires a much smaller quantity nanogram or ng 10g of DNA. For cloning, DNA is needed, as the starting material, in microgram (ug) quantities.

Different schemes of PCR

- In the previous section, we discussed the basic outline of the PCR technique, outlining how billions of copies of a DNA sequence can be obtained within a few hours, without the use of a vector and a host as done in gene cloning.
- However, there are several variations to this basic PCR technique, which allow varied applications of this technique.

Inverse PCR

- In this technique, the amplification of those DNA sequences takes place, which is away from the primers and not those flanked by the primers.
- For instance, if the border sequences of a DNA segment are not known and those of a vector arc known, there the sequence to be amplified may be cloned in the vector and sequences of the vector may be used as polymerization sequence flanked by the primers and towards the DNA inserted segment.

Anchored PCR

- In the basic PCR technique and the inverse PCR, one has to use two primers representing the sequences lying at the ends of sequences to be amplified.
- But sometimes, we may have knowledge about sequence at only one of the two ends of the DNA sequence to be amplified.
- In such cases, anchored PCR may be used, which will utilize only one primer instead of two primers. In this technique, due to the use of one primer, only one strand will be copied first, after which a poly G tail will be attached at the end of the newly synthesized strand.
- This newly synthesized strand with poly G tail at its 3′ end will then become a template for the daughter strand synthesis utilizing an anchor primer with which a poly C sequence is linked to complement with poly G of the template.
- In the next cycle, both the original primer and anchored primer will be used for gene amplification.

Cloning and expression of the PCR product

- The amplified PCR product can be cloned if restriction sites for specific enzymes are introduced at the 5′ end of each primer.

- These restriction sites do not interfere with amplification and can be used for cloning with a variety of vectors in the usual manner.
- Similarly, incorporation of a T7 promoter at the 5′ end of one primer will allow copies of PCR product on the use of RNA polymerase enzyme to get RNA copies of the PCR product.

Asymmetric PCR for DNA sequencing

- PCR products can also be used as templates for direct DNA sequencing (ligation-mediated PCR - LMPCR).
- Single-stranded DNA can be produced for this purpose using asymmetric PCR, in which the two primers are used in a 100:1 ratio, so that after 20-25 cycles of amplification, one primer is exhausted so that single-stranded DNA is produced in the next 5-10 cycles.
- The morphological markers generally correspond to the qualitative traits that can be scored visually.
- They have been found in nature or as the result of mutagenesis experiments.
- Morphological markers are usually dominant or recessive. Until recently, however, the genetic markers are used to develop maps in plants have been those affecting morphological characters, including genes for dwarfism, albinism and altered leaf morphology etc.
- Scientists have long theorized about the use of genetic maps and markers to speed up the process of plant and animal breeding. In 1923, **Sax** proposed the identification and selection of minor genes of interest by linkage with major genes, which could be scored more easily.
- There are constrainers on the use of morphological markers.
- It causes such large effects on phenotypes that they are undesirable in breeding programs.
- The effects of linked minor genes, making it nearly impossible to identify desirable linkages for selection.
- It is highly influenced by the environment. The morphology phenotype is the result of the genetic constitution and its interaction with the environment.

- Due to varying levels of G × E interaction, it is not appropriate to compare the morphological data of varieties that have been collected across different years and or locations.
- It exhibits dominant intragenic interactions. Therefore homozygous dominant and heterozygous individuals cannot be distinguished.
- The traits show intergenic interactions leading to deviated dihybrid segregation ratios that are not useful for linkage analysis.
- It covers only a limited fraction of the genome of an organism.

There are two types of markers

- Biochemical markers
- Molecular Markers

Biochemical markers

- There are proteins produced by gene expression.
- These proteins can be isolated and identified by electrophoresis and staining.
- Isozymes the different molecular forms of the same enzyme that catalyze the same reaction and proteins.
- They are revealed on electrophoresis through a colored reaction associated with enzymatic activity.
- They are the products of various alleles of one or several genes.
- Monomer and dimer isozymes are most often used because analysis of their segregation is easier. Isozymes are generally codominant.

Moelecular Markers

- The molecular markers are DNA sequences that are readily detected and whose inheritance can be easily monitored.
- The use of molecular markers is based on naturally occurring DNA polymorphism, which forms the basis for designing strategies to exploit for applied purposes.
- A marker must be polymorphic, that chromosome carrying the mutant gene can be distinguished from the chromosome with the normal gene by markers it also carries.

Table 1. Molecular markers

Acronym	**Nomenclature**
RFLP	Restriction Fragment Length Polymorphism
VNTR	Variable Number Tandem Repeat
SSCP	Single-stranded Conformational Polymorphism
STS	Sequence Tagged Sites
STMS	Sequence Tagged Microsatellite sites
RAPD	Random Amplified Polymorphic DNA
AP-PCR	Arbitrarily Primed Polymerase Chain Reaction
DAF	DNA Amplification Finger Printing
SSR	Simple Sequenced Repeats
CAPS	Cleaved Amplified Polymorphic Sequence
MAAP	Multiple Arbitrary Amplicon Prefilling
SCAR	Sequenced Characterized Amplified Region
SNP	Single Nucleotide Polymorphism
SAMPL	Selective Amplification of Microsatellite Polymorphic Loci
ISSR	Inter-Simple Sequence Repeat
ALFP	Amplified Fragment Length Polymorphism
S-SAP	Sequence specific Amplified Polymorphism
RBIP	Retrotransposon-Based Insertional
IRAP	Inter-Retrotranspososon Amplified Polymorphism
REMAP	Retrotransposon-Microsatellite Amplified Polymorphism

The moleculars markers have some desirable properties:

- It must be polymorphic as it is the polymorphism that is measured for genetic diversity studies.
- *Codominant inheritance:* The different forms of a marker should be detectable in diploid organisms to allow discrimination of homo- and heterozygous.
- The marker should be evenly and frequently distributed throughout the genome.

- It should be easy, fast cheap to detect.
- It should be reproducible.
- It is high exchange of data between laboratories. Unfortunately, there is no single molecular marker that meets all these requirements.

Non-PCR based approaches

- Those are based on DNA-DNA hybridization between a DNA or RNA probe and total genomic DNA, e.g., RFLP (Restriction Fragment Length Polymorphism).

PCR based approaches

- Those based on the PCR amplification of genomic DNA fragments: Random amplified polymorphic DNA (PARD), microsatellites or simples sequences repeat polymorphic (SSRP), amplified fragment length polymorphism (AFLP), arbitrarily primed PCR (AP-PCR), SCAR, SNP etc.
- However some assays combine features from both (e.g., retrotransposon based insertional polymorphism-RBIP).

Targetted PCR and Sequencing

Those based on sequencing: sequenced tagged sites (STS), sequence characterized amplified region (SCARs) sequence tagged microsatellites (STMs), cleaved amplified polymorphic sequences (CAPS) etc.

2.2.2 NON-PCR based Approaches

Restriction fragment length polymorphism (RFLP)

Procedure

- Isolate the DNA.
- Cut the DNA into smaller fragments using restriction enzymes (s).
- Separation of DNA fragments by gel electrophoresis.
- Transfering DNA fragments to a nylon or nitrocellulose membrane filter.
- Visualization of specific DNA fragments using labeled probes (Incubation with clined probe and allow for hybridization).

- Southern Hybridization (Autoradiography)
- Analysis of results (RFLP pattern with positive bands).

Uses of RFLP

- It permits direct identification of a genotype or cultivar in any tissue at any developmental stage in an environment independent manner.
- RFLPs are codominant markers enabling heterozygotes to be distinguished from homozygous.
- It has discriminating power that can be at the species/ population (single locus probes).
- The method is simple as no sequence-specific information is required.
- Indirect selection using qualitative traits.
- Indirect selection using quantitative trait loci.

2.2.3 PCR Based Approaches

Random amplified polymorphic DNA (RAPD) markers

Procedure

- Isolation of DNA.
- Keep the tubes in PCR thermocycler (Taq DNA polymerase, primer, dNTPs and buffer).
- Denature the DNA (94°C, 1 min).
- DNA strands separated.
- Annealing of primer (36°, 2 min).
- Primer annealed to template DNA strands (DNA synthesis 72°C, 1.5 min).
- Complementary strand synthesis (35 to 45 cycles).
- Amplified products are separated by gel electrophoresis.

Application of RAPD

- Construction of genetic maps.
- Mapping of traits.
- Analysis of the genetic structure of populations.
- Fingerprinting of individuals.

- Targeting markers to specific regions of the genome.
- RAPDs have been utilized for the identification of somatic hybrids.
- RAPDs have been recommended as a useful system for the evaluation and characterization of genetic resources.
- Some success has already been achieved in the characterization of collections from tree genera Gliricidia and Thenobroma.

Limitations of RAPD

RAPD polymorphisms are inherited as dominant-recessive characters. This causes a loss of information relative to markers that show codominance.

RAPD primers are relatively short, a mismatch of even a single nucleotide can often prevent the primer from annealing, hence there is a loss of band.

The production of non-parental bands in the offspring of known pedigrees warrants its use with caution and extreme care.

RAPD is sensitive to changes in PCR conditions, resulting in changes to some of the amplified fragments.

2.2.4 Amplified Fragment Length Polymorphism (AFLP).

Procedure

High quality genomic DNA is cut with restriction enzymes (generally by two enzymes) and double-stranded (ds) oligonucleotide adaptors are ligated to the ends of the DNA fragments.

Selective amplification of sets of restriction fragments is usually carried out with 32P labeled primers designed according to the sequence of adaptors plus 1-3 additional nucleotides. Only fragments containing the restriction site sequence plus the additional nucleotides will be amplified.

Gel analysis of the amplified fragments. The amplification products are separated on highly resolving sequencing gels and visualized using autoradiography.

Fluorescent or silver staining techniques can be used to visualize the products in cases where radiolabeled nucleotides are not used in the PCR.

Advantages of AFLP

This technique is extremely sensitive.

It has high reproducibility, rendering it superior to RAPD.

It has wide-scale applicability, proving extremely proficient in revealing diversity.

It discriminates heterozygotes from homozygotes when a gel scanner is used.

It permits the detection of restriction fragments in any background or complexity, including pooled DNA samples and cloned DNA segments.

It is not a single fingerprinting technique but it can be used for mapping also.

Disadvantages

It is highly expensive and requires more DNA that is needed in RAPD (1 µg per reaction).

It is technically more demanding than RAPDs as it requires the experience in sequencing gels.

AFLPs are expensive to generate as silver staining, fluorescent dye or radioactivity detects the bands.

2.2.5 DNA Polymorphism using PCR

Using PCR, DNA polymorphism can be studied sequence at loci with (using primers based on known sequence) or at random site primers having unknown but random DNA sequences these are called random primers). In either case, DNA may be amplified from one or more genomic DNAs and subjected to digestion with one or mo to electrophoresis and the restriction patterns may be photographed t reveal polymorphism. Sequences of the prolamin gene and phytochrome gene have been used for the study of polymorphism for these genes in different species of Oryza (wild and cultivated rice). Similarly, storage protein genes in wheat and many genes in other animal and plant species have been analyzed by specific restriction endonuclcases. The digested DNA may be subjected used, for the study of DNA polymorphism. The non-availability of sequences of specific genes for designing primers for DNA amplification has been overcome by a variety of approaches including the following (i) In the case of multigene families like those for RNA, tRNA and heat shock proteins, there are conserved consensus sequences known, which are used for designing universal primers. Similarly for amplification of antibody genes also, universal primers have been used in PCR (ii) Gene bank Sequence Data have also been used for designing primers for amplification of microsatellites like poly (A), poly (G), pol (TC) and poly (TG), etc.

RAPD Markers

The most important use of PCR for e study of DNA polymorphism, however, involves the use of random prime giving Random Amplified Polymorphic DNA. This technique is considered simpler and offers several advantages over more commonly used techniques known as RFLP and RAPDs have actually been used for a study of polymorphism chromosome mapping in maize, soybean, mouse, humans, etc.

VNTR loci

Alec Jeffreys and his colleagues in the U.K suggested the use of PCR for the study of DNA polymorphism at VNTR (variable number tandem repeats, also known as 'minisatellites loci in humans. Primers were developed for the conserved flanking regions of VNTR loci so that all available VNTR loci could be amplified. The PCR products differ according to the number of repeat units present in the different VNTR loci and this variation is visualized in the positions occupied by different bands after electrophoresis.

SSR

The VNTR loci discussed above consist of repeat electrophoresis units in the range of 11-60 base pairs in length. A much simpler type of repeat units exists in the form of dinucleotides, trinucleotides or tetra nucleotides, e.g. (CA)n, (GT)n (AAT)n or (AGAT)n. These repeat units were described as microsatellites, but later also called short tandem repeats or simple sequence repeats (SSR). Some of these SSR loci, such as (CA)n or (GT) n occur in the human with n varying from 10 to 60. The DNA sequences flanking SSRs are conserved, allowing the selection of PCR primers that will amplify the intervening SSR in all genotypes of the target species PCR reaction includes a small amount of 3P labeled nucleotide (only one of the four nucleotides is labeled) or it may include primers, one or both of them being end-labeled. This allows visualization of PCR products using autoradiography after electrophoresis on a standard sequencing gel. Variation in the length of products is a function of the number of

SSR units. The principle underlying is detection polymorphism using two parents and their F1 hybrid. Markers resulting from SSR length polymorphisms are placed on genetic maps relation RFLP's, RAPD's, VNTR's, other SSR's and even the morphological markers.

A comparison of very limited sequence data from plants (including algae) those of vertebrates (including humans) indicated that SSR's plants are found with the same frequency as in vertebrates. Recently DNA sequence data for five tree species and Zea mays have been screened for the presence of (AC)n and (AG)n and the frequency in these six species was found to be close to one microsatellite (SSR) per 5x3x 10 bp. Other plant and animal species have also been examined for the presence of a number of SSR's and a wide occurrence of these markers in plants and animals preparing saturated was suggesting their utility in preparing saturated molecular maps. Molecular mapping using PCR Genetic and physical chromosome maps have also been prepared, both in plants and animals, using PCR. Once RAPD's are detected, recombinant inbreeds (RI's) or F2/backcross segregating populations, or doubled haploids (DH) derived from haploids (in plants) can be used to detect linkage and recombination frequencies. The molecular marker which cosegregates RI's or F,'s or DH's suggests linkage and the frequency of cosegregation among RIs or Fa/backcross plants will be a measure of the strength of linkage.

- The molecular markers identified through PCR can also be assigned to specific chromosomes using aneuploids or alien addition lines e.g. wheat-barley additions -21 pairs of wheat chromosomes +1 pair of a specific barley chromosome). A number of molecular markers have been mapped using PCR in maize, soybean, barley, mouse, humans, etc.

2.2.6 Sequence Tagged Microsatellite Sites (STMS) or SSR loci

It has also been examined and mapped using PCR. The gene sequence databases are screened for a microsatellite and the flanking sequences are used for designing primers for PCR. Several STMS or SSR loci using this technique have been mapped in mice, where as many as 200 microsatellites have been detected in the genome database. Besides SSR's or STMS's, other markers including sequence tagged sites or STS's (a short DNA sequence in a molecular genomic DNA marker) and expression sequence tags or ESTs (a short DNA sequence from a cDNA clone representing an expressed gene) have also been for genetic mapping.In both these cases, PCR can be profitably utilized for screening RI's (mouse), somatic hybrid cell lines (humans) or F2 individuals (in plants).This will allow detection of linkage and calculation of recombination frequencies leading to the preparation of genetic maps. Complete molecular maps of several human chromosomes could be prepared using STS's and PCR.

Gene Tagging Using PCR

PCR has also been utilized for developing molecular m to specific genes of economic importance. For instance in tomato 144 random primers were developed to produce 625 PCR products from a set at markers closely link instance in tomato, near isogenic lines (NIL's, which differ only for the presence and absence of a specific gene) for *Pseudomonas* resistance. At least three PCR products were present in one and not in the other of a pair NIR's showing linkage with *Pseudomonas* resistance gene.

2.2.7 Plant diversity

Variation in the genome and epigenetic activities such as gene silencing and altered chromatin structure within or across a crop species is referred to as "plant genetic diversity". This diversity provides the reason why for e.g., some crop varieties are tall and some are short, why some species can survive insect attacks while other closely related ones cannot, etc. For millennia farmers have taken advantage of genetic diversity by selecting the most favorable material for the following reason. Crop breeders need plant genetic diversity in their breeding programs to achieve genetic gain. They can exploit the allelic diversity present in the breeding materials by chromosome reassortment and/or recombination, and they can import new alleles by hybridizing elite germplasm with exotic materials. The amount of genetic diversity present in a particular collection of individuals can be divided from comparisons between DNA fingerprints. Such estimates of the genetic diversity present can be used to define the phylogenetic relationship among germplasm is particularly useful as it indicates the combinations of breeding parents, which is most likely to offer a high degree of allelic complementary contributing to a large potential for genetic gain in the offspring. Thus one of the most powerful applications of markers in breeding is the guiding of parental combination before making crosses. Although most fingerprinting assays are based on anonymous markers whose function is not known. The increasing pace of gene diversity and elaboration of gene function is driving a shift towards markers associated with specific genes. The advantages of such markers is that they can allow the selection of parental lines based on specific traits. Fingerprinting is also valuable for establishing varietal identity and purity and variety protection which are real issues for the breeder both in the commercial and the public sector.

REFERENCES

1. Brendan P. Lucey, Walter A. Nelson-Rees, Grover M. Hutchins; Henrietta Lacks, HeLa Cells, and Cell Culture Contamination. Arch Pathol Lab Med 1 September 2009; 133 (9): 1463–1467.
2. Harris AR, Peter L, Bellis J, Baum B, Kabla AJ, Charras GT (October 2012). "Characterizing the mechanics of cultured cell monolayers". Proceedings of the National Academy of Sciences of the United States of America. 109 (41): 16449–16454.
3. Lagziel S, Lee WD, Shlomi T (April 2019). "Inferring cancer dependencies on metabolic genes from large-scale genetic screens". BMC Biology. 17 (1): 37.
4. Nguyen HT, Geens M, Spits C (2012). "Genetic and epigenetic instability in human pluripotent stem cells". Human Reproduction Update. 19 (2): 187–205.
5. Taylor MW (2014). "A History of Cell Culture". Viruses and Man: A History of Interactions. Cham: Springer International Publishing. pp. 41–52.

CHAPTER 3

BASIC MOLECULAR BIOLOGY TECHNIQUES

INTRODUCTION

Molecular Biology methods used to study the molecular basis of biological activity. Most commonly used methods are protein methods, immunostaining methods, nucleic acid methods. These methods used to explore cells, their characteristics, parts, and chemical processes, and pays special attention to how molecules control a cell's activities and growth.

Molecular Biology Techniques include DNA cloning, cut and paste DNA, bacterial transformation , transfection, chromosome integration, cellular screening, cellular culture, extraction of DNA, DNA polymerase DNA dependent, reading and writing DNA, DNA sequencing, DNA synthesis, molecular hybridization, rewriting DNA: mutations, random mutagenesis, point mutation, chromosome mutation. Most important techniques are Polymerase Chain Reaction (PCR), Expression cloning, Gel electrophoresis, Macromolecule blotting and probing, Arrays (DNA array and protein array).

3.1 STRUCTURE AND FUNCTION OF NUCLEIC ACIDS

Nucleic acids are the most important macromolecules for the continuity of life. They carry the genetic blueprint of a cell and carry instructions for the functioning of the cell. The two main types of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).

3.1.1 The Structure of Deoxyribonucleic Acid

Deoxyribonucleic acid (DNA) is one of the most important molecules in living cells. It encodes the instruction manual for life. Genome is the complete set of DNA molecules within the organism, so in humans this would be the DNA present in the 23 pairs of chromosomes in the nucleus plus the relatively small mitochondrial genome. Humans have a diploid genome, inheriting one set of chromosomes from each parent. A complete and functioning diploid genome is required for normal development and to maintain life.

Discovery and Chemical Characterization of DNA

DNA was discovered in 1869 by a Swiss biochemist, Friedrich Miescher. He wanted to determine the chemical composition of leucocytes (white blood cells), his source of leucocytes was pus from fresh surgical bandages. Although initially interested in all the components of the cell, Miescher quickly focused on the nucleus because he observed that when treated with acid, a precipitate was formed which he called 'nuclein'. Almost all molecular bioscience graduates would have repeated a form of this experiment in laboratory classes where DNA is isolated from cells. Miescher, Richard Altmann and Albrecht Kossel further characterized 'nuclein' and the name was changed to nucleic acid by Altmann. Kossel went on to show that nucleic acid contained purine and pyrimidine bases, a sugar and phosphate. Work in the 1930s from many scientists further characterized nucleic acids including the identification of the four bases and the presence of deoxyribose, hence the name deoxyribonucleic acid (DNA). Erwin Chargaff had found that DNA molecules from a particular species always contained the same amount of the bases cytosine (C) and guanine (G) and the same amount of adenosine (A) and thymine (T). So, for example, the human genome contains 20% C, 20% G, 30% A and 30% T.

DNA is a polymer made of monomeric units called nucleotides (Figure 1A), a nucleotide comprises a 5-carbon sugar, deoxyribose, a nitrogenous base and one or more phosphate groups. The building blocks for DNA synthesis contain three phosphate groups, two are lost during this process, so the DNA strand contains one phosphate group per nucleotide.

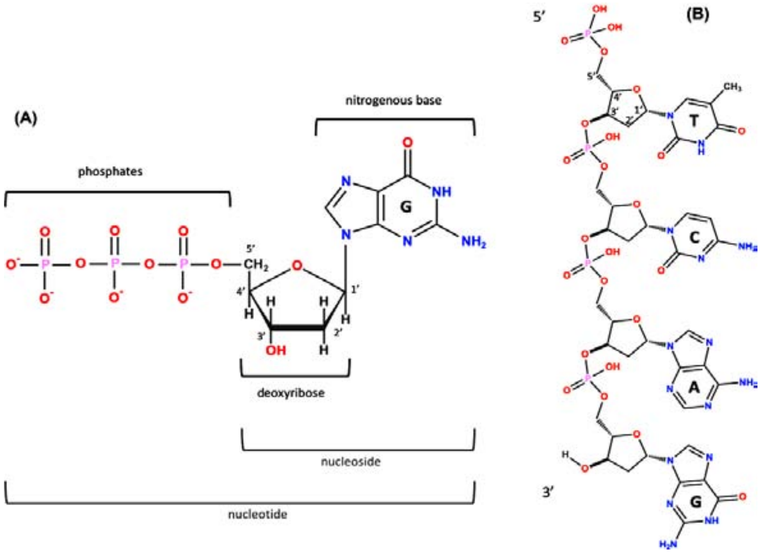


Figure 1. The structure of DNA. (A) A nucleotide (guanosine triphosphate). The nitrogenous base (guanine in this example) is linked to the 1′ carbon of the deoxyribose and the phosphate groups are linked to the 5′ carbon. A nucleoside is a base linked to a sugar. A nucleotide is a nucleoside with one or more phosphate groups. (B) A DNA strand containing four nucleotides with the nitrogenous bases thymine (T), cytosine (C), adenine (A) and guanine (G) respectively. The 3′ carbon of one nucleotide is linked to the 5′ carbon of the next via a phosphodiester bond. The 5′ end is at the top and the 3′ end at the bottom.

There are four different bases in DNA, the double-ring purine bases: adenine and guanine; and the single-ring pyrimidine bases: cytosine and thymine (Figure 1B). The carbon within the deoxyribose ring are numbered 1′ to 5′. Within each monomer the phosphate is linked to the 5′ carbon of deoxyribose and the nitrogenous base is linked to the 1′ carbon, this is called an N-glyosidic bond. The phosphate group is acidic, hence the name nucleic acid. In the DNA chain (Figure 1B), the phosphate residue forms a link between the 3′-hydroxyl of one deoxyribose and the 5′-hydroxyl of the next. This linkage is called a phosphodiester bond. DNA strands have a ‘sense of direction’. The deoxyribose at the top of the diagram in Figure 1B is not linked to another deoxyribose; it terminates with a 5′ phosphate group. At the other end the chain terminates with a 3′ hydroxyl.

DNA is the Genetic Material

Although many scientists, including Miescher, had observed that prior to cell division the amount of nucleic acid increased, it was not believed to be the genetic material until the work of Fredrick Griffith, Oswald Avery, Colin MacLeod and Maclyn McCarty. In 1928, Griffith showed that living cells could be transformed by extracts from heat-killed cells and that this transformation had the potential to permanently change the genetic makeup of the recipient cell. Griffith was working with two strains of the bacterium Streptococcus pneumoniae. The encapsulated so-called S strain is virulent, whereas the non-capsulated R strain is nonvirulent. If the S strain is injected subcutaneously into mice, the mice die, whereas, if either live R strain is injected or heat-killed S strain is injected, the mouse lives. However, if a mixture of live R strain and heat-killed S strain is injected into a mouse, the mouse will die, and live S strain can be isolated from the blood.

So, in the Griffith experiment a component of the heat-killed S strain is transforming the R strain. In 1944, Avery, MacLeod and McCarty went on to show that it was DNA that could transform the avirulent bacterium.

They isolated a crude DNA extract from the S strain and destroyed any protein, lipid, carbohydrate and ribonucleic acid (RNA) component and showed that this purified DNA could still transform the R strain. However, when the purified DNA was treated with DNAse, an enzyme that degrades DNA, transformation was lost.

Alfred Hershey and Martha Chase confirmed that DNA was the genetic material. They used a virus that infects bacteria called a bacteriophage. The bacteriophage contains a protein capsid surrounding a DNA molecule. They showed that when bacteriophage T2 infects *Escherichia coli*, it is the phage DNA, not protein, that enters the bacterial cell.

Determining the Structure of DNA

Once it had been shown that DNA was the genetic material, there was a race to determine the three-dimensional structure of the DNA molecule. At King's College London, Rosalind Franklin and Maurice Wilkins, having obtained data using X-ray diffraction, had proposed that DNA had a helical structure and Franklin had obtained a particularly good X-ray diffraction pattern. In Cambridge, James Watson and Francis Crick used model building together with data from a variety of sources including Franklin's X-ray diffraction pattern and Chargaff's base composition data to work out the now well-known double helix structure of DNA. Their work was published in *Nature* in 1953. The Watson–Crick structure is shown in Figure 2A.

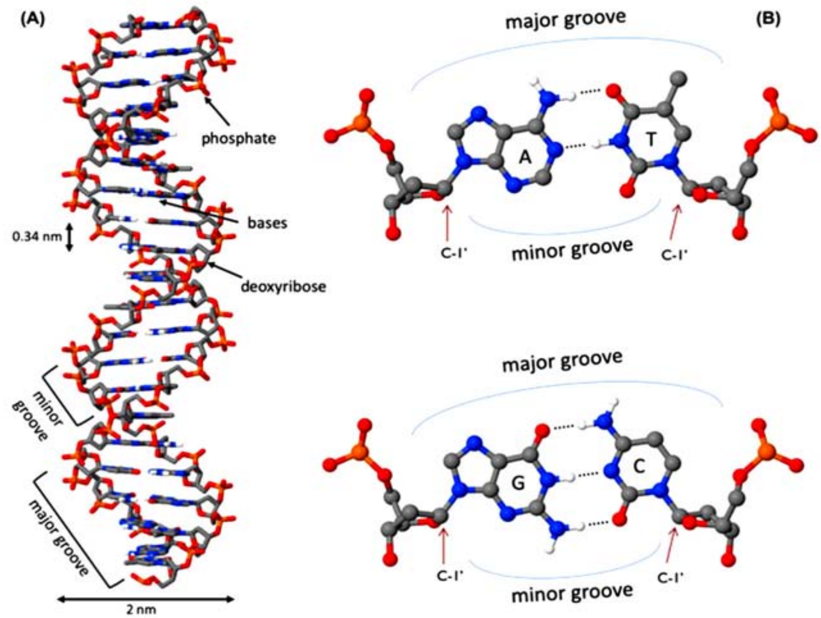


Figure 2. DNA structure (A) The DNA double helix, with the sugar phosphate backbone on the outside and the nitrogenous bases in the middle. (B) An A:T and a G:C base pair with the C1 of the deoxyribose indicated by the arrow. Note that the C1 of the deoxyribose is in the same position in all base pairs. In this figure, the atoms on the upper edge of the base pair face into the major groove and those facing lower edge face into the minor groove. The hydrogen bonds between the base pairs are indicated by the dotted line.

DNA is a two-stranded helical structure, the two strands run in opposite directions. In Figure 2A, one strand is running 5′ to 3′ top to bottom, whereas the other strand is running 3′ to 5′ top to bottom. The helix is right-handed which means that if you are looking down the axis, the helix turns clockwise as it gets further away from you. The two chains interact via hydrogen bonds between pairs of bases with adenine always pairing with thymine, and guanine always pairing with cytosine. The Watson–Crick structure therefore accounts for and explains the Chargaff data which showed that there was always an equal amount of C and G and of A and T. The regular nature of the double helix comes about because the distance between the 1′ carbon of the deoxyribose on one strand and 1′ carbon of the opposite deoxyribose is always the same irrespective of the base pair (Figure 2B). The 1′ carbons of the deoxyribose opposing nucleotides do not lie directly opposite each other on the helical axis, this means that the two sugar–phosphate backbones are not equally spaced along the helical axis resulting in major and minor grooves.

The diameter of the helix is 2 nm, adjacent bases are separated by 0.34 nm (0.34×10^{-9} m) and related by a rotation of 36°, this results in the helical structure repeating every 10 residues. DNA molecules are normally very long and the sequence of bases along the DNA chain is not restricted. For example, the genome of the bacterium *E. coli* is a single circular chromosome which contains 4.6 million base pairs (4.6×10^6 bp), this is therefore 1.6 mm long ($4.6 \times 10^6 \times 0.34 \times 10^{-9}$ m). The human genome is made up of 24 distinct chromosomes, chromosomes 1–22 and the X and Y chromosomes present in the nucleus plus mitochondrial DNA. The nuclear chromosomes vary in size from approximately $50–250 \times 10^6$ bp, the mitochondrial DNA is 17×10^3 bp. The total length of a haploid human genome is 3×10^9 bp. Within a single human diploid cell, which contains 23 chromosome pairs there is 2 m of DNA. Based on the assumption that humans contain 3 trillion cells with a nucleus, if all the DNA from a single human individual was put end to end, it would reach to the sun and back approximately 20 times.

RNA

Another important class of nucleic acids is RNA, the roles of RNA molecules in the cell will be discussed below. Chemically RNA is similar to DNA, it is a chain of similar monomers. The building blocks are nucleotides containing the 5-carbon sugar ribose, a phosphate and a nitrogenous base. The phosphate is attached to the 5′ carbon of the ribose and the nitrogenous base to the 1′ carbon (Figure 3). RNA contains four bases adenine, guanine, cytosine and uracil. RNA is more labile (easily broken down) than DNA and most RNA molecules do not form stable secondary structures, some notable exceptions will be discussed below. The properties of RNA make it ideal as a genetic messenger during protein synthesis, the idea of this genetic messenger, mRNA, was proposed by François Jacob and Jacques Monod.

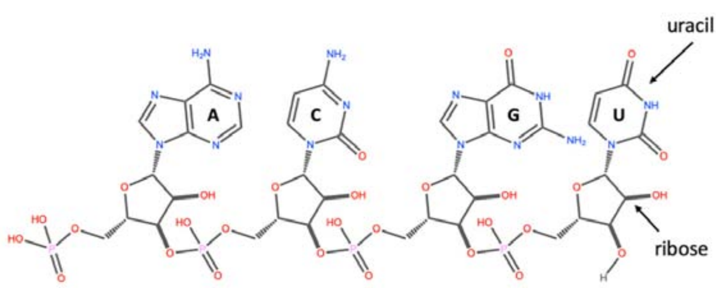


Figure 3. The structure of RNA. An RNA strand containing the four nucleotides with the nitrogenous bases: adenine (A), cytosine (C), guanine (G) and uracil

(U) respectively. The 3′ carbon of the ribose of one nucleotide is linked to the 5′ carbon of the next via a phosphodiester bond. The 5′ end on the left and the 3′ end on the right.

Packaging of DNA into Eukaryotic Cells

DNA has to be highly condensed to fit into the bacterial cell or eukaryotic nucleus. In eukaryotes, histone proteins are used to condense the DNA into chromatin. The basic structure of chromatin is the nucleosome, a nucleosome contains DNA wrapped almost two times around the histone octamer (comprising two copies each of the histone proteins H2A, H2B, H3 and H4) (Figure 4). Further levels of compaction are required to fit the DNA into the nucleus (Figure 4), the nucleosomes are folded upon themselves to form the 30-nm fiber, this is then folded again to form the 300-nm fiber and during mitosis further compaction can occur forming the chromatid which is 700 nm in diameter.

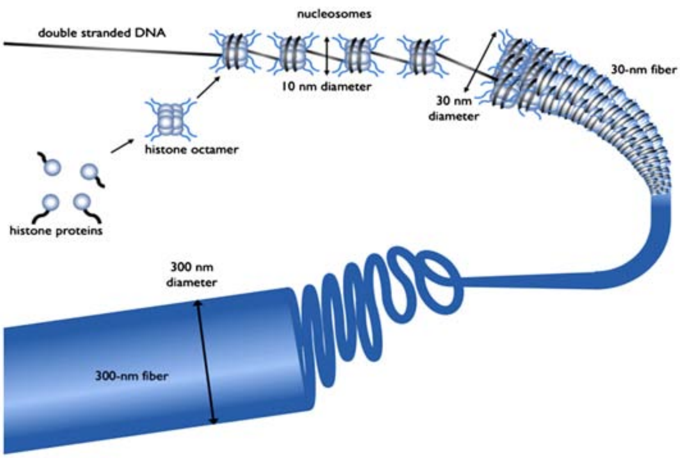


Figure 4. The different levels of chromatin structure. Histone proteins (H2A, H2B, H3 and H4) associate to form a histone octamer. Approximately 147 bp of DNA wraps around histone octamer to form a nucleosome, generating a 'beads on a string' structure, the nucleosome together with histone H1 condense into the 30-nm fibre, there is further condensation to form the 300-nm fibre. During mitosis there is further compaction

Processes such as DNA replication and DNA transcription need to occur in the chromatin environment and because of the level of compaction, this acts as a barrier to proteins that need to interact with DNA. Therefore, chromatin structure plays an important role in processes such as regulation of gene expression in eukaryotes. DNA and the histone proteins can be chemically modified, these are called epigenetic modifications as they do not change the DNA sequence, however, they can be passed on during cell division and to subsequent generations, a process known as epigenetic inheritance. As these epigenetic modifications can alter the chromatin structure they regulate gene transcription and can affect the phenotype. Epigenetics plays key roles in many processes, including development, cancer and behavior and addiction. This will be discussed further later in this article.

Nuclear organization plays an important role in many biological processes including regulation of gene transcription. In recent years the development of several techniques, including microscopy, have allowed us to gain an understanding of the way the genome is organized in 3D. Individual chromosomes are not randomly spaced within the nucleus; each chromosome has a distinct territory. Actively transcribed regions from different chromosomes are often close to each other and near the interior of the nucleus, whereas, inactive genes are on the periphery or near a special area called the nucleolus where ribosomal RNA is transcribed.

3.1.2 DNA Replication

Whenever a cell divides there is a need to synthesize two copies of each chromosome present within the cell. For example in a human, prior to cell division, all 23 pairs of chromosomes need to be replicated to form 46 pairs, so that following cell division each daughter cell has a full complement (23 pairs) of chromosomes. The structure of DNA gives us a clue to how it is replicated, this was eloquently postulated by Watson and Crick in their 1953 paper: "It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material". Each strand can act as a template for the synthesis of the complementary strand, so the replication machinery would 'unzip' the double helix and read along the two existing 'parent' strands, synthesizing a complementary new 'daughter' strand with A opposite T, C opposite G etc. This is described as semi-conservative, since each 'new' double-stranded DNA molecule has one original parent strand and one newly made daughter 'strand'.

The evidence that DNA replication was semi-conservative came from an elegant experiment completed by Matthew Meselson and Franklin Stahl. They labelled the parental DNA with a heavy isotope of nitrogen (^{15}N) by growing bacteria in a growth medium that contained $^{15}NH_4Cl$. They then grew the bacteria, in a medium that contained $^{14}NH_4Cl$, in conditions such that any newly synthesized DNA would contain ^{14}N. Since DNA replication is semi-conservative, after one round of DNA replication, each cell would have a DNA molecule that contains one 'old' parental strand labelled with ^{15}N and one 'new' daughter strand labelled with ^{14}N. This was shown by analyzing the density of the DNA using density-gradient centrifugation. As predicted, they observed that the new daughter DNA molecule had a density consistent with the fact that it contained both ^{15}N and ^{14}N and that this daughter DNA contained one strand with ^{15}N and another strand with ^{14}N.

DNA Polymerase and DNA Synthesis

The enzyme, DNA polymerase, is responsible for DNA synthesis. DNA polymerase is a template-driven enzyme, so it will use the parental DNA strand as a template. It cannot synthesize DNA in the absence of a template. In addition, it will only add nucleotides on to the 3′ end of an existing nucleic acid chain. The building blocks for DNA synthesis are deoxynucleoside triphosphates (dATP, dTTP, dCTP and dGTP). During DNA synthesis, the base within the incoming deoxynucleoside triphosphate pairs with the complementary base on the template strand, a phosphodiester bond is formed between the 5′ phosphate on the incoming nucleotide and the free 3′ hydroxyl on the existing nucleic acid chain; pyrophosphate is released (Figure 5).

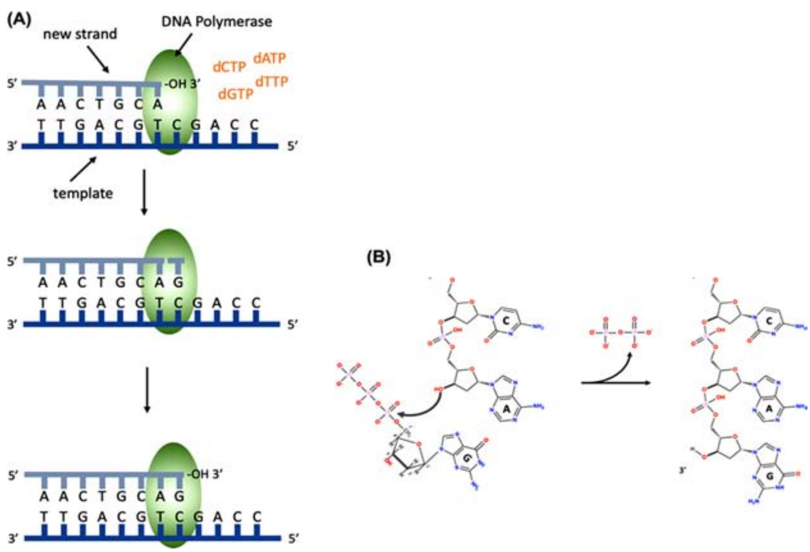


Figure 5. DNA synthesis. (**A**) DNA polymerase binds the template DNA and the new strand. The next nucleotide to be added to the 3′ end of the growing

chain will contain guanine (G), this is complementary to the C on the template strand. DNA polymerase catalyses the formation of a phosphodiester bond. (**B**) The chemical reaction during the formation of a phosphodiester bond, showing the addition of a nucleotide containing guanine and the release of pyrophosphate.

Pyrophosphate is the two phosphate residues within the deoxynucleoside triphosphate building block that are not incorporated into the DNA chain. DNA polymerase synthesizes DNA in the 5′ to 3′ direction, because it can only add nucleotides on to the 3′ end of the chain. DNA polymerase has proofreading activity, so after the phosphodiester bond has been formed, the base pairing is checked and if a nucleotide with an incorrect base has been added, DNA polymerase will remove the nucleotide using a 3′ to 5′ exonuclease activity.

Exonucleases are enzymes that can remove nucleotides from the ends of a DNA molecule, 3′ to 5′ exonucleases remove nucleotides from the 3′ end of a DNA molecule and therefore can remove the last nucleotide that was added during DNA replication. This is analogous to using the delete key to remove a letter that you have typed incorrectly before adding the correct one and continuing typing.

DNA polymerase requires a short double-stranded region with a free 3′ hydroxyl in order to start making a copy of the template; this ensures that DNA is synthesized in a controlled way. Initiation of DNA synthesis uses a small RNA primer (8–12 bases) made by the enzyme primase. DNA polymerase will then extend from the primer copying the template and synthesizing the daughter DNA strand. This means that when DNA synthesis first starts each DNA molecule actually contains a small piece of RNA at its 5′ end.

The Origin of Replication and the Replisome

A large multiprotein complex, called the replisome, is responsible for DNA replication. In prokaryotes, two replisomes form at a specific point on the chromosome called the Origin of Replication (*ori*).

The DNA in this region will be opened up, 'unzipped' so that the replication machinery can gain access to single-stranded parental DNA, which will act as template for synthesis of the new daughter strands. The two replisomes then travel in opposite directions around the circular prokaryotic chromosome, each replisome forming a replication fork, a schematic representation of one replication fork is shown in Figure 6.

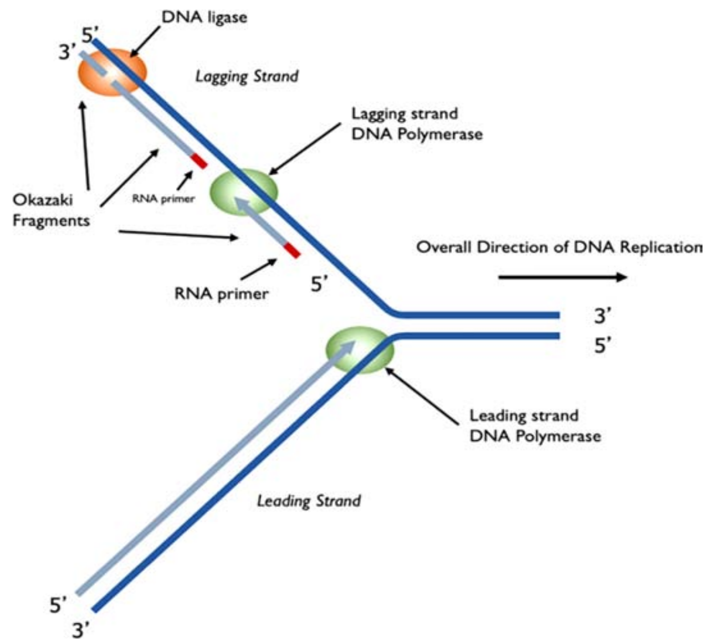


Figure 6. DNA synthesis at a replication fork. A single replication fork showing the leading and lagging strands. The leading strand is synthesized continuously, reading the template 3′ to 5′, synthesizing DNA in the 5′ to 3′ direction. The lagging strand is synthesized discontinuously, in short Okazaki fragments (1000 bases in prokaryotes and 100 bases in eukaryotes).

The Replication Fork

Within the replication fork, on the so-called leading strand, DNA polymerase moves 3′ to 5′ with respect to the template and synthesizes DNA in the 5′ to 3′ direction as it moves in the same direction as the replication fork. Although overall the lagging strand is synthesized in the 3′ to 5′ direction, it is actually synthesized discontinuously in small segments called Okazaki fragments, which are synthesized 5′ to 3′ (Figure 6). Each Okazaki fragment will be started with an RNA primer and is synthesized in the opposite direction to the movement of the replication fork. In prokaryotes, Okazaki fragments are 1000–2000 bases in length. In Figure 6 you will see that the DNA polymerase synthesizing the Okazaki fragment will eventually reach the primer for the previous Okazaki fragment. When this happens the primer for the previous fragment is removed by a DNA polymerase using 5′ to 3′ exonuclease activity. DNA polymerase then replaces the missing nucleotides by adding them to the 3′ end of the last Okazaki fragment. When all the primer has been removed, there will be two DNA strands adjacent to each other but not joined by a phosphodiester bond, these two strands are joined together by the enzyme DNA ligase.

The replisome contains a number of other important proteins required for DNA replication. The double-stranded DNA needs to be separated, 'unzipped', by a helicase to generate the single-stranded DNA templates for DNA polymerase. As the replication fork moves along the helical DNA, the coils in the DNA in front of the fork become compressed so the DNA is described as being overwound; a topoisomerase is required to 'relax' it by remove the over-winding. Single-stranded binding proteins (SSBs) bind the lagging strand template to stabilize and protect the single-stranded DNA.

The two replication forks that form at the *ori* will move in opposite directions around the circular prokaryotic genome until they reach the terminator sequence, *ter*, which is on the opposite side of the genome compared with the *ori*, i.e. it is at 6 o'clock compared with 12 o'clock. This results in the complete replication of the genome. Once DNA replication has been completed a post-replication DNA repair process will correct errors that were not corrected by the proofreading activity of DNA polymerase. The fidelity of DNA replication is extremely high, resulting in an error rate of 1 mistake per 10^9–10^{10} nucleotides added.

DNA Replication in Eukaryotes

DNA replication is essentially the same in eukaryotes and prokaryotes. In both cases two replisomes form at an *ori* and generate two replication forks moving in opposite directions away from the origin. In each replication fork there are leading and lagging strands. There are two major differences. The first is that, due to the larger genome size, each chromosome has multiple origins of replication, so there will be a large number of replication forks on each chromosome.

The second difference is that, with the exception of mitochondrial DNA, eukaryotic chromosomes are linear and this results in an issue because of lagging strand synthesis. Replication of a linear chromosome results in shortening of one 5′ end of each daughter DNA molecule. This is because when the primer required for the last Okazaki fragment is removed, DNA polymerase cannot fill the gap (Figure 7A). Repeated rounds of DNA replication results in shorter and shorter DNA molecules. If this is not corrected, eukaryotes would have become extinct as their chromosomes get shorter with each generation. Eukaryotes have a mechanism to preserve the ends of chromosomes when it counts; that is in the gametes.

The terminal ends of chromosomes, telomeres, contain a highly repeated sequence, for example, in humans the sequence TTAGGG is repeated in tandem 100 to over 1000-times. Repeated rounds of DNA replication will result in the shortening of these telomeric sequences that is the number of repeats will reduce. Telomerase, an RNA containing enzyme, can add additional copies of the repeat sequence to the 3′ end, replacing those lost during DNA replication (see Figure 7).

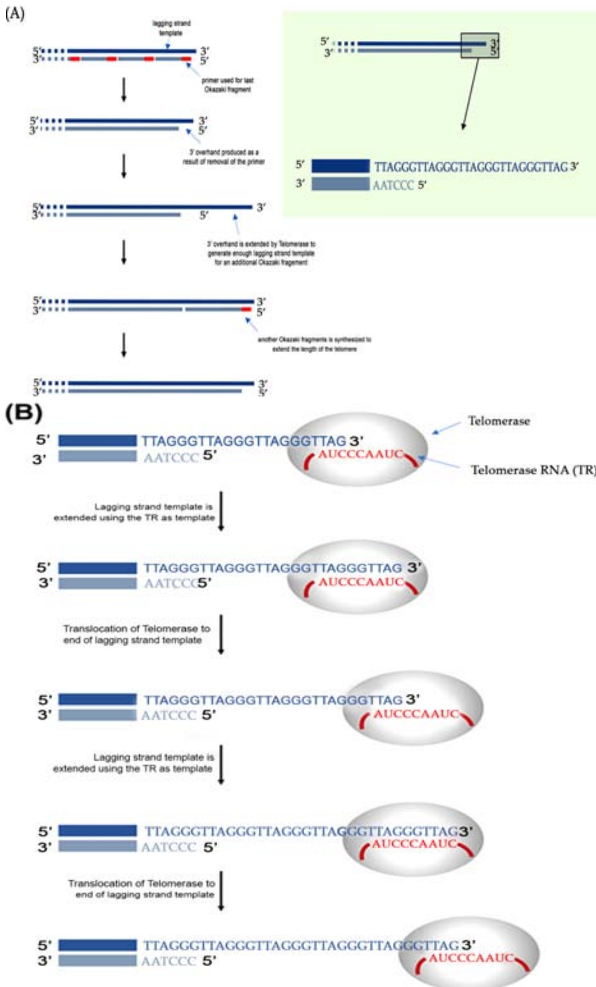


Figure 7. Telomeres and telomerase. (A) Following DNA replication and removal of the primer for the last Okazaki fragment of the lagging strand, there will be a region at the 3′ end that is not base paired, called a 3′ overhang. (B) Telomerase binds and uses the RNA it contains to act as a template to extend the 3′ overhang. This extends the 3′ end sufficiently for a new RNA primer to bind and the final Okazaki fragment to be made.

This actually extends the 3′ end of the telomere rather than extending the 5′ that is initially lost during DNA replication. The RNA sequence within telomerase is complementary to the 3′ telomeric sequence and so can bind and act as a template for synthesis of a short DNA sequence. Telomerase then moves along the newly synthesized strand and the process is repeated. Multiple rounds of elongation and translocation ultimately results in the 3′ end being extended so that it is long enough for it to act as template for synthesis of another Okazaki fragment, hence extending both strands of the telomere. Only germ cells and a few other actively dividing cells (e.g. haematopoietic cells) have sufficient levels of telomerase activity to counteract the loss of repeat sequences during DNA replication. At birth, telomeres are over 10000 base pairs in length and there are enough repeats to allow DNA replication and somatic cell division during the lifetime of the organism. If telomeres become too short this will trigger programmed cell death (a process called apoptosis). The lack of telomerase activity in somatic cells limits the number of cell divisions that can occur, and this is a ‘problem’ that needs to be overcome by cancer cells. Telomerase activity is reactivated in most cancers, allowing these cells to divide indefinitely and therefore this activity is a potential target for cancer therapies.

An understanding of DNA synthesis is central to many experimental approaches in molecular biosciences, it allows us to determine DNA sequences including that of the human genome, to analyse environmental samples to better understand the living world around us and to analyse minute biological samples from crime scenes to identify offenders. It is exploited in medicine, for example several drugs used to treat HIV infection or exposure are nucleoside analogues that inhibit DNA synthesis. Many chemotherapy agents used to treat cancer target DNA replication.

3.2 ENZYMES USED IN MOLECULAR BIOLOGY

The discovery and characterization of a number of key enzymes have permitted the development of various techniques for the analysis and manipulation of DNA. In particular, the enzymes termed type II restriction endonucleases have come to play a key role in all aspects of molecular biology. These enzymes recognize specific DNA sequences, usually 4–6 base pairs (bp) in length, and cleave them in a defined manner. The sequences recognized are palindromic or of an inverted repeat nature, that is, they read the same in both directions on each strand. When cleaved they leave a flush-ended or staggered (also termed a cohesive-ended) fragment depending on the particular enzyme used (Figure 8).

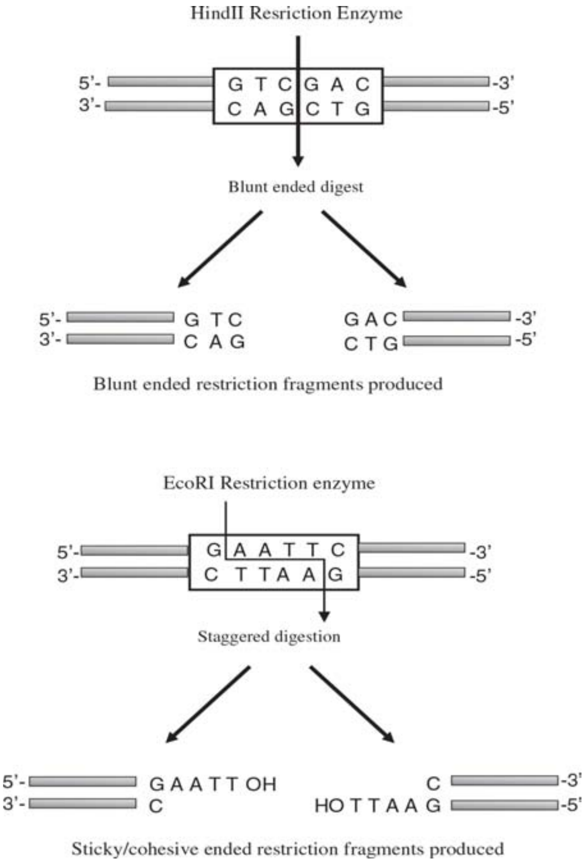


Figure 8. Examples of digestion of DNA by restriction endonucleases. The upper panel indicates the result of a restriction digestion forming blunt fragments with the enzyme HindIII. The bottom panel indicates the cohesive fragments produced by digestion with the enzyme EcoR1.

An important property of staggered ends is that those produced from different molecules by the same enzyme are complementary (or 'sticky') and so will anneal to each other (Table 1). The annealed strands are held together only by hydrogen bonding between complementary bases on opposite strands. Covalent joining of ends on each of the two strands may be brought about by the enzyme DNA ligase. This is widely exploited in molecular biology to allow the construction of recombinant DNA, i.e. the joining of DNA fragments from different sources.

Table 1. Examples of restriction endonucleases that recognize different target sequences and the resulting fragments following digestion

Name	Recognition Sequence	Digestion Products	
	Four Nucleotide Recognition Sequence		
*Hae*III	5'-GGCC-3' 3'-CCGG-5'	5'-GG CC-3' 3'-CC GG-5'	Blunt End Digestion
*Hpa*II	5'-CCGG-3' 3'-GGCC-5'	5'-C CGG-3' 3'-GGC C-5'	Cohesive End Digestion
	Six Nucleotide Recognition Sequence		
*Bam*HI	5'-GGATTC-3' 3'-GGCC-5'	5'-G GATCC-3' 3'-CCTAG G-5'	
*Eco*RI	5'-GAATTC-3' 3'-CTTAAG-5'	5'-G AATCC-3' 3'-CTTAA G-5'	
*Hin*dIII	5'-AAGCTT-3' 3'-TTCGAA-5'	5'-A AGCTT-3' 3'-TTCGA A-5'	
	Eight Nucleotide Recognition Sequence		
*Not*I	5'-GCGGCCGC-3' 3'-CGCCGGCG-5'	5'-GC GGCCGC-3' 3'-CGCCGG CG-5'	

Approximately 500 restriction enzymes have been characterized that recognize over 100 different target sequences. A number of these, termed isoschizomers, recognize different target sequences but produce the same staggered ends or overhangs. A number of other enzymes have proved to be of value in the manipulation of DNA, as summarized in Table 2.

Table 2. Comparison of the various labelling methods for DNA.

Labelling method	*Enzyme*	*Probe Type*	*Specific Activity*
5' end labelling	Alkaline Phosphatase Polynucleotide Kinase	DNA	Low
3' end labelling	Terminal Transferase	DNA	Low
Nick Translation	DNase I DNA Polymerase I	DNA	High
Random Hexamer	DNA Polymerase I	DNA	High
PCR	*Taq* DNA Polymerase	DNA	High
Riboprobes (cRNA)	RNA Polymerase	RNA	High

3.3 ISOLATION AND SEPARATION OF NUCLEIC ACIDS

3.3.1 Isolation of DNA

The use of DNA for analysis or manipulation usually requires that it is isolated and purified to a certain extent. DNA is recovered from cells by the gentlest possible method of cell rupture to prevent the DNA from fragmenting by mechanical shearing. This is usually in the presence of EDTA which chelate the Mg^{2+} ions needed for enzymes that degrade DNA termed DNase. Ideally, cell walls, if present, should be digested enzymatically (e.g. lysozyme treatment of bacteria) and the cell membrane should be solubilized using detergent. If physical disruption is necessary, it should be kept to a minimum and should involve cutting or squashing of cells, rather than the use of shear forces. Cell disruption (and most subsequent steps) should be performed at 4 °C, using glassware and solutions which have been autoclaved to destroy DNase activity (Figure 9).

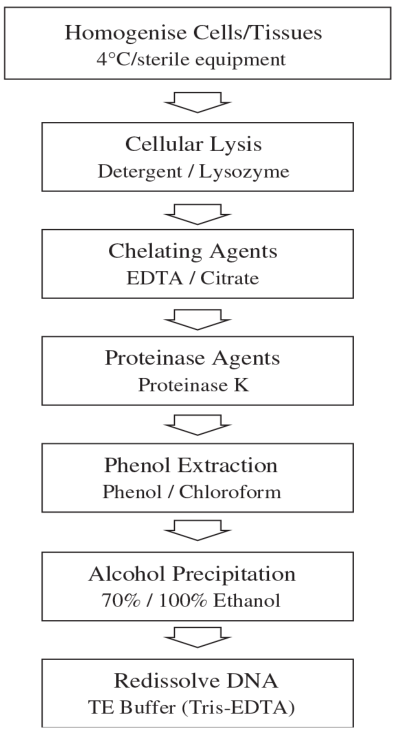


Figure 9. Flow diagram of the main steps involved in the extraction of DNA.

After release of nucleic acids from the cells, RNA can be removed by treatment with ribonuclease (RNase) which has been heat treated to inactivate any DNase contaminants; RNase is relatively stable to heat as a result of its disulfide bonds, which ensure rapid renaturation of the molecule on cooling. The other major contaminant, protein, is removed by shaking the solution gently with water-saturated phenol or with a phenol–chloroform mixture, either of which will denature proteins but not nucleic acids. Centrifugation of the emulsion formed by this mixing produces a lower, organic phase, separated from the upper, aqueous phase by an interface of denatured protein. The aqueous solution is recovered and deproteinised repeatedly, until no more material is seen at the interface. Finally, the deproteinised DNA preparation is mixed with two volumes of absolute ethanol and the DNA allowed to precipitate out of solution in a freezer. After centrifugation, the DNA pellet is redissolved in a buffer containing EDTA to inactivate any DNases present. This solution can be stored at 4 °C for at least 1 month. DNA solutions can be stored frozen, although repeated freezing and thawing tend to damage long DNA molecules by shearing.

The procedure described above is suitable for total cellular DNA. If the DNA from a specific organelle or viral particle is needed, it is best to isolate the organelle or virus before extracting its DNA, since the recovery of a particular type of DNA from a mixture is usually rather difficult. Where a high degree of purity is required, DNA may be subjected to density gradient ultracentrifugation through caesium chloride, which is particularly useful for the preparation of plasmid DNA. It is possible to check the integrity of the DNA by agarose gel electrophoresis and determine the concentration of the DNA by using the fact that 1 absorbance unit equates to 50 μ mg ml^1 of DNA and so

$$50 \times A_{260} = \text{concentration of DNA sample } (\mu g\, ml^{-1})$$

Contaminants may also be identified by scanning UV spectrophotometry from 200 to 300 nm. A ratio of 260 nm:280 nm of approximately 1.8 indicates that the sample is free of protein contamination, which absorbs strongly at 280 nm.

3.3.2 Isolation of RNA

The methods used for RNA isolation are very similar to those described above for DNA; however, RNA molecules are relatively short and therefore less easily damaged by shearing, so cell disruption can be more vigorous. RNA is, however, very vulnerable to digestion by RNases which are present endogenously in various concentrations in certain cell types and exogenously on fingers. Gloves should therefore be worn and a strong detergent should be included in the isolation medium to denature immediately any RNases. Subsequent deproteinisation should be particularly rigorous, since RNA is often tightly associated with proteins. DNase treatment can be used to remove DNA and RNA can be precipitated by ethanol. One reagent in which is commonly used in RNA extraction is guanadinium thiocyanate, which is both a strong inhibitor of RNase and a protein denaturant. It is possible to check the integrity of an RNA extract by analyzing it by agarose gel electrophoresis. The most abundant RNA species are rRNA molecules, 23S and 16S for prokaryotes and 18S and 28S for eukaryotes. These appear as discrete bands on the agarose gel and indicate that the other RNA components are likely to be intact. This is usually carried out under denaturing conditions to prevent secondary structure formation in the RNA. The concentration of the RNA may be estimated by using UV spectrophotometry. At 260 nm, 1 absorbance unit equates to 40 μg ml[1] of RNA and therefore

$$40 \times A_{260} = \text{concentration of RNA sample } (\mu g\, ml^{-1}) \tag{2}$$

Contaminants may also be identified in the same way as for DNA by scanning UV spectrophotometry; however, in the case of RNA a 260 nm:280 nm ratio of approximately 2 would be expected for a sample containing no contamination.

In many cases, it is desirable to isolate eukaryotic mRNA, which constitutes only 2–5% of cellular RNA, from a mixture of total RNA molecules. This may be carried out by affinity chromatography on oligo (dT)-cellulose columns. At high salt concentrations, the mRNA containing poly(A) tails binds to the complementary oligo(dT) molecules of the affinity column and so mRNA will be retained; all other RNA molecules can be washed through the column with further high-salt solution. Finally, the bound mRNA can be eluted using a low concentration of salt. Nucleic acid species may also be subfractionated by more physical means such as electrophoretic or chromatographic separations based on differences in nucleic acid fragment sizes or physicochemical characteristics.

3.4 ELECTROPHORESIS OF NUCLEIC ACIDS

Electrophoresis in agarose or polyacrylamide gels is the most usual way to separate DNA molecules according to size (Figure 10). The technique can be used analytically or preparatively and can be qualitative or quantitative. Large fragments of DNA such as chromosomes may also be separated by a modification of electrophoresis termed pulsed field gel electrophoresis (PFGE). The easiest and most widely applicable method is electrophoresis in horizontal agarose gels, followed by staining with ethidium bromide. This dye binds to DNA by insertion between stacked base pairs (intercalation) and it exhibits a strong orange/red fluorescence when illuminated with ultraviolet light. Very often electrophoresis is used to check the purity and intactness of a DNA preparation or to assess the extent of an enzymatic reaction during, for example, the steps involved in the cloning of DNA. For such checks ‘mini-gels’ are particularly convenient, since they need little preparation, use small samples and give results quickly. Agarose gels can be used to separate molecules larger than about 100 bp. For higher resolution or for the effective separation of shorter DNA molecules, polyacrylamide gels are the preferred method.

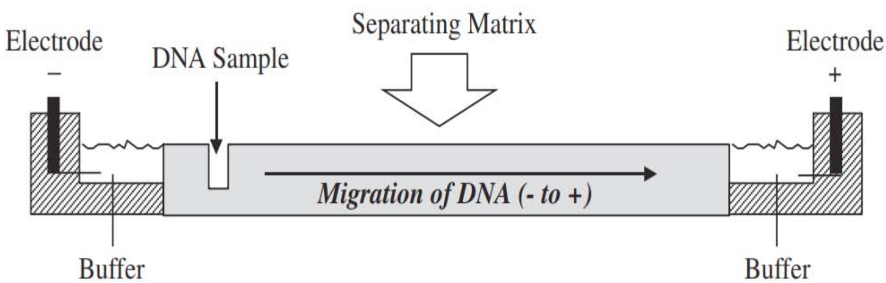


Figure 10. A typical setup required for agarose gel electrophoresis of DNA. The upper panel indicates a cross-section of the unit used for gel electrophoresis.

When electrophoresis is used preparatively, the piece of gel containing the desired DNA fragment is physically removed with a scalpel. The DNA may be recovered from the gel fragment in various ways. This may include crushing with a glass rod in a small volume of buffer, using agarase to digest the agarose thus leaving the DNA, or by the process of electroelution. In the latter method, the piece of gel is sealed in a length of dialysis tubing containing buffer and is then placed between two electrodes in a tank containing more buffer. Passage of an electric current between the electrodes causes DNA to migrate out of the gel piece, but it remains trapped within the dialysis tubing and can therefore be recovered easily. More commonly, commercial spin columns can be used which contain an isolating matrix used in conjunction with a bench-top microcentrifuge. The use of such standardized ‘kits’ in molecular biology is now commonplace. An alternative to conventional analysis of nucleic acids by electrophoresis is through the use of microfluidic systems. These are carefully manufactured chip-based units where microlitre volumes may be used and with the aid of computer analysis provide much of the data necessary for analysis. Their advantage lies in the fact that the sample volume is very small, allowing much of an extract to be used for further analysis.

3.5 RESTRICTION MAPPING OF DNA FRAGMENTS

Restriction mapping involves the size analysis of restriction fragments produced by several restriction enzymes individually and in combination. The principle of this mapping is illustrated, in which the restriction sites of two enzymes, A and B, are being mapped. Cleavage with A gives fragments 2 and 7 kilobases (kb) from a 9 kb molecule, hence we can position the single A site 2 kb from one end. Similarly, B gives fragments 3 and 6 kb, so it has a single site 3 kb from one end; but it is not possible at this stage to say if it is near to A's site or at the opposite end of the DNA. This can be resolved by a double digestion. If the resultant fragments are 2, 3 and 4 kb, then A and B cut at opposite ends of the molecule; if they are 1, 2 and 6 kb, the sites are near each other. Not surprisingly, the mapping of real molecules is rarely as simple as this and computer analysis of the restriction fragment lengths is usually needed to construct a map (Figure 11).

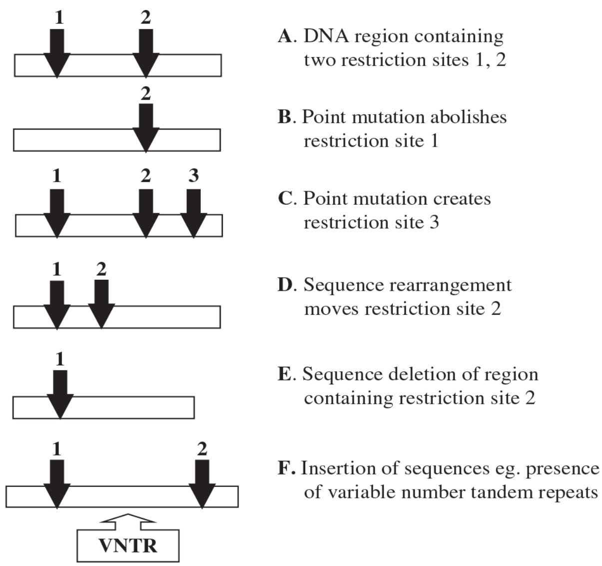


Figure 11. Restriction fragment length polymorphisms (RFLP). The schematic panels A–F indicate the various fragments obtained following digestion as a result of differences in the position of restriction endonuclease target sequences.

3.6 NUCLEIC ACID ANALYSIS METHODS

There are numerous methods for analyzing DNA and RNA; however, many of them are solution based or more recently include the use of chip-based array systems. Indeed, the lab-on-a-chip approach is developing rapidly and it is possible to envisage many detection and analysis methods being developed in this format in the future.7 However, traditional methods are still employed in many laboratories and much is still made of producing a hard copy of digested and separated single stranded DNA fragments attached to a matrix such as nylon for analysis with an appropriate labelled probe.

3.6.1 DNA Blotting

Electrophoresis of DNA restriction fragments allows separation based on size to be carried out; however, it provides no indication as to the presence of a specific, desired fragment among the complex sample (Figure 12).

This can be achieved by transferring the DNA from the intact gel on to a piece of nitrocellulose or nylon membrane placed in contact with it.8 This provides a more permanent record of the sample since DNA begins to diffuse out of a gel that is left for a few hours. First the gel is soaked in alkali to render the DNA single stranded. It is then transferred to the membrane so that the DNA becomes bound to the it in exactly the same pattern as that originally on the gel.

This transfer, named a Southern blot after its inventor Ed Southern, can be performed electrophoretically or by drawing large volumes of buffer through both gel and membrane, thus transferring DNA from one to the other by capillary action. The point of this operation is that the membrane can now be treated with a labelled DNA molecule, for example a gene probe. This single-stranded DNA probe will hybridize under the right conditions to complementary fragments immobilized on the membrane.

The conditions of hybridization, including the temperature and salt concentration, are critical for this process to take place effectively. This is usually referred to as the stringency of the hybridization and it is particular for each individual gene probe and for each sample of DNA.

A series of washing steps with buffer are then carried out to remove any unbound probe and the membrane is developed, after which the precise location of the probe and its target may be visualized. It is also possible to analyse DNA from different species or organisms by blotting the DNA and then using a gene probe representing a protein or enzyme from one of the organisms. In this way, it is possible to search for related genes in different species. This technique is generally termed Zoo blotting.

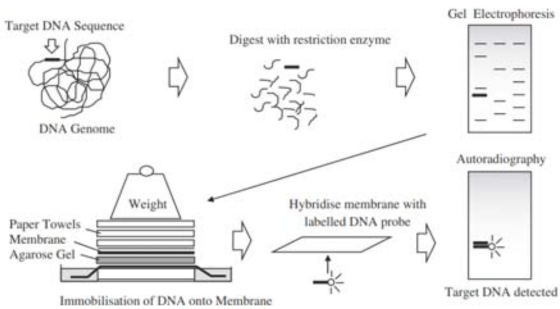


Figure 12. The steps involved in the production of a Southern blot and the subsequent detection of a specific DNA sequence following hybridization with a complementary labelled gene probe.

3.6.2 RNA Blotting

The same basic process of nucleic acid blotting can be used to transfer RNA from gels on to similar membranes. This allows the identification of specific mRNA sequences of a defined length by hybridization to a labelled gene probe and is known as Northern blotting. With this technique it is not only possible to detect specific mRNA molecules but it may also be used to quantify the relative amounts of the specific mRNA. It is usual to separate the mRNA transcripts by gel electrophoresis under denaturing conditions since this improves resolution and allows a more accurate estimation of the sizes of the transcripts. The format of the blotting may be altered from transfer from a gel to direct application to slots on a specific blotting apparatus containing the nylon membrane. This is termed slot or dot blotting and provides a convenient means of measuring the abundance of specific mRNA transcripts without the need for gel electrophoresis; it does not, however, provide information regarding the size of the fragments.

A further method of RNA analysis that overcomes the problems of RNA blotting is termed the ribonuclease protection assay. Here the RNA from a sample is extracted and then mixed with a probe representing the sequence of interest in solution. The probe and the appropriate RNA fragment hybridize to form a double-stranded sequence. RNase is then added, which cleaves any single-stranded RNA present but leaves the double-stranded RNA intact. The intact RNA can then be separated by electrophoresis and an indication of the size of the fragment generated. The efficient removal of the background of RNA and the improved sensitivity make the ribonuclease protection assay a popular choice for the analysis of specific RNA molecules. An important step in the field of RNA analysis was the development of RNAi (RNA interference), which inhibits gene expression. Here double-stranded DNA promotes the degradation of mRNA. Doublestranded RNA in the cell is cleaved by a dicer enzyme, resulting in the formation of small 21–25 bp interfering RNAs (siRNA). The siRNA are complementary to a target RNA strand. Small RNAi proteins are guided by the siRNA to the appropriate mRNA, where the target is then cleaved and is unable to be translated. Many areas are now benefiting from the adoption of this technique in the molecular biology and biotechnology fields.

3.7 GENE PROBE DERIVATION

The availability of a gene probe is essential in many molecular biology techniques, yet in many cases is one of the most difficult steps (Figure 13). The information needed to produce a gene probe may come from many sources, but with the development and sophistication of genetic databases this is usually one of the first stages. There are a number of genetic databases throughout the world and it is possible to search these over the internet and identify particular sequences relating to a specific gene or protein. In some cases it is possible to use related proteins from the same gene family to gain information on the most useful DNA sequence. Similar proteins or DNA sequences but from different species may also provide a starting point with which to produce a so-called heterologous gene probe. Although in some cases probes have already been produced and cloned, it is possible, armed with a DNA sequence from a DNA database, to synthesise chemically a single-stranded oligonucleotide probe. This is usually undertaken by computer-controlled gene synthesizers which link dNTPs together based on a desired sequence. It is essential to carry out certain checks before probe production to determine that the probe is unique, is not able to self-anneal or is self-complementary, all of which may compromise its use.

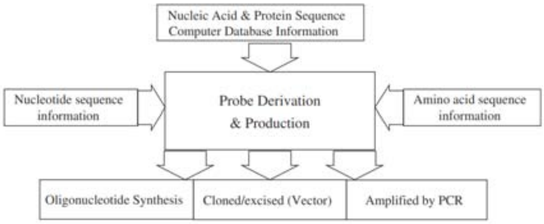


Figure 13. Alternative strategies for designing and producing a gene probe.

Where little DNA information is available to prepare a gene probe, it is possible in some cases to use the knowledge gained from analysis of the corresponding protein. Thus it is possible to isolate and purify proteins and sequence part of the N-terminal end of the protein. From our knowledge of the genetic code, it is possible to predict the various DNA sequences that could code for the protein and then synthesize appropriate oligonucleotide sequences chemically. Due to the degeneracy of the genetic code, most amino acids are coded for by more than one codon, hence there will be more than one possible nucleotide sequence which could code for a given polypeptide. The longer the polypeptide, the greater is the number of possible oligonucleotides which must be synthesized. Fortunately, there is no need to synthesize a sequence longer than about 20 bases, since this should hybridize efficiently with any complementary sequences and should be specific for one gene. Ideally, a section of the protein should be chosen which contains as many tryptophan and methionine residues as possible, since these have unique codons and there will therefore be fewer possible base sequences which could code for that part of the protein. The synthetic oligonucleotides can then be used as probes in a number of molecular biology methods.

3.8 LABELLING DNA GENE PROBE MOLECULES

An essential feature of a gene probe is that it can be visualized by some means. In this way, a gene probe that hybridizes to a complementary sequence may be detected and identify that desired sequence from a complex mixture. There are two main ways of labelling gene probes; traditionally it has been carried out using radioactive labels, but gaining in popularity are non-radioactive labels. Perhaps the most often used radioactive label is phosphorus-32 (^{32}P), although for certain techniques sulfur-35 (^{35}S) and tritium (^{3}H) are used. These may be detected by the process of autoradiography, where the labelled probe molecule, bound to sample DNA, located for example on a nylon membrane, is placed in contact with an X-ray-sensitive film. Following exposure, the film is developed and fixed just as a black and white negative and reveals the precise location of the labelled probe and therefore the DNA to which it has hybridized.

Non-radioactive labels are increasingly being used to label DNA gene probes. Until recently, radioactive labels were more sensitive than their non-radioactive counterparts. However, recent developments have led to similar sensitivities, which, when combined with their improved safety, have led to their greater acceptance. The labelling systems are termed either direct or indirect. Direct labelling allows an enzyme reporter such as alkaline phosphatase to be coupled directly to the DNA. Although this may alter the characteristics of the DNA gene probe, they offer the advantage of rapid analysis since no intermediate steps are needed. However, indirect labelling is at present more popular. This relies on the incorporation of a nucleotide which has a label attached. At present, three of the main labels in use are biotin, fluorescein and digoxygenin. These molecules are covalently linked to nucleotides using a carbon spacer arm of 7, 14 or 21 atoms.

Specific binding proteins may then be used as a bridge between the nucleotide and a reporter protein such as an enzyme. For example, biotin incorporated into a DNA fragment is recognized with a very high affinity by the protein streptavidin. This may be either coupled or conjugated to a reporter enzyme molecule such as alkaline phosphatase. This is able to convert a colorless substrate, p-nitrophenol phosphate (PNPP), into a yellow compound, p-nitrophenol (PNP), and also offers a means of signal amplification.

Alternatively labels such as digoxygenin incorporated into DNA sequences may be detected by monoclonal antibodies, again conjugated to reporter molecules including alkaline phosphatase. Thus, rather than the detection system relying on autoradiography, which is necessary for radiolabels, a series of reactions resulting in either a color or a light or chemiluminescent reaction takes place. This has important practical implications since autoradiography may take 1–3 days, whereas color and chemiluminescent reactions take minutes.

3.8.1 End Labelling of DNA Molecules

The simplest form of labelling DNA is by 5' or 3' end labelling; 5' end labelling involves a phosphate transfer or exchange reaction where the 50 phosphate of the DNA to be used as the probe is removed and in its place a labelled phosphate, usually ^{32}P, is added. This is usually carried out by using two enzymes; the first, alkaline phosphatase, is used to remove the existing phosphate group from the DNA. Following removal of the released phosphate from the DNA, a second enzyme, polynucleotide kinase, is added, which catalyses the transfer of a phosphate group (^{32}P-labelled) to the 5' end of the DNA. The newly labelled probe is then purified, usually by chromatography through a Sephadex column, and may be used directly (Figure 14).

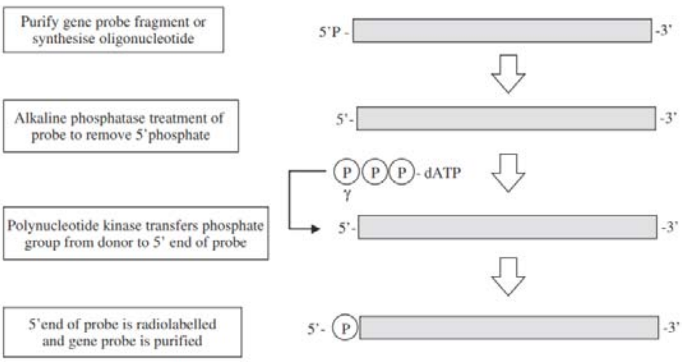


Figure 14. The steps involved in the production of a 5'-labelled gene probe.

Using the other end of the DNA molecule, the 3' end, is slightly less complex. Here a new dNTP which is labelled (e.g. [^{32}P]adATP or biotinlabelled dNTP) is added to the 3' end of the DNA by the enzyme terminal transferase. Although this is a simpler reaction, a potential problem exists because a new nucleotide is added to the existing sequence and so the complete sequence of the DNA is altered, which may affect its hybridization to its target sequence. End labelling methods also suffer from the fact that only one label is added to the DNA so they are of a lower activity in comparison with methods that incorporate label along the length of the DNA (Figure 15).

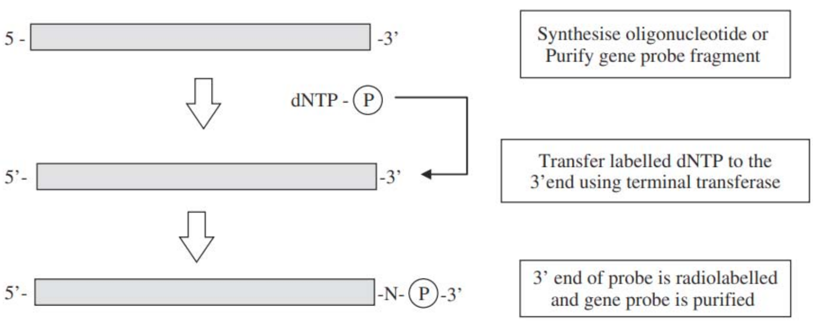


Figure 15. The steps involved in the production of 3'-labelled gene probe.

3.8.2 Random Primer Labelling

In random primer labelling the DNA to be labelled is first denatured and then placed under renaturing conditions in the presence of a mixture of many different random sequences of hexamers or hexanucleotides. These hexamers will, by chance, bind to the DNA sample wherever they encounter a complementary sequence and so the DNA will rapidly acquire an approximately random sprinkling of hexanucleotides annealed to it. Each of the hexamers can act as a primer for the synthesis of a fresh strand of DNA catalyzed by DNA polymerase since it has an exposed 3' -hydroxyl group. The Klenow fragment of DNA polymerase is used for random primer labelling because it lacks a 5' –3' exonuclease activity. This is prepared by cleavage of DNA polymerase with subtilisin, giving a large enzyme fragment which has no 5' to 3' exonuclease activity, but which still acts as a 5' to 3' polymerase. Thus, when the Klenow enzyme is mixed with the annealed DNA sample in the presence of dNTPs, including at least one which is labelled, many short stretches of labelled DNA will be generated (Figure 16). In a similar way to random primer labelling, the polymerase chain reaction may also be used to incorporate radioactive or non-radioactive labels.

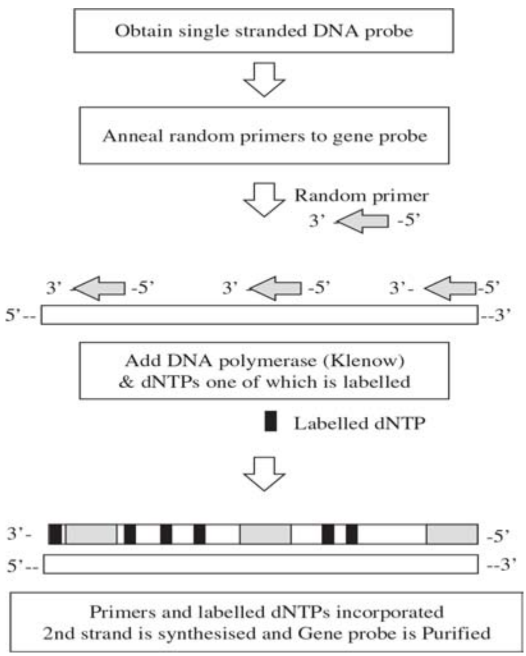


Figure 16. The steps involved in the production of a gene probe produced by the random hexamer method.

3.8.3 Nick Translation

A traditional method of labelling DNA is by the process of nick translation. Low concentrations of DNase I are used to make occasional single-strand nicks in the double-stranded DNA that is to be used as the gene probe. DNA polymerase then fills in the nicks, using an appropriate deoxyribonucleoside triphosphate (dNTP), at the same time making a new nick to the 3' side of the previous one. In this way, the nick is translated along the DNA. If labelled dNTPs are added to the reaction mixture, they will be used to fill in the nicks and so the DNA can be labelled to a very high specific activity (Figure 17).

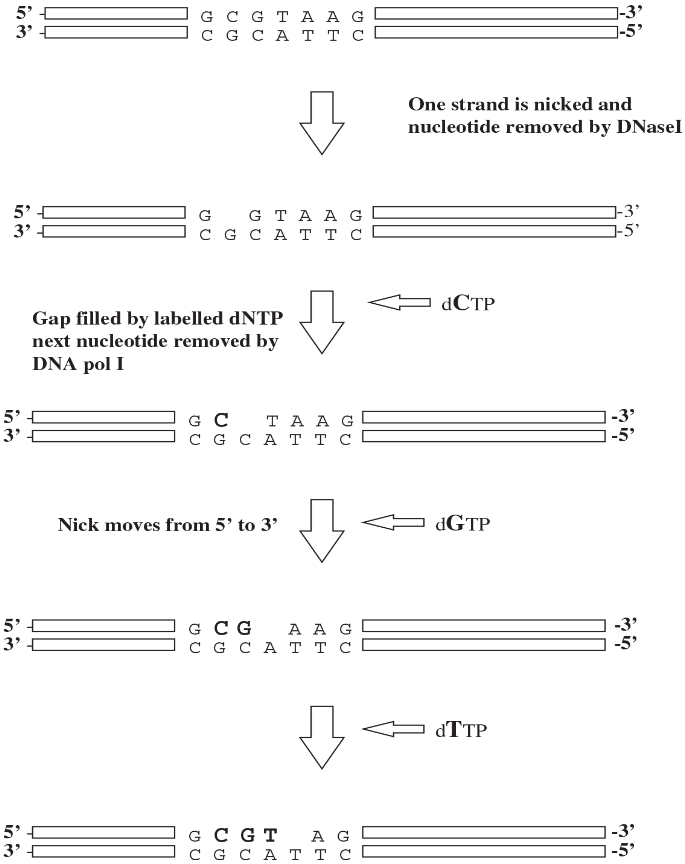
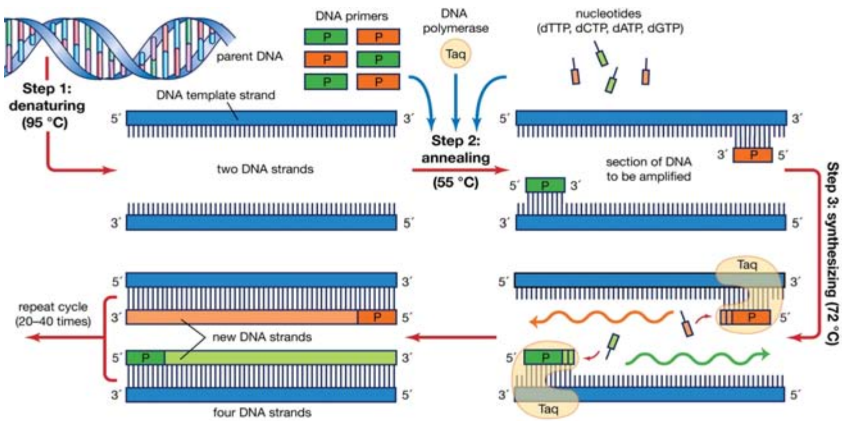


Figure 17. The steps involved in the production of a gene probe by the nick translation method.

3.9 POLYMERASE CHAIN REACTION

Polymerase chain reaction (PCR), a technique used to make numerous copies of a specific segment of DNA quickly and accurately. The polymerase chain reaction enables investigators to obtain the large quantities of DNA that are required for various experiments and procedures in molecular biology, forensic analysis, evolutionary biology, and medical diagnostics. PCR was developed in 1983 by Kary B. Mullis, an American biochemist who won the Nobel Prize for Chemistry in 1993 for his invention. Before the development of PCR, the methods used to amplify, or generate copies of, recombinant DNA fragments were time-consuming and labor-intensive. In contrast, a machine designed to carry out PCR reactions can complete many rounds of replication, producing billions of copies of a DNA fragment, in only a few hours.



The PCR technique is based on the natural processes a cell uses to replicate a new DNA strand. Only a few biological ingredients are needed for PCR. The integral component is the template DNA—i.e., the DNA that contains the region to be copied, such as a gene. As little as one DNA molecule can serve as a template. The only information needed for this fragment to be replicated is the sequence of two short regions of nucleotides (the subunits of DNA) at either end of the region of interest. These two short template sequences must be known so that two primers—short stretches of nucleotides that correspond to the template sequences—can be synthesized. The primers bind, or anneal, to the template at their complementary sites and serve as the starting point for copying. DNA synthesis at one primer is directed toward the other, resulting in replication of the desired intervening sequence. Also needed are free nucleotides used to build the new DNA strands and a DNA polymerase, an enzyme that does the building by sequentially adding on free nucleotides according to the instructions of the template.

PCR is a three-step process that is carried out in repeated cycles. The initial step is the denaturation, or separation, of the two strands of the DNA molecule. This is accomplished by heating the starting material to temperatures of about 95 °C (203 °F). Each strand is a template on which a new strand is built. In the second step the temperature is reduced to about 55 °C (131 °F) so that the primers can anneal to the template. In the third step the temperature is raised to about 72 °C (162 °F), and the DNA polymerase begins adding nucleotides onto the ends of the annealed primers. At the end of the cycle, which lasts about five minutes, the temperature is raised and the process begins again. The number of copies doubles after each cycle. Usually 25 to 30 cycles produce a sufficient amount of DNA. In the original PCR procedure, one problem was that the DNA polymerase had to be replenished after every cycle because it is not stable at the high temperatures needed for denaturation. This problem was solved in 1987 with the discovery of a heat-stable DNA polymerase called *Taq,* an enzyme isolated from the thermophilic bacterium *Thermus aquaticus*, which inhabits hot springs. *Taq* polymerase also led to the invention of the PCR machine. Because DNA from a wide range of sources can be amplified, the technique has been applied to many fields. PCR is used to diagnose genetic disease and to detect low levels of viral infection. In forensic medicine it is used to analyze minute traces of blood and other tissues in order to identify the donor by his genetic "fingerprint." The technique has also been used to amplify DNA fragments found in preserved tissues, such as those of a 40,000-year-old frozen woolly mammoth or of a 7,500-year-old human found in a peat bog.

REFERENCES

1. D. Baxevanis and B. F. Ouelette, Bioinformatics. A Practical Guide to the Analysis of Genes and Proteins, Wiley, Chichester, 2004.
2. D. W. Mount, Bioinformatics Sequence and Genome Analysis, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 2004.
3. G. J. Hannon, RNAi. A Guide to Gene Silencing, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 2003.
4. G. J. S. Jenkins, in Molecular Biomethods Handbook, 2nd edn, ed. R. Rapley and J. M. Walker, Humana Press, Totowa, NJ, 2008, pp. 17–27.
5. K. Wilson and J. M. Walker, Principles and Techniques of Practical Biochemistry, Cambridge University Press, Cambridge, 2005.
6. M. Aquino de Muro, in Molecular Biomethods Handbook, 2nd edn, ed. R. Rapley and J. M. Walker, Humana Press, Totowa, NJ, 2008, pp. 41–54.
7. R. E. Farrell, RNA Methodologies. A Laboratory Guide for Isolation and Characterisation, Academic Press, New York, 2005.

CHAPTER

4

RECOMBINANT DNA AND GENETIC ANALYSIS

INTRODUCTION

The considerable advances made in microarray, sequencing technologies and bioinformatics analysis are now beginning to provide true insights into the development and maintenance of cells and tissues. Indeed areas of analysis such as metabolomics, transcriptomics and systems biology are now well established and allow analysis of vast numbers of samples simultaneously. This type of large-scale parallel analysis is now the main driving force of biological discovery and analysis. However, the techniques of molecular biology and genetic analysis have their foundations in methods developed a number of decades ago. One of the main cornerstones on which molecular biology analysis was developed was the discovery of restriction endonucleases in the early 1970s which not only led to the possibility of analyzing DNA more effectively but also provided the ability to cut different DNA molecules so that they could later be joined together to create new recombinant DNA fragments. The newly created DNA molecules heralded a new era in the manipulation, analysis and exploitation of biological molecules. This process, termed gene cloning, has enabled numerous discoveries and insights into gene structure, function and regulation. Since their initial use the methods for the production of gene libraries have been steadily refined and developed. Although microarray analysis and the polymerase chain reaction (PCR) have provided short cuts to gene analysis there are still many cases where gene cloning methods are not only useful but are an absolute requirement. The following provides an account of the process of gene cloning and other methods based on recombinant DNA technology.

4.1 CONSTRUCTING GENE LIBRARIES

A gene library is a collection of different DNA sequences from an organism, which has been cloned into a vector for ease of purification, storage and analysis. Gene library is a collection of cloned DNA designed so that there is a high probability of finding any particular piece of the source DNA in the collection.

4.1.1 Digesting Genomic DNA Molecules

Following the isolation and purification of genomic DNA it is possible to specifically fragment it with enzymes termed restriction endonucleases. These enzymes are the key to molecular cloning because of the specificity they have for particular DNA sequences. It is important to note that every copy of a given DNA molecule from a specific organism will give the same set of fragments when digested with a particular enzyme. DNA from different organisms will, in general, give different sets of fragments when treated with the same enzyme. By digesting complex genomic DNA from an organism it is possible to reproducibly divide its genome into a large number of small fragments, each approximately the size of a single gene. Some enzymes cut straight across the DNA to give flush or blunt ends. Other restriction enzymes make staggered single-strand cuts, producing short single-stranded projections at each end of the digested DNA. These ends are not only identical, but complementary, and will base-pair with each other; they are therefore known as cohesive or sticky ends. In addition the 5′ end projection of the DNA always retains the phosphate groups.

Over 600 enzymes, recognizing more than 200 different restriction sites, have been characterized. The choice of which enzyme to use depends on a number of factors. For example, the recognition sequence of 6 bp will occur, on average, every 4096 (46) bases assuming a random sequence of each of the four bases. This means that digesting genomic DNA with EcoR1, which recognizes the sequence 5′-GAATTC-3′, will produce fragments each of which is on average just over 4 kb. Enzymes with 8 bp recognition sequences produce much longer fragments. Therefore very large genomes, such as human DNA, are usually digested with enzymes that produce long DNA fragments. This makes subsequent steps more manageable, since a smaller number of those fragments need to be cloned and subsequently analyzed (Table 1).

Table 1. Numbers of clones required for representation of DNA in a genome library

		No. of clones required	
Species	Genome size (kb)	17 kb fragments	35 kb fragments
Bacteria (*E. coli*)	4 000	700	340
Yeast	20 000	3 500	1 700
Fruit fly	165 000	29 000	14 500
Man	3 000 000	535 000	258 250
Maize	15 000 000	2 700 000	1 350 000

4.1.2 Ligating DNA Molecules

The DNA products resulting from restriction digestion to form sticky ends may be joined to any other DNA fragments treated with the same restriction enzyme. Thus, when the two sets of fragments are mixed, base-pairing between sticky ends will result in the annealing together of fragments that were derived from different starting DNA. There will, of course, also be pairing of fragments derived from the same starting DNA molecules, termed reannealing. All these pairings are transient, owing to the weakness of hydrogen bonding between the few bases in the sticky ends, but they can be stabilized by use of an enzyme termed DNA ligase in a process termed ligation.

This enzyme, usually isolated from bacteriophage T4 and termed T4 DNA ligase, forms a covalent bond between the 5′ phosphate at the end of one strand and the 3′ hydroxyl of the adjacent strand (Fig. 1). The reaction, which is ATP dependent, is often carried out at 10°C to lower the kinetic energy of molecules, and so reduce the chances of base-paired sticky ends parting before they have been stabilized by ligation. However, long reaction times are needed to compensate for the low activity of DNA ligase in the cold. It is also possible to join blunt ends of DNA molecules, although the efficiency of this reaction is much lower than sticky-ended ligations.

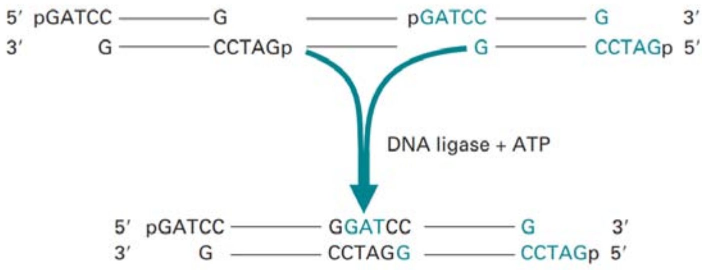


Figure 1. Ligation molecules with cohesive ends.

Since ligation reconstructs the site of cleavage, recombinant molecules produced by ligation of sticky ends can be cleaved again at the 'joins', using the same restriction enzyme that was used to generate the fragments initially. In order to propagate digested DNA from an organism it is necessary to join or ligate that DNA with a specialized DNA carrier molecule termed a vector. Thus each DNA fragment is inserted by ligation into the vector DNA molecule, which allows the whole recombined DNA to then be replicated indefinitely within microbial cells (Fig. 2). In this way a DNA fragment can be cloned to provide sufficient material for further detailed analysis, or for further manipulation. Thus, all of the DNA extracted from an organism and digested with a restriction enzyme will result in a collection of clones. This collection of clones is known as a gene library.

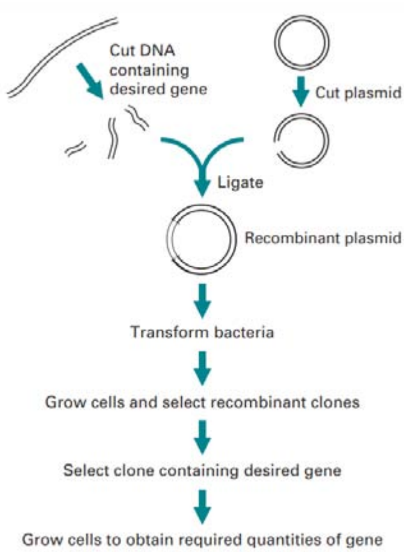


Figure 2. Outline of gene cloning.

4.1.3 Aspects of Gene Libraries

There are two general types of gene library: a genomic library which consists of the total chromosomal DNA of an organism and a cDNA library which represents only the mRNA from a particular cell or tissue at a specific point in time (Fig.3). The choice of the particular type of gene library depends on a number of factors, the most important being the final application of any DNA fragment derived from the library. If the ultimate aim understands the control of protein production for a particular gene or the analysis of its architecture, then genomic libraries must be used. However, if the goal is the production of new or modified proteins, or the determination of the tissue-specific expression and timing patterns, cDNA libraries are more appropriate. The main consideration in the construction of genomic or cDNA libraries is therefore the nucleic acid starting material. Since the genome of an organism is fixed, chromosomal DNA may be isolated from almost any cell type in order to prepare genomic libraries. In contrast, however, cDNA libraries only represent the mRNA being produced from a specific cell type at a particular time. Thus, it is important to consider carefully the cell or tissue type from which the mRNA is to be derived in the construction of cDNA libraries.

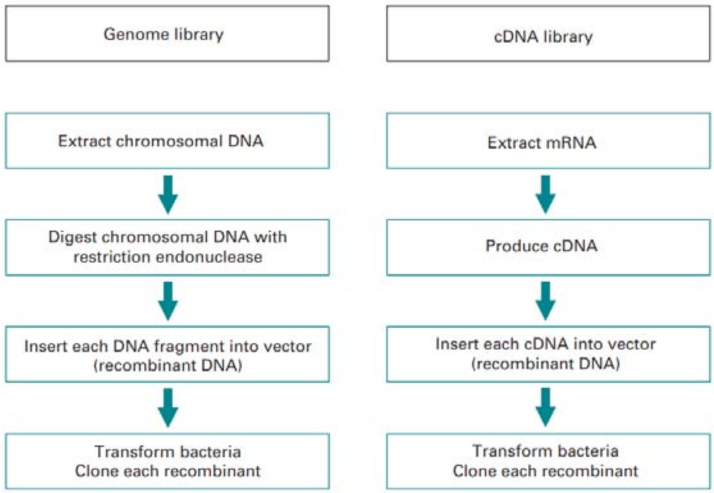


Figure 3. Comparison of the general steps involved in the construction of genomic and complementary DNA (cDNA) libraries.

There are a variety of cloning vectors available, many based on naturally occurring molecules such as bacterial plasmids or bacteria-infecting viruses. The choice of vector depends on whether a genomic library or cDNA library is constructed.

4.1.4 Genomic DNA Libraries

Genomic libraries are constructed by isolating the complete chromosomal DNA from a cell, then digesting it into fragments of the desired average length with restriction endonucleases. This can be achieved by partial restriction digestion using an enzyme that recognizes tetranucleotide sequences. Complete digestion with such an enzyme would produce a large number of very short fragments, but if the enzyme is allowed to cleave only a few of its potential restriction sites before the reaction is stopped, each DNA molecule will be cut into relatively large fragments. Average fragment size will depend on the relative concentrations of DNA and restriction enzyme, and in particular, on the conditions and duration of incubation (Fig. 4). It is also possible to produce fragments of DNA by physical shearing although the ends of the fragments may need to be repaired to make them flush-ended. This can be achieved by using a modified DNA polymerase termed Klenow polymerase. This is prepared by cleavage of DNA polymerase with subtilizing, giving a large enzyme fragment which has no 5′ to 3′ exonuclease activity, but which still acts as a 5′ to 3′ polymerase. This will fill in any recessed 3′ ends on the sheared DNA using the appropriate dNTPs.

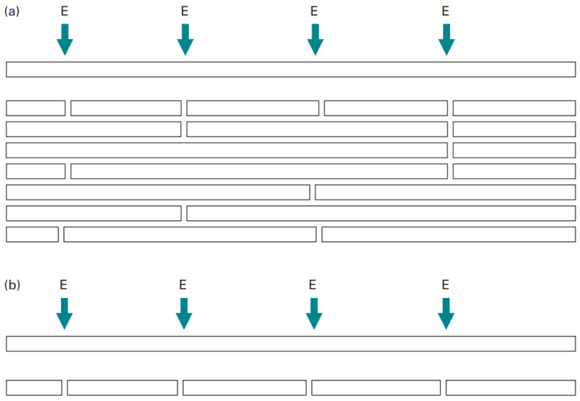


Figure 4. Comparison of (a) partial and (b) complete digestion of DNA molecules at restriction enzyme sites (E).

The mixture of DNA fragments is then ligated with a vector, and subsequently cloned. If enough clones are produced there will be a very high chance that any particular DNA fragment such as a gene will be present in at least one of the clones. To keep the number of clones to a manageable size, fragments about 10 kb in length are needed for prokaryotic libraries, but the length must be increased to about 40 kb for mammalian libraries. It is possible to calculate the number of clones that must be present in a gene library to give a probability of obtaining a particular DNA sequence. This formula is:

$$N = \frac{\ln(1-P)}{\ln(1-f)}$$

where N is the number of recombinants, P is the probability and f is the fraction of the genome in one insert. Thus for the E. coli DNA chromosome of 5 10^6 bp and with an insert size of 20 kb the number of clones needed (N) would be 1 10^3 with a probability of 0.99.

4.1.5 cDNA Libraries

There may be several thousand different proteins being produced in a cell at any one time, all of which have associated mRNA molecules. To identify any one of those mRNA molecules the clones of each individual mRNA have to be synthesized. Libraries that represent the mRNA in a particular cell or tissue are termed cDNA libraries. mRNA cannot be used directly in cloning since it is too unstable. However it is possible to synthesize complementary DNA molecules (cDNAs) to all the mRNAs from the selected tissue. The cDNA may be inserted into vectors and then cloned. The production of cDNA (complementary DNA) is carried out using an enzyme termed reverse transcriptase which is isolated from RNA-containing retroviruses. Reverse transcriptase is an RNA-dependent DNA polymerase, and will synthesize a first-strand DNA complementary to an mRNA template, using a mixture of the four dNTPs. There is also a requirement (as with all polymerase enzymes) for a short oligonucleotide primer to be present (Fig. 5). With eukaryotic mRNA bearing a poly(A) tail, a complementary oligo(dT) primer may be used. Alternatively random hexamers may be used which randomly anneal to the mRNAs in the complex. Such primers provide a free 3′ hydroxyl group which is used as the starting point for the reverse transcriptase. Regardless of the method used to prepare the first-strand cDNA one absolute requirement is high-quality undegraded mRNA. It is usual to check the integrity of the RNA by gel electrophoresis. Alternatively a fraction of the extract may be used in a cell-free translation system, which, if intact mRNA is present, will direct the synthesis of proteins represented by the mRNA molecules in the sample.

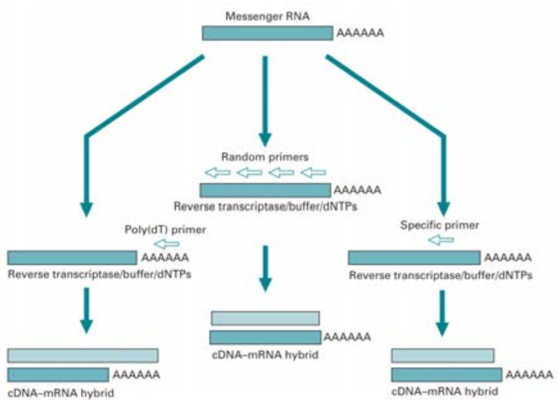


Figure 5. Strategies for producing first-strand cDNA from mRNA.

Following the synthesis of the first DNA strand, a poly(dC) tail is added to its 3′ end, using terminal transferase and dCTP. This will also, incidentally, put a poly(dC) tail on the poly(A) of mRNA. Alkaline hydrolysis is then used to remove the RNA strand, leaving single-stranded DNA which can be used, like the mRNA, to direct the synthesis of a complementary DNA strand. The second-strand synthesis requires an oligo(dG) primer, base-paired with the poly(dC) tail, which is catalyzed by the Klenow fragment of DNA polymerase I. The final product is double-stranded DNA, one of the strands being complementary to the mRNA. One further method of cDNA synthesis involves the use of RNase H. Here the first-strand cDNA is carried out as above with reverse transcriptase but the resulting mRNA-CDNA hybrid is retained. RNase H is then used at low concentrations to nick the RNA strand. The resulting nicks expose 3′ hydroxyl groups which are used by DNA polymerase as a primer to replace the RNA with a second strand of cDNA (Fig. 6).

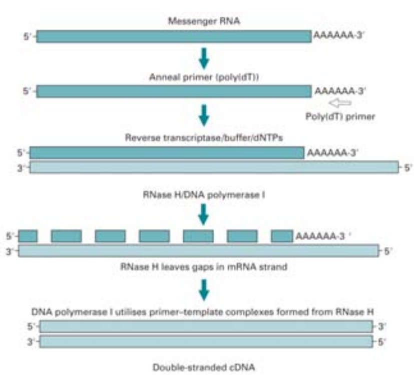


Figure 6. Second-strand cDNA synthesis using the RNase H method.

4.1.6 Treatment of Blunt cDNA Ends

Ligation of blunt-ended DNA fragments is not as efficient as ligation of sticky ends, therefore with cDNA molecules additional procedures are undertaken before ligation with cloning vectors. One approach is to add small double-stranded molecules with one internal site for a restriction end nuclease, termed nucleic acid linkers, to the cDNA. Numerous linkers are commercially available with internal restriction sites for many of the most commonly used restriction enzymes. Linkers are blunt-end ligated to the cDNA but since they are added much in excess of the cDNA the ligation process is reasonably successful. Subsequently the linkers are digested with the appropriate restriction enzyme which provides the sticky ends for efficient ligation to a vector digested with the same enzyme. This process may be made easier by the addition of adaptors rather than linkers which are identical except that the sticky ends are preformed and so there is no need for restriction digestion following ligation (Fig. 7).

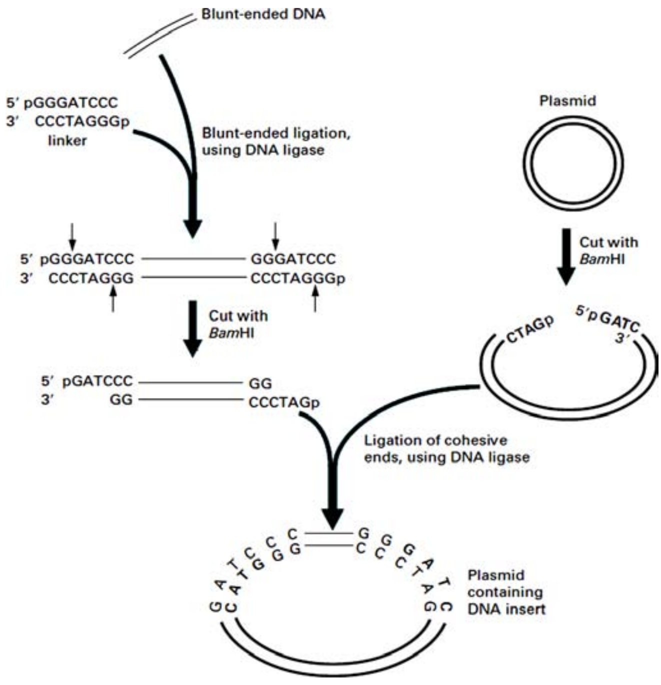


Figure 7. Use of linkers. In this example, blunt-ended DNA is inserted into a specific restriction site on a plasmid, after ligation to a linker containing the same restriction site.

4.1.7 Enrichment Methods for RNA

Frequently an attempt is made to isolate the mRNA transcribed from a desired gene within a particular cell or tissue that produces the protein in high amounts. Thus if the cell or tissue produces a major protein of the cell a large fraction of the total mRNA will code for the protein. An example of this are the B cells of the pancreas, which contain high levels of pro-insulin mRNA. In such cases it is possible to precipitate polysomes which are actively translating the mRNA, by using antibodies to the ribosomal proteins; mRNA can then be dissociated from the precipitated ribosomes. More usually the mRNA required is only a minor component of the total cellular mRNA. In such cases total mRNA may be fractionated by size using sucrose density gradient centrifugation. Then each fraction is used to direct the synthesis of proteins using an in vitro translation system.

4.1.8 Subtractive Hybridization

It is often the case that genes are transcribed in a specific cell type or differentially activated during a particular stage of cellular growth, often at very low levels. It is possible to isolate those mRNA transcripts by subtractive hybridization. Usually the mRNA species common to the different cell types are removed, leaving the cell type or tissue-specific mRNAs for analysis (Fig. 8). This may be undertaken by isolating the mRNA from the so-called subtractor cells and producing a first-strand CDNA. The original mRNA from the subtractor cells is then degraded and the mRNA from the target cells isolated and mixed with the cDNA. All the complementary mRNA-CDNA molecules common to both cell types will hybridize leaving the unbound mRNA which may be isolated and further analyzed. A more rapid approach of analyzing the differential expression of genes has been developed using the PCR.

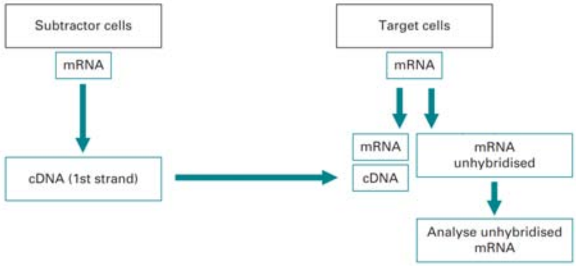


Figure 8. Scheme of analyzing specific mRNA molecules by subtractive hybridization.

4.1.9 Cloning PCR Products

While PCR has to some extent replaced cloning as a method for the generation of large quantities of a desired DNA fragment there is, in certain circumstances, still a requirement for the cloning of PCR-amplified DNA. For example certain techniques such as in vitro protein synthesis are best achieved with the DNA fragment inserted into an appropriate plasmid or phage cloning vector. Cloning methods for PCR follow closely the cloning of DNA fragments derived from the conventional manipulation of DNA. The techniques with which this may be achieved are through one of two ways, blunt-ended or cohesive-ended cloning. Certain thermostable DNA polymerases such as Taq DNA polymerase and 7th DNA polymerase give rise to PCR products having a 3′ overhanging A residue. It is possible to clone the PCR product into dT vectors termed dA: dT cloning. This makes use of the fact that the terminal additions of A residues may be successfully ligated to vectors prepared with T residue overhangs to allow efficient ligation of the PCR product (Fig. 9). The reaction is catalyzed by DNA ligase as in conventional ligation reactions.

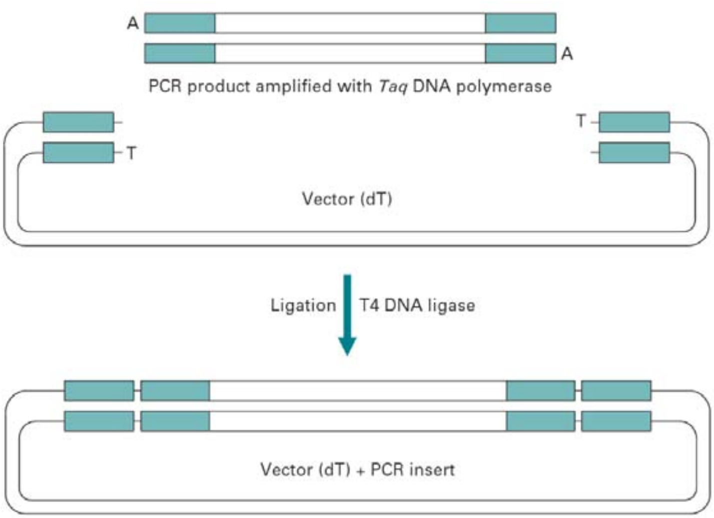


Figure 9. Cloning of PCR products using dA: dT cloning.

It is also possible to carry out cohesive ended cloning with PCR products. In this case oligonucleotide primers are designed with a restriction endonuclease site incorporated into them. Since the complementarity of the primers needs to be absolute at the 3′ end the 5′ end of the primer is usually the region for the location of the restriction site. This needs to be designed with care since the efficiency of digestion with certain restriction endonuclease decreases if extra nucleotides, not involved in recognition, are absent at the 5′ end. In this case the digestion and ligation reactions are the same as those undertaken for conventional reactions.

4.2 CLONING VECTORS

For the cloning of any molecule of DNA it is necessary for that DNA to be incorporated into a cloning vector. These are DNA elements that may be stably maintained and propagated in a host organism for which the vector has replication functions. A typical host organism is a bacterium such as E. coli which grows and divides rapidly. Thus any vector with a replication origin in E. coli will replicate (together with any incorporated DNA) efficiently. Thus, any DNA cloned into a vector will enable the amplification of the inserted foreign DNA fragment and also allow any subsequent analysis to be undertaken. In this way the cloning process resembles the PCR although there are some major differences between the two techniques. By cloning, it is possible to not only store a copy of any particular fragment of DNA, but also produce unlimited amounts of it (Fig. 10).

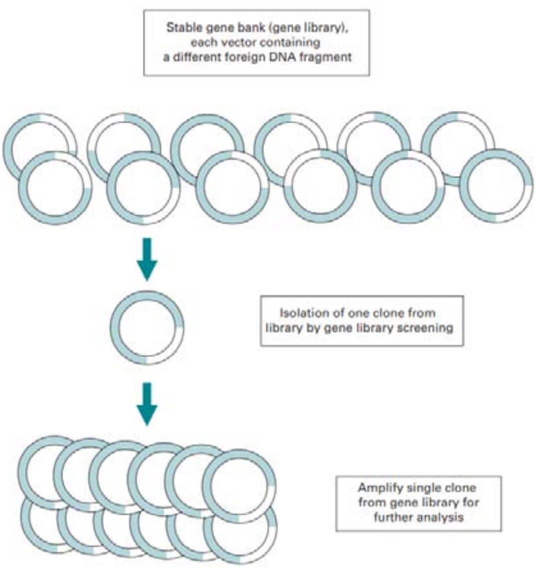


Figure 10. Production of multiple copies of a single clone from a stable gene bank or library.

The vectors used for cloning vary in their complexity, their ease of manipulation, their selection and the amount of DNA sequence they can accommodate (the insert capacity). Vectors have in general been developed from naturally occurring molecules such as bacterial plasmids, bacteriophages or combinations of the elements that make them up, such as cosmids. For gene library constructions there is a choice and trade-off between various vector types, usually related to ease of the manipulations needed to construct the library and the maximum size of foreign DNA insert of the vector (Table 2). Thus, vectors with the advantage of large insert capacities are usually more difficult to manipulate, although there are many more factors to be considered, which are indicated in the following treatment of vector systems.

Table 2. Comparison of vectors generally available for cloning DNA fragments

Vector	Host cell	Vector structure	Insert range (kb)
M13	*E. coli*	Circular virus	1–4
Plasmid	*E. coli*	Circular plasmid	1–5
Phage λ	*E. coli*	Linear virus	2–25
Cosmids	*E. coli*	Circular plasmid	35–45
BACs	*E. coli*	Circular plasmid	50–300
YACs	*S. cerevisiae*	Linear chromosome	100–2000

4.2.1 Plasmids

Many bacteria contain an extra-chromosomal element of DNA, termed a plasmid, which is a relatively small, covalently closed circular molecule, carrying genes for antibiotic resistance, conjugation or the metabolism of ‘unusual’ substrates. Some plasmids are replicated at a high rate by bacteria such as E. coli and so are excellent potential vectors. In the early 1970s a number of natural plasmids were artificially modified and constructed as cloning vectors, by a complex series of digestion and ligation reactions. One of the most notable plasmids, termed pBR322 after its developers Bolivar and Rodriguez (pBR), was widely adopted and illustrates the desirable features of a cloning vector as indicated below (Fig. 11).

- The plasmid is much smaller than a natural plasmid, which makes it more resistant to damage by shearing, and increases the efficiency of uptake by bacteria, a process termed transformation.

- A bacterial origin of DNA replication ensures that the plasmid will be replicated by the host cell. Some replication origins display stringent regulation of replication, in which rounds of replication are initiated at the same frequency as cell division. Most plasmids, including pBR322, have a relaxed origin of replication, whose activity is not tightly linked to cell division, and so plasmid replication will be initiated far more frequently than chromosomal replication. Hence a large number of plasmid molecules will be produced per cell.
- Two genes coding for resistance to antibiotics have been introduced. One of these allows the selection of cells which contain plasmid: if cells are plated on medium containing an appropriate antibiotic, only those that contain plasmid will grow to form colonies. The other resistance gene can be used for detection of those plasmids that contain inserted DNA.
- There are single recognition sites for a number of restriction enzymes at various points around the plasmid, which can be used to open or linearize the circular plasmid. Linearizing a plasmid allows a fragment of DNA to be inserted and the circle closed. The variety of sites not only makes it easier to find a restriction enzyme that is suitable for both the vector and the foreign DNA to be inserted, but, since some of the sites are placed within an antibiotic resistance gene, the presence of an insert can be detected by loss of resistance to that antibiotic. This is termed insertional inactivation.

Insertional inactivation is a useful selection method for identifying recombinant vectors with inserts. For example, a fragment of chromosomal DNA digested with BamH1 would be isolated and purified. The plasmid pBR322 would also be digested at a single site, using BamH1, and both samples would then be deproteinized to inactivate the restriction enzyme. BamH1 cleaves to give sticky ends, and so it is possible to obtain ligation between the plasmid and digested DNA fragments in the presence of T4 DNA ligase. The products of this ligation will include plasmid containing a single fragment of the DNA as an insert, but there will also be unwanted products, such as plasmid that has recircularized without an insert, dimers of plasmid, fragments joined to each other, and plasmid with an insert composed of more than one fragment. Most of these unwanted molecules can be eliminated during subsequent steps. The products of such reactions are usually identified by agarose gel electrophoresis.

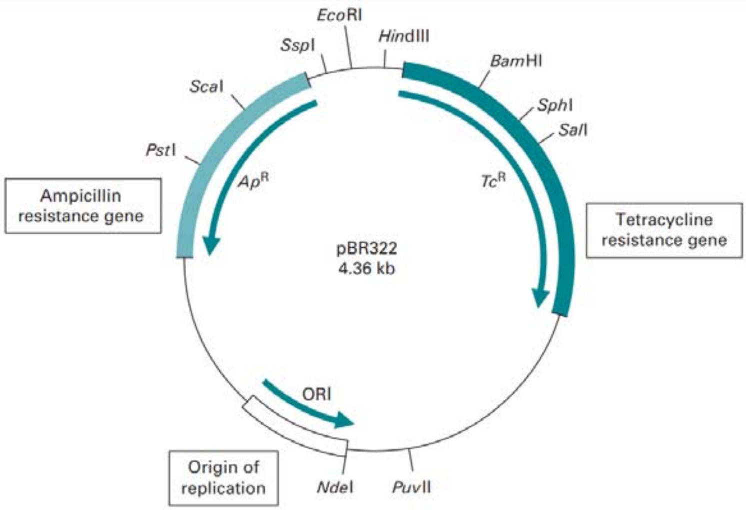


Figure 11. Map and important features of pBR322.

The ligated DNA must now be used to transform E. coli. Bacteria do not normally take up DNA from their surroundings, but can be induced to do so by prior treatment with Ca^{2+} at 4°C; they are then termed competent, since DNA added to the suspension of competent cells will be taken up during a brief increase in temperature termed heat shock. Small, circular molecules are taken up most efficiently, whereas long, linear molecules will not enter the bacteria.

After a brief incubation to allow expression of the antibiotic resistance genes the cells are plated onto medium containing the antibiotic, e.g. ampicillin. Colonies that grow on these plates must be derived from cells that contain plasmid, since this carries the gene for resistance to ampicillin. It is not, at this stage, possible to distinguish between those colonies containing plasmids with inserts and those that simply contain recircularized plasmids. To do this, the colonies are replica plated, using a sterile velvet pad, onto plates containing tetracycline in their medium. Since the BamHI site lies within the tetracycline resistance gene, this gene will be inactivated by the presence of insert, but will be intact in those plasmids that have merely recircularized (Fig. 12). Thus colonies that grow on ampicillin but not on tetracycline must contain plasmids with inserts. Since replica plating gives an identical pattern of colonies on both sets of plates, it is straightforward to recognize the colonies with inserts, and to recover them from the ampicillin plate for further growth. This illustrates the importance of a second gene for antibiotic resistance in a vector.

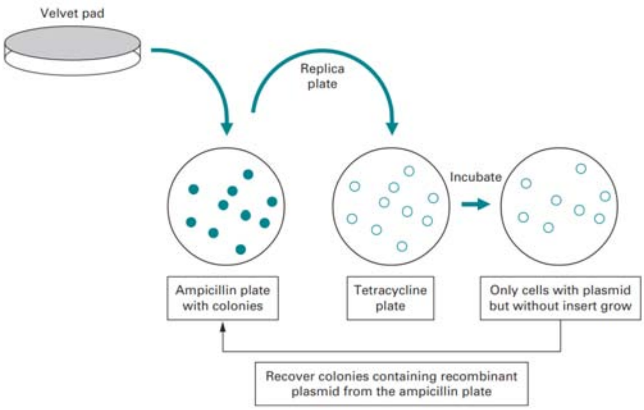


Figure 12. Replica plating to detect recombinant plasmids. A sterile velvet pad is pressed onto the surface of an agar plate, picking up some cells from each colony growing on that plate.

Although recircularized plasmid can be selected against, its presence decreases the yield of recombinant plasmid containing inserts. If the digested plasmid is treated with the enzyme alkaline phosphatase prior to ligation, recircularization will be prevented, since this enzyme removes the 5′ phosphate groups that are essential for ligation. Links can still be made between the 5′ phosphate of insert and the 3′ hydroxyl of plasmid, so only recombinant plasmids and chains of linked DNA fragments will be formed. It does not matter that only one strand of the recombinant DNA is ligated, since the nick will be repaired by bacteria transformed with these molecules.

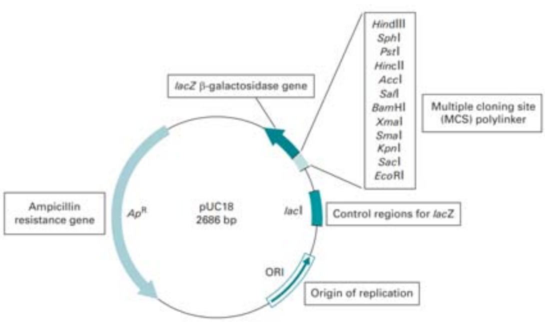


Figure 13. Map and important features of pUC18.

The valuable features of pBR322 have been enhanced by the construction of a series of plasmids termed pUC (produced at the University of California) (Fig. 13). There is an antibiotic resistance gene for tetracycline and origin of replication for E. coli. In addition the most popular restriction sites are concentrated into a region termed the multiple cloning site (MCS). In addition the MCS is part of a gene in its own right and codes for a portion of a polypeptide called β-galactosidase. When the pUC plasmid has been used to transform the host cell E. coli the gene may be switched on by adding the inducer IPTG (isopropyl-β-$_D$-thiogalactopyranoside). Its presence causes the enzyme b-galactosidase to be produced. The functional enzyme is able to hydrolyse a colorless substance called X-gal (5-bromo-4-chloro-3-indolylb-galactopyranoside) into a blue insoluble material (5,5′-dibromo-4,4′–dichloro indigo) (Fig. 14). However if the gene is disrupted by the insertion of a foreign fragment of DNA, a non-functional enzyme results which is unable to carry out hydrolysis of X-gal. Thus, a recombinant pUC plasmid may be easily detected since it is white or colorless in the presence of X-gal, whereas an intact non-recombinant pUC plasmid will be blue since its gene is fully functional and not disrupted. This elegant system, termed blue/white selection, allows the initial identification of recombinants to be undertaken very quickly and has been included in a number of subsequent vector systems. This selection method and insertional inactivation of antibiotic resistance genes do not, however, provide any information on the character of the DNA insert, just the status of the vector. To screen gene libraries for a desired insert hybridization to gene probes is required.

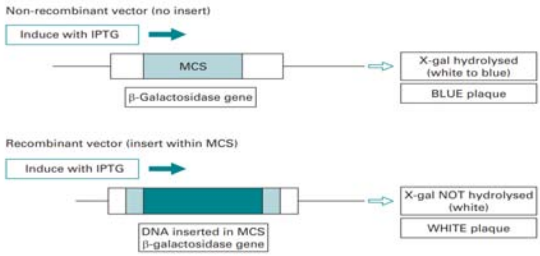


Figure 14. Principle of blue/white selection for the detection of recombinant vectors.

4.2.2 Virus-based Vectors

A useful feature of any cloning vector is the amount of DNA it may accept or have inserted before it becomes unviable. Inserts greater than 5 kb increase plasmid size to the point at which efficient transformation of bacterial cells decreases markedly, and so bacteriophages (bacterial viruses) have been adapted as vectors in order to propagate larger fragments of DNA in bacterial cells. Cloning vectors derived from λ bacteriophage are commonly used since they offer an approximately 16-fold advantage in cloning efficiency in comparison with the most efficient plasmid cloning vectors.

Phage λ is a linear double-stranded phage approximately 49 kb in length (Fig. 15). It infects E. coli with great efficiency by injecting its DNA through the cell membrane. In the wild-type phage λ the DNA follows one of two possible modes of replication. Firstly the DNA may either become stably integrated into the E. coli chromosome where it lies dormant until a signal triggers its excision. This is termed the lysogenic life cycle. Alternatively, it may follow a lytic life cycle where the DNA is replicated upon entry to the cell, phage head and tail proteins synthesized rapidly and new functional phage assembled. The phage are subsequently released from the cell by lysing the cell membrane to infect further E. coli cells nearby. At the extreme ends of the phage λ are 12 bp sequences termed cos (cohesive) sites. Although they are asymmetric they are similar to restriction sites and allow the phage DNA to be circularized. Phage may be replicated very efficiently in this way, the result of which are concatemers of many phage genomes which are cleaved at the cos sites and inserted into newly formed phage protein heads.

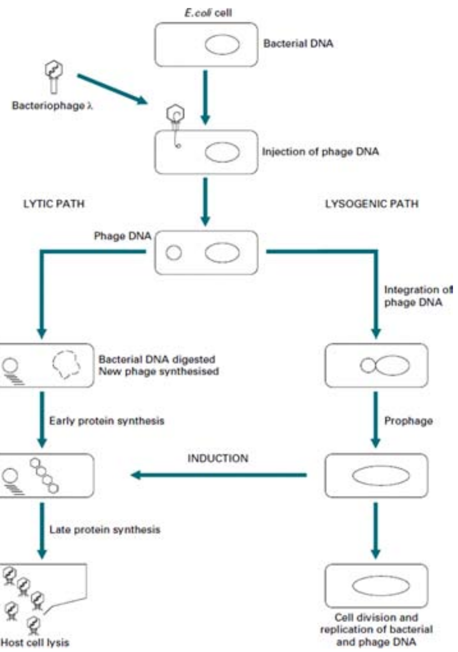


Figure 15. The lysogenic and lytic cycles of bacteriophage λ.

Much use of phage λ has been made in the production of gene libraries mainly because of its efficient entry into the E. coli cell and the fact that larger fragments of DNA may be stably integrated. For the cloning of long DNA fragments, up to approximately 25 kb, much of the non-essential λ DNA that codes for the lysogenic life cycle is removed and replaced by the foreign DNA insert. The recombinant phage is then assembled into pre-formed viral protein particles, a process termed in vitro packaging. These newly formed phage are used to infect bacterial cells that have been plated out on agar (Fig. 16).

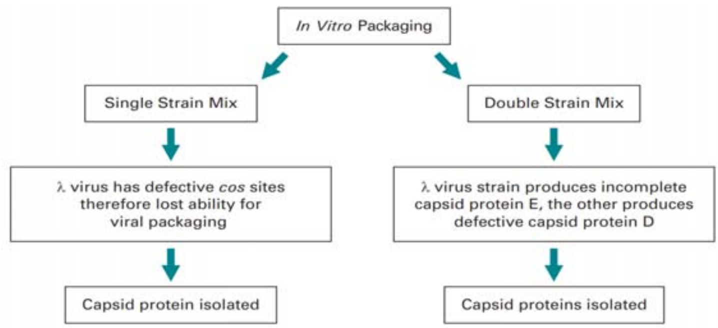


Figure 16. Two strategies for producing in vitro packaging extracts for bacteriophage λ.

Once inside the host cells, the recombinant viral DNA is replicated. All the genes needed for normal lytic growth are still present in the phage DNA, and so multiplication of the virus takes place by cycles of cell lysis and infection of surrounding cells, giving rise to plaques of lysed cells on a background, or lawn, of bacterial cells. The viral DNA including the cloned foreign DNA can be recovered from the viruses from these plaques and analyzed further by restriction mapping and agarose gel electrophoresis.

In general two types of λ phage vectors have been developed, λ insertion vectors and λ replacement vectors (Fig. 17). The λ insertion vectors accept less DNA than the replacement type since the foreign DNA is merely inserted into a region of the phage genome with appropriate restriction sites; common examples are λgt10 and λcharon16A. With a replacement vector a central region of DNA not essential for lytic growth is removed (a stuffer fragment) by a double digestion with for example EcoRI and BamHI. This leaves two DNA fragments termed right and left arms. The central stuffer fragment is replaced by inserting foreign DNA between the arms to form a functional recombinant λ phage. The most notable examples of λ replacement vectors are λEMBL and λZap.

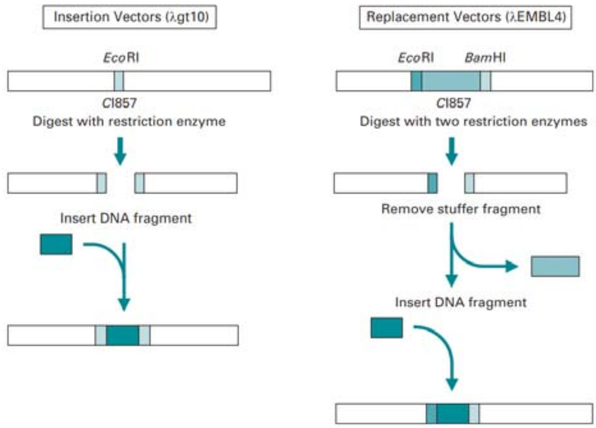


Figure 17. General schemes used for cloning in λ insertion and λ replacement vectors. Cl857 is a temperature-sensitive mutation that promotes lysis at 42°C after incubation at 37°C.

λZap is a commercially produced cloning vector that includes unique cloning sites clustered into a multiple cloning site (MCS) (Fig. 18). Furthermore the MCS is located within a lacZ region providing a blue/white screening system based on insertional inactivation. It is also possible to express foreign cloned DNA from this vector. This is a very useful feature of some λ vectors since it is then possible to screen for protein product rather than the DNA inserted into the vector. This screening is therefore undertaken with antibody probes directed against the protein of interest. Other features that make this a useful cloning vector are the ability to produce RNA transcripts termed cRNA or riboprobes. This is possible because two promoters for RNA polymerase enzymes exist in the vector, a T7 and a T3 promoter which flank the MCS.

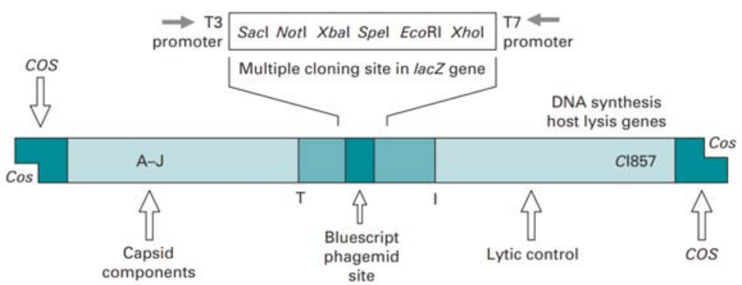


Figure 18. General map of λZap cloning vector, indicating important areas of the vector. The multiple cloning site is based on the lacZ gene, providing blue/

white selection based on the β-galactosidase gene. In between the initiator (I) site and terminator (T) site lie sequences encoding the phagemid Bluescript.

One of the most useful features of λZap is that it has been designed to allow automatic excision in vivo of a small 2.9 kb colony-producing vector termed a phagemid, pBluescript SK. This technique is sometimes termed single-stranded DNA rescue and occurs as the result of a process termed super-infection where helper phage are added to the cells which are grown for an additional period of approximately 4 h (Fig. 19).

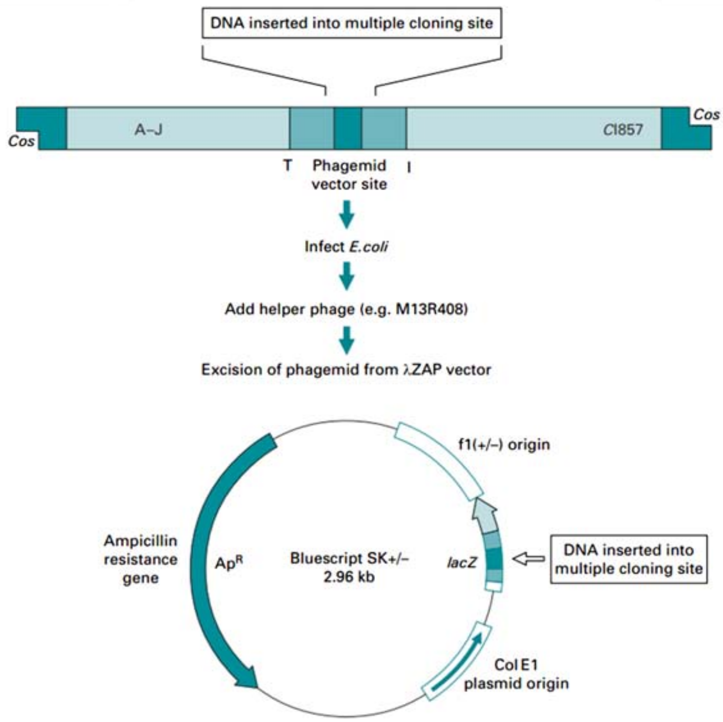


Figure 19. Single-stranded DNA rescue of phagemid from λZap. The single-stranded phagemid pBluescript SK may be excised from λZap by addition of helper phage. This provides the necessary proteins and factors for transcription between the I and T sites in the parent phage to produce the phagemid with the DNA cloned into the parent vector.

The helper phage displaces a strand within the λZap which contains the foreign DNA insert. This is circularized and packaged as a filamentous phage similar to M13. The packaged phagemid is secreted from the E. coli cell and may be recovered from the supernatant. Thus the λZap vector allows a number of diverse manipulations to be undertaken without the necessity of recloning or subcloning foreign DNA fragments. The process of subcloning is sometimes necessary when the manipulation of a gene fragment cloned in a general purpose vector needs to be inserted into a more specialized vector for the application of techniques such as in vitro mutagenesis or protein production.

4.2.3 M13 and Phagemid-based Vectors

Much use has been made of single-stranded bacteriophage vectors such as M13 and vectors which have the combined properties of phage and plasmids, termed phagemids. M13 is a filamentous coliphage with a single-stranded circular DNA genome. Upon infection of E. coli, the DNA replicates initially as a double-stranded molecule but subsequently produces single-stranded virions for infection of further bacterial cells (lytic growth). The nature of these vectors makes them ideal for techniques such as chain termination sequencing and in vitro mutagenesis since both require single-stranded DNA.

M13 or phagemids such as pBluescript SK infect E. coli harboring a male-specific structure termed the F-pilus. They enter the cell by adsorption to this structure and once inside the phage DNA is converted to a double-stranded replicative form or RF DNA. Replication then proceeds rapidly until some 100 RF molecules are produced within the E. coli cell. DNA synthesis then switches to the production of single strands and the DNA is assembled and packaged into the capsid at the bacterial periplasm. The bacteriophage DNA is then encapsulated by the major coat protein, gene VIII protein, of which there are approximately 2800 copies with three to six copies of the gene III protein at one end of the particle. The extrusion of the bacteriophage through the bacterial periplasm results in a decreased growth rate of the E. coli cell rather than host cell lysis and is visible on a bacterial lawn as an area of clearing. Approximately 1000 packaged phage particles may be released into the medium in one cell division.

In addition to producing single-stranded DNA the coliphage vectors have a number of other features that make them attractive as cloning vectors. Since the bacteriophage DNA is replicated as a double-stranded RF DNA intermediate a number of regular DNA manipulations may be performed such as restriction digestion, mapping and DNA ligation. RF DNA is prepared by lysing infected E. coli cells and purifying the supercoiled circular phage DNA with the same methods used for plasmid isolation. Intact single-stranded DNA packaged in the phage protein coat located in the supernatant may be precipitated with reagents such as polyethylene glycol, and the DNA purified with phenol/chloroform. Thus the bacteriophage may act as a plasmid under certain circumstances and at other times produce DNA in the fashion of a virus. A family of vectors derived from M13 are currently widely used termed M13mp8/9, mp18/19, etc., all of which have a number of highly useful features. All contain a synthetic MCS, which is located in the lacZ gene without disruption of the reading frame of the gene. This allows efficient selection to be undertaken based on the technique of blue/white screening. As the series of vectors were developed the number of restriction sites was increased in an asymmetric fashion. Thus M13mp8, mp12, mp18 and sister vectors which have the same MCS but in reverse orientation, M13mp9, mp13 and mp19 respectively have more restriction sites in the MCS making the vector more useful since greater choice of restriction enzymes is available. However, one problem frequently encountered with M13 is the instability and spontaneous loss of inserts that are greater than 6 kb.

Phagemids are very similar to M13 and replicate in a similar fashion. One of the first phagemid vectors, pEMBL, was constructed by inserting a fragment of another phage termed f1 containing a phage origin of replication and elements for its morphogenesis into a pUC8 plasmid. Following super-infection with helper phage the f1 origin is activated allowing single-stranded DNA to be produced. The phage is assembled into a phage coat extruded through the periplasm and secreted into the culture medium in a similar way to M13. Without superinfection the phagemid replicates as a pUC type plasmid and in the replicative form (RF) the DNA isolated is double-stranded. This allows further manipulations such as restriction digestion, ligation and mapping analysis to be performed. The pBluescript SK vector is also a phagemid and can be used in its own right as a cloning vector and manipulated as if it were a plasmid. It may, like M13, be used in nucleotide sequencing and site-directed mutagenesis, and it is also possible to produce RNA transcripts that may be used in the production of labelled cRNA probes or riboprobes.

4.2.4 Cosmid Based Vectors

The way in which the phage λ DNA is replicated is of particular interest in the development of larger insert cloning vectors termed cosmids. These are especially useful for the analysis of highly complex genomes and are an important part of various genome mapping projects.

The upper limit of the insert capacity of phage λ is approximately 21 kb. This is because of the requirement for essential genes and the fact that the maximum length between the cos sites is 52 kb. Consequently cosmid vectors have been constructed that incorporate the cos sites from phage λ and also the essential features of a plasmid, such as the plasmid origin of replication, a gene for drug resistance, and several unique restriction sites for insertion of the DNA to be cloned. When a cosmid preparation is linearised by restriction digestion, and ligated to DNA for cloning, the products will include concatamers of alternating cosmid vector and insert. Thus the only requirement for a length of DNA to be packaged into viral heads is that it should contain cos sites spaced the correct distance apart; in practice this spacing can range between 37 and 52 kb. Such DNA can be packaged in vitro if phage head precursors, tails and packaging proteins are provided. Since the cosmid is very small, inserts of about 40 kb in length will be most readily packaged. Once inside the cell, the DNA recircularises through its cos sites, and from then onwards behaves exactly like a plasmid.

4.2.5 Large Insert Capacity Vectors

The advantage of vectors that accept larger fragments of DNA than phage λ or cosmids is that fewer clones need to be screened when searching for the foreign DNA of interest. They have also had an enormous impact in the mapping of the genomes of organisms such as the mouse and are used extensively in the human genome mapping project. Recent developments have allowed the production of large insert capacity vectors based on human artificial chromosomes, bacterial artificial chromosomes (BACs), mammalian artificial chromosomes (MACs) and on the virus P1 (PACs), P1 artificial chromosomes. However, perhaps the most significant development are vectors based on yeast artificial chromosomes (YACs).

4.2.6 Yeast Artificial Chromosome (YAC) Vectors

Yeast artificial chromosomes (YACs) are linear molecules composed of a centromere, telomere and a replication origin termed an ARS element (autonomous replicating sequence). The YAC is digested with restriction enzymes at the SUP4 site (a suppressor tRNA gene marker) and BamHI sites separating the telomere sequences. This produces two arms and the foreign genomic DNA is ligated to produce a functional YAC construct. YACs are replicated in yeast cells; however, the external cell wall of the yeast needs to be removed to leave a spheroplast. These are osmotically unstable and also need to be embedded in a solid matrix such as agar. Once the yeast cells are transformed only correctly constructed YACs with associated selectable markers are replicated in the yeast strains. DNA fragments with repeat sequences are sometimes difficult to clone in bacterial-based vectors but may be successfully cloned in YAC systems. The main advantage of YAC-based vectors, however, is the ability to clone very large fragments of DNA. Thus the stable maintenance and replication of foreign DNA fragments of up to 2000 kb have been carried out in YAC vectors and they are the main vector of choice in the various genome mapping and sequencing projects.

REFERENCES

1. Augen, J. (2005). Bioinformatics in the Post-Genomic Era. Reading, MA: Addison-Wesley.
2. Brooker, R. J. (2005). Genetics Analysis and Principles, 2nd edn. McGraw-Hill.
3. Brown, T. A. (2006). Gene Cloning and DNA Analysis. Oxford, UK: Wiley–Blackwell.
4. Cross, D. M. and Bayliss, M. K. (2000). A commentary on the use of hepatocytes in drug metabolism studies during drug discovery. Drug Metabolism Reviews, 32, 219–240.
5. Eichler, H.-G., Pinatti, F., Flamion, B., Leufkens, H. and Breckenridge, A. (2008). Balancing early market access to new drugs with the need for benefit/risk data: a mounting dilemma. Nature Reviews Drug Discovery, 7, 818–826.
6. Ekins, S., Mestres, J. and Tasta, B. (2007). In silico pharmacology for drug discovery: methods for virtual ligand screening and profiling. British Journal of Pharmacology, 152, 9–20.
7. Gomez-Hens, A. and Aguillar-Caballos, M. P. (2007). Modern analytical approaches to highthroughput drug discovery. Trends in Analytical Chemistry, 26, 171–180.
8. Imming, P., Sinning, C. and Meyer, A. (2006). Drugs, their targets and the nature and number of drug targets. Nature Reviews Drug Discovery, 5, 821–832.
9. Joy, T. and Hegele, R. A. (2008). Is raising HDL a futile strategy for atheroprotection? Nature Reviews Drug Discovery, 7, 143–154.
10. Karlberg, J. P. E. (2008). Trends in disease focus of drug development. Nature Reviews Drug Discovery, 7, 639–640.
11. Primrose, S. B. and Twyman, R. (2006). Principles of Gene Manipulation and Genomics. Oxford, UK: Wiley–Blackwell.

12. Rader, D. J. and Daugherty, A. (2008). Translating molecular discoveries into new therapies for atherosclerosis. Nature, 451, 904–912.
13. Rognan, D. (2007). Chemogenomic approaches to rational drug design. British Journal of Pharmacology, 152, 38–52.
14. Rose, A. D. (2008). Pharmacogenetics in drug discovery and development: a translational perspective. Nature Reviews Drug Discovery, 7, 807–817.
15. Starkuviene, V. and Pepperkok, R. (2007). The potential of high-content high-throughput microscopy in drug discovery. British Journal of Pharmacology, 152, 62–71.
16. Strachan, T. and Read, A. P. (2004). Human Molecular Genetics, 3rd edn. Oxford, UK: Bios.
17. Walker, J. M. and Rapley, R. (2008). Molecular Biomethods Handbook, 2nd edn. Totowa, NJ: Humana Press.
18. Wang, C.-Y., Liu, P. Y. and Liao, J. K. (2008). Pleiotropic effects of statin therapy: molecular mechanisms and clinical results. Trends in Molecular Medicine, 14, 32–44.
19. Watson, J. D., Caudy, A. A., Myers, R. M. and Witkowski, J. A. (2007). Recombinant DNA: Genes and Genomes. San Francisco, CA: W. H. Freeman.

CHAPTER

5

IMMUNOCHEMICAL TECHNIQUES

INTRODUCTION

All vertebrates have advanced immune system. The more complex the organism the more advanced the immune system. The immune system of mammals has evolved over a million years; it provides an incredible protection system capable of responding to infective challenges that arise in the body. Immunity in our body is monitored or controlled by specific cells from stem cells in bone marrow.

The most important cell types are B and T lymphocytes, which have the ability to act against bacterial and other viruses. B-cells releases antibody against stimulation of a foreign substance into the body.

An antigen is a foreign substance capable of an immune response leading to the production of antibodies. They are the targets to which antibodies bind. So antibodies are antigen specific (bind to the antigen that initiated its production).

This unit focuses on the identification and diagnostic chemical techniques on the response of an antibody to a specific antigen. Immunochemical methods are based on the selective, reversible and non-covalent binding of antigens by antibodies. These methods are employed to detect or quantify either antigens or antibodies.

5.1 OVERVIEW OF IMMUNOCHEMICAL TECHNIQUES

Immunochemistry is an advanced area of immunology. It deals with the chemical components and chemistry (chemical reactions) of immunological phenomena, that is of antibody and antigen. Immunochemical methods are processes utilizing the highly specific affinity of an antibody for its antigen. It detects the distribution of a given protein or antigen in tissues or cells. The methods used for the immunochemical analysis are called Immunochemical techniques; they are highly important in diagnostic and clinical context, as now even normal cell with many proteins are altered in diseased state (in cancer). Immunisation of an animal with mixture of antigens results in production of polyspecific antibodies, containing immunoglobulins against many antigens (e.g. antiserum against human serum proteins used in immunoelectrophoresis).

5.1.1 Characteristics and Role

Characteristics

- Simple, rapid and robust
- Highly sensitive
- Easily automated – applicable to regular clinical diagnostic laboratories
- Does not require extensive and easily destructible sample preparation
- Do not require expensive instrumentation
- Mostly based on simple photo-, fluoro- and luminometric detection
- Measurements may be either qualitative or quantitative.

Roles

The characteristic features of these techniques are helpful for the following:

- Function of newly identified or novel proteins can be identified
- Importance of uncharacterized protein in their natural environment can be analyzed
- Determination of species or tissues in which the protein or residues is expressed

- The cell type or sub-cellular compartment in which the protein can be found
- Detect whether there is any variation in expression of protein during development of the organism
- Some alteration in the normal expression pattern of a particular protein may indicate a particular disease state; Gives information that may be useful for diagnosis and treatment.

5.2 METHODS OF ANALYSIS

All immunochemical methods are based on a highly specific and sensitive reaction between an antigen and an antibody. Antibodies are immunoglobins, belonging to a family of glycoproteins – IgG, IgA, IgD, IgM and IgE. Structurally, antibodies are often visualized as Y-shaped molecules, each containing 4 polypeptides – 2 identical polypeptide units called heavy chains and another 2 called light chains. It has a domain called Fab, the site where it binds to an antigen. The region of an antigen binding to an antibody is called an epitope.

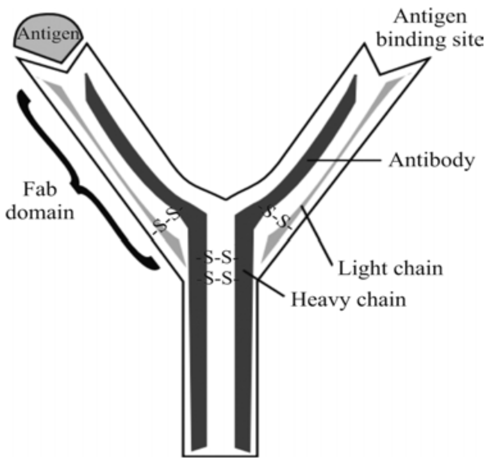


Figure 1. Antigen-Antibody reaction.

The measure of the strength of the binding is called affinity, and it is usually expressed in terms of the concentration of an antibody-antigen complex measured at equilibrium. It is measured by quantitative precipitin curve (basis for many immunochemical techniques) proposed by Heidelberger and Kendall in 1935. Quantitative precipitin curve: it describes the relationship between the antigen concentration and the amount of precipitate for a constant quantity of an antibody. Three zones can be distinguished from the precipitin curve:

- antibody excess zone – first phase where less antigen is present in sample
- equivalence zone – both antigen and antibody are cross-linked forming precipitate; no free antigen or antigen is present
- antigen excess zone – amount of precipitate reduces due to high antigen concentration

The precipitin curve forms the basis of most of the immunochemical techniques that can be performed in clinical laboratories.

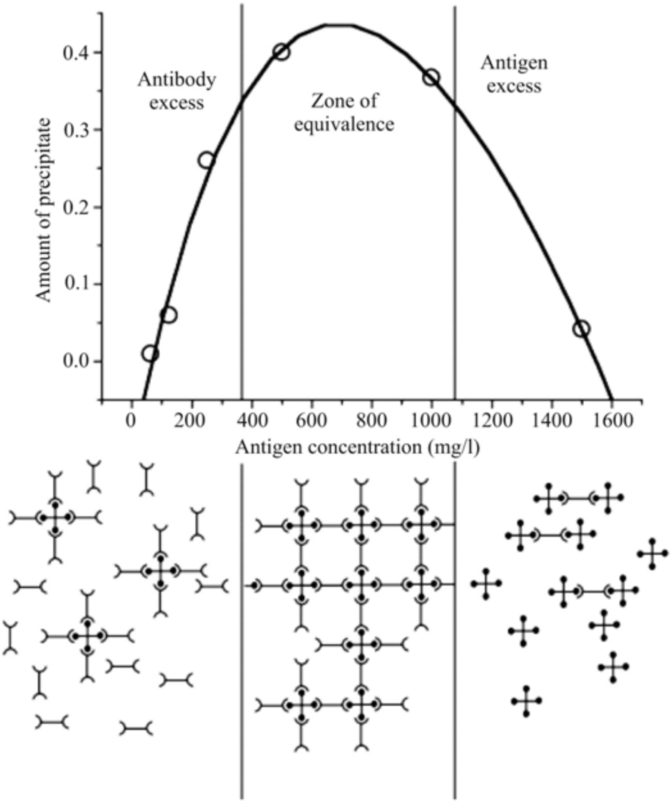


Figure 2. Precipitin curve.

5.2.1 Antigens

Antigens are macromolecules of natural or synthetic origin; chemically they consist of various polymers – proteins, polypeptides, polysaccharides or nucleoproteins. Antigens display two essential properties: first, they are able to evoke a specific immune response, either cellular or humoral type; and, second, they specifically interact with products of this immune response, i.e. antibodies or immunocompetent cells. A complete antigen – immunogen – consists of a macromolecule that bears antigenic determinants (epitopes) on its surface (Fig. 3). The antigenic determinant (epitope) is a certain group of atoms on the antigen surface that actually interacts with the binding site on the antibody or lymphocyte receptor for the antigen. Number of epitopes on the antigen surface determines its valency. Low-molecular-weight compound that cannot as such elicit production of antibodies, but is able to react specifically with the products of immune response, is called hapten (incomplete antigen).

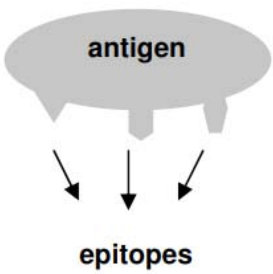


Figure 3. Antigen and epitopes.

5.2.2 Antibodies

Antibodies are produced by plasma cells that result from differentiation of B lymphocytes following stimulation with antigen. Antibodies are heterogeneous group of animal glycoproteins with electrophoretic mobility β - γ, and are also called immunoglobulins (Ig). Every immunoglobulin molecule contains at least two light (L) and two heavy (H) chains connected with disulphidic bridges (Fig. 4). One antibody molecule contains only one type of light as well as heavy chain. There are two types of light chains - κ and λ - that determine type of immunoglobulin molecule; while heavy chains exist in 5 isotypes - γ, μ, α, δ, ε; and determine class of immunoglobulins - IgG, IgM, IgA, IgD and IgE. The C terminal parts of both chains form constant regions on an antibody molecule, while the N-terminal parts are designated as variable regions and constitute the portion of antibody molecule that binds antigen – binding site or paratope. Except for IgM that has 10 binding sites and IgA with 4 binding sites, immunoglobulin molecules possess 2 antigen binding sites.

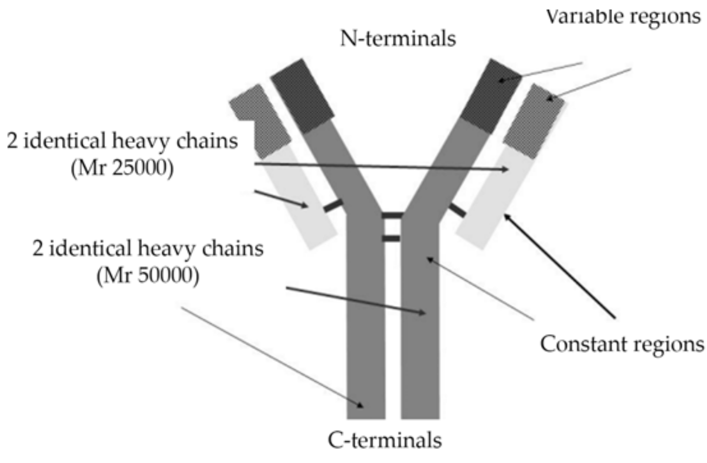


Figure 4. Structure of antibody of IgG class.

Types of Antibody Used

The antibodies used for the methods are produced by various ways:

i. *Monoclonal antibody* – products of single clone of plasma cells by Blymphocytes; mostly prepared in laboratory. They are directed against single epitope – identical copies with same structure and antigen specificity. They have excellent specificity but poor ability to precipitate antigen.

ii. *Polyclonal antibody* – they are conventional, i.e., produced by immunization of animals with antigen. Thus antibody consists of mixture of monoclonal antibodies having specificity for complex antigens.

Sometimes monoclonal is called as “monovalent” and polyclonal as “polyvalent” which indicates the antigen specificity

5.3 PRECIPITATION METHODS IN GEL

As a support matrix, agar or agarose gels are used most often. In single immunodiffusion only one component (i.e. antigen or antibody) diffuses from the place of sample application, while the other reaction partner is dispersed evenly in the gel. If both components of the immunochemical reaction diffuse in the gel against each other from places of their application, the technique is called double immunodiffusion. In the area of antigenantibody reaction a precipitation zone appears as line, crescent or circle. The immunodiffusion methods in gel are represented e.g. by the Mancini's single radial immunodiffusion and the Ouchterlony's double immunodiffusion. Other techniques combine an immunochemical reaction with electrophoretic separation, such as immunoelectrophoresis, Laurell's (rocket) immunoelectrophoresis, and immunofixation.

Single Radial Immunodiffusion

Single radial immunodiffusion represents a simple method without requirements for expensive instruments. The antigen is applied into wells that are cut in the agarose gel containing dispersed corresponding monospecific antibody. The agarose plate is incubated at room temperature for 48-72 hours depending on the specific protein in question. The antigen from the sample diffuses out from the wells into the agarose where the antibody concentration is constant. In a distance from the well where antigen concentration is equivalent to the antibody concentration, the complex antigen-antibody precipitates and appears as a strong white ring around the well. The square of the ring diameter is directly proportional to the antigen concentration.

The protocol involves several steps:

- Boiled agarose is cooled to 50°C and monospecific serum is added to it. The gel is poured onto a glass plate or a Petri dish.
- After the agarose has solidified, round-shaped wells (starts) are cut out. Samples and standards are applied to the wells.
- The plate is incubated in a wet chamber. The antigen diffuses radially to the gel and reacts with antibody.
- Then, the plate is stained in order to visualise the precipitate rings, and dried.
- Diameters of the rings are measured and squared values are calculated. Calibration curve is plotted using standards. Finally, concentration of antigen in samples is read from the plot.

The sensitivity of this technique is about 10-200 mg/L, and therefore it is suitable for estimation of most serum proteins (e.g. immunoglobulin IgG, IgM, IgA, prealbumin, transferrin, α_2-macroglobulin, ceruloplasmin). This technique has been, however, currently largely replaced with immunoturbidimetry or nephelometry.

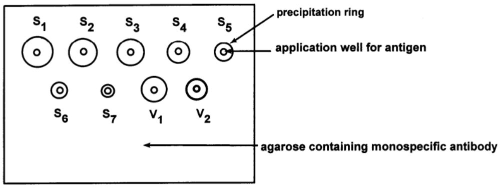


Figure 5. Single radial imunodiffusion (S1 – S7 – standards, V1-V2 – samples)

Double immunodiffusion

In double immunodiffusion reaction, the antigen and the antibody diffuse towards each other. The Ouchterlony's modification is the one most often used. It is based on wells punched into the agarose gel in a rosette pattern that are filled with antigen or antibody solutions, respectively. Both antigen and antibody molecules are allowed to diffuse radially into the gel surrounding the wells; and where the antigen and specific reactive antibody meet, a precipitin line forms. If antiserum to several possible antigens is placed into the central well and the outer wells are filled by different antigens, precipitin lines of various shapes can arise. For instance, if two antigenic mixtures are applied into two adjacent wells, the following patterns of precipitin lines can be observed, in dependence on the relationship between the two antigenic mixtures:

- *a reaction of identity* – if two identical antigens are applied into two adjacent wells, the precipitin bands form a continuous arc.
- *a reaction of non-identity* – the precipitin bands form lines that intersect.
- *a reaction of partial identity* – it is characterized by a formation of a spur. The common hooked precipitation line arises from the reaction of the common antigenic determinants on both antigens with the antibody. The spur means that the second antigen lacks an epitope present in the first antigen that is recognized by one of the antibodies in the antiserum.

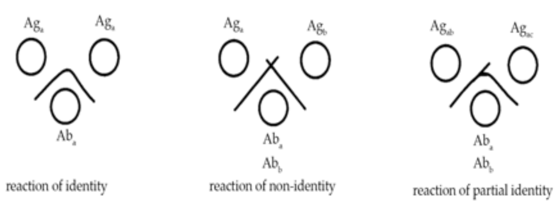


Figure 6. Double immunodiffusion according to Ouchterlony (Ag$_a$, Ag$_b$, Ag$_c$ – antigens with epitopes a, b, c; Ab$_a$, Ab$_b$ – antibodies against epitopes a and b)

Double immunodiffusion is a qualitative technique suitable for the identification of antigens and the estimation of their mutual immunochemical relationships if the specific antisera are available. On the other hand, this method may be used for characterization of antibodies using known purified antigens.

5.3.1 Agglutination

Agglutination (from Latin, agglutino – to glue/ attach) is a process of formation of clumping of cells; it occurs due to reaction of antibody on a particulate antigen (Fig. 7). Agglutination reaction is based on the formation of bridges between bivalent (IgG) or multivalent (IgM) antibodies and antigenic particles with multiple epitopes. Bivalency or multivalency of the used antibody and multiple antigenic determinants on the surface of particles are necessary for the creation of cross-linking and the formation of a high-molecular-weight lattice that is observable macroscopically. IgM antibodies with ten antigenic-combining sites permit a more effective bridging than IgG. Some antibodies react with the corpuscular antigen, but may not produce agglutination. In this case, the agglutination may be achieved if an antiimmunoglobulin is added into the reaction mixture (the Coombs test).

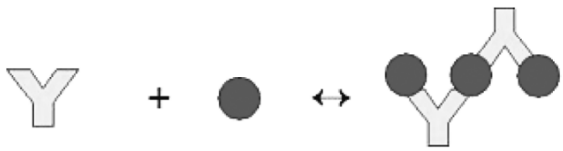


Figure 7. Agglutination reaction.

Direct Agglutination

In a direct agglutination test, the antigen is an integral part of the cell surface (red blood cells, bacteria). A suspension of particles is directly agglutinated by specific antibodies present in the examined sample. This assay is frequently used in the hematology for the determination of blood group or in the immunological diagnostics for detection of specific antibodies directed against naturally occurring antigens on the surface of some microbes (for example against Salmonella typhi – the Widal test). For the examination of antibodies, the test is usually performed with serial dilutions of the sample. The highest dilution of serum that still causes agglutination is denoted as a titer of the antibody.

Indirect Agglutination

Indirect agglutination assay utilises particles with the antigens that have been passively attached to their surface. Originally, red blood cells were used as carriers for antigens; lately, inert particles such as latex, colloid gold and other substances have been shown to be more versatile for the agglutination technique. Many proteins, bacterial and viral antigens are easily adsorbed onto the particle, while other substances require modification by tannic acid or chromium chloride.

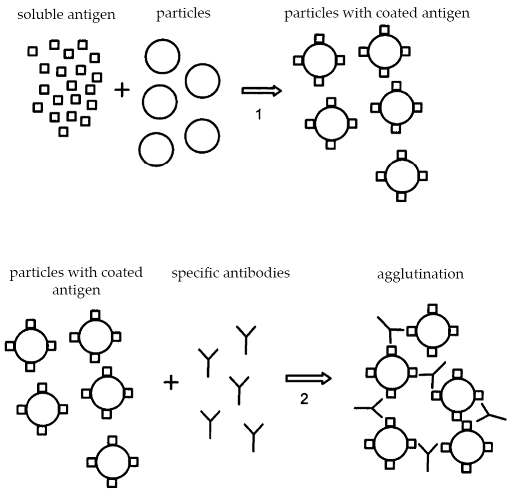


Figure 8. Indirect agglutination for the determination of antibodies.

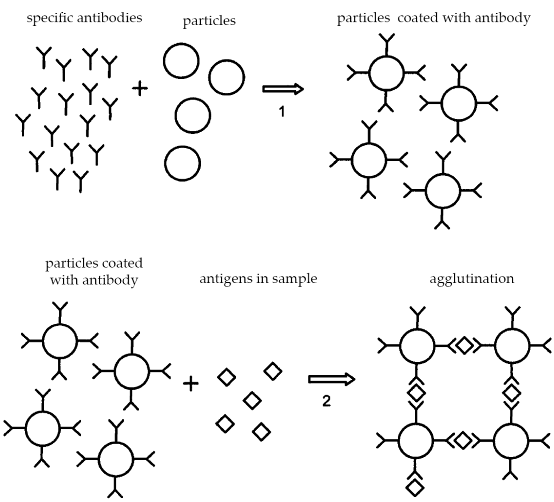


Figure 9. Indirect agglutination for the determination of antigens.

Agglutination Inhibition Test

Agglutination inhibition test is another form of agglutination reaction that permits the determination of soluble antigens. It is based on the competition between the antigens in solution and the same antigens on the particle surface for limited amount of antibodies. The specific antibodies are incubated with the test solution containing the soluble antigens. Following the addition of the particles coated with the antigens, the agglutination does not occur because most of the antibodies have been already saturated with a soluble form of the same antigen from the sample. Therefore, antibody binding sites are unavailable for bridging the coated particles. A lack of agglutination indicates a positive result (Fig. 10). On the other hand, if the soluble antigen is not present in the tested sample, after the addition of the corpuscular antigen the agglutination develops.

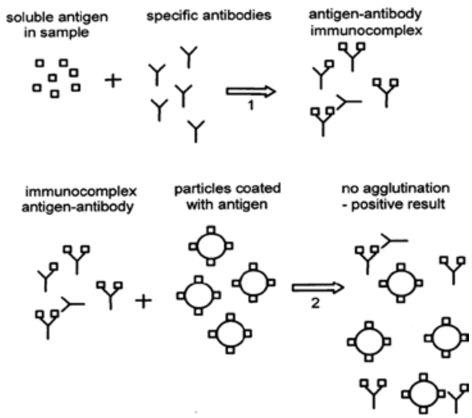


Figure 10. Agglutination inhibition.

5.3.2 Immunoprecipitation

Immunoprecipitation methods include flocculation and precipitation reactions. When a solution of an antigen is mixed with its corresponding antibody under suitable conditions, the reactants form flocculating or precipitating aggregates. They can be assessed visually by the formation of precipitin line at the region of equivalence – where equivalent amount of antigen and antibody are present.

It may be either:

- Simple – reaction of one antigen and an Precipitin line antibody

- Complex – when many unrelated reactants are used

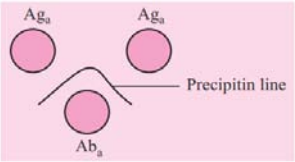


Figure 11. Precipitation reaction.

In simple methods, a concentration gradient is established between the antigen and antibody; it includes:

- Single radial immune diffusion (SRID) – developed by Mancini; simple technique where the circular precipitin area originating from antigen well in gel is equal to amount of antigen concentration.
- Double immune diffusion – developed by Orjan Ouchterlony; both antigen and antibody diffuse from separate wells in a gel to form precipitin line.

In complex methods, different antigens are compared with an antibody and vice versa. The result is based on the presence or absence of precipitin line.

5.3.3 Immunoelectrophoresis

Immunoelectrophoresis is a qualitative technique that combines of two methods one after another:

- Gel electrophoresis – separation of components by charge (Fig. 12a)
- Immunodiffusion – diffusion of both antibody and antigen toward each other produces precipitin line (Fig 12b)

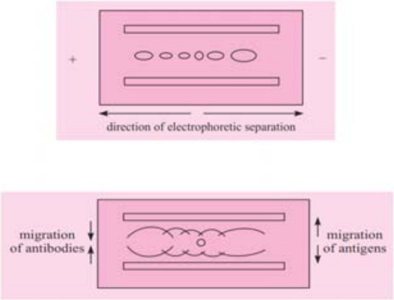


Figure 12. (a) electrophoresis, (b) immunodiffusion.

It is strictly qualitatively technique to detect antibody and antigen in sample. But also a quantitative technique exist called rocket immunoelectrophoresis to measure the antigen level in a sample. Another variation called as counter-current immunoelectrophoresis where it is similar to double immunodiffusion.

5.3.4 Immunofixation

Immunofixation is used for detection and identifying antibody in serum, urine and other body fluids (Fig. 13). The principle is similar to immunoelectrophoresis, involves two stages:

- Electrophoresis separation of antibody proteins
- Immunoprecipitation separation with specific antibodies.

It is easy to perform, more sensitive and easily evaluated. In clinical laboratories, is used for detection for cancerous myeloma through specific antibody.

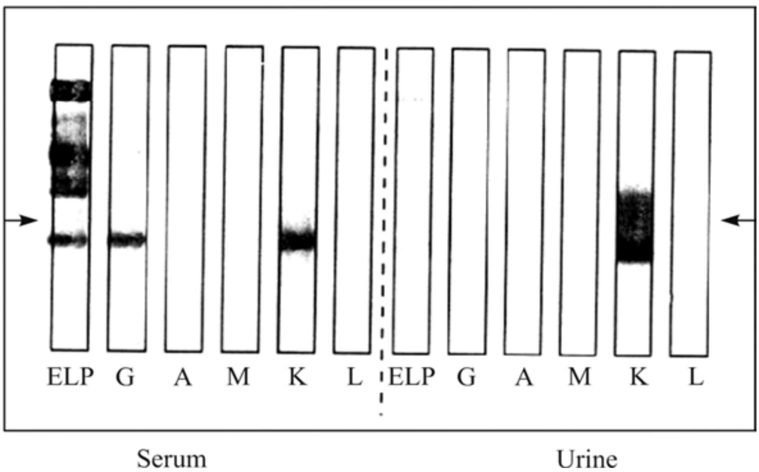


Figure 13. Immunofixation – serum and urine samples.

5.3.5 Immunoturbidimetry

The precipitin line or curve that forms in a gel, can also be formed in a solution. When an antigen solution is combined with a specific antibody solution, the clumping formed results in a cloudy appearance. This is called turbidity; it forms the principle of immunoturbidimetry. It is measured based on the intensity/ amount of light that pass through the sample. It is performed by an instrument called spectrophotometer (Fig. 14-A).

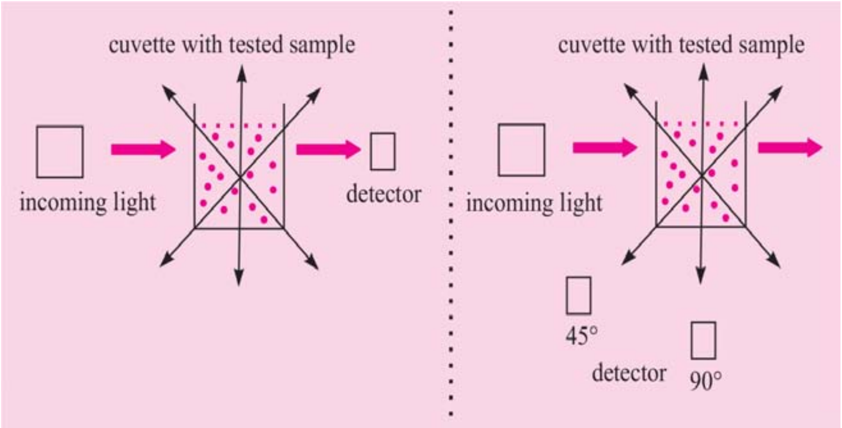


Figure 14. A – turbidimetry, B – nephelometry.

5.3.6 Immunonephelometry

Immunonephelometry works on the principle similar to turbidimetry, but this method measures the intensity of scattered light from sample directly. It uses laser light and is measured by using a nephelometer (Fig. 14-B). Both the methods are faster but are more expensive. Increased sensitivity of antigen and antibody can be got by automation of these techniques.

5.3.7 Immunoassay

Immunoassay technique works on the principle of labeling either the antigen or antibody before the reaction. This increases the sensitivity of detection in the experiment than the formation of immunoprecipitate.

The label can be of:

- Radio isotope – called as Radio Immunoassay (RIA)
- Enzyme – reaction known as Enzyme Immunoassay (EIA); it also includes ELISA (Enzyme Linked Immuno Sorbent Assay)
- Fluorescent substance
- Chemilumniscent substance

RIA – first immunoassay technique; based on the measurement of radioactivity associated with antigen-antibody complexes.

ELISA – safer and efficient method; based on the measurement of an enzyme action associated with antigen-antibody complexes (Fig. 15) Most commonly used enzymes – biotin, alkaline phosphatase, etc.

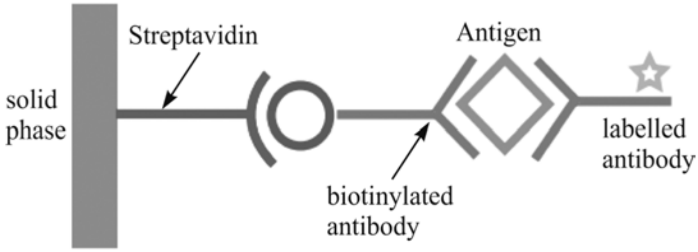


Figure 15. ELISA – sandwich type.

5.3.8 Competitive Binding

Another method to measure the amounts of antigen is competitive binding; in this technique, it works on the principle of competitive binding:

- Antibody is first incubated in solution having antigen
- This mixture is added to antigen coated reaction well

The more the antigen present in sample, the less the antibody will bind to the coated antigen. To this, when labeled antibody is added, we can measure the amount of antigen present in sample.

5.4 FLUORESCENCE ACTIVATED CELL SORTING

Fluorescence activated cell sorting (FACS) is a cutting-edge technique used to isolate individual cells from a sample and analyses their properties. The basics of FACS sorting protocol and how it can be used in a variety of applications. Fluorescence activated cell sorting (FACS) is a technique that can be used to isolate specific cell types from a mixed population. FACS is based on the principle that cells can be labeled with fluorescent dyes and sorted according to their fluorescence intensity. FACS is carried out using a flow cytometer, which is a machine that can measure the fluorescence of cells as they pass through a laser beam. The main benefits of FACS are cell separation, cell labeling, and cell analysis. Cell separation is the most common application of FACS, which is carried out based on a variety of parameters, including cell size, shape, and surface markers. Commonly used surface markers include CD markers such as CD45 (a leukocyte marker), CD14 (a monocyte marker), and HLA-DR (a T-cell marker).

Cell labelling is another common process performed using FACS. Cells can be labeled with fluorescent dyes that bind to specific proteins or

DNA sequences. This allows for the detection and quantification of specific proteins or DNA sequences. Cell analysis is the final benefit of FACS. Cells can be analyzed for their size, shape, fluorescence intensity, and other parameters. This allows for the characterization of cell populations.

FACS offers many benefits over traditional cell separation techniques, such as centrifugation and filtration. FACS is much faster than these methods and can often be completed in minutes. Secondly, FACS can be used to purify cells that are difficult to separate using other methods. Finally, FACS can be used to sort cells into multiple groups, allowing for a more detailed analysis of cell populations.

- Cell mixture containing fluorescently labelled cells exits via nozel
- Laser beam strikes droplets
- FSC detector identifies the size of the cell
- SCC detector identifies the granularity or fluorescence of the cell
- Electrode assign positive or negative charge
- Positive charged cells are drawn to the negative plate while the negative cells are drawn to the poisitive plate
- The seperated cells are collected in different collection tubes

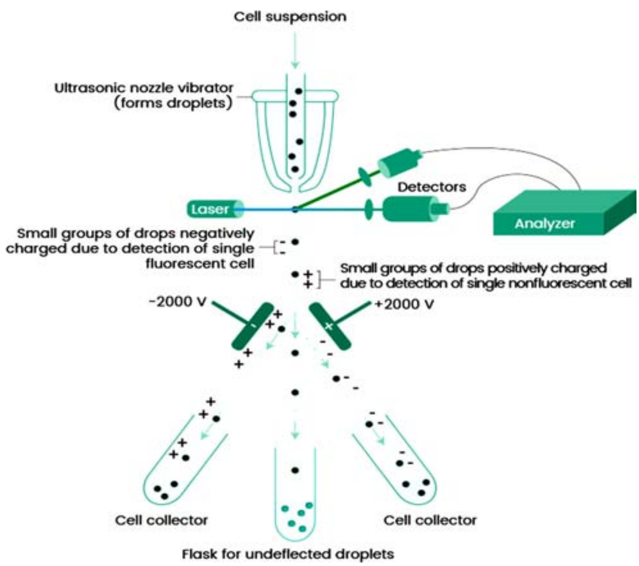


Figure 16. A fluorescence-activated cell sorter (FACS)

5.4.1 Fluorescence Activated Cell Sorting Protocol

Fluorescent activated cell sorting is a complex procedure and requires trained personnel and specialized equipment.

1. The first step in FACS sorting is to label the cells with fluorescent dyes. The fluorescent dyes are usually attached to antibodies that bind to specific cell surface markers. Cells are routinely labelled with PE or FITC conjugated antibodies. FITC is a green fluorescent dye that is excited by blue light and PE is a red fluorescent dye that is excited by green light. The cells are incubated with the fluorescent dyes for 30-60 minutes.
2. After incubation, the cells are washed and diluted using a buffer. The cell sorting buffer contains salts, proteins, and other molecules that help to keep the cells healthy during the sorting process. The cells are then resuspended in phosphate buffered saline (PBS) to remove unbound dye.
3. The next step is to pass the labeled cells through a flow cytometer. The flow cytometer uses lasers to excited the fluorescent dyes and detectors to measure the fluorescence intensity of the dyes. Based on the fluorescence intensity, the flow cytometer sorts the cells into different groups.
4. The cells are sorted into groups according to their fluorescence intensity and size. The most common sorting method is called forward scatter (FSC) and side scatter (SSC). The FSC channel sorts cells based on their size, with larger cells being sorted into the high FSC channel and smaller cells being sorted into the low FSC channel. The SSC channel sorts cells based on their granularity, with more granular cells being sorted into the high SSC channel and less granular cells being sorted into the low SSC channel.
5. After sorting, the cell groups are collected in tubes or plates for further analysis. The cell purity can be checked using a microscope or by flow cytometry. Purity can be calculated by dividing the number of cells in the desired group by the total number of cells sorted.

5.4.2 Applications of Fluorescence Activated Cell Sorting

Fluorescence activated cell sorting is a powerful tool that has many applications in biomedical research. Cancer research, immunology research, stem cell research, and developmental biology research are all areas where FACS sorting can be used to isolate specific cell types from a mixed population. FACS is also a valuable tool in identify different subtypes of cells.

Cancer Research

FACS can be used to isolate cancer cells from a mixed population of cells. FACS can also be used to label cancer cells with multiple dyes, allowing for the detection of multiple cell surface markers. This is important because it allows for the identification of different types of cancer cells. FACS can also be used to isolate cancer cells from healthy cells, allowing for the study of specific cancer types.

Immunology Research

FACS can be used to study the immune system. It can be used to isolate specific types of immune cells, such as T-cells and B-cells, from a mixed population. FACS can be used to sort immune cells based on their surface markers. This is important because it allows for the isolation of specific immune cell types that are involved in a particular disease process.

Stem Cell Research

FACS can be used to isolate specific stem cell types from a mixed population. Stem cells can be labelled with fluorescent dyes and sorted according to their fluorescence intensity. This is important because it allows researchers to study the properties of stem cells and develop new treatments for diseases.

Developmental Biology Research

Developmental biology is the study of how organisms develop from fertilization to adulthood. Fluorescence activated cell sorting is a powerful tool that can be used to sort embryonic cells based on their surface markers. This is important because it allows for the isolation of specific embryonic cell types that are involved in a particular stage of development.

5.5 IMMUNOAFFINITY CHROMATOGRAPHY

Immunoaffinity chromatography (IAC) is a type of LC in which the stationary phase consists of an antibody or antibody-related reagent. This technique represents a special sub category of affinity chromatography, in which a biologically related binding agent is used for the selective purification or analysis of a target compound. The selective and strong binding of antibodies for their given targets has made these agents of great interest for many years as immobilized ligands in affinity chromatography. Other methods sometimes included under the heading of IAC are those that use immobilized targets for antibody purification. However, in this review, the emphasis will be placed on methods that use antibodies as the immobilized ligands.

Affinity chromatography has the advantage of specific binding interactions between the analyte of interest (normally dissolved in the mobile phase), and a binding partner or ligand (immobilized on the stationary phase). In a typical affinity chromatography experiment, the ligand is attached to a solid, insoluble matrix—usually a polymer such as agarose or polyacrylamide—chemically modified to introduce reactive functional groups with which the ligand can react, forming stable covalent bonds. The stationary phase is first loaded into a column to which the mobile phase is introduced. Molecules that bind to the ligand will remain associated with the stationary phase. A wash buffer is then applied to remove non-target biomolecules by disrupting their weaker interactions with the stationary phase, while the biomolecules of interest will remain bound. Target biomolecules may then be removed by applying a so-called elution buffer, which disrupts interactions between the bound target biomolecules and the ligand. The target molecule is thus recovered in the eluting solution. Affinity chromatography does not require the molecular weight, charge, hydrophobicity, or other physical properties of the analyte of interest to be known, although knowledge of its binding properties is useful in the design of a separation protocol.

5.5.1 Structure and Properties of Antibodies

The basis for IAC relies on the selective binding of antibodies. This binding is a result of a large variety of noncovalent interactions that can occur between an antibody and an antigen and can result in association equilibrium constants in the range of 10^5–10^{12} M^{-1}. It has been estimated that the human body is able to produce between 10^7 and 10^8 different types of antibodies, with each capable of binding to a separate antigen. The typical structure of an antibody, using IgG as an example, consists of four polypeptide chains. These four chains consist of two identical heavy chains and two identical light chains that are linked by disulfide bonds to from a Y-shaped structure. The lower stem region of an antibody is referred to as the F_c *region* and is highly conserved from one antibody class to the next. The upper arms of the antibody are called the F_{ab} *regions*. The amino acid sequences in the F_{ab} regions are identical within a single type of antibody but are highly variable between different antibodies. It is this variability that allows antibodies to bind a wide range of antigens.

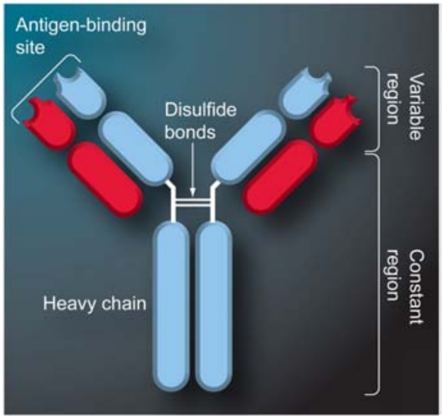


Figure 17. Typical structure of an IgG-class antibody.

A foreign agent that is capable of initiating antibody production is called an antigen. Common antigens include viruses, bacteria and foreign proteins from animals and plants that are capable of producing an immune response. Due to the large size of naturally occurring antigens, antibodies that bind to several different regions of the antigen with a range of binding affinities are often generated. Each individual location on an antigen that can bind to a given antibody is called an epitope. In order for a substance to be recognized by the body's immune system and to lead to the production of antibodies, this substance must have a size that corresponds to a mass of several thousand Daltons. Antibodies can also be produced against smaller substances, but these substances first must be coupled to a larger species (e.g., a carrier protein) before antibody production can occur. A small substance that is used to produce antibodies after being linked to a carrier agent is known as a hapten. The two main types of antibodies that are used in IAC are polyclonal antibodies and monoclonal antibodies. Polyclonal antibodies are produced from multiple cell lines within the body and as a population can bind a variety of epitopes on a single antigen with a range of binding strengths. Monoclonal antibodies are produced by the fusion of a myeloma cell line with spleen cells obtained from an animal that has been immunized with the desired antigen. Because monoclonal antibodies are generated from a single cell line, they bind to a single epitope with identical binding affinities. Two other types of antibodies that can be used in IAC are autoantibodies, which are polyclonal antibodies obtained from patients with auto immune diseases, and anti-idiotypic antibodies, which are antibodies that can mimic the interactions of antigens, hormones or substrates for cell receptors.

5.5.2 Production of Antibodies

Polyclonal antibodies can be produced against a given target by injecting the antigen or a hapten–carrier conjugate into a laboratory animal (e.g., a mouse, rabbit, goat or sheep). Often this solution of the antigen or hapten–carrier conjugate contains an enhancing agent called an adjuvant. After this initial injection, blood samples from the animal are collected after approximately a month (e.g., 3–4 weeks, although the exact times used in this sequence may vary) and tested for the presence of antibodies that are specific for the desired target. Another injection of the antigen or hapten conjugate (called a 'booster') is then made into the animal. The animal's blood is then retested later (e.g., after 10 days) for the presence of antibodies. Based on the antibody levels that are detected, the animal can be allowed to rest for a period of time (e.g., a few weeks) before being administered another booster injection.

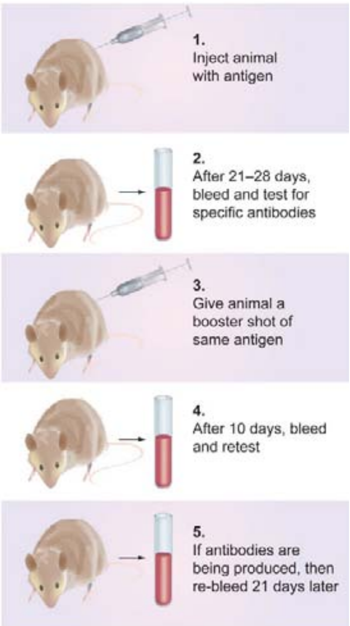


Figure 18. Process for polyclonal antibody production.

This booster/bleed routine can be repeated several times until the antibody concentrations for the required antigen reach the desired level (i.e., as determined by an assay of the blood). At this point, antibody-containing serum can be collected from the animal and stored for later use.

The antibodies that are produced upon the first exposure of an animal to a foreign agent are typically IgM class antibodies. After repeated exposure, IgG class antibodies will also be produced. It is this secondary immune response that produces antibodies that are best suited for IAC applications.

Antibodies that are produced through the normal immune response are polyclonal antibodies that can bind with various strengths and to a variety of epitopes on the antigen.

Before these antibodies can be used in IAC, some further purification is often required. This purification may involve the use of ion-exchange chromatography, precipitation with ammonium or dextran sulfate or isolation on a protein A or protein G column.

The isolated products from each of these techniques will contain some antibodies that do not bind to the desired antigen, but these other antibodies can later be removed by using an immobilized antigen column.

Monoclonal antibodies can be produced by isolating a single antibody-producing cell and combining this cell with a carcinoma or myeloma cell. The resulting hybrid cell line, called a hybridoma, is relatively easy to culture and grow for long-term antibody production.

In this method, a solution of the antigen or the hapten–carrier conjugate is mixed with adjuvant and injected subcutaneously into an animal. The animal is later given a booster shot, killed and the spleen harvested. The lymphocyte cells are mixed with myeloma cells in the presence of polyethylene glycol, which is added to promote cell fusion.

The cells are then grown in the presence of drugs that kill myeloma cells and unfused lymphocytes, but permit the growth of hybridoma cells.

Individual cultures of hybridomas are examined for the production of specific antibodies and those that make the desired antibody are cloned to produce a homogenous culture of cells making a monoclonal antibody.

Although this process can be tedious and difficult, it has the advantage of producing antibodies in relatively large quantities that have well-defined specificity.

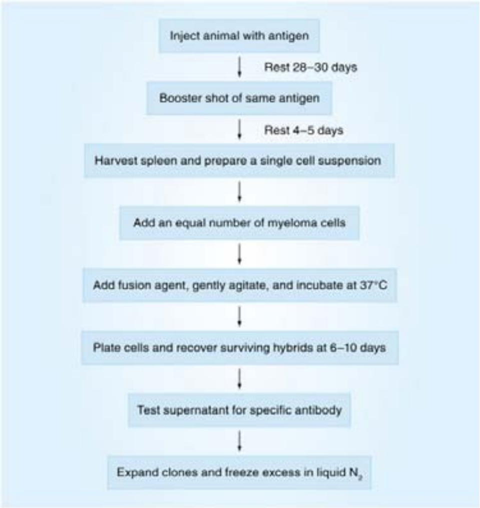


Figure 19. Process for monoclonal antibody production.

5.5.3 Supports for IAC

There are several types of supports that can be utilized to place antibodies within columns for use in IAC. Traditional immunoaffinity supports have been based on low-performance materials such as carbohydrate-related media (e.g., agarose and cellulose) or synthetic organic supports (e.g., acrylamide polymers, copolymers or derivatives, polymethacrylate derivatives and polyethersulfone matrixes). Table 1 lists several commercial supports in these categories that can be used to immobilize antibodies for IAC. The low cost of these materials has made these supports popular for IAC applications involving target purification or offline immunoextraction. However, most of these materials can be used at only relatively low back pressures and are best suited for work under gravity flow or with a peristaltic pump. These supports also can have slow mass-transfer properties.

Table 1. Commercially available supports that can be used for immunoaffinity chromatography.

Support type	Supplier
Low- or medium-performance supports	
Affi-Gel	BioRad
Affinica Agarose/Polymeric Supports	Schleicher and Schuell
AvidGel	BioProbe

Bio-Gel	BioRad
Fractogel	EM Separations
HEMA-AFC	Alltech
Reacti-Gel	Pierce
Sephacryl	Pharmacia
Sepharose	Pharmacia
Superose	Pharmacia
Trisacryl	IBF
TSK Gel Toyopearl	TosoHaas
Ultragel	IBF
High-performance supports	
AvidGel CPG	BioProbe
HiPAC	ChromatoChem
Protein-Pak Affinity Packing	Waters
Ultraaffinity-EP	Bodman
Emphaze	3M Corp./Pierce
POROS	ABI/PerSeptive Biosystems
Data from.	

These disadvantages have limited the use of the low-performance immunoaffinity supports in applications requiring their direct use with HPLC. When antibodies are used within an HPLC column, the resulting method is referred to as HPIAC. Immunoaffinity supports that are used for HPIAC must be rigid and have higher efficiency than typical low-performance supports. Examples of materials that have been used in HPIAC include derivatized silica, glass, azalactone beads, methacrylate polymeric supports and polystyrene-based perfusion media.

In addition to good efficiency and mechanical stability, the ideal support for IAC should have low nonspecific binding and be easily modified for antibody attachment. Another item to consider for a porous material is the size of its pores for immobilization of antibodies and binding by antibodies to the target. Supports with small pores have the largest amount of total surface area, but much of this area may not be accessible for antibody immobilization. By contrast, supports with larger pores do not have accessibility problems, but their lower surface area can result in small amounts of immobilized antibodies. As a result, the maximum coverage for antibodies is often observed for supports with pore sizes of 300–500 Å, which is approximately three- to five-times the diameter of an antibody. This size range is also suitable for the binding of immobilized antibodies to many small- or medium-sized targets (i.e., agents with sizes less than about 100–150 kDa), although larger pore sizes may be needed for larger targets. Other support materials that have recently been used in IAC are disks, fibers and monolithic rods. Unique features offered by these newer support materials include their good mass-transfer and flow properties. A general procedure for immobilizing antibodies onto monoliths can be seen in Figure 20.

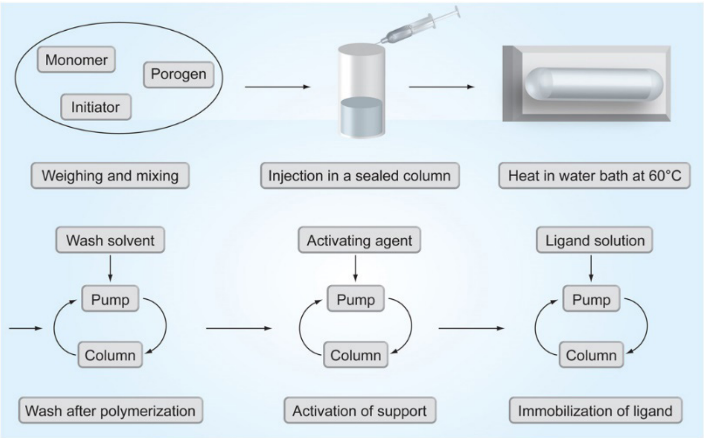


Figure 20. General procedure for immobilizing antibodies within a monolithic column.

5.5.4 Antibody-immobilization Methods

Antibodies can be immobilized onto supports by using a variety of techniques that range from covalent attachment to adsorption-based methods. Of these techniques, those that make use of covalent attachment are the most common, but even these methods can range from the use of random attachment via amino or carboxyl groups to more site-selective immobilization approaches that make use of modified carbohydrate residues or thiol groups. The ideal situation in any of these immobilization methods is to have antibodies attached to the support in a way that does not affect the activity of the binding sites or the accessibility of these sites to their target compound.

Antibodies can be immobilized through free amine groups by using supports that have been activated with agents such as N,N′-carbonyldiimidazole, cyanogen bromide, N-hydroxysuccinimide or tresyl chloride/tosyl chloride. Antibodies can also be immobilized through amine groups using a support that has been treated to produce reactive epoxy or aldehyde groups on its surface. The use of amine groups is one of the easiest ways to immobilize antibodies but can cause a decrease in activity if the antibodies have some of these amine groups in their antigen-binding sites. In addition, this approach can cause the antibodies to be immobilized in a random orientation, leading to steric hindrance and a decrease in binding efficiency.

Antibodies or Fab fragments can be covalently linked to IAC supports through more site-selective methods. This can be achieved by utilizing the free sulfhydryl groups that are created when Fab fragments are generated. These groups can be used for immobilization by using techniques such as the divinylsulfone, epoxy, iodoacetyl/bromoacetyl, maleimide or tresyl chloride/tosyl chloride methods. In addition, site-selective immobilization can be accomplished by coupling antibodies through the carbohydrate residues that are located in their Fc regions. This process is carried out by first oxidizing the carbohydrate residues under mild conditions to generate aldehyde residues. These aldehyde groups are then reacted with a hydrazide- or amine-containing support.

A third way antibodies can be immobilized onto supports is by using a secondary ligand to adsorb these antibodies. This can be accomplished by using antibodies that have been reacted with biotin or biotinylated, and then adsorbed to a support that contains immobilized avidin or streptavidin. The most popular biotinylation technique is to incubate antibodies with N-hydroxysuccinimide-D-biotin at pH 9. The strong noncovalent linkage of biotin to strepavidin or avidin can then be used to immobilize these antibodies. These linkages have association equilibrium constants in the range of 1013–1015 M−1 and can resist many types of elution conditions without dissociation. In addition, a modified version of avidin called neutravidin (from Pierce) can be used for the immobilization of biotinylated antibodies. If amine-based coupling of biotin to the antibodies is employed, the biotin can attach at or near the antibody's binding sites and cause a decrease in binding. There will also be random orientiation of the resulting biotinylated antibodies on the streptavidin support. These problems can instead be minimized by using hydrazide-biotin, which is reacted with the carbohydrate residues of antibodies after these regions have been oxidized under mild conditions to produce some aldehyde groups, as discussed in the previous paragraph.

Another approach for antibody immobilization utilizes the strong binding of protein A or protein G to the F_c regions of many antibodies. Protein A and protein G are bacterial cell wall proteins that have strong binding for many types of antibodies under physiological conditions. However, if the pH of the surrounding solution is decreased to approximately pH 2–3, this binding will be weakened and the retained antibodies can be eluted. Protein A and protein G supports are useful when high antibody activity is needed and it is desirable to have frequent antibody replacement in an IAC column. Long-term reproducibility of the IAC binding capacity is possible when using protein A or protein G supports, but a much larger amount of antibodies will be needed than is required for traditional immobilization methods if the antibodies are eluted and replaced on a regular basis. If desired, protein A or protein G supports can be prepared with a permanent coating of antibodies by cross-linking the antibodies to the immobilized protein A or protein G by using carbodiimide or dimethyl pimelimidate.

5.5.5 Chromatographic Separation

The separation of the analyte using IAC involves three steps, viz. binding, washing and elution as depicted in general scheme for IAC in figure 21.

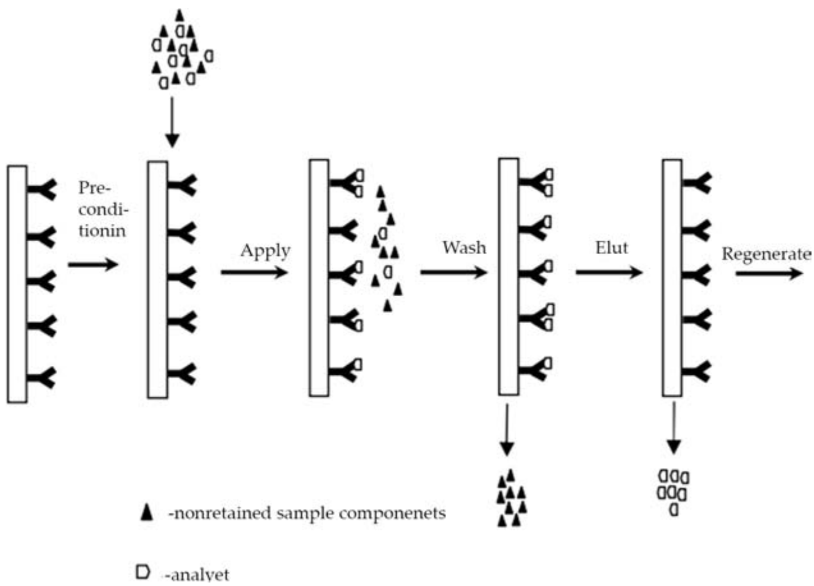


Figure 21. General Scheme for IAC.

Binding

Binding is an essential step that has a profound influence on the purity of the final product of an IAC process. Binding is performed at or slightly above the neutral pH in a phosphate or borate buffer containing NaCl and a nonionic detergent. This facilitates only the binding of analyte of interest and the non-specific absorption on the stationary phase is minimized. Presence of salt in buffer is essential to reduce the ionic interactions of the charged surface groups of immobilized MAb. Detergents reduce hydrophobic interactions. Use of detergents for potential therapeutic products is irrational owing to the inherent difficulties in removing detergents from proteins. Inclusion of polyethylene glycol (PEG) in the binding buffer helps reduce nonspecific binding.

Washing

Washing is essential in the chromatography to remove any material bound nonspecifically to the matrix. Hydrophobic forces are more likely the cause of the presence of unwanted molecules. The low molecular weight compounds which may be retained in the gel beads will elute in 1 to 2 column volumes. The compounds retained through the hydrophobic forces may be removed by washing with lowionic-strength buffers or with water.

Elution

Antigens and antibodies are bound to each other by a web of forces, which include ionic bonding, hydrophobic interactions, hydrogen bonding, and van der Waals attractions. The strength of Ag-Ab complexes depends on the relative affinities and avidities of the antibodies. In addition, steric orientation, coupling density, and nonspecific interactions can also influence the binding. The objective of the elution step is to recover the specifically bound protein at a high yield, purity, and stability. Elution conditions, which might denature the protein product, have to be avoided. A wide variety of elution conditions and the choice of an eluant seems empirical. However, a logical sequence of available elution strategies can be considered when selecting an appropriate elution protocol.

Specific elution: Certain antibodies bind to their respective antigens under high pH or in the presence of metals like calcium or magnesium or in the presence of chelating agent like EDTA. Antigens bound to such antibodies can be eluted under gentle conditions where lowering the pH or adding EDTA to the elution buffer or adding divalent metals to the elution buffer causes the Ag-Ab complex to dissociate.

Use of extreme pH:

- *Acid elution*: This is the most widely used method of desorption and is normally very effective. The commonly used acid eluants are glycine-HCl, pH 2.5; 0.02 M HCl and sodium citrate, pH 2.5. Upon elution, the pH of the eluant sample is quickly neutralized to 7.0 with 2 M Tris base, pH 8.5, to avoid acid-induced denaturation. In some cases increased hydrophobic interactions between antigen and antibody gives low recovery with acid elution. Incorporation of 1 M propionic acid, or adding 10% dioxane or addition of ethylene glycol to the acid eluant, is effective in dissociating such complexes.
- *Base elution*: It is less frequently employed than acid elution. Typically, 1 M NH4OH, or 0.05 M diethylamine, pH 11.5 have been employed to elute membrane proteins (i.e., hydrophobic character) and other antigens that precipitate in acid but are stable in basic conditions.

Use of Chaotropic agents: These agents disrupt the tertiary structure of proteins and, therefore, can be used to disrupt the Ag:Ab complexes. Chaotropic salts are particularly useful as they disrupt ionic interactions, hydrogen bonding, and sometimes hydrophobic interactions. The relative order of the effectiveness of chaotropic anions is $SCN^- > ClO^{4-} > I^- > Br^- > Cl^-$. Chaotropic cations are effective in the order of Mg>K>Na. Eluants such as 8 M urea, 6 M guanidine-HCl, and 4 M NaSCN are effective in disrupting most Ag-Ab interactions. To avoid and minimize chaotropic salt-induced protein denaturation, rapid desalting or dialysis of the eluant is essential.

Use of denaturants: The denaturants most frequently used in IAC are 6-8M urea and 3- 4M guanidine hydrochloride. These reagents were more commonly used in conjunction with polyclonal antibodies with a very high affinity for the antigen and in cases where the antigen is stable to the treatment.

Use of organic solvents: Organic solvents have found favor as eluants in some circumstances but with the possible exception of ethylene glycol are infrequently employed in IAC.

Changes in ionic strength: Raising the ionic strength of the solvent can be a mild method of elution but is rarely effective when using NaC1. Of more use are the salts of divalent cations such as Mg^{2+} and Ca^{2+}. The ionic strength of a 4.5M $MgC1_2$ solution is 13.5M and the chloride concentration is 9M. This may in some way explain the effectiveness of such solutions as eluants in terms of very high concentrations of a weakly chaotropic ion in combination with an ability to disrupt ionic interactions. In addition Ca^{2+} and Mg^{2+} are more chaotropic than either Na^{+} or K^{+}.

5.5.6 Application and Elution Conditions

The application and elution conditions are another important set of factors to consider in the design and use of an IAC method. The application buffer used in IAC is generally chosen for its ability to promote fast and efficient binding of the desired analyte or target compound to the immobilized antibodies. Optimum binding for antibodies typically occurs under physiological conditions, so IAC generally makes use of a neutral pH buffer (i.e., pH 7.0–7.4). Under these conditions, the equilibrium constants for antibody binding is usually in the range of 10^{6}–10^{12} M^{-1}. Due to this strong binding between the antibody and its target, isocratic elution is often not feasible unless the IAC method is using low-affinity antibodies (i.e., those with association equilibrium constants of less than 10^{6} M^{-1}). The elution conditions for IAC need to allow for fast elution of the analyte while still allowing later regeneration of the immobilized antibodies. The need for fast but reversible binding and regeneration is especially important when an IAC column is to be used for a large number of samples. Elution is often carried out by temporarily lowering the effective strength of antibody binding to the target. The most-common approaches for elution in IAC include changing the mobile phase pH or adding a chaotropic agent to the mobile phase. Other, less-common IAC elution methods include adding a competing agent, organic modifier or denaturing agent to the mobile phase, or changing the temperature of the column during elution. Usually the elution buffer is applied in a step gradient, but gradual or nonlinear gradients can be used as well. Figure 22 shows a common scheme by which an IAC column can be used to selectively bind and elute analytes from a sample.

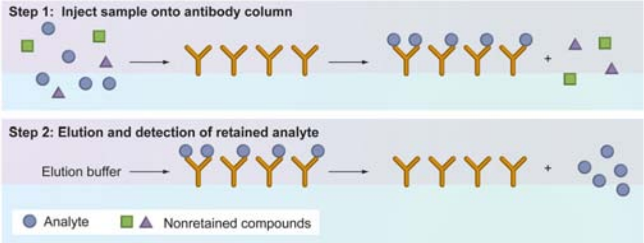


Figure 22. Typical format in which a sample containing the analyte is applied to an immunoaffinity column and nonretained sample components are allowed to pass through.

Changing the mobile phase pH is the most popular method for eluting retained compounds from IAC columns. This approach is usually conducted by applying an acidic buffer (pH 1–3) to the column. Alkaline elution conditions have been used in conjunction with low-performance IAC supports, but cannot be used with common HPIAC supports, such as silica or glass beads, due to the instability of these supports at a pH greater than 8.0. One difficulty with changing the pH of the mobile phase is the possibility of denaturing the immobilized antibodies or any retained compounds that are susceptible to variations in pH. However, many IAC columns have been shown to be quite stable when moderate pH changes in the range of 7.0–7.5 to 2.5–3.0 are used for elution.

To avoid denaturing effects that are caused by lowering the pH, the elution of retained compounds in IAC can alternatively be performed by adding chaotropic agents such as thiocyanate (SCN^-), trifluoroacetate (CF_3COO^-), perchlorate (ClO_4^-), iodide (I^-) or chloride (Cl^-) to the elution buffer. The elution strength of these agents follows the approximate order $SCN^- > CF_3COO^- > ClO_4^- > I^- > Cl^-$. These agents are typically used at concentrations of 1.5–8 M and have been shown to be effective in dissociating high-affinity antibody–antigen complexes. When an organic modifier is used in the mobile phase, care must be taken to ensure that the concentration of the organic additive does not permanently denature the antibodies. When methanol or other organic modifiers are used, the capacity of the immunoaffinity column has been shown to decrease. This denaturation is not irreversible, but kinetic regeneration can take a few days. In one study, an anticlenbuterol immunoaffinity column was regenerated 20-times after offline elution with 2 ml of ethanol 80% in water. However, this column was shown to lose half its activity over these 20 regenerations. Therefore, if harsh elution conditions are required, the capacity of the immunoaffinity column should be much larger than the actual amounts of analyte that are to be measured or isolated.

REFERENCES

1. Carpenter AB. Antibody-based methods. In: Rose NR, Hamilton RG, Detrick B, eds. Manual of Clinical Laboratory Immunology. Washington, DC: ASM Press; 2002:6-25.
2. Crowther JR. The ELISA guidebook. Totowa, NJ: Humana Press, 2001:421.
3. Dessars B, Cotton F, Thiry P, et al. Comparison of automated ACMIA and EMIT immunoassays for whole blood cyclosporin monitoring. Clin Lab. 2003;49:135-140.

4. Devey ME, Steward MW. The role of antibody affinity in the performance of solid phase assays. In: Kemeny DM, Challacombe SJ, eds. ELISA and other solid phase immunoassays. Theoretical and practical aspects. Chichester, UK: John Wiley & Sons, 2000:135-154.
5. Hage DS, Nelson MA. Chromatographic immunoassays. Anal. Chem. 2001;73 198A–205A
6. Jiang T, Mallik R, Hage DS. Affinity monoliths for ultrafast immunoextraction. Anal. Chem. 2005;77:2362–2372.
7. Koivunen ME, Gee SJ, Park E-K, et al. Application of an enzyme-linked immunosorbent assay for the analysis of paraquat in human-exposure samples. Arch Environ Contam Toxicol. 2005;48:184-190.
8. Lerner RA. Manufacturing immunity to disease in a test tube: the magic bullet realized. Angew. Chem. Int. Ed. Engl. 2006;45:8106–8125.
9. Peoples MC, Karnes HT. Microfluidic immunoaffinity separations for bioanalysis. J. Chromatogr. B. 2008;866:14–25.
10. Ridker PM, Morrow DA. C-reactive protein, inflammation, and coronary risk. Cardiol Clin. 2003;21:315-325.
11. Ridker PM. High-sensitivity C-reactive protein, inflammation, and cardiovascular risk: from concept to clinical practice to clinical benefit. Am Heart J. 2004;148:S19-S26.
12. Weller MG. Immunochromatographic techiques – a critical review. Fres. J. Anal. Chem. 2000;366:635–645.
13. Williams BAR, Diehnelt CW, Belcher P, et al. Creating protein affinity reagents by combining peptide ligands on synthetic DNA scaffolds. J. Am. Chem. Soc. 2009;131:17233–17241.
14. Zolotarjova N, Boyes B, Martosella J, et al. Immunoaffinity depletion of high-abundant proteins for proteomic sample preparation. In: Smejkal GB, Lazarev A, editors. Separation Methods in Proteomics. FL, USA: CRC Press; 2006. pp. 63–79.

CHAPTER

6

PROTEIN STRUCTURE, PURIFICATION, CHARACTERISATION AND FUNCTION ANALYSIS

INTRODUCTION

Proteins are formed by the condensation of the a-amino group of one amino acid with the a-carboxyl of the adjacent amino acid. With the exception of the two terminal amino acids, therefore, the a-amino and carboxyl groups are all involved in peptide bonds and are no longer ionisable in the protein. Amino, carboxyl, imidazolyl, guanidino, phenolic and sulphydryl groups in the side chains are, however, free to ionise and of course there will be many of these. Proteins fold in such a manner that the majority of these ionisable groups are on the outside of the molecule, where they can interact with the surrounding aqueous medium. Some of these groups are located within the structure and may be involved in electrostatic attractions that help to stabilise the three-dimensional structure of the protein molecule. The relative numbers of positive and negative groups in a protein molecule influence aspects of its physical behaviour, such as solubility and electrophoresis mobility.

6.1 PROPERTIES OF AMINO ACIDS AND PROTEINS

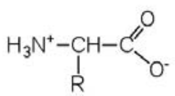
The proteins in all living species, from bacteria to humans, are constructed from the same set of 20 amino acids, so called because each contains an amino group attached to a carboxylic acid. The amino acids in proteins are α-amino acids, which means the amino group is attached to the α-carbon of the carboxylic acid.

Humans can synthesize only about half of the needed amino acids; the remainder must be obtained from the diet and are known as essential amino acids.

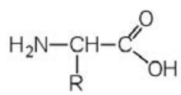
However, two additional amino acids have been found in limited quantities in proteins: Selenocysteine was discovered in 1986, while pyrrolysine was discovered in 2002.

The amino acids are colorless, nonvolatile, crystalline solids, melting and decomposing at temperatures above 200°C.

These melting temperatures are more like those of inorganic salts than those of amines or organic acids and indicate that the structures of the amino acids in the solid state and in neutral solution are best represented as having both a negatively charged group and a positively charged group. Such a species is known as a zwitterion.



α-Amino acid drawn as a zwitterion



α-Amino acid drawn as an uncharged molecule; not an accurate respresentation of amino acid structure

6.1.1 Classification

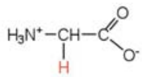
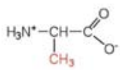
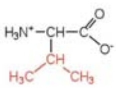
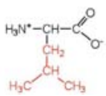
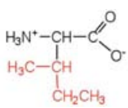
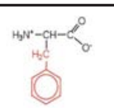
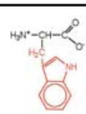
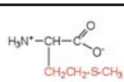
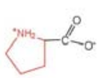
In addition to the amino and carboxyl groups, amino acids have a side chain or R group attached to the α-carbon. Each amino acid has unique characteristics arising from the size, shape, solubility, and ionization properties of its R group.

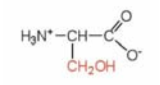
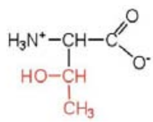
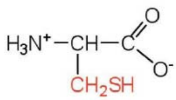
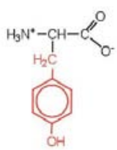
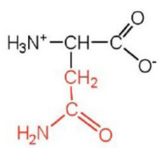
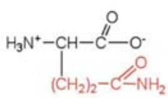
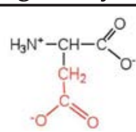
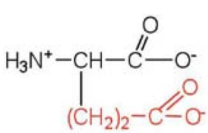
As a result, the side chains of amino acids exert a profound effect on the structure and biological activity of proteins.

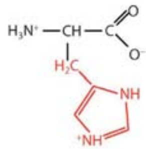
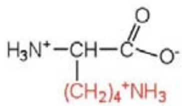
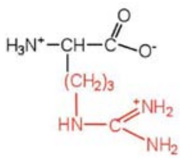
Although amino acids can be classified in various ways, one common approach is to classify them according to whether the functional group on the side chain at neutral pH is nonpolar, polar but uncharged, negatively charged, or positively charged.

The structures and names of the 20 amino acids, their one- and three-letter abbreviations, and some of their distinctive features are given in Table 1.

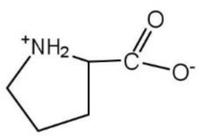
Table 1: Common Amino Acids Found in Proteins

Common Name	Abbreviation	Structural Formula (at pH 6)	Molar Mass	Distinctive Feature
Amino acids with a nonpolar R group				
glycine	gly (G)		75	the only amino acid lacking a chiral carbon
alanine	ala (A)		89	—
valine	val (V)		117	a branched-chain amino acid
leucine	leu (L)		131	a branched-chain amino acid
isoleucine	ile (I)		131	an essential amino acid because most animals cannot synthesize branched-chain amino acids
phenylalanine	phe (F)		165	also classified as an aromatic amino acid
tryptophan	trp (W)		204	also classified as an aromatic amino acid
methionine	met (M)		149	side chain functions as a methyl group donor
proline	pro (P)		115	contains a secondary amine group; referred to as an *α-imino acid*

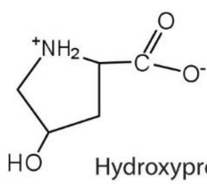
Amino acids with a polar but neutral R group				
serine	ser (S)		105	found at the active site of many enzymes
threo-nine	thr (T)		119	named for its similarity to the sugar threose
cysteine	cys (C)		121	oxidation of two cysteine molecules yields *cystine*
tyro-sine	tyr (Y)		181	also classified as an aro-matic amino acid
aspara-gine	asn (N)		132	the amide of aspartic acid
gluta-mine	gln (Q)		146	the amide of glutamic acid
Amino acids with a negatively charged R group				
aspartic acid	asp (D)		132	carboxyl groups are ion-ized at physiological pH; also known as aspartate
glutam-ic acid	glu (E)		146	carboxyl groups are ion-ized at physiological pH; also known as glutamate
Amino acids with a positively charged R group				

histi-dine	his (H)		155	the only amino acid whose R group has a pK_a (6.0) near physiological pH
lysine	lys (K)		147	—
argi-nine	arg (R)		175	almost as strong a base as sodium hydroxide

The first amino acid to be isolated was asparagine in 1806. It was obtained from protein found in asparagus juice (hence the name). Glycine, the major amino acid found in gelatin, was named for its sweet taste (Greek *glykys*, meaning "sweet"). In some cases an amino acid found in a protein is actually a derivative of one of the common 20 amino acids (one such derivative is hydroxyproline). The modification occurs *after* the amino acid has been assembled into a protein.



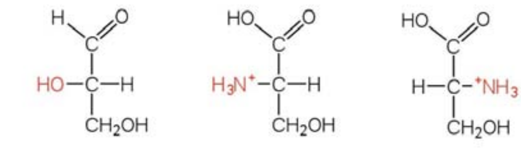
Proline



Hydroxyproline

6.1.2 Configuration

Notice in Table 1 that glycine is the only amino acid whose α-carbon is not chiral. Therefore, with the exception of glycine, the amino acids could theoretically exist in either the D- or the L-enantiomeric form and rotate plane-polarized light. As with sugars, chemists used L-glyceraldehyde as the reference compound for the assignment of absolute configuration to amino acids. Its structure closely resembles an amino acid structure except that in the latter, an amino group takes the place of the OH group on the chiral carbon of the L-glyceraldehyde and a carboxylic acid replaces the aldehyde. Modern stereochemistry assignments using the Cahn-Ingold-Prelog priority rules used ubiquitously in chemistry show that all of the naturally occurring chiral amino acids are S except Cys which is R.



We learned that all naturally occurring sugars belong to the D series. It is interesting, therefore, that nearly all known plant and animal proteins are composed entirely of L-amino acids. However, certain bacteria contain D-amino acids in their cell walls, and several antibiotics (e.g., actinomycin D and the gramicidins) contain varying amounts of D-leucine, D-phenylalanine, and D-valine.

6.2 PROTEIN STRUCTURE

Proteins are the end products of the decoding process that starts with the information in cellular DNA. As workhorses of the cell, proteins compose structural and motor elements in the cell, and they serve as the catalysts for virtually every biochemical reaction that occurs in living things. This incredible array of functions derives from a startlingly simple code that specifies a hugely diverse set of structures.

In fact, each gene in cellular DNA contains the code for a unique protein structure. Not only are these proteins assembled with different amino acid sequences, but they also are held together by different bonds and folded into a variety of three-dimensional structures. The folded shape, or conformation, depends directly on the linear amino acid sequence of the protein.

To understand how a protein gets its final shape or conformation, we need to understand the four levels of protein structure: primary, secondary, tertiary, and quaternary.

6.2.1 Primary Structure

The simplest level of protein structure, primary structure, is simply the sequence of amino acids in a polypeptide chain. For example, the hormone insulin has two polypeptide chains, A and B, shown in diagram below. (The insulin molecule shown here is cow insulin, although its structure is similar to that of human insulin.) Each chain has its own set of amino acids, assembled in a particular order. For instance, the sequence of the A chain starts with glycine at the N-terminus and ends with asparagine at the C-terminus, and is different from the sequence of the B chain.

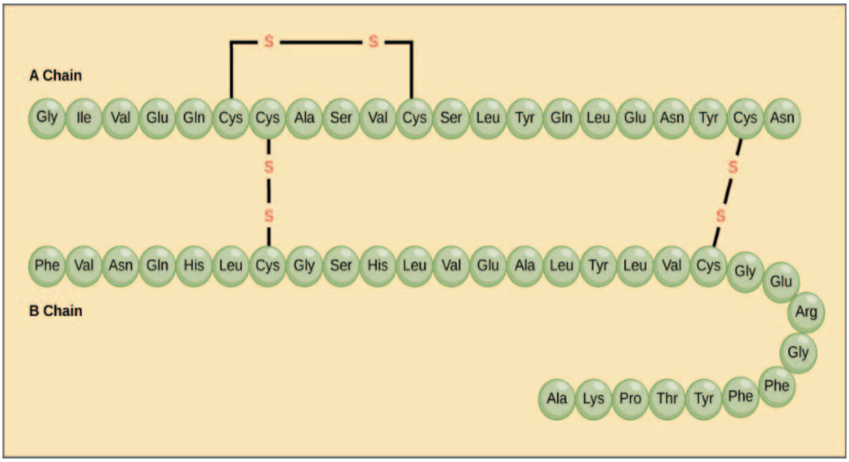


Figure 1: Insulin consists of an A chain and a B chain. They are connected to one another by disulfide bonds (sulfur-sulfur bonds between cysteines). The A chain also contains an internal disulfide bond. The amino acids that make up each chain of insulin are represented as connected circles, each with the three-letter abbreviation of the amino acid's name.

The sequence of a protein is determined by the DNA of the gene that encodes the protein (or that encodes a portion of the protein, for multi-subunit proteins).

A change in the gene's DNA sequence may lead to a change in the amino acid sequence of the protein. Even changing just one amino acid in a protein's sequence can affect the protein's overall structure and function.

For instance, a single amino acid change is associated with sickle cell anemia, an inherited disease that affects red blood cells. In sickle cell anemia, one of the polypeptide chains that make up hemoglobin, the protein that carries oxygen in the blood, has a slight sequence change. The glutamic acid that is normally the sixth amino acid of the hemoglobin β chain (one of two types of protein chains that make up hemoglobin) is replaced by a valine. This substitution is shown for a fragment of the β chain in the diagram below.

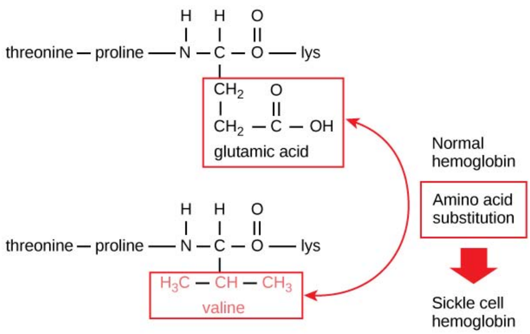


Figure 2: Normal and sickle cell mutant hemoglobin chains, showing substitution of valine for glutamic acid in the sickle cell version.

What is most remarkable to consider is that a hemoglobin molecule is made up of two α chains and two β chains, each consisting of about 150 amino acids, for a total of about 600 amino acids in the whole protein. The difference between a normal hemoglobin molecule and a sickle cell molecule is just 2 amino acids out of the approximately 600.

A person whose body makes only sickle cell hemoglobin will suffer symptoms of sickle cell anemia. These occur because the glutamic acid-to-valine amino acid change makes the hemoglobin molecules assemble into long fibers. The fibers distort disc-shaped red blood cells into crescent shapes. Examples of "sickled" cells can be seen mixed with normal, disc-like cells in the blood sample below.

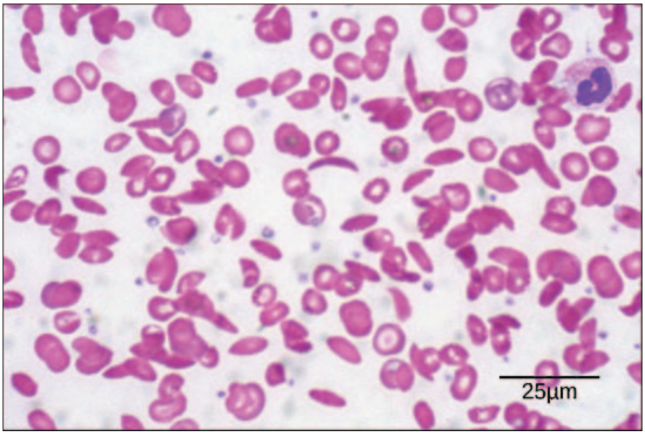


Figure 3: The sickled cells get stuck as they try to pass through blood vessels. The stuck cells impair blood flow and can cause serious health problems for people with sickle cell anemia, including breathlessness, dizziness, headaches, and abdominal pain.

6.2.2 Secondary Structure

The next level of protein structure, secondary structure, refers to local folded structures that form within a polypeptide due to interactions between atoms of the backbone. (The backbone just refers to the polypeptide chain apart from the R groups – so all we mean here is that secondary structure does not involve R group atoms.) The most common types of secondary structures are the *α* helix and the β pleated sheet. Both structures are held in shape by hydrogen bonds, which form between the carbonyl O of one amino acid and the amino H of another.

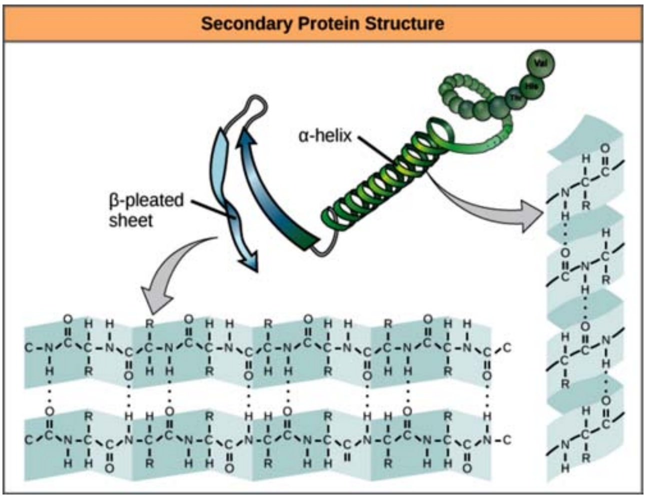


Figure 4: Hydrogen bonding patterns in beta pleated sheets and alpha helices.

In an ***α*** helix, the carbonyl (C=O) of one amino acid is hydrogen bonded to the amino H (N-H) of an amino acid that is four down the chain. (E.g., the carbonyl of amino acid 1 would form a hydrogen bond to the N-H of amino acid 5.) This pattern of bonding pulls the polypeptide chain into a helical structure that resembles a curled ribbon, with each turn of the helix containing 3.6 amino acids. The R groups of the amino acids stick outward from the *α* helix, where they are free to interact.

In a **β pleated sheet**, two or more segments of a polypeptide chain line up next to each other, forming a sheet-like structure held together by hydrogen bonds.

The hydrogen bonds form between carbonyl and amino groups of backbone, while the R groups extend above and below the plane of the sheet33cubed. The strands of a β pleated sheet may be parallel, pointing in the same direction (meaning that their N- and C-termini match up), or antiparallel, pointing in opposite directions (meaning that the N-terminus of one strand is positioned next to the C-terminus of the other).

Certain amino acids are more or less likely to be found in α-helices or β pleated sheets. For instance, the amino acid proline is sometimes called a "helix breaker" because its unusual R group (which bonds to the amino group to form a ring) creates a bend in the chain and is not compatible with helix formation. Proline is typically found in bends, unstructured regions between secondary structures. Similarly, amino acids such as tryptophan, tyrosine, and phenylalanine, which have large ring structures in their R groups, are often found in β pleated sheets, perhaps because the β pleated sheet structure provides plenty of space for the side chains.

Many proteins contain both α helices and β pleated sheets, though some contain just one type of secondary structure (or do not form either type).

6.2.3 TERTIARY STRUCTURE

The overall three-dimensional structure of a polypeptide is called its **tertiary structure**. The tertiary structure is primarily due to interactions between the R groups of the amino acids that make up the protein.

R group interactions that contribute to tertiary structure include hydrogen bonding, ionic bonding, dipole-dipole interactions, and London dispersion forces – basically, the whole gamut of non-covalent bonds.

For example, R groups with like charges repel one another, while those with opposite charges can form an ionic bond. Similarly, polar R groups can form hydrogen bonds and other dipole-dipole interactions.

Also important to tertiary structure are hydrophobic interactions, in which amino acids with nonpolar, hydrophobic R groups cluster together on the inside of the protein, leaving hydrophilic amino acids on the outside to interact with surrounding water molecules.

Finally, there's one special type of covalent bond that can contribute to tertiary structure: the disulfide bond.

Disulfide bonds, covalent linkages between the sulfur-containing side chains of cysteines, are much stronger than the other types of bonds that contribute to tertiary structure. They act like molecular "safety pins," keeping parts of the polypeptide firmly attached to one another.

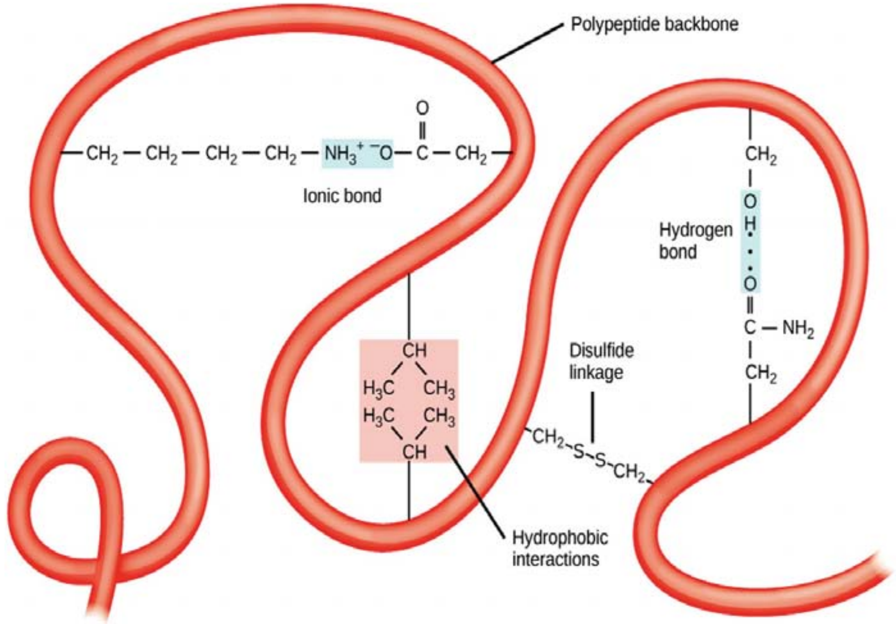


Figure 5: A hypothetical polypeptide chain, depicting different types of side chain interactions that can contribute to tertiary structure. These include hydrophobic interactions, ionic bonds, hydrogen bonds, and disulfide bridge formation.

6.2.4 Quaternary Structure

Many proteins are made up of a single polypeptide chain and have only three levels of structure (the ones we've just discussed). However, some proteins are made up of multiple polypeptide chains, also known as subunits.

When these subunits come together, they give the protein its quaternary structure. We've already encountered one example of a protein with quaternary structure: hemoglobin.

As mentioned earlier, hemoglobin carries oxygen in the blood and is made up of four subunits, two each of the α and β types.

Another example is DNA polymerase, an enzyme that synthesizes new strands of DNA and is composed of ten subunits.

In general, the same types of interactions that contribute to tertiary structure (mostly weak interactions, such as hydrogen bonding and London dispersion forces) also hold the subunits together to give quaternary structure.

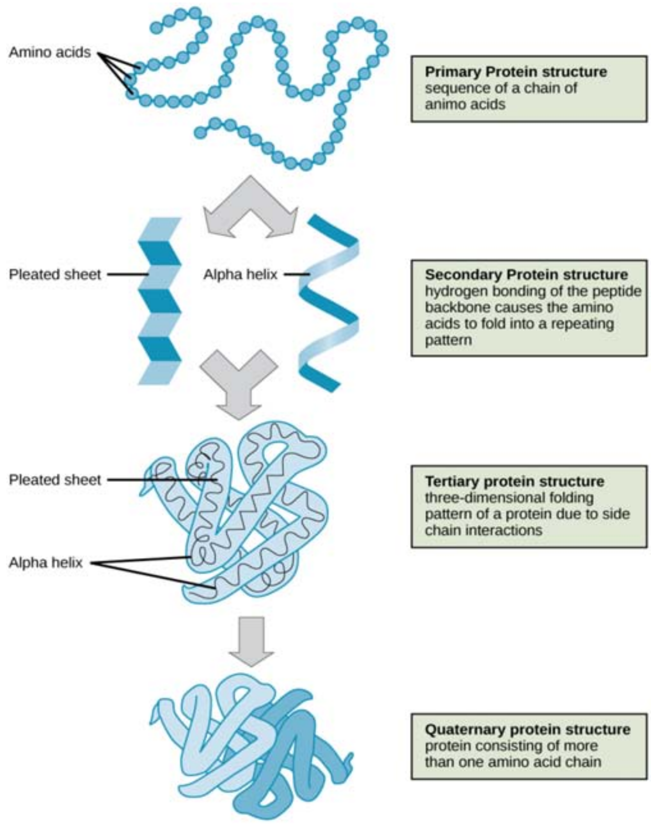


Figure 6: Flowchart depicting the four orders of protein structure.

6.2.5 Denaturation and Protein Folding

Each protein has its own unique shape. If the temperature or pH of a protein's environment is changed, or if it is exposed to chemicals, these interactions may be disrupted, causing the protein to lose its three-dimensional structure and turn back into an unstructured string of amino acids. When a protein loses its higher-order structure, but not its primary sequence, it is said to be denatured. Denatured proteins are usually non-functional.

For some proteins, denaturation can be reversed. Since the primary structure of the polypeptide is still intact (the amino acids haven't split up), it may be able to re-fold into its functional form if it's returned to its normal environment. Other times, however, denaturation is permanent. One example of irreversible protein denaturation is when an egg is fried. The albumin protein in the liquid egg white becomes opaque and solid as it is denatured by the heat of the stove, and will not return to its original, raw-egg state even when cooled down. Researchers have found that some proteins can re-fold after denaturation even when they are alone in a test tube. Since these proteins can go from unstructured to folded all by themselves, their amino acid sequences must contain all the information needed for folding. However, not all proteins are able to pull off this trick, and how proteins normally fold in a cell appears to be more complicated. Many proteins don't fold by themselves, but instead get assistance from chaperone proteins (chaperonins).

6.3 PROTEIN PURIFICATION

Protein purification is a series of processes intended to isolate one or a few proteins from a complex mixture, usually cells, tissues or whole organisms. Protein purification is vital for the characterization of the function, structure and interactions of the protein of interest. The purification process may separate the protein and non-protein parts of the mixture, and finally separate the desired protein from all other proteins. Separation of one protein from all others is typically the most laborious aspect of protein purification. Separation steps usually exploit differences in protein size, physico-chemical properties, binding affinity and biological activity. The pure result may be termed protein isolate.

Protein purification is either *preparative* or *analytical*. Preparative purifications aim to produce a relatively large quantity of purified proteins for subsequent use. Examples include the preparation of commercial products such as enzymes (e.g. lactase), nutritional proteins (e.g. soy protein isolate), and certain biopharmaceuticals (e.g. insulin). Several preparative purifications steps are often deployed to remove bi-products, such as host cell proteins, which poses as a potential threat to the patient's health. Analytical purification produces a relatively small amount of a protein for a variety of research or analytical purposes, including identification, quantification, and studies of the protein's structure, post-translational modifications and function. Pepsin and urease were the first proteins purified to the point that they could be crystallized.

6.3.1 Extraction

If the protein of interest is not secreted by the organism into the surrounding solution, the first step of each purification process is the disruption of the cells containing the protein. Depending on how fragile the protein is and how stable the cells are, one could, for instance, use one of the following methods: i) repeated freezing and thawing, ii) sonication, iii) homogenization by high pressure (French press), iv) homogenization by grinding (bead mill), and v) permeabilization by detergents (e.g. Triton X-100) and/or enzymes (e.g. lysozyme). Finally, the cell debris can be removed by centrifugation so that the proteins and other soluble compounds remain in the supernatant.

Also proteases are released during cell lysis, which will start digesting the proteins in the solution. If the protein of interest is sensitive to proteolysis, it is recommended to proceed quickly, and to keep the extract cooled, to slow down the digestion. Alternatively, one or more protease inhibitors can be added to the lysis buffer immediately before cell disruption. Sometimes it is also necessary to add DNAse in order to reduce the viscosity of the cell lysate caused by a high DNA content.

6.3.2 Precipitation and Differential Solubilization

In bulk protein purification, a common first step to isolate proteins is precipitation using a salt such as ammonium sulfate $(NH_4)_2SO_4$. This process is called *Salting In* or *Salting Out* (Figure 7) This is performed by adding increasing amounts of ammonium sulfate and collecting the different fractions of precipitate protein. Ammonium sulfate is often used as it is highly soluble in water, has relative freedom from temperature effects and typically is not harmful to most proteins. Furthermore, ammonium sulfate can be removed by dialysis (Figure 8). The hydrophobic groups on the proteins get exposed to the atmosphere, attract other protein hydrophobic groups and get aggregated. Protein precipitated will be large enough to be visible. One advantage of this method is that it can be performed inexpensively with very large volumes.

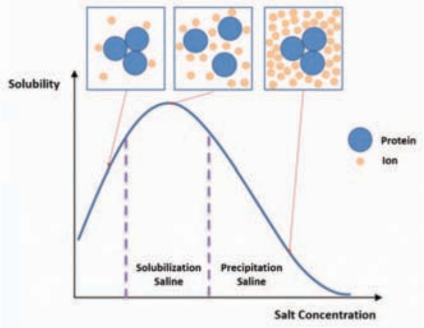


Figure 7: Salting In and Salting Out. During the salting in process, salt molecules increase the solubility of proteins by reducing the electrostatic interactions be-

tween protein molecules. As the salt concentration is increased, protein-protein interactions become more energetically favorable than protein-solvent interactions and the proteins precipitate from solution.

The first proteins to be purified are water-soluble proteins. Purification of integral membrane proteins requires disruption of the cell membrane in order to isolate any one particular protein from others that are in the same membrane compartment. Sometimes a particular membrane fraction can be isolated first, such as isolating mitochondria from cells before purifying a protein located in a mitochondrial membrane. A detergent such as sodium dodecyl sulfate (SDS) can be used to dissolve cell membranes and keep membrane proteins in solution during purification; however, because SDS causes denaturation, milder detergents such as Triton X-100 or CHAPS can be used to retain the protein's native conformation during complete purification.

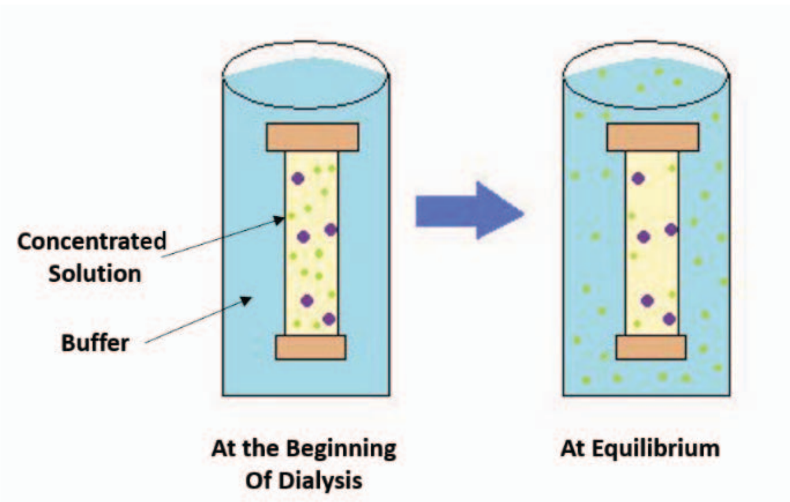


Figure 8: Dialysis**.** The process of dialysis separates dissolved molecules by their size. The biological sample is placed inside a closed membrane, where the protein of interest is too large to pass through the pores of the membrane, but through which smaller ions can easily pass. As the solution comes to equilibrium, the ions become evenly distributed throughout the entire solution, while the protein remains concentrated in the membrane. This reduces the overall salt concentration of the suspension.

6.3.3 Ultracentrifugation

Centrifugation is a process that uses centrifugal force to separate mixtures of particles of varying masses or densities suspended in a liquid. When a vessel (typically a tube or bottle) containing a mixture of proteins or other particulate matter, such as bacterial cells, is rotated at high speeds, the inertia of each particle yields a force in the direction of the particles velocity that is proportional to its mass.

The tendency of a given particle to move through the liquid because of this force is offset by the resistance the liquid exerts on the particle. The net effect of "spinning" the sample in a centrifuge is that massive, small, and dense particles move outward faster than less massive particles or particles with more "drag" in the liquid. When suspensions of particles are "spun" in a centrifuge, a "pellet" may form at the bottom of the vessel that is enriched for the most massive particles with low drag in the liquid.

Non-compacted particles remain mostly in the liquid called "supernatant" and can be removed from the vessel thereby separating the supernatant from the pellet.

The rate of centrifugation is determined by the angular acceleration applied to the sample, typically measured in comparison to the *g*. If samples are centrifuged long enough, the particles in the vessel will reach equilibrium wherein the particles accumulate specifically at a point in the vessel where their buoyant density is balanced with centrifugal force. Such an "equilibrium" centrifugation can allow extensive purification of a given particle.

In *sucrose gradient centrifugation*, a linear concentration gradient of sugar (typically sucrose, glycerol, or a silica based density gradient media, like Percoll) is generated in a tube such that the highest concentration is on the bottom and lowest on top. Percoll is a trademark owned by GE Healthcare companies.

A protein sample is then layered on top of the gradient and spun at high speeds in an ultracentrifuge. This causes heavy macromolecules to migrate towards the bottom of the tube faster than lighter material. During centrifugation in the absence of sucrose, as particles move farther and farther from the center of rotation, they experience more and more centrifugal force (the further they move, the faster they move). The problem with this is that the useful separation range of within the vessel is restricted to a small observable window. A properly designed sucrose gradient will counteract the increasing centrifugal force so the particles move in close proportion to the time they have been in the centrifugal field. Samples separated by these gradients are referred to as "rate zonal" centrifugations. After separating the protein/particles, the gradient is then fractionated and collected.

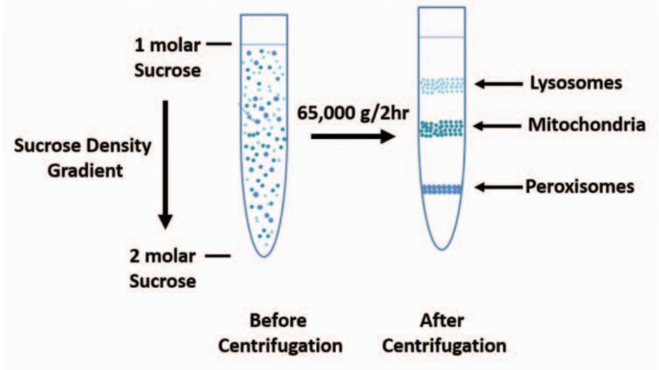


Figure 9: Sucrose Density Gradient.

6.3.4 Purification Strategy

Choice of a starting material is key to the design of a purification process. In a plant or animal, a particular protein usually isn't distributed homogeneously throughout the body; different organs or tissues have higher or lower concentrations of the protein.

Use of only the tissues or organs with the highest concentration decreases the volumes needed to produce a given amount of purified protein. If the protein is present in low abundance, or if it has a high value, scientists may use recombinant DNA technology to develop cells that will produce large quantities of the desired protein (this is known as an expression system). Recombinant expression allows the protein to be tagged, e.g. by a His-tag or Strep-tag to facilitate purification, reducing the number of purification steps required.

An analytical purification generally utilizes three properties to separate proteins. First, proteins may be purified according to their isoelectric points by running them through a pH graded gel or an ion exchange column. Second, proteins can be separated according to their size or molecular weight via size exclusion chromatography or by SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis) analysis. Proteins are often purified by using 2D-PAGE and are then analysed by peptide mass fingerprinting to establish the protein identity. This is very useful for scientific purposes and the detection limits for protein are nowadays very low and nanogram amounts of protein are sufficient for their analysis. Thirdly, proteins may be separated by polarity/hydrophobicity via high performance liquid chromatography or reversed-phase chromatography.

For preparative protein purification, the purification protocol generally contains one or more chromatographic steps. The basic procedure in chromatography is to flow the solution containing the protein through a column packed with various materials. Different proteins interact differently with the column material, and can thus be separated by the time required to pass the column, or the conditions required to elute the protein from the column. Usually proteins are detected as they are coming off the column by their absorbance at 280 nm. Many different chromatographic methods exist, with the most common described below:

Size Exclusion Chromatography (also known as Gel Filtration Chromatography)

Chromatography can be used to separate protein in solution or under denaturing conditions by using porous gels. This technique is known as size exclusion chromatography. The principle is that smaller molecules have to traverse a larger volume in a porous matrix. Consequentially, proteins of a certain range in size will require a variable volume of eluent (solvent) before being collected at the other end of the column of gel. Thus, proteins will be separated based on their size (Figure 10).

In the context of protein purification, the eluent is usually pooled in different test tubes. All test tubes containing no measurable trace of the protein to purify are discarded. The remaining solution is thus made of the protein to purify and any other similarly-sized proteins.

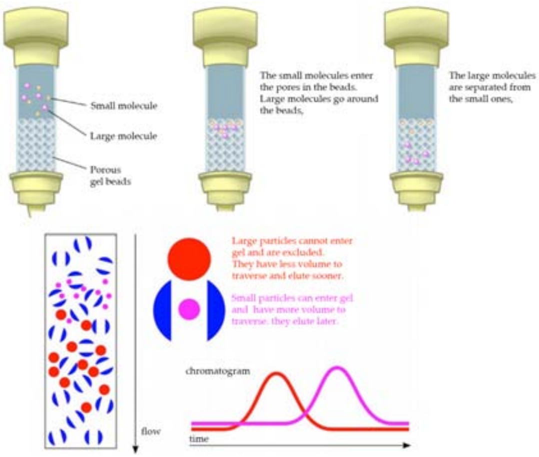


Figure 10: Size Exclusion Chromatography. Also known as Gel Filtration Chromatography, is a low resolution isolation method that involves the use of

beads that have tiny "tunnels" in them that each have a precise size. The size is referred to as an "exclusion limit," which means that molecules above a certain molecular weight will not fit into the tunnels. Molecules with sizes larger than the exclusion limit do not enter the tunnels and pass through the column relatively quickly by making their way between the beads. Smaller molecules, which can enter the tunnels, do so, and thus, have a longer path that they take in passing through the column. Because of this, molecules larger than the exclusion limit will leave the column earlier, while smaller molecules that pass through the beads will elute from the column later. This method allows separation of molecules by their size.

Hydrophobic Interaction Chromatography (HIC)

HIC media is amphiphilic, with both hydrophobic and hydrophilic regions, allowing for separation of proteins based on their surface hydrophobicity. Target proteins and their product aggregate species tend to have different hydrophobic properties and removing them via HIC further purifies the protein of interest. Additionally, the environment used typically employs less harsh denaturing conditions than other chromatography techniques, thus helping to preserve the protein of interest in its native and functional state. In pure water, the interactions between the resin and the hydrophobic regions of protein would be very weak, but this interaction is enhanced by applying a protein sample to HIC resin in high ionic strength buffer. The ionic strength of the buffer is then reduced to elute proteins in order of decreasing hydrophobicity (Figure 11).

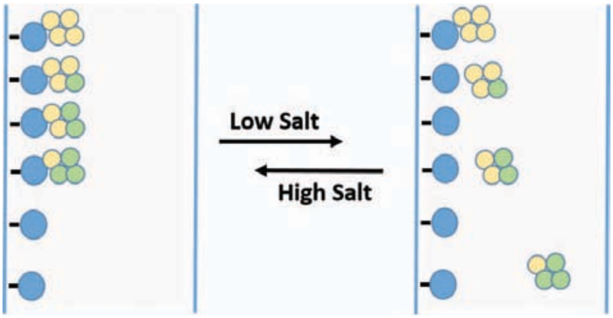


Figure 11: Hydrophobic Interaction Chromatography. The column matrix, shown in blue has a hydrophobic ligand covalently attached. In high salt conditions, proteins will bind to the matrix with differing affinity, with more hydrophobic proteins (shown in yellow) binding more tightly than more hydrophilic proteins (shown in green) When the salt concentration is decreased, proteins that are more hydrophilic will be released first, followed more hydrophobic proteins.

Ion Exchange Chromatography

Ion exchange chromatography separates compounds according to the nature and degree of their ionic charge. The column to be used is selected according to its type and strength of charge. Anion exchange resins have a positive charge and are used to retain and separate negatively charged compounds (anions), while cation exchange resins have a negative charge and are used to separate positively charged molecules (cations).

Before the separation begins a buffer is pumped through the column to equilibrate the opposing charged ions. Upon injection of the sample, solute molecules will exchange with the buffer ions as each competes for the binding sites on the resin. The length of retention for each solute depends upon the strength of its charge. The most weakly charged compounds will elute first, followed by those with successively stronger charges. Because of the nature of the separating mechanism, pH, buffer type, buffer concentration, and temperature all play important roles in controlling the separation. Figure 12 demonstrates a type of ion exchange column known as a *cation exchange column*. In this case, the support consists of tiny beads to which are attached chemicals possessing a charge. Each charged molecule has a counter-ion. The figure shows the beads (blue) with negatively charged groups (red) attached. In this example, the counter-ion is sodium, which is positively charged. The negatively charged groups are unable to leave the beads, due to their covalent attachment, but the counter- ions can be "exchanged" for molecules of the same charge. Thus, *a cation exchange column* will have positively charged counter-ions and positively charged compounds present in a mixture passed through the column will exchange with the counter-ions and "stick" to the negatively charged groups on the beads. Molecules in the sample that are neutral or negatively charged will pass quickly through the column. On the other hand, in *anion exchange chromatography*, the chemical groups attached to the beads are positively charged and the counter-ions are negatively charged. Molecules in the sample that are negatively charged will "stick" and other molecules will pass through quickly. To remove the molecules "stuck" to a column, one simply needs to add a high concentration of the appropriate counter-ions to displace and release them. This method allows the recovery of all components of the mixture that share the same charge.

Ion exchange chromatography is a very powerful tool for use in protein purification and is frequently used in both analytical and preparative separations.

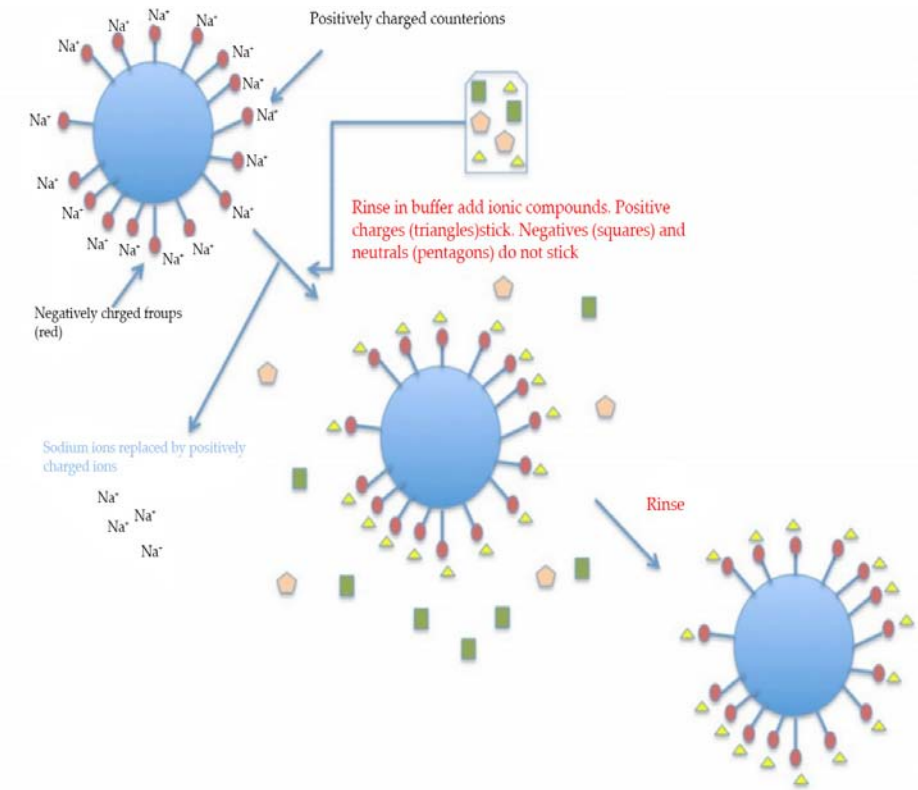


Figure 12: Cation Exchange Chromatography. In this diagram the negatively charged molecules (shown in red) are covalently attached to the column matrix beads (shown in blue). Sodium ions (Na+) are the counter ions that are replaced by positively charged proteins within the protein mixture. Neutral and negatively charged proteins do not stick and will pass through the column. The positively charged proteins can then be eluted from the column by adding higher concentrations of the counter ion (in this case the sodium ions).

Affinity Chromatography

Affinity Chromatography is a separation technique based upon molecular conformation, which frequently utilizes application specific resins. These resins have ligands (small molecules) attached to their surfaces which are specific for and will bind with the compounds to be separated. Most frequently, these ligands function in a fashion similar to that of antibody-antigen interactions. This “lock and key” fit between the ligand and its target compound makes it highly specific, frequently generating a single peak, while all else in the sample is unretained (Figure 13).

For example, many membrane proteins are glycoproteins and can be purified by lectin affinity chromatography. Detergent-solubilized proteins can be allowed to bind to a chromatography resin that has been modified to have a covalently attached lectin. Proteins that do not bind to the lectin are washed away and then specifically bound glycoproteins can be eluted by adding a high concentration of a sugar that competes with the bound glycoproteins at the lectin binding site. Some lectins have high affinity binding to oligosaccharides of glycoproteins that is hard to compete with sugars, and bound glycoproteins need to be released by denaturing the lectin.

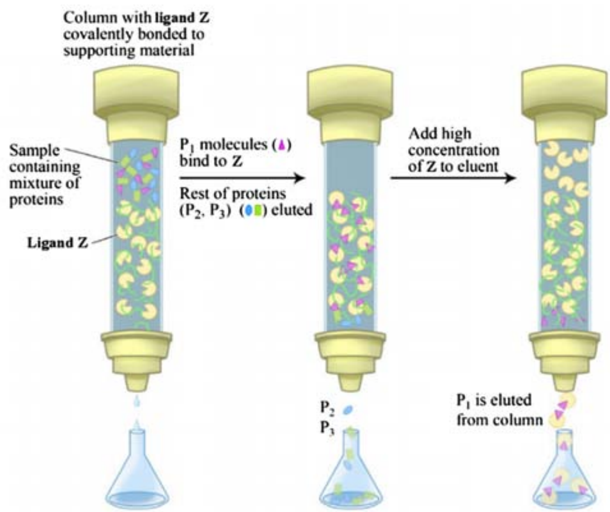


Figure 13: Example of Affinity Chromatography. In this example, protein P1 has affinity for ligand Z and will bind to the column while proteins P2 and P3 will pass through the column. Protein P1 can then be eluted from the column using high concentrations of free ligand Z.

A common technique involves engineering a sequence of 6 to 8 histidine residues into the N- or C-terminal of a recombinant protein. The polyhistidine binds strongly to divalent metal ions such as nickel and cobalt. The protein can be passed through a column containing immobilized nickel ions, which binds the polyhistidine tag. All untagged proteins pass through the column. The protein can be eluted with imidazole, which competes with the polyhistidine tag for binding to the column, or by a decrease in pH (typically to 4.5), which decreases the affinity of the tag for the resin. While this procedure is generally used for the purification of recombinant proteins with an engineered affinity tag (such as a 6xHis tag), it can also be used for natural proteins with an inherent affinity for divalent cations.

Immunoaffinity Chromatography

A special type of affinity chromatography is Immunoaffinity chromatography (Figure 14). This technique uses the specific binding of an antibody with its antigen (target molecule that the antibody will bind with selectively) to purify the protein of interest. The procedure involves immobilizing an antibody to a solid substrate (e.g. a porous bead or a membrane), which then selectively binds the target, while everything else flows through. The target protein can be eluted by changing the pH or the salinity. The immobilized ligand can be an antibody (such as Immunoglobulin G) or it can be a protein (such as Protein A). Because this method does not involve engineering in a tag, it can be used for proteins from natural sources.

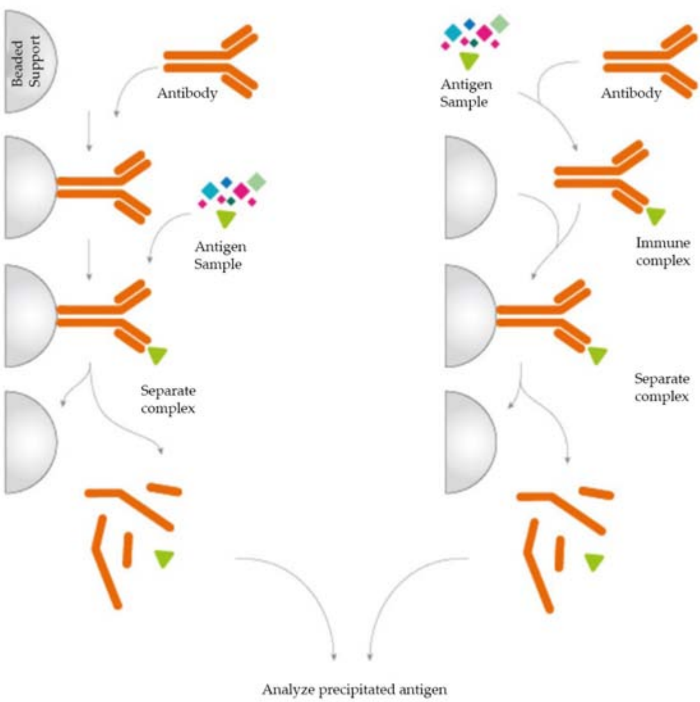


Figure 14: An Antigen Immunoprecipitation Experiment. The antibody is either pre-immobilized to a solid support (left) or immobilized using antibody binding proteins after incubation with the sample (right). Immobilization allows the immune complex to be extracted from the complex sample, washed and eluted providing a high enrichment of the protein under investigation

High Performance Liquid Chromatography (HPLC) and Fast Protein Liquid Chromatography (FPLC)

High performance liquid chromatography or high pressure liquid chromatography (HPLC) is a form of chromatography applying high pressure to drive the solutes through the column faster. This means that the diffusion is limited and the resolution is improved. The most common form is "reversed phase" HPLC, where the column material is hydrophobic. The proteins are eluted by a gradient of water and increasing amounts of an organic solvent, such as acetonitrile. The proteins elute according to their hydrophobicity. After purification by HPLC the protein is in a solution that only contains volatile compounds, and can easily be *lyophilized (freeze dried)*. HPLC purification frequently results in denaturation of the purified proteins and is thus not applicable to proteins that do not spontaneously refold.

Due to the drawbacks of HPLC, an alternative technique using a lower pressure system was developed and is called *Fast protein liquid chromatography (FPLC). FPLC* is a form of liquid chromatography that is often used to analyze or purify mixtures of proteins. As in other forms of chromatography, separation is possible because the different components of a mixture have different affinities for two materials, a moving fluid (the "mobile phase") and a porous solid (the stationary phase). In FPLC the mobile phase is an aqueous solution, or "buffer". The buffer flow rate is controlled by a positive-displacement pump and is normally kept constant, while the composition of the buffer can be varied by drawing fluids in different proportions from two or more external reservoirs. The stationary phase is a resin composed of beads, usually of cross-linked agarose, packed into a cylindrical glass or plastic column. FPLC resins are available in a wide range of bead sizes and surface ligands depending on the application.

In the most common FPLC strategy, an ion exchange resin is typically chosen (Figure 15). A mixture containing one or more proteins of interest is dissolved in 100% buffer A and pumped into the column. The proteins of interest bind to the resin while other components are carried out in the buffer. The total flow rate of the buffer is kept constant; however, the proportion of Buffer B (the "elution" buffer) is gradually increased from 0% to 100% according to a programmed change in concentration (the "gradient"). Buffer B contains high concentrations of the exchanger ion. Thus as the concentration of the Buffer B gradually increases, bound proteins will dissociate depending on their ionic interactions with the column matrix and appear in the eluant. The eluant passes through two detectors which measure salt concentration (by conductivity) and protein concentration (by absorption of ultraviolet light at a wavelength of 280nm). As each protein is eluted it appears in the eluant as a "peak" in protein concentration and can be collected for further use.

FPLC was developed and marketed in Sweden by Pharmacia in 1982 and was originally called *fast performance liquid chromatography* to contrast it with HPLC or high-performance liquid chromatography. FPLC is generally applied only to proteins; however, because of the wide choice of resins and buffers it has broad applications. In contrast to HPLC the buffer pressure used is relatively low, typically less than 5 bar, but the flow rate is relatively high, typically 1-5 ml/min. FPLC can be readily scaled from analysis of milligrams of mixtures in columns with a total volume of 5 ml or less to industrial production of kilograms of purified protein in columns with volumes of many liters.

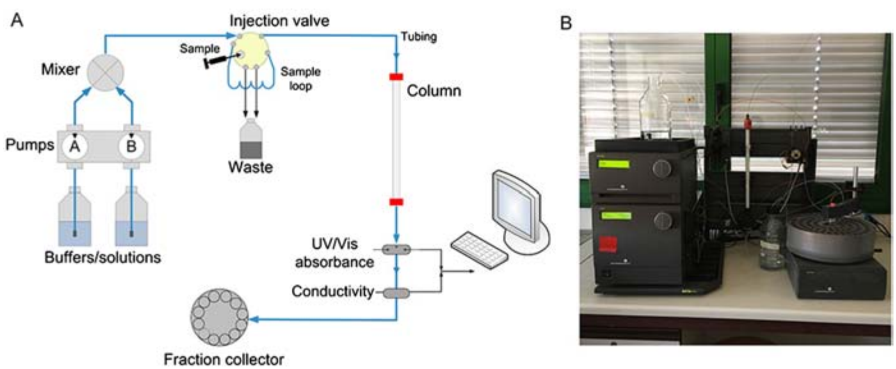


Figure 15: Typical FPLC System. A. Scheme of basic compounents and typical flow path for a chromatography system. B. Picrue of GE Healthcare AKTA FPLC apparatus.

6.3.5 Purification Scheme

During the protein purification process it is necessary to have a quantitative system to determine how much protein has been purified, what concentration the protein represents from the original mixture, how biologically active the purified protein is, and the overall purity of the protein. This will help guide and optimize the purification method being developed. Ineffective separation techniques can be disregarded and other techniques that give higher yield or that retain biologically activity of the protein can be adopted.

Thus, each step in the purification scheme is quantitatively evaluated for the following parameters: total protein, total activity, specific activity, yield, purification level. Each of these parameters will be defined within the sample protocol given below.

Pretend you are a researcher that wants to isolate a novel, unknown protein from a bacterial culture. You grow 500 ml of the bacteria overnight at 37°C and harvest the bacteria by centrifugation. You remove the culture broth and retain the bacterial pellet. You then lyse the bacteria using freeze/thaw in 10 mL of reaction buffer. You then centrifuge the lysed bacteria to remove the insoluble materials and retain the supernatent that contains the soluble proteins. Your protein of interest has a biological activity that you can measure using a simple assay that causes a color change in the reaction mixture (Figure 16). You also note that this reaction rate increases with increasing concentrations of your protein supernatent (Figure 16)

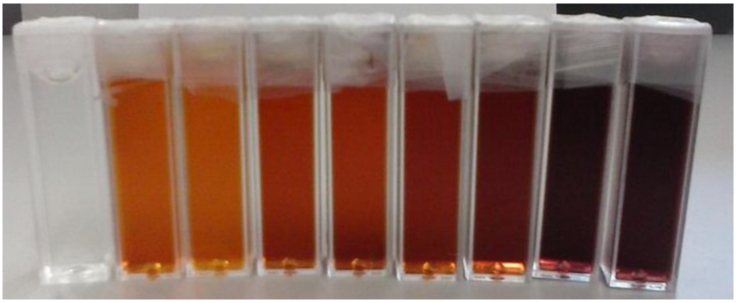


Figure 16: Example of a Chemical Reaction that causes a color change from orange to brown depending on increasing concentration.

At this point, you can measure your baseline concentrations for the first purification level (bacterial lysis and removal of insoluble proteins and other cellular debris by centrifugation).

Total Protein is calculated by measuring the concentration in a fraction of your sample, and then multiplying that by the total volume of your sample. In this case, you are starting with 10 mL of supernatent. In a typical assay to measure protein concentration, you will use 50 – 200 µL of sample to determine the protein concentration. For example, if you calculate that there is 7.5 µg/µL in your initial assay, you would need to convert that value into mg/mL and then multiply it by 10 mL for a total of 75 mg of protein in 10 mL of supernatant (Table 2)

Total Activity is measured as the enzyme activity within the assay, multiplied by the total volume of the sample. For example, in the initial sample, you might use 5 to 50 µL of sample in your biological reaction (Figure 16). If you calculated the activity in your assay to be 2.5 units/

µL, this would be equivalent to 2,500 units/mL or 25,000 units/10 mL of supernatant. Note that, the enzyme unit, or international unit for enzyme (symbol U, sometimes also IU) is a unit of enzyme's catalytic activity. 1 U (µmol/min) is defined as the amount of the enzyme that catalyzes the conversion of one micromole of substrate per minute under the specified conditions of the assay method.

Specific Activity is measured by dividing the Total Activity by the Total Protein. In our example, 25,000 units divided by 75 mg of protein = 333.3 units/mg.

Yield is a measure of the biological activity retained in the sample after each purification step. The amount in the first step is set to be 100%. All subsequent yield steps will be evaluated using the first purification step. It is calculated by dividing the total activity of the current step, by the total activity of the first step and then multiplying by 100.

Purification level evaluates the purity of the protein of interest by dividing the specific activity calculated after each purification step by the specific activity of the first purification step. Thus, the first step always has a value of 1.

Table 2: Typical Protein Purification Scheme

Purification Step	Purification Method	Total Protein (mg)	Total Activity (units)	Specific Activity (units/mg)	Yield (%)	Purification Level
1	Cell Lysis - Centrifugation	75	25,000	333.3	100	1
2	Salting Out	54	23,416	433.6	93.7	1.3
3	Ion Exchange Chromatography	12.3	21,227	1725.8	84.9	5.2
4	Size Exclusion Chromatography	0.93	18,633	20,035	74.5	60.1

Note that after each purification step that the *Total Protein* goes down, as you are purifying your protein away from other proteins in the mixture. *Total Activity* also goes down with each purification step, as some of your protein of interest is also lost at each purification step, because (1) some protein will stick to the test tubes and glassware, (2) some protein won't bind with 100% efficiency to your column matrix, (3) some protein may bind too tightly to be removed from the column matrix during elution, and (4) some protein may be denatured or degraded during the purification process.

The amount of your protein of interest that is lost is represented within the overall percent yield for each purification step. If the percent yield is too low alternative purification methods should be explored.

Note that in a good protein purification scheme that the *specific activity* should go up substantially with each level of purification as the amount of your protein of interest makes up a greater percentage of the total protein within that fraction. If the specific activity only increases modestly within a purification step, or if it decreases during a purification step, this could indicate that (1) your protein of interest is being substantially lost at that step, (2) that your protein of interest is being denatured or degraded and is no longer biologically active, or (3) that a required cofactor or binding protein is being reduced at that purification step. Additional experiments may need to be conducted to determine which of the causes predominates, so that steps can be taken to reduce protein inactivation. For example, many proteins are temperature sensitive and will degrade or denature at room temperature. Completing purification steps on ice can often reduce degradation.

Overall, the fold increase in *purification level* should increase exponentially during the purification process. Note that in our example, if after 4 steps of purification our proteins is close to 95% pure, this would indicate that our protein of interest makes up approximately 1.24% of the total protein within the sample.

6.4 PROTEIN STRUCTURE DETERMINATION

Proteins play a crucial role in the world of research and their structures are, therefore, of great interest to the researcher.

Many techniques and methods have been used to determine this, but one of the most important has been the discovery and development of X-ray crystallography.

The tertiary structure of a protein is responsible for its properties and behavior. The primary methods in use to study this include:

6.4.1 X-Ray Crystallography

This method provides the maximum data regarding the tertiary structure of a protein. X-rays are passed through the rotating crystal and the diffracted rays are collected on a target and analyzed by computerized systems.

Crystal quality is crucial; thus it results in very detailed data on atomic arrangement within a rigid protein with other neighboring ions, ligands and molecules, but may not work so well with proteins which have large flexible domains and therefore do not form precisely ordered crystalline structures. It requires relatively large amounts of the protein. For this reason recombinant proteins are often produced in the necessary quantity, and then impurities removed, followed by refolding of the protein and crystallographic study. It provides very high resolution.

6.4.2 Electron Microscopy and Cryo-Electron Microscopy

These methods are suited to the study of large macromolecule complexes or even cellular organelles, which are relatively bigger on the molecular scale, and can help to reconstruct the tertiary structure of a single particle. This has the great advantage of obviating the challenging prerequisite of protein crystallization.

Cryoelectron microscopy is a variant at temperatures at or below that of liquid nitrogen and can visualize protein structures at very high resolution, though is less than that of methods like NMR spectroscopy or crystallography.

It works with minute amounts of protein as well and reduces the artefacts due to radiation damage.

6.4.3 Nuclear Magnetic Resonance (NMR) Spectroscopy

This method depends on the effect of varying radiofrequency waves on nuclear resonance of various atoms within the protein.

It needs larger quantities of protein in a stable soluble form at room temperature, and to remain stable for the long duration of data acquisition.

The proteins should be of small size as well to avoid overlapping peaks.

However, it gives a higher resolution. It is most suitable when protein crystallization is not feasible as with flexible proteins, or when system dynamics are to be detailed.

6.4.4 Small-Angle X-Ray Scattering and Small-Angle Neutron Scattering

These methods are valuable for studying protein structure when limited resolution is sufficient. Experimental conditions can be better controlled because the protein is usually in solution.

6.4.5 Homology Modeling

This enables researchers to obtain a three-dimensional picture of a protein. It needs prior knowledge of the close homologue or template which has a very high degree of identical amino acid sequence to the analyte.

Protein modeling based on this template depends on the fact that protein structure is highly conserved in most cases.

6.4.6 Partial Structural Study Methods

These include ultracentrifugation, mass spectrometry and fluorescence spectrometry, which may be used to complement other techniques.

6.5 PROTEOMICS AND PROTEIN FUNCTION

An organism's proteome is the collection of all proteins that the organism makes. This differs from the genome in a number of ways. There are pre-translational events such as alternative splicing (in which several coding regions of DNA ["exons"] are joined together, but it is known that this can be done in more than one way, and that which way the exons are spliced is one way that protein expression is regulated) and post-translational events such as phosphorylation (which may sometimes determine whether a protein is active or not). Which proteins are present and in what quantities in a given cell at a given time is highly variable, whereas the genome is static.

Proteomics necessarily involves a greater level of complexity than genomics. The human proteome — the collection of all proteins generated by the human genome — is estimated to contain 10 million to 20 million proteins — about 2-3 orders of magnitude higher than the number of human genes.

The practical distinction between genomics and proteomics is what you focus on — whether DNA is the center of attention, or whether proteins are the central object. The basic reason why genomics took off ahead of proteomics is that techniques for high-throughput sequencing became available — this involved a convergence of innovative biotechnologies and ingenious fast algorithms. Because DNA is just a linear sequence of 4 symbols, high-throughput comparisons of DNA sequences with each other gave an added dimension to the subject. This has generated a lot of really interesting problems and there continues to be active progress in many directions as mathematical abstraction, primarily in the form of new algorithms and statistical methods, meets biological reality. New technologies, such as microarrays, the subject of IPAM's first long program, continue to stir things up. This area of research is so highly active that many proteomic strategies incorporate a genomic component.

Proteins direct almost all biological functions, but how any given protein acts is rarely transparent. Proteins do not function independently, but interact in highly complex networks, which in turn influence the intricate network of regulatory mechanisms by which the amount of each protein that is produced is regulated. These regulatory mechanisms, despite many tantalizing clues, remain mysterious and are difficult to model. It is a fundamental problem to determine which proteins interact, and how they fit into a network of interactions, and this becomes even more challenging if one wants to do it in an automated way. Many current approaches attempt to integrate together information from a number of sources — data from microarrays that give information about which proteins are up and down-regulated together, cross-genome analysis about whether the genes for two proteins have evolved together, etc.

The sequence of amino acids in each protein is determined by genomic data, plus some variation coming from alternative splicing. The secondary structure of the protein (alpha-helices, etc.) and the 3-D configuration of the protein are determined by the amino acid sequence, but we are very far from being able to predict one from the other. Approaches vary widely, ranging from attempting to do an ab initio computation from quantum mechanical forces to homology modeling, which operates by using data-mining to make comparisons with sequences of proteins whose secondary or 3-D structure is known.

The 3-D configuration of a protein is important in determining its function. Certain regions of the protein mainly serve to hold certain active regions in the right position, so that they can interact correctly with active regions of other proteins. Finding the right way to determine the active region of a protein, even once its shape is known, is a topic of active research.

There are two basic ways to learn of a protein's existence — by finding the nucleotides that code for it in the organism's DNA, or by finding it already assembled in the organism's cells. The first method is much less direct, and less certain, because the identification of exons within the DNA is still an art, and reconstructing how the exons are spliced is uncertain. A second disadvantage is that one does not have the protein itself in hand, to study its chemistry and structure. The big advantage of this approach is that it is high-throughput, so that one can find the sequences of vast numbers of hitherto unknown proteins in this way. A second advantage of this genetic approach is that one can study the effect of the protein indirectly by producing "knockout" organisms, which differ from the wild type in having exactly one gene disabled.

New technologies now hold out the possibility of a high-throughput approach that starts with the protein rather than with the DNA that codes for it.

Proteomics is of importance to basic science, for elucidating the fundamental mechanisms of biology. It is also of great interest to the biotech industry, since proteins whose function and active regions are known provide potential targets for drugs.

The human proteome — the collection of all proteins generated by the human genome — is estimated to contain 10 million to 20 million proteins — about 2-3 orders of magnitude higher than the number of human genes. Proteins direct almost all biological functions, but how any given protein acts is rarely transparent. Moreover, proteins do not function independently, but interact in highly complex networks, which in turn influence the intricate network of regulatory mechanisms by which the amount of each protein that is produced is regulated. These regulatory mechanisms, despite many tantalizing clues, remain mysterious and are difficult to model.

REFERENCES

1. Ahern, K. and Rajagopal, I. (2019) Biochemistry Free and Easy. Published by Libretexts. Available at: https://bio.libretexts.org/Bookshelves/Biochemistry/Book%3A_Biochemistry_Free_and_Easy_(Ahern_and_Rajagopal)/09%3A_Techniques/9.04%3A_Gel_Exclusion_Chromatography.
2. Bhella, D. (2019) Cryo-electron microscopy: an introduction to the technique, and considerations when working to establish a national facility. *Biochysical Reviews* 11: 515-519. Available at: https://link.springer.com/article/10.1007/s12551-019-00571-w
3. Magdeldin, S. (2012) Gel Electrophoresis – Principles and Basics. Published by InTech under Creative Commons Attribution 3.0. Available at: https://pdfs.semanticscholar.org/4b93/70ac3946cec6e12c369679c4178a5ef38e61.pdf
4. Molnar, C. and Gair, J. (2013) Antibodies. Chapter in Concepts in Biology, Published by B.C. Open Textbook Project. Available at: https://opentextbc.ca/biology/chapter/23-3-antibodies/
5. Thul PJ et al, 2017. A subcellular map of the human proteome. *Science.* PubMed: 28495876 DOI: 10.1126/science.aal3321
6. Uhlén M et al, 2015. Tissue-based map of the human proteome. *Science*

PubMed: 25613900 DOI: 10.1126/science.1260419

7. Uhlen M et al, 2017. A pathology atlas of the human cancer transcriptome. *Science.* PubMed: 28818916 DOI: 10.1126/science.aan2507

CHAPTER

7

TECHNIQUES USED IN BIOCHEMISTRY AND MOLECULAR BIOLOGY

INTRODUCTION

Biochemistry and molecular biology rely on a range of techniques to investigate the structure, function, and interactions of biomolecules. DNA sequencing methods, such as Sanger sequencing and next-generation sequencing, have enabled the comprehensive analysis of genomes. Polymerase Chain Reaction (PCR) amplifies specific DNA sequences and is widely used in genetic testing and gene expression analysis. Gel electrophoresis separates DNA, RNA, and proteins based on size and charge, facilitating applications like DNA fingerprinting and protein variant identification. Western blotting detects and quantifies specific proteins using antibodies, providing insights into protein expression and interactions. Mass spectrometry analyzes the mass and charge of molecules, aiding in the identification and characterization of biomolecules. X-ray crystallography determines the three-dimensional structure of molecules, including proteins and nucleic acids. Nuclear Magnetic Resonance (NMR) spectroscopy examines molecular structure, dynamics, and interactions in solution. These techniques, among others, play crucial roles in advancing our understanding of biological systems, offering insights into fundamental processes and facilitating the development of diagnostic tools and therapeutics.

7.1 MASS SPECTROSCOPIC TECHNIQUES

Mass spectrometry is a powerful analytical technique widely used in biochemistry and molecular biology for the identification, characterization, and quantification of biomolecules. It provides information on the mass-to-charge ratio (m/z) of ions, enabling the determination of molecular weight, structural elucidation, and analysis of molecular interactions. Mass spectrometry encompasses a variety of techniques, each with its own applications and capabilities. Here, we will provide an overview of some commonly used mass spectroscopic techniques:

1. Electrospray Ionization (ESI): ESI is a soft ionization technique primarily used for the analysis of large biomolecules such as proteins, peptides, and nucleic acids. It involves the formation of ions in the gas phase by the application of a high voltage to a liquid containing the analyte. ESI allows for gentle ionization and preservation of the native structure of biomolecules, making it suitable for studying non-covalent interactions and post-translational modifications.
2. Matrix-Assisted Laser Desorption/Ionization (MALDI): MALDI is another soft ionization technique that is commonly used for the analysis of large biomolecules. It involves embedding the analyte in a matrix compound, which absorbs laser energy and facilitates the transfer of analyte molecules into the gas phase. MALDI is particularly useful for the analysis of proteins, peptides, and other biomolecules, allowing the generation of intact molecular ions with minimal fragmentation.
3. Tandem Mass Spectrometry (MS/MS or MS^n): Tandem mass spectrometry involves the sequential analysis of ions generated in a mass spectrometer. It allows for structural elucidation of complex molecules by fragmenting precursor ions and analyzing the resulting fragments. MS/MS techniques, such as collision-induced dissociation (CID) or electron capture dissociation (ECD), provide valuable information about the sequence, modifications, and interactions of biomolecules.
4. High-Resolution Mass Spectrometry (HRMS): HRMS is characterized by its ability to accurately measure the mass-to-charge ratio of ions with high precision and resolution. It allows for the discrimination of closely related ions and

provides enhanced accuracy in determining molecular weights. HRMS techniques, such as Fourier transform ion cyclotron resonance (FT-ICR) and Orbitrap, are valuable for the analysis of complex mixtures and the identification of unknown compounds.

5. Quantitative Mass Spectrometry: Mass spectrometry can also be used for quantitative analysis of biomolecules. Stable isotope labeling, such as stable isotope labeling by amino acids in cell culture (SILAC) or isobaric tags for relative and absolute quantitation (iTRAQ), enables the comparison of protein or peptide abundance between different samples. Selected reaction monitoring (SRM) or parallel reaction monitoring (PRM) techniques allow for targeted quantification of specific peptides or proteins.
6. Imaging Mass Spectrometry: Imaging mass spectrometry combines mass spectrometry with spatial information, allowing the visualization and mapping of molecules within tissues or cells. It provides insights into the spatial distribution of biomolecules, such as metabolites, lipids, and peptides, and enables the identification of molecular signatures associated with specific regions or diseases.

Mass spectrometry continues to evolve, with advancements in instrumentation, data analysis, and sample preparation techniques. It has become an indispensable tool in biochemistry and molecular biology, providing valuable information on molecular structure, interactions, and function. The versatility and sensitivity of mass spectrometry make it an essential technique for a wide range of applications, including proteomics, metabolomics, lipidomics, and drug discovery.

7.1.1 Electrospray Ionization (ESI)

Electrospray Ionization (ESI) is a widely used ionization technique in mass spectrometry that enables the analysis of large biomolecules such as proteins, peptides, nucleic acids, carbohydrates, and lipids. ESI is particularly valuable in the field of biochemistry and molecular biology because it allows the transfer of labile biomolecules into the gas phase while preserving their native structure and noncovalent interactions.

The ESI process involves the creation of ions from a liquid sample by applying a high voltage to a nebulizing gas. The sample is typically dissolved in a volatile solvent, such as methanol or acetonitrile, and introduced into the mass spectrometer through a capillary emitter. As the liquid flows through the emitter, a high voltage is applied, leading to the formation of a Taylor cone at the emitter tip.

The high electric field at the Taylor cone generates tiny droplets from the liquid, which undergo a series of events leading to ionization. As the droplets continue to decrease in size through the process of solvent evaporation, charges accumulate on their surface. Eventually, the droplets become highly charged and reach a critical point where coulombic repulsion overcomes surface tension, resulting in the ejection of ions into the gas phase.

The generated ions are then drawn into the mass spectrometer, where they are separated based on their mass-to-charge ratio (m/z) using various mass analyzers, such as quadrupoles, time-of-flight (TOF) analyzers, or ion traps. The resulting mass spectrum provides information about the molecular weight of the ions, their charge state, and their relative abundance.

ESI offers several advantages in the analysis of biomolecules. Firstly, it allows the analysis of large and complex molecules that are difficult to volatilize and ionize using other ionization methods. Secondly, ESI can preserve noncovalent interactions, enabling the investigation of protein-ligand or protein-protein complexes and the determination of their stoichiometry and binding affinities. Additionally, ESI facilitates the analysis of post-translational modifications, such as phosphorylation, glycosylation, or acetylation, providing insights into protein function and signaling pathways.

ESI has found numerous applications in various areas of research, including proteomics, metabolomics, lipidomics, and drug discovery. It has revolutionized the field of mass spectrometry, enabling the analysis of biomolecules with high sensitivity and accuracy. The development of hybrid mass spectrometry platforms and advancements in ESI interface technologies have further expanded the capabilities of ESI in the analysis of complex biological samples.

7.1.2 Matrix-Assisted Laser Desorption/Ionization (MALDI)

Matrix-Assisted Laser Desorption/Ionization (MALDI) is a powerful ionization technique used in mass spectrometry for the analysis of biomolecules, particularly large proteins and peptides. MALDI allows the generation of intact molecular ions from a sample by incorporating it into a matrix material, which facilitates the desorption and ionization of the analyte molecules upon laser irradiation. The MALDI process involves the formation of a solid-state matrix, typically composed of an organic compound such as a small molecule or a polymer, mixed with the analyte of interest. The matrix absorbs laser energy at a specific wavelength, which is typically in the ultraviolet (UV) or near-infrared (NIR) range. Upon laser irradiation, the matrix material absorbs the energy and becomes excited, transferring energy to the analyte molecules in close proximity.

As a result of this energy transfer, the analyte molecules undergo desorption from the solid matrix and become ionized. The desorbed and ionized analyte ions are then accelerated into the mass spectrometer, where they are separated based on their mass-to-charge ratio (m/z) using various mass analyzers, such as time-of-flight (TOF) analyzers or ion traps. The resulting mass spectrum provides information on the molecular weight and abundance of the analyte ions.

MALDI offers several advantages in the analysis of biomolecules. One major advantage is the ability to analyze large biomolecules, including proteins and peptides, while preserving their structural integrity. The incorporation of the analyte into a matrix material allows for gentle ionization, minimizing fragmentation and preserving noncovalent interactions within the molecule. This is particularly important in protein analysis, where maintaining the native conformation is crucial for studying protein-protein interactions, post-translational modifications, and structural characterization. MALDI has been widely applied in various fields, including proteomics, lipidomics, metabolomics, and clinical diagnostics. It has revolutionized the analysis of complex biological samples by providing high sensitivity, mass accuracy, and the ability to analyze a wide range of biomolecules. MALDI imaging mass spectrometry, in particular, has enabled the visualization and mapping of biomolecules within tissues, allowing researchers to study spatial distributions and identify molecular signatures associated with specific regions or diseases.

Matrix-Assisted Laser Desorption/Ionization (MALDI) is a powerful ionization technique used in mass spectrometry for the analysis of large biomolecules. It involves incorporating the analyte into a matrix material, which facilitates the desorption and ionization of the analyte upon laser irradiation. MALDI has revolutionized the analysis of proteins, peptides, and other biomolecules, providing valuable insights into their structure, function, and interactions. It has found extensive applications in various fields and continues to contribute to advancements in biochemistry and molecular biology research.

7.1.3 Tandem Mass Spectrometry

Tandem mass spectrometry (MS/MS) is a powerful technique used in analytical chemistry and proteomics to obtain detailed structural information about biomolecules. It involves the sequential analysis of ions, where a selected precursor ion is isolated and fragmented, and the resulting fragments are then analyzed to provide insights into the molecule's structure, composition, and interactions.

The process of tandem mass spectrometry typically involves three main steps: precursor ion selection, fragmentation, and product ion analysis.

First, a precursor ion of interest is selected from the mass spectrum based on its mass-to-charge ratio (m/z). This can be done using various mass analyzers, such as quadrupoles or ion traps, which allow the isolation of a specific ion from the mixture.

Once the precursor ion is isolated, it undergoes fragmentation to generate product ions. Different fragmentation techniques can be employed, including collision-induced dissociation (CID), electron capture dissociation (ECD), electron transfer dissociation (ETD), and higher-energy collisional dissociation (HCD). These techniques induce the breaking of chemical bonds within the precursor ion, resulting in the formation of fragments with specific m/z ratios.

Finally, the generated product ions are analyzed to obtain structural information. This analysis is typically performed using a mass analyzer, such as a quadrupole or a time-of-flight (TOF) analyzer, which measures the m/z ratios of the fragments. The resulting tandem mass spectrum provides information about the fragment ions and their relative abundance, allowing for the determination of the molecular sequence, post-translational modifications, and other structural features.

Tandem mass spectrometry has numerous applications in biochemistry and molecular biology. In proteomics, it is commonly used for peptide sequencing, enabling the identification and characterization of proteins and their modifications. By comparing the obtained tandem mass spectra with known peptide databases, proteins can be identified with high confidence. Tandem mass spectrometry also plays a vital role in the study of metabolomics, lipidomics, and the analysis of small organic molecules, allowing for the determination of their structures and elucidation of metabolic pathways.

Furthermore, tandem mass spectrometry can be combined with other techniques to enhance its capabilities. For example, liquid chromatography-tandem mass spectrometry (LC-MS/MS) combines the separation power of liquid chromatography with the structural information provided by MS/MS, enabling the identification and quantification of complex mixtures of biomolecules.

Tandem mass spectrometry is a versatile technique used in biochemistry and molecular biology to obtain detailed structural information about biomolecules. By sequentially isolating and fragmenting ions, tandem mass spectrometry allows for the determination of molecular sequence, modifications, and other structural features. Its applications range from protein identification and characterization to metabolite profiling, contributing significantly to our understanding of biological systems.

7.1.4 High-Resolution Mass Spectrometry (HRMS)

High-Resolution Mass Spectrometry (HRMS) is a powerful technique used in analytical chemistry and biochemistry for the precise determination of the mass-to-charge ratio (m/z) of ions. Unlike conventional mass spectrometry techniques, HRMS offers increased mass accuracy and resolution, allowing for the accurate measurement of molecular masses and the differentiation of closely related ions.

HRMS utilizes advanced mass analyzers, such as Fourier transform ion cyclotron resonance (FT-ICR) or Orbitrap analyzers, which provide high-resolution capabilities. These analyzers operate based on the principles of ion motion in a magnetic field or an electrostatic trap, allowing for the accurate determination of ion mass.

One of the main advantages of HRMS is its exceptional mass accuracy. By achieving mass accuracies within parts per million (ppm), HRMS can differentiate between ions that differ by a few atomic masses. This is particularly valuable in the analysis of complex mixtures where various ions may have similar masses but different compositions. High mass accuracy improves the reliability of compound identification and helps in reducing false-positive and false-negative results.

In addition to mass accuracy, HRMS offers high-resolution capabilities. Resolution refers to the ability to separate ions with closely spaced m/z ratios. With high resolution, HRMS can distinguish between ions that differ by a small mass increment, providing more detailed information about the sample. This is particularly useful in metabolomics, proteomics, and other omics studies where the analysis of complex mixtures requires the separation and identification of numerous compounds.

HRMS has numerous applications in biochemistry and molecular biology. In proteomics, HRMS is used for the identification and characterization of proteins, including the determination of post-translational modifications, protein variants, and protein-protein interactions. In metabolomics, HRMS enables the profiling and identification of small molecules, including metabolites and lipids, providing insights into metabolic pathways and disease biomarkers. HRMS is also used in drug discovery and development, environmental analysis, and forensic science.

The advancements in HRMS technology, including increased mass resolving power, improved mass accuracy, and enhanced sensitivity, have expanded its capabilities and made it an indispensable tool in modern research. Additionally, HRMS can be combined with other separation techniques such as liquid chromatography (LC) or gas chromatography (GC), further enhancing its analytical capabilities.

7.1.5 Quantitative Mass Spectrometry

Quantitative mass spectrometry is a technique used to determine the abundance or concentration of specific molecules in a sample. It provides a quantitative assessment of analytes based on their mass-to-charge ratio (m/z) and allows for the measurement of the relative or absolute amounts of target molecules in complex mixtures.

There are different approaches to quantitative mass spectrometry, each with its own advantages and limitations. Here are a few commonly used techniques:

1. Label-based quantification: This method involves introducing stable isotopic labels into the analyte molecules before or during sample preparation. Isotope labeling can be achieved using isotopically labeled reagents or through metabolic labeling techniques. The labeled and unlabeled forms of the analyte can then be distinguished based on their mass shift, allowing for the relative quantification of the target molecule.
2. Label-free quantification: Label-free quantification does not rely on isotopic labels but instead compares the intensity or abundance of analyte signals in different samples. This approach is often based on peak intensities or spectral counting and is useful for relative quantification when comparing multiple samples.
3. Selected reaction monitoring (SRM) or multiple reaction monitoring (MRM): SRM/MRM is a targeted quantification method used to measure specific analytes in a complex mixture.

It involves selecting a specific precursor ion and monitoring the fragmentation pattern of a predetermined product ion for each analyte. This technique offers high sensitivity and selectivity, allowing for the absolute quantification of target molecules.

4. Absolute quantification using stable isotope-labeled internal standards: This approach involves the addition of known amounts of stable isotope-labeled internal standards to the sample. The internal standards have the same chemical properties as the analyte but can be distinguished by their mass shift. By comparing the intensity or abundance of the analyte and the internal standard, the absolute concentration of the analyte can be determined.

Quantitative mass spectrometry has a wide range of applications in various fields, including proteomics, metabolomics, pharmacokinetics, and environmental analysis. It enables researchers to study changes in protein expression, identify biomarkers, measure drug concentrations, and assess the impact of environmental pollutants, among many other applications.

It is worth noting that quantitative mass spectrometry requires careful experimental design, data analysis, and the use of appropriate internal standards or reference materials to ensure accurate and reliable quantification. Factors such as sample preparation, instrument calibration, and data normalization techniques must be considered to achieve accurate and reproducible quantitative results.

7.1.6 Imaging Mass Spectrometry

Imaging mass spectrometry (IMS) is a powerful analytical technique that combines the spatial information of microscopy with the molecular information provided by mass spectrometry. It allows for the visualization and mapping of biomolecules directly within tissues or other complex samples, providing valuable insights into their distribution, localization, and molecular composition.

In IMS, a sample, such as a tissue section or a biological specimen, is analyzed by a mass spectrometer in a raster scanning mode. The technique can be performed using different mass spectrometry platforms, including matrix-assisted laser desorption/ionization (MALDI) and secondary ion mass spectrometry (SIMS), among others. The imaging process begins by preparing the sample, typically by applying a matrix compound in the case of MALDI-IMS. The matrix assists in desorbing and ionizing the molecules of interest when subjected to laser irradiation. For SIMS-IMS, a focused ion beam is used to sputter and ionize the sample's surface.

Once the sample is prepared, the mass spectrometer scans across the sample surface in a systematic manner, acquiring mass spectra at each position.

The acquired mass spectra contain information about the molecular ions present at each location, allowing for the generation of an ion intensity map that represents the spatial distribution of specific biomolecules within the sample.

IMS enables the visualization of a wide range of biomolecules, including proteins, lipids, metabolites, and drugs, among others.

By comparing the ion intensity maps obtained from different molecules, researchers can correlate the spatial distribution of various biomolecules and gain insights into their interactions and roles within biological systems.

The applications of IMS are diverse and span multiple fields. In biological and medical research, IMS has been used to study disease pathology, identify biomarkers, investigate drug distribution within tissues, and understand the spatial organization of cellular processes. It has found applications in oncology, neurobiology, pharmacology, and microbiology, among others.

IMS also has applications in fields such as environmental science, forensics, and materials science. It allows for the characterization and mapping of compounds within environmental samples, the analysis of trace evidence in forensic investigations, and the investigation of material surfaces and interfaces.

However, IMS has certain challenges and limitations. It can be time-consuming and requires careful sample preparation and optimization. The interpretation of IMS data can be complex due to the high dimensionality of the acquired datasets and the need for appropriate data analysis methods.

Imaging mass spectrometry is a powerful technique that combines the spatial information of microscopy with the molecular information provided by mass spectrometry. It allows for the visualization and mapping of biomolecules within tissues or other complex samples, providing valuable insights into their distribution and molecular composition. IMS has diverse applications in biological, medical, environmental, and forensic research, and continues to advance our understanding of complex systems at the molecular level.

7.2 ELECTROPHORETIC TECHNIQUES

Electrophoretic techniques are widely used in biochemistry and molecular biology to separate and analyze biomolecules based on their charge and size. These techniques involve the migration of charged molecules in an electric field, allowing for their separation and characterization. Here are some commonly employed electrophoretic techniques:

1. Agarose Gel Electrophoresis: Agarose gel electrophoresis is primarily used to separate and analyze nucleic acids, such as DNA and RNA. Agarose, a polysaccharide derived from seaweed, forms a gel matrix through which the nucleic acids migrate. Smaller nucleic acid fragments move faster and travel farther through the gel, resulting in separation based on size. The separated fragments can be visualized using DNA-binding dyes or by transferring them onto a membrane for further analysis.
2. Polyacrylamide Gel Electrophoresis (PAGE): Polyacrylamide gel electrophoresis is a higher resolution technique commonly used for separating proteins and nucleic acids. Polyacrylamide gels can be customized with different acrylamide concentrations to achieve the desired resolution. Proteins are typically denatured and treated with SDS (sodium dodecyl sulfate) to give them a uniform negative charge before separation based on size. The separated proteins can be visualized using protein stains or transferred to a membrane for Western blotting.
3. Capillary Electrophoresis (CE): Capillary electrophoresis is a high-resolution technique that employs narrow-bore capillaries filled with an electrolyte solution. Analytes are separated based on their charge-to-size ratio and migrate through the capillary under the influence of an electric field. CE offers excellent separation efficiency, short analysis times, and low sample consumption. It is used for a variety of applications, including DNA sequencing, protein analysis, and small molecule analysis.
4. Isoelectric Focusing (IEF): Isoelectric focusing separates proteins based on their isoelectric points (pI), which is the pH at which a protein has no net charge. The technique utilizes a pH gradient generated within a gel or capillary. Proteins migrate until they reach the pH region in the gel where their

net charge is zero, resulting in their separation based on pI. IEF is often used as the first dimension in two-dimensional gel electrophoresis (2D-PAGE) for comprehensive protein analysis.

5. SDS-PAGE: Sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) is a technique used to separate proteins based on their molecular weight. SDS, a detergent, denatures the proteins and imparts a negative charge to them. As a result, proteins migrate through the gel in proportion to their molecular weight, with smaller proteins moving faster and farther. SDS-PAGE is commonly used for protein separation in biochemical and molecular biology research and is often followed by immunoblotting or protein staining for visualization and further analysis.

These are just a few examples of electrophoretic techniques used in biochemistry and molecular biology. Each technique offers specific advantages and is suitable for different types of biomolecules and research applications. Electrophoretic techniques continue to play a crucial role in the separation, analysis, and characterization of biomolecules, contributing to our understanding of their structure and function.

7.2.1 Agarose Gel Electrophoresis

Agarose gel electrophoresis is a commonly used technique in biochemistry and molecular biology for separating and analyzing nucleic acids, such as DNA and RNA, based on their size. Agarose, a polysaccharide derived from seaweed, is used to create a gel matrix through which the nucleic acids can migrate when subjected to an electric field.

The process of agarose gel electrophoresis involves several steps. First, agarose powder is dissolved in a buffer solution and heated to create a molten gel mixture. The gel mixture is then poured into a gel tray or a horizontal gel apparatus, and a comb is inserted to create wells for loading the samples.

Once the gel solidifies, the comb is removed, and the gel is placed in an electrophoresis chamber filled with a running buffer. The samples, which usually contain DNA or RNA fragments of different sizes, are mixed with a loading dye that contains a tracking dye for visualization during electrophoresis. The samples are then carefully loaded into the wells using a micropipette.

When an electric current is applied, the negatively charged nucleic acid molecules migrate through the gel toward the positive electrode. Smaller molecules move more easily through the gel matrix and travel faster, while larger molecules encounter more resistance and migrate slower.

After the electrophoresis run, the gel is stained with a DNA-specific dye, such as ethidium bromide or a fluorescent dye, which binds to the nucleic acids and allows for their visualization under UV light. The DNA bands appear as distinct bands or smears on the gel, with smaller fragments located near the bottom and larger fragments near the top.

Agarose gel electrophoresis is used for a variety of purposes, including DNA fragment analysis, DNA purification, and analysis of PCR products. It is commonly employed in molecular biology research, genetic analysis, and diagnostics. The technique allows researchers to determine the size distribution of DNA or RNA fragments, assess the success of DNA amplification reactions, and analyze the results of genetic engineering experiments.

Overall, agarose gel electrophoresis is a versatile and widely used technique that provides valuable information about the size and integrity of nucleic acid molecules. It is a fundamental tool in the field of molecular biology and has contributed significantly to our understanding of genetic material and the development of various genetic engineering techniques.

7.2.2 Polyacrylamide Gel Electrophoresis (PAGE)

Polyacrylamide gel electrophoresis (PAGE) is a widely used technique in biochemistry and molecular biology for separating and analyzing proteins and nucleic acids based on their size and charge. Unlike agarose gel electrophoresis, which is mainly used for larger nucleic acids, PAGE provides higher resolution and is suitable for the separation of smaller molecules, such as proteins and short nucleic acids. The principle of PAGE is similar to agarose gel electrophoresis, but the gel matrix is made of polyacrylamide, a synthetic polymer. Polyacrylamide gels can be prepared with different concentrations, which can be adjusted to achieve the desired resolution for the specific application.

To perform PAGE, a mixture of acrylamide, bisacrylamide (a cross-linking agent), and a buffer system is polymerized using a free-radical initiator. The polymerization is initiated by adding a catalyst, such as ammonium persulfate (APS), and a chemical called N,N,N′,N′-tetramethylethylenediamine (TEMED). The resulting polyacrylamide gel forms a network of pores that act as a sieving matrix.

For protein separation, the samples are usually denatured and treated with a detergent, such as sodium dodecyl sulfate (SDS), which binds to the proteins and imparts a negative charge. SDS-PAGE, the most common form of PAGE, allows for the separation of proteins based on their molecular weight. The SDS-bound proteins are loaded into wells created in the gel, and an electric current is applied. The negatively charged proteins migrate through the gel matrix towards the positively charged electrode, with smaller proteins moving faster and larger proteins moving more slowly.

Nucleic acid separation by PAGE is often performed under native or denaturing conditions. In native PAGE, the nucleic acids retain their natural structure, and separation is based on size and charge. Denaturing PAGE involves the use of chemical denaturants, such as urea or formamide, to disrupt the secondary structure of nucleic acids, allowing for separation based on size alone.

After electrophoresis, the separated molecules can be visualized by staining with specific dyes or by using labeled probes or antibodies for detection. The bands or spots representing the separated molecules can be quantified, allowing for the analysis of relative abundance, molecular weight, and other characteristics.

PAGE has a wide range of applications in biochemistry and molecular biology. It is used for protein analysis, including protein purification, protein characterization, and protein-protein interaction studies. PAGE is also commonly employed for nucleic acid analysis, such as DNA sequencing, DNA fragment analysis, and the analysis of RNA transcripts.

In summary, polyacrylamide gel electrophoresis (PAGE) is a versatile and widely used technique for separating and analyzing proteins and nucleic acids based on their size and charge. It provides higher resolution than agarose gel electrophoresis and is applicable to a wide range of biological molecules. PAGE has been instrumental in advancing our understanding of biomolecules and has contributed significantly to various fields of research, including molecular biology, genetics, and protein biochemistry.

7.2.3 Capillary Electrophoresis (CE)

Capillary electrophoresis (CE) is a powerful analytical technique used in biochemistry and molecular biology for the separation and analysis of biomolecules, including nucleic acids, proteins, peptides, carbohydrates, and small molecules. It offers high separation efficiency, short analysis times, and low sample consumption compared to traditional gel electrophoresis methods.

In CE, the separation occurs in a narrow-bore capillary filled with an electrolyte solution. The capillary is made of fused silica, a material that provides high resolution and minimizes interactions with the analytes. The capillary is typically several tens of centimeters long and has an inner diameter of 10-100 micrometers.

The separation in CE is based on the differential migration of charged analytes in an electric field. When the electric field is applied, the analytes migrate through the capillary based on their charge-to-size ratio. The separation can be achieved by differences in charge, size, or both.

There are several modes of capillary electrophoresis, including:

1. Capillary Zone Electrophoresis (CZE): CZE is the most common mode of CE and is used for the separation of charged molecules, such as ions, peptides, and small proteins. The separation is based on the differential migration of analytes according to their charge and size.
2. Capillary Gel Electrophoresis (CGE): CGE involves the incorporation of a sieving matrix, typically a polymer solution, within the capillary. The sieving matrix helps to separate analytes based on their size. CGE is useful for the separation of large nucleic acids, proteins, and complex mixtures.
3. Capillary Electrochromatography (CEC): CEC combines the principles of electrophoresis and liquid chromatography. It involves the use of a stationary phase within the capillary to enhance the separation based on both charge and size.

CE offers several advantages over other separation techniques. It allows for faster separations, typically within minutes, due to the high electric field strength and the small dimensions of the capillary. CE also requires small sample volumes, making it suitable for analysis of limited sample quantities or precious samples. Furthermore, CE provides high resolution and sensitivity, allowing for the separation and detection of closely related analytes.

CE is commonly coupled with various detection methods, including UV absorbance, fluorescence, mass spectrometry, and electrochemical detection, for the identification and quantification of the separated analytes. The detection sensitivity can be enhanced through the use of derivatization techniques or the incorporation of specific probes or labels.

In biochemistry and molecular biology, CE is widely used for applications such as DNA sequencing, fragment analysis, protein analysis, drug analysis, and environmental analysis. It has significantly contributed to the understanding of biological molecules and has found applications in research, clinical diagnostics, and quality control in industries.

Overall, capillary electrophoresis is a versatile and powerful technique that offers high-resolution separation and analysis of a wide range of biomolecules. Its speed, sensitivity, and low sample consumption make it a valuable tool in various areas of biochemistry and molecular biology research.

7.2.4 Isoelectric Focusing (IEF)

Isoelectric focusing (IEF) is a powerful technique used in biochemistry and molecular biology to separate and analyze proteins based on their isoelectric points (pI). The pI is the pH at which a protein has no net charge, and it is determined by the amino acid composition and sequence of the protein.

In IEF, a pH gradient is created within a gel matrix, typically using a mixture of buffering compounds with different pKa values. The gel can be made of polyacrylamide or other materials suitable for electrophoresis. The pH gradient is established by applying an electric field across the gel, with a positive electrode at one end and a negative electrode at the other.

When proteins are loaded onto the gel, they migrate through the pH gradient until they reach a region where the pH matches their pI. At this point, the proteins become electrically neutral and cease to migrate. This creates a series of bands or spots along the gel, with each band representing proteins with different pI values.

The separation in IEF is based on the principle of electrophoresis, where charged molecules migrate in response to an electric field. However, unlike other electrophoretic techniques, IEF takes advantage of the fact that proteins have different isoelectric points and will migrate to their specific pH regions in the gel.

IEF can be performed in different formats, including tube gels, slab gels, or capillaries, depending on the specific application and desired resolution. The technique is often combined with other methods, such as sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE), to separate and analyze proteins based on both their pI and molecular weight.

After IEF separation, the proteins can be visualized by staining with specific dyes or using immunoblotting techniques. Additionally, the separated proteins can be further analyzed by cutting out specific bands or spots from the gel for subsequent protein identification using mass spectrometry or other techniques.

IEF has a wide range of applications in biochemistry and molecular biology. It is commonly used for protein characterization, determination of protein pI values, protein purification, and identification of post-translational modifications. It is also valuable in studying protein variants, such as isoforms or genetic variants, and in analyzing complex protein mixtures.

Overall, isoelectric focusing (IEF) is a versatile and powerful technique for protein separation based on their isoelectric points. It provides high resolution and specificity, allowing for the separation and analysis of proteins with different pI values. IEF has contributed significantly to our understanding of protein structure, function, and diversity, and it remains a valuable tool in various areas of research, including proteomics, biotechnology, and clinical diagnostics.

7.2.5 Protein Blotting Techniques (Western Blotting)

Protein blotting techniques, also known as Western blotting, are widely used in biochemistry and molecular biology to detect and analyze specific proteins within complex biological samples. Western blotting allows for the identification of target proteins based on their size, abundance, and post-translational modifications.

The Western blotting technique involves several key steps. First, proteins within a sample, such as cell lysates or tissue extracts, are separated using gel electrophoresis, typically sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). The proteins are then transferred from the gel onto a solid membrane, usually nitrocellulose or polyvinylidene difluoride (PVDF).

After transfer, the membrane is blocked with a protein-based solution, such as bovine serum albumin (BSA) or non-fat dry milk, to prevent non-specific binding of antibodies. The membrane is then incubated with a primary antibody specific to the target protein of interest. The primary antibody recognizes and binds to the target protein, forming an antigen-antibody complex.

Next, the membrane is washed to remove any unbound primary antibody. This is followed by the addition of a secondary antibody, which is linked to a detection marker, such as an enzyme or fluorescent dye. The secondary antibody binds to the primary antibody-antigen complex, enabling the detection of the target protein.

Finally, the presence of the target protein is visualized using various methods, depending on the detection marker. Common methods include enzyme-linked chemiluminescence (ECL) for enzyme detection or fluorescence imaging for fluorescent detection. The signal generated is captured using specialized imaging systems or X-ray film.

Western blotting provides several advantages for protein analysis. It allows for the detection of specific proteins within a complex mixture, providing information on protein expression levels, protein size, and potential modifications. It also enables the assessment of protein interactions, protein degradation, and changes in protein expression under different conditions or treatments.

Western blotting has a wide range of applications in research, diagnostics, and biotechnology. It is commonly used to study protein expression in cell and molecular biology, investigate disease markers, validate antibody specificity, and assess protein-protein interactions. It is a valuable tool in fields such as cancer research, immunology, and drug development.

Protein blotting techniques, particularly Western blotting, are essential methods in biochemistry and molecular biology for the detection and analysis of specific proteins. They provide valuable information on protein expression, size, and modifications within complex biological samples. Western blotting has played a crucial role in advancing our understanding of protein biology and continues to be an indispensable tool in various areas of scientific research and clinical diagnostics.

7.2.6 DNA Gel Electrophoresis

DNA gel electrophoresis is a fundamental technique used in molecular biology to separate and analyze DNA molecules based on their size. It is a widely employed method for various applications, including DNA fragment analysis, DNA sequencing, and DNA purification.

The process of DNA gel electrophoresis involves several key steps. First, a gel matrix is prepared, typically using agarose or polyacrylamide. Agarose gels are commonly used for separating large DNA fragments, while polyacrylamide gels offer higher resolution for separating smaller DNA fragments. The gel is cast in a horizontal or vertical electrophoresis apparatus, forming wells or slots for loading the DNA samples. The

DNA samples are mixed with a loading buffer containing a tracking dye, which helps to monitor the progress of the electrophoresis.

Next, the DNA samples are loaded into the wells of the gel, and an electric current is applied across the gel. The negatively charged DNA molecules migrate towards the positively charged electrode. Smaller DNA fragments move more quickly through the gel matrix, while larger fragments move more slowly.

After a suitable period of time, the electrophoresis is stopped, and the DNA fragments are visualized. This can be achieved by staining the DNA with fluorescent dyes, such as ethidium bromide or SYBR Green, which intercalate into the DNA and emit fluorescence when exposed to UV light. Alternatively, fluorescently labeled DNA probes or specific antibodies can be used to detect target DNA sequences.

The separated DNA fragments appear as distinct bands or smears on the gel, with the size of the fragments inversely correlated with their migration distance. By comparing the DNA fragment sizes to a DNA ladder, which consists of DNA fragments of known sizes, the size of the unknown DNA fragments can be estimated.

DNA gel electrophoresis is a versatile technique used for various applications. It is commonly employed in molecular biology research to analyze DNA fragments generated by restriction enzyme digestion, polymerase chain reaction (PCR), or DNA sequencing reactions. It is also used in forensic science for DNA profiling and in diagnostic laboratories for genetic testing.

In recent years, advancements in DNA gel electrophoresis techniques have led to the development of capillary electrophoresis-based systems, such as capillary gel electrophoresis (CGE) and capillary array electrophoresis (CAE). These techniques offer higher resolution, increased throughput, and automated analysis capabilities.

7.2.7 RNA Gel Electrophoresis

RNA gel electrophoresis is a commonly used technique in molecular biology for the separation and analysis of RNA molecules based on their size and/or secondary structure. It is a fundamental method that allows researchers to study RNA expression, identify RNA species, and analyze RNA integrity.

The process of RNA gel electrophoresis is similar to DNA gel electrophoresis, with a few modifications. First, an agarose or polyacrylamide gel is prepared, typically using denaturing conditions to ensure the separation of RNA molecules based on their size rather than their secondary structure. Polyacrylamide gels are commonly used for RNA electrophoresis due to their higher resolution.

Before loading the RNA samples onto the gel, they are often treated with denaturing agents, such as heat or chemical agents like formaldehyde or urea, to disrupt secondary structures and ensure that the RNA molecules adopt a single-stranded conformation. Additionally, a denaturing loading buffer containing intercalating dyes, such as ethidium bromide or SYBR Green, may be added to the RNA samples.

The RNA samples are loaded into wells in the gel, and an electric current is applied across the gel. The negatively charged RNA molecules migrate through the gel towards the positively charged electrode. Smaller RNA molecules move more quickly through the gel matrix, while larger RNA molecules move more slowly.

After electrophoresis, the RNA molecules are visualized by staining the gel with dyes that bind specifically to RNA, such as ethidium bromide or SYBR Green. These dyes fluoresce when exposed to UV light, allowing the visualization of RNA bands.

In addition to size separation, RNA gel electrophoresis can also be performed under native conditions to preserve RNA secondary structures. This is particularly useful for studying RNA folding, RNA-protein interactions, or noncoding RNA molecules that rely on their secondary structure for function.

RNA gel electrophoresis is a versatile technique used in various molecular biology applications. It is commonly employed for the analysis of total RNA extracted from cells or tissues, the separation and purification of specific RNA species, and the analysis of RNA integrity. It is an essential tool in gene expression studies, RNA sequencing, and functional analysis of noncoding RNAs.

Advancements in technology have also led to the development of capillary electrophoresis-based systems for RNA analysis, offering higher resolution and increased sensitivity.

RNA gel electrophoresis is a fundamental technique in molecular biology used for the separation and analysis of RNA molecules based on their size or secondary structure. It provides valuable information about RNA expression, integrity, and interactions, and it is widely employed in gene expression studies, RNA characterization, and functional analysis of noncoding RNAs.

7.3 CHROMATOGRAPHIC TECHNIQUES

Chromatographic techniques are widely used in biochemistry and molecular biology to separate and analyze complex mixtures of molecules based on their physical and chemical properties. These techniques are based on the principle of differential migration of components in a mixture between a stationary phase and a mobile phase.

There are several types of chromatographic techniques commonly employed in biochemistry and molecular biology, including:

1. Liquid Chromatography (LC): In liquid chromatography, the stationary phase is typically a solid material or a liquid immobilized on a solid support, while the mobile phase is a liquid solvent. LC can be further classified into different modes, such as reversed-phase chromatography (RP-HPLC) and size exclusion chromatography (SEC), depending on the nature of the stationary phase and the separation mechanism.
2. Gas Chromatography (GC): Gas chromatography involves the separation of volatile compounds based on their partitioning between a stationary phase (often a liquid coated on a solid support) and a mobile phase (an inert gas). GC is commonly used for the analysis of volatile organic compounds and small molecules.
3. Ion Exchange Chromatography (IEC): Ion exchange chromatography separates molecules based on their charge. The stationary phase consists of an ion exchange resin that selectively binds ions of opposite charge to the resin. IEC is often used for the separation of proteins, nucleic acids, and other charged molecules.
4. Affinity Chromatography: Affinity chromatography utilizes the specific interaction between a target molecule and a ligand immobilized on the stationary phase. This technique allows for highly selective purification and isolation of biomolecules, such as proteins, based on their affinity for a specific ligand or antibody.
5. Thin-Layer Chromatography (TLC): Thin-layer chromatography involves the separation of compounds on a thin layer of an adsorbent material, typically coated on a glass plate or a plastic sheet. The separation is achieved by the differential migration of compounds in a solvent system. TLC is often used for rapid qualitative analysis and can be a preliminary step for other chromatographic techniques.

6. High-Performance Liquid Chromatography (HPLC): HPLC is a powerful variant of liquid chromatography that utilizes high-pressure pumps to improve separation efficiency and speed. It is widely used for the separation and quantification of complex mixtures, such as protein or peptide mixtures, in research and analytical laboratories.

Chromatographic techniques are versatile and widely applied in various fields, including pharmaceutical analysis, protein purification, environmental analysis, and metabolomics. They offer excellent resolution, sensitivity, and selectivity for separating and analyzing complex mixtures of molecules. By using different types of stationary phases and adjusting the mobile phase composition, chromatographic techniques can be optimized for specific applications, providing valuable insights into the composition and properties of biological samples.

7.3.1 Reversed-Phase Chromatography

Reversed-phase chromatography (RPC) is a widely used chromatographic technique in biochemistry and molecular biology for the separation and analysis of biomolecules, particularly hydrophobic compounds such as proteins, peptides, nucleic acids, and small organic molecules. It operates on the principle of differential partitioning of analytes between a nonpolar stationary phase and a polar mobile phase.

In reversed-phase chromatography, the stationary phase is typically a nonpolar hydrocarbon chain, such as octadecylsilane (C18) or octyldecylsilane (C8), bonded to a solid support. The nonpolar stationary phase promotes hydrophobic interactions with the analytes. On the other hand, the mobile phase is usually an aqueous solvent containing an organic modifier, such as acetonitrile or methanol. The addition of the organic modifier reduces the polarity of the mobile phase.

During the chromatographic process, the sample mixture is introduced onto the column, and the nonpolar analytes interact with the nonpolar stationary phase, leading to their retention. The more hydrophobic the analyte, the longer it remains in the stationary phase. Meanwhile, more polar components elute faster as they have weaker interactions with the nonpolar stationary phase.

Reversed-phase chromatography is often performed using high-performance liquid chromatography (HPLC) systems, which allow for precise control of solvent composition, flow rate, and temperature. The separation is typically monitored using a UV-visible detector, which measures the absorbance of the eluting compounds at specific wavelengths. The advantages of reversed-phase chromatography include its compatibility with a wide range of analytes, high resolution, and versatility in mobile phase composition. It is particularly useful for the separation and purification of hydrophobic compounds, such as lipids, pharmaceutical drugs, and natural products. Reversed-phase chromatography is extensively utilized in drug discovery, proteomics, metabolomics, and quality control in the pharmaceutical and biotechnology industries.

To optimize the separation in reversed-phase chromatography, factors such as the composition of the mobile phase, column temperature, and flow rate can be adjusted. Gradient elution, where the composition of the mobile phase is gradually changed during the separation, is commonly employed to enhance resolution and separation efficiency.

Reversed-phase chromatography is a powerful technique in biochemistry and molecular biology for the separation and analysis of hydrophobic biomolecules. It offers excellent resolution, high sample capacity, and compatibility with various detection methods. Reversed-phase chromatography has become an indispensable tool in many research areas, aiding in the characterization, purification, and quantification of diverse analytes in complex mixtures.

7.3.2 Size Exclusion Chromatography

Size exclusion chromatography (SEC), also known as gel filtration chromatography or gel permeation chromatography, is a widely used chromatographic technique for the separation and analysis of biomolecules based on their size and molecular weight. It is particularly useful for the separation of macromolecules such as proteins, nucleic acids, polysaccharides, and synthetic polymers.

In size exclusion chromatography, a porous gel matrix is used as the stationary phase. The gel matrix consists of spherical beads made of a cross-linked polymer, such as agarose or polyacrylamide. These beads contain pores of various sizes that allow the passage of mobile phase and analyte molecules.

The separation principle in size exclusion chromatography is based on the differential exclusion or inclusion of analyte molecules in the pores of the gel matrix. Larger molecules cannot enter the smaller pores and therefore elute quickly, while smaller molecules penetrate the pores and spend more time within the gel matrix, resulting in a longer elution time. Thus, the elution order is inverse to the size of the analyte, with larger molecules eluting first and smaller molecules eluting later.

To perform size exclusion chromatography, a sample containing a mixture of molecules of different sizes is injected into the chromatographic column. The mobile phase, usually a buffer or an appropriate solvent, is then passed through the column at a constant flow rate. As the sample components travel through the column, they encounter the gel matrix and undergo differential partitioning based on their size. Larger molecules bypass the pores and pass through the column more quickly, while smaller molecules penetrate the pores and experience a longer retention time.

Detection in size exclusion chromatography can be performed using various techniques, such as UV absorption, refractive index detection, fluorescence, or multi-angle light scattering (MALS). These detection methods provide information about the elution profile and concentration of the analytes.

The advantages of size exclusion chromatography include its simplicity, versatility, and the ability to work under physiological conditions. It does not require prior knowledge of analyte properties and can be used for both analytical and preparative purposes. Size exclusion chromatography is commonly employed in protein purification, protein structure analysis, determination of molecular weight, and the assessment of sample purity and aggregation state.

Optimization of size exclusion chromatography involves selecting the appropriate gel matrix, column size, and mobile phase conditions to achieve the desired separation and resolution. Different gel matrices with varying pore sizes are available, allowing for the customization of the separation range.

Size exclusion chromatography is a valuable technique in biochemistry and molecular biology for the separation and analysis of biomolecules based on their size and molecular weight. It provides information about the relative size distribution of analytes in a mixture and is widely used for protein purification, determination of molecular weight, and assessing sample integrity. Size exclusion chromatography is a versatile and reliable method that contributes to the characterization and understanding of biomolecular interactions and structures.

Size exclusion chromatography (SEC) is considered as the most common, efficient, and fastest method that provides information about the polymer's MW distribution. This method separates the molecules in the sample according to their hydrodynamic volumes instead of their MW. In the SEC system, a stationary phase that contains a porous packing material is placed in a column and the mobile phase that contains the sample solution is injected into the stationary phase. The molecules that are larger than the pore volume will be eluted first due to their inability to penetrate the pores. Whereas the smaller molecules will take a longer time to be eluted because they can penetrate into the pores (Fig. 1). However, molecules that can freely diffuse through the pores (in and out) cannot be separated as they will be eluted at the total permeation point.

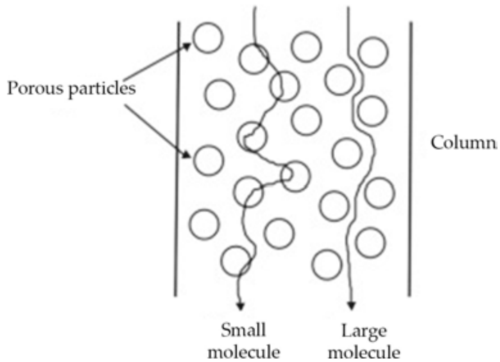


Figure 1. Illustration of the separation process by size exclusion chromatography.

The main limitation of the SEC method is that it is poorly reproducible in terms of molar mass analysis due to the difficulty to obtain a pure size-exclusion separation for various factors such as concentration effects, osmotic effects, side processes, secondary retention mechanisms, secondary exclusion, SEC band broadening, parasitic processes, and preferential interactions.

7.3.3 Ion Exchange Chromatography

Ion exchange chromatography (IEC) is a widely used chromatographic technique that separates molecules based on their charge properties. It is particularly effective for the separation and purification of charged molecules, such as proteins, nucleic acids, peptides, and ions.

In ion exchange chromatography, the stationary phase consists of a solid support material, usually a resin or gel, containing charged functional groups. These functional groups can be either positively charged (cation exchange chromatography) or negatively charged (anion exchange chromatography). The choice of the ion exchange resin depends on the desired separation and the charge characteristics of the target molecules.

The separation process in ion exchange chromatography occurs as the sample mixture is passed through the ion exchange column. The charged molecules in the sample interact with the oppositely charged functional groups on the stationary phase. Positively charged molecules bind to negatively charged functional groups (in anion exchange chromatography), while negatively charged molecules bind to positively charged functional groups (in cation exchange chromatography). The strength of the interaction between the charged molecules and the stationary phase depends on the ionic strength and pH of the mobile phase.

To elute the bound molecules, a change in the ionic strength or pH of the mobile phase is typically required. This can be achieved by gradually increasing the salt concentration or adjusting the pH to disrupt the ionic interactions between the sample molecules and the stationary phase. Elution is carried out in a step-wise or gradient manner, allowing for the separation and collection of the bound molecules based on their differential affinity for the stationary phase.

Detection in ion exchange chromatography can be achieved using various methods, such as UV-visible absorption, conductivity, or fluorescence, depending on the nature of the analyte and the specific detection requirements.

Ion exchange chromatography offers several advantages, including high selectivity, versatility, and the ability to operate under a wide range of pH and salt conditions. It can be used for both analytical and preparative purposes, allowing for the separation, purification, and analysis of charged molecules in complex mixtures. Furthermore, it can be easily integrated with other chromatographic techniques for multidimensional separations.

Optimization of ion exchange chromatography involves selecting the appropriate ion exchange resin, pH, salt concentration, and elution conditions to achieve the desired separation and resolution. The choice of mobile phase composition and column dimensions also plays a critical role in the efficiency and selectivity of the separation.

7.3.4 Hydrophobic Interaction Chromatography

Hydrophobic interaction chromatography (HIC) is a chromatographic technique commonly used for the separation and purification of proteins, peptides, and other biomolecules based on their hydrophobic properties. It operates on the principle of differential affinity between hydrophobic regions of the analytes and a hydrophobic stationary phase. In hydrophobic interaction chromatography, the stationary phase typically consists of a hydrophobic resin or gel, such as Sepharose or Sephadex, derivatized with alkyl chains, such as butyl or phenyl groups. The hydrophobic stationary phase promotes the binding of hydrophobic regions on the surface of the analyte molecules, while hydrophilic regions are solvated by the aqueous mobile phase.

The separation process in HIC occurs as the sample mixture is applied to the column containing the hydrophobic stationary phase. Initially, the analyte molecules bind to the hydrophobic surface through hydrophobic interactions. As the hydrophobic interaction is weaker than covalent or ionic interactions, the binding can be easily reversed or modulated by adjusting the mobile phase conditions.

Elution in HIC is typically achieved by applying a decreasing gradient of a water-miscible organic solvent, such as ethanol, isopropanol, or acetonitrile. The increase in the organic solvent concentration reduces the hydrophobic interactions between the analyte and the stationary phase, leading to the release and elution of the bound molecules. The elution order is generally determined by the strength of hydrophobic interactions, with more hydrophobic molecules requiring a higher concentration of organic solvent for elution.

Detection in hydrophobic interaction chromatography can be performed using various techniques, including UV-visible absorption, refractive index detection, or fluorescence. These detection methods provide information about the elution profile and concentration of the analytes.

Hydrophobic interaction chromatography offers several advantages in biochemistry and molecular biology. It allows for the separation of biomolecules based on their hydrophobic characteristics, which is particularly useful for proteins and peptides with similar charges or sizes. It is also a gentle separation technique that preserves the native structure and activity of the analytes.

Optimization of hydrophobic interaction chromatography involves selecting the appropriate hydrophobic stationary phase, mobile phase composition, gradient conditions, and column dimensions to achieve the desired separation and resolution. The hydrophobicity of the stationary phase can be adjusted by modifying the alkyl chain length or adding specific functional groups.

7.3.5 Gas Chromatography

Gas chromatography (GC) is a widely used analytical technique that separates and analyzes volatile and semi-volatile compounds in a sample mixture. It is based on the principles of partitioning and differential migration of analytes between a stationary phase and a mobile phase.

In gas chromatography, the stationary phase is typically a high-boiling liquid or solid immobilized on a solid support, known as the packed column, or a thin film coated on the inner surface of a capillary column, known as the capillary column.

The choice of stationary phase depends on the properties of the analytes and the specific separation requirements.

The mobile phase in gas chromatography is an inert gas, such as helium or nitrogen, which carries the sample vapor through the column. The sample mixture is injected into the GC system, and as it passes through the column, the different components interact with the stationary phase based on their affinity. Analytes with stronger interactions with the stationary phase spend more time in the column and elute later, while those with weaker interactions elute earlier.

Detection in gas chromatography is typically achieved by a detector located at the end of the column. Common detectors include flame ionization detector (FID), thermal conductivity detector (TCD), electron capture detector (ECD), and mass spectrometry (MS) detector. These detectors measure different properties of the analytes, such as ionization, heat conductivity, electron capture, or mass-to-charge ratio, to generate signals that are used to identify and quantify the separated components.

Gas chromatography has numerous applications in various fields, including environmental analysis, pharmaceuticals, forensics, food and beverage testing, and petrochemical analysis. It is particularly suitable for the analysis of volatile organic compounds (VOCs), such as hydrocarbons, alcohols, esters, and pesticides. GC allows for high-resolution separations, excellent sensitivity, and rapid analysis times.

Optimization of gas chromatography involves selecting the appropriate stationary phase, column temperature, flow rate, and injection technique to achieve optimal separation and resolution of the analytes. The choice of detector depends on the specific requirements of the analysis, such as sensitivity, selectivity, and the type of compounds to be detected.

How does gas chromatography work?

As the name implies, GC uses a carrier gas in the separation, this plays the part of the mobile phase (Figure 2(1)). The carrier gas transports the sample molecules through the GC system, ideally without reacting with the sample or damaging the instrument components.

- The sample is first introduced into the gas chromatograph (GC), either with a syringe or transferred from an autosampler (Figure 2 (2)) that may also extract the chemical components from solid or liquid sample matrices. The sample is injected into the GC inlet (Figure 2 (3)) through a septum which enables the injection of the sample mixture without losing the mobile phase.
- Connected to the inlet is the analytical column (Figure 2 (4)), a long (10 – 150 m), narrow (0.1 – 0.53 mm internal diameter) fused silica or metal tube which contains the stationary phase coated on the inside walls.
- The analytical column is held in the column oven which is heated during the analysis to elute the less volatile components.
- The outlet of the column is inserted into the detector (Figure 2 (5)) which responds to the chemical components eluting from the column to produce a signal.
- The signal is recorded by the acquisition software on a computer to produce a chromatogram (Figure 2 (6)).

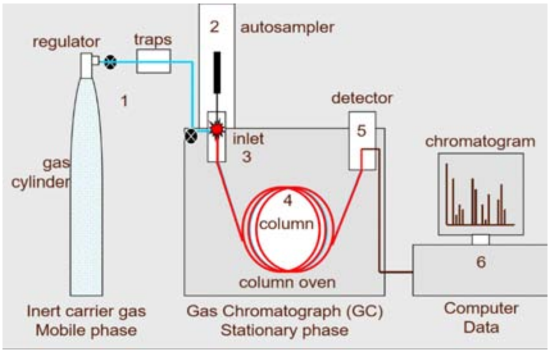


Figure 2. A simplified diagram of a gas chromatograph showing: (1) carrier gas, (2) autosampler, (3) inlet, (4) analytical column, (5) detector and (6) PC. Credit: Anthias Consulting.

After injection into the GC inlet, the chemical components of the sample mixture are first vaporized, if they aren't already in the gas phase. For low concentration samples the whole vapour cloud is transferred into the analytical column by the carrier gas in what is known as splitless mode. For high concentration samples only a portion of the sample is transferred to the analytical column in split mode, the remainder is flushed from the system through the split line to prevent overloading of the analytical column.

Once in the analytical column, the sample components are separated by their different interactions with the stationary phase. Therefore, when selecting the type of column to use, the volatility and functional groups of the analytes should be considered to match them to the stationary phase. Liquid stationary phases mainly fall into two types: polyethylene glycol (PEG) or polydimethylsiloxane (PDMS) based, the latter with varying percentages of dimethyl, diphenyl or mid-polar functional groups, for example cyanopropylphenyl. Like separates like, therefore non-polar columns with dimethyl or a low percentage of diphenyl are good for separating non-polar analytes. Those molecules capable of π-π interactions can be separated on stationary phases containing phenyl groups. Those capable of hydrogen bonding, for example acids and alcohols, are best separated with PEG columns, unless they have undergone derivatization to make them less polar.

The final step is the detection of the analyte molecules when they elute from the column. There are many types of GC detectors, for example: those that respond to C-H bonds like the flame ionization detector (FID); those that respond to specific elements for example sulfur, nitrogen or phosphorus; and those that respond to specific properties of the molecule, like the ability to capture an electron, as is used with the electron capture detector (ECD).

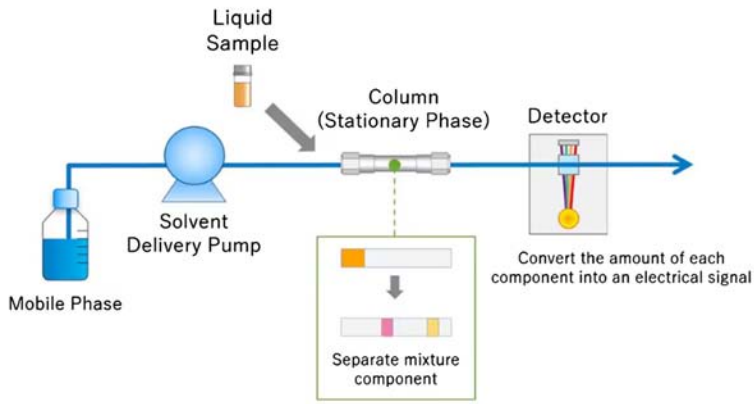
7.3.6 High-Performance Liquid Chromatography (HPLC)

High-performance liquid chromatography (HPLC) is a powerful analytical technique used for the separation, identification, and quantification of compounds in a sample mixture. It is widely employed in various fields, including pharmaceuticals, food and beverage analysis, environmental monitoring, forensics, and biochemistry.

In HPLC, the separation is based on the interaction between the sample components and the stationary phase, which is typically a packed column filled with porous particles coated with a specific material. The stationary phase can be categorized into different types, such as reversed-phase, normal-phase, ion-exchange, size exclusion, and affinity chromatography, depending on the nature of the sample and the desired separation mechanism.

The separation process in HPLC involves the continuous flow of a liquid mobile phase (also known as the eluent or solvent) through the column under high pressure. The sample mixture is injected into the mobile phase, and as it passes through the column, the components interact with the stationary phase to different extents. Compounds with stronger interactions with the stationary phase will be retained longer in the column, resulting in a slower elution time, while those with weaker interactions will elute faster.

Detection in HPLC is typically performed using various types of detectors, including ultraviolet-visible (UV-Vis) absorbance, fluorescence, refractive index (RI), electrochemical (EC), and mass spectrometry (MS) detectors. These detectors measure different properties of the analytes to generate signals that are used for identification and quantification.



HPLC offers several advantages over other chromatographic techniques, including high resolution, sensitivity, and versatility. It can handle a wide range of sample types, from small molecules to large biomolecules, and provides excellent separation efficiency. HPLC also allows for the analysis of complex mixtures and the determination of compound concentrations.

Optimization of HPLC involves selecting the appropriate stationary phase, mobile phase composition, flow rate, column temperature, and detection method to achieve the desired separation and resolution. The choice of parameters depends on the properties of the analytes, such as their polarity, molecular weight, and stability.

7.4 SPECTROSCOPIC TECHNIQUES

Spectroscopic techniques are powerful tools used in the field of science to study the interaction of light with matter. These techniques provide valuable information about the composition, structure, and behavior of molecules, helping scientists understand the properties and processes occurring within various materials.

There are several types of spectroscopic techniques commonly employed in scientific research:

1. Ultraviolet-Visible Spectroscopy (UV-Vis): UV-Vis spectroscopy involves the measurement of the absorption or transmission of ultraviolet and visible light by a sample. It provides information about the electronic transitions within molecules, allowing scientists to determine the presence of certain compounds, quantify their concentrations, and investigate the nature of their chemical bonding.
2. Infrared Spectroscopy (IR): IR spectroscopy measures the absorption or reflection of infrared light by molecules. It provides information about the vibrational and rotational modes of molecules, enabling scientists to identify functional groups, analyze chemical bonds, and determine the structure of compounds.
3. Nuclear Magnetic Resonance (NMR) Spectroscopy: NMR spectroscopy exploits the magnetic properties of atomic nuclei, particularly hydrogen and carbon nuclei. By subjecting a sample to a magnetic field and applying radiofrequency pulses, NMR spectroscopy provides detailed information about the molecular structure, dynamics, and interactions. It is widely used for compound identification, structural elucidation, and studying molecular processes in solution.
4. Mass Spectrometry (MS): Mass spectrometry measures the mass-to-charge ratio of ions generated from molecules. It enables the determination of molecular weights, identification of compounds, analysis of molecular structures, and characterization of chemical and isotopic compositions. Mass spectrometry is widely used in proteomics, metabolomics, lipidomics, and drug discovery.
5. Fluorescence Spectroscopy: Fluorescence spectroscopy measures the emission of light by molecules following their excitation with light of a specific wavelength. It provides information about the energy states and electronic transitions within molecules. Fluorescence spectroscopy is commonly used to study biomolecules, such as proteins and nucleic acids, as well as for cellular imaging and molecular labeling.
6. Raman Spectroscopy: Raman spectroscopy involves the measurement of the inelastic scattering of light by molecules. It provides information about molecular vibrations, rotational

modes, and structural properties. Raman spectroscopy is used for chemical identification, analysis of materials, characterization of crystals, and studying molecular dynamics.

These spectroscopic techniques offer valuable insights into the physical, chemical, and structural properties of materials, ranging from small molecules to complex biological systems. By utilizing different spectroscopic methods, researchers can obtain a comprehensive understanding of the composition, behavior, and interactions of molecules, aiding in the advancement of fields such as chemistry, biochemistry, materials science, and pharmaceutical research.

7.4.1 Ultraviolet-Visible Spectroscopy (UV-Vis)

Ultraviolet-visible spectroscopy (UV-Vis spectroscopy) is a widely used technique in analytical chemistry and biochemistry for studying the absorption and transmission of light in the ultraviolet and visible regions of the electromagnetic spectrum. It provides valuable information about the electronic transitions and energy levels of molecules, allowing researchers to analyze the composition, concentration, and chemical properties of a sample.

UV-Vis spectroscopy relies on the principle that molecules absorb light at specific wavelengths corresponding to the energy differences between their electronic energy levels. The sample is exposed to a range of wavelengths, and the amount of light absorbed or transmitted is measured. The resulting UV-Vis spectrum provides a graphical representation of the absorption or transmission intensity as a function of wavelength.

Key features and applications of UV-Vis spectroscopy include:

1. Quantitative Analysis: UV-Vis spectroscopy is commonly used for quantitative analysis, as the absorption intensity is directly proportional to the concentration of the absorbing species. By measuring the absorbance at a specific wavelength, researchers can determine the concentration of a compound in a sample using Beer-Lambert's Law.
2. Compound Identification: UV-Vis spectra can be used for compound identification by comparing the characteristic absorption patterns of known compounds with those of an unknown sample. Each compound has a unique UV-Vis spectrum, enabling the identification and differentiation of similar compounds.

3. Determination of Purity: UV-Vis spectroscopy is employed to assess the purity of a substance. Impurities or contaminants in a sample can alter the absorption profile, allowing researchers to quantify the extent of impurities or monitor the progress of purification processes.
4. Kinetic Studies: UV-Vis spectroscopy is valuable for investigating chemical reactions and studying reaction rates. By monitoring the absorbance changes over time, researchers can determine reaction kinetics, including reaction rates, rate constants, and reaction mechanisms.
5. Protein and Nucleic Acid Analysis: UV-Vis spectroscopy is widely used in biochemistry for the analysis of proteins and nucleic acids. The absorption characteristics of these biomolecules in the UV-Vis range provide insights into their concentration, structure, and interactions.
6. Environmental Analysis: UV-Vis spectroscopy finds applications in environmental analysis, such as the detection and quantification of pollutants, monitoring water quality, and assessing the presence of contaminants in air and soil samples.

UV-Vis spectroscopy is a versatile and widely accessible technique that offers valuable insights into the electronic properties and composition of various samples. Its simplicity, speed, and broad applications make it an essential tool in numerous scientific disciplines, ranging from chemistry and biochemistry to pharmaceutical analysis and environmental monitoring.

7.4.2 Infrared Spectroscopy (IR)

Infrared spectroscopy (IR spectroscopy) is a powerful analytical technique used to study the interaction of infrared radiation with molecules. It provides valuable information about the vibrational and rotational energy levels of molecules, enabling the identification of functional groups, characterization of chemical bonds, and determination of molecular structure.

IR spectroscopy operates in the infrared region of the electromagnetic spectrum, which spans wavelengths longer than visible light. It is based on the principle that molecules absorb infrared radiation at specific frequencies corresponding to the energy differences between their vibrational and rotational states. The resulting IR spectrum provides a unique fingerprint of the sample's molecular composition and structure.

Key features and applications of IR spectroscopy include:

1. Functional Group Analysis: IR spectroscopy is widely used for the identification and characterization of functional groups in organic and inorganic compounds. Different functional groups exhibit characteristic absorption peaks in the IR spectrum, allowing for their rapid identification.
2. Chemical Bond Analysis: IR spectroscopy provides information about the types of chemical bonds present in a molecule. It allows researchers to determine the presence of single or multiple bonds, as well as the presence of functional groups such as alcohols, amines, carbonyls, and carboxylic acids.
3. Structural Elucidation: IR spectroscopy helps determine the structure and connectivity of molecules. By analyzing the positions and intensities of absorption peaks, researchers can deduce the presence and arrangement of atoms within a molecule.
4. Quantitative Analysis: IR spectroscopy can be used for quantitative analysis, such as determining the concentration of a specific compound in a mixture. This is achieved by measuring the intensity of absorption at a specific wavelength and correlating it to the concentration using calibration curves.
5. Polymer Analysis: IR spectroscopy is widely used in the analysis of polymers. It can provide information about polymer composition, structure, and degree of polymerization. It is also used to monitor polymerization reactions and assess the presence of impurities or additives in polymer samples.
6. Pharmaceutical Analysis: IR spectroscopy plays a vital role in pharmaceutical analysis. It is used for the identification and characterization of drug substances and formulations, assessment of drug purity, and detection of impurities or degradation products.
7. Environmental and Forensic Analysis: IR spectroscopy finds applications in environmental and forensic analysis. It can be used to identify pollutants, analyze environmental samples, detect counterfeit drugs, and analyze forensic evidence.

IR spectroscopy is a versatile and widely utilized technique in various scientific fields, including chemistry, biochemistry, materials science, pharmaceuticals, and environmental analysis. Its ability to provide detailed structural and compositional information makes it an indispensable tool for both qualitative and quantitative analysis.

7.4.3 Nuclear Magnetic Resonance (NMR) Spectroscopy

Nuclear Magnetic Resonance (NMR) spectroscopy is a powerful analytical technique used to study the interactions of atomic nuclei with a strong magnetic field. It provides detailed information about the structure, dynamics, and chemical environment of molecules, making it a fundamental tool in the fields of chemistry, biochemistry, and materials science.

NMR spectroscopy is based on the principle of nuclear spin and the magnetic properties of atomic nuclei. When placed in a strong magnetic field and exposed to radiofrequency pulses, atomic nuclei absorb and re-emit energy at specific frequencies. By measuring these frequencies, researchers can obtain valuable insights into the chemical and physical properties of the sample.

Key features and applications of NMR spectroscopy include:

1. Structural Elucidation: NMR spectroscopy is widely used to determine the structure of organic and inorganic compounds. It provides information about the connectivity of atoms, stereochemistry, and molecular conformation. By analyzing the chemical shifts and splitting patterns in the NMR spectrum, researchers can deduce the arrangement of atoms in a molecule.
2. Compound Identification: NMR spectroscopy is used for compound identification by comparing the NMR spectra of unknown compounds with those of known reference compounds. Each compound has a unique NMR fingerprint, allowing for accurate identification and differentiation of chemical species.
3. Quantitative Analysis: NMR spectroscopy can be used for quantitative analysis, enabling researchers to determine the concentration of a specific compound in a mixture. By integrating the area under the NMR peaks, the relative concentrations of different components can be determined.
4. Determination of Molecular Dynamics: NMR spectroscopy provides insights into the dynamics and motion of molecules. It can be used to study molecular rotations, vibrations, and conformational changes, providing valuable information

about chemical reactions, protein folding, and molecular interactions.

5. Biomolecular Structure and Interactions: NMR spectroscopy is extensively used in structural biology to determine the three-dimensional structures of proteins, nucleic acids, and other biomolecules. It can also study protein-ligand interactions, protein-protein interactions, and protein-nucleic acid interactions.
6. Metabolomics: NMR spectroscopy is employed in metabolomics, the study of small molecule metabolites in biological systems. It allows researchers to identify and quantify metabolites, providing insights into metabolic pathways, disease biomarkers, and drug metabolism.
7. Solid-State NMR: In addition to solution NMR, solid-state NMR spectroscopy is used to study the structure and dynamics of solids, including materials, catalysts, and complex biomolecular assemblies.

NMR spectroscopy is a versatile technique with applications in various scientific disciplines. Its ability to provide detailed structural and dynamic information about molecules and materials makes it an essential tool for research, drug discovery, materials characterization, and understanding biological processes.

7.4.4 Fluorescence Spectroscopy

Fluorescence spectroscopy is an investigative method based on the fluorescence properties of the sample under study, and is used for quantitative measurements of chemical products.

Fluorescence spectroscopy analyzes fluorescence from a molecule based on its fluorescent properties.

Fluorescence is a type of luminescence caused by photons exciting a molecule, raising it to an electronic excited state.

Fluorescence spectroscopy uses a beam of light that excites the electrons in molecules of certain compounds, and causes them to emit light. That light is directed towards a filter and onto a detector for measurement and identification of the molecule or changes in the molecule.

The term fluorescence refers to one type of luminescence. Luminescence, broadly defined, is light emission from a molecule. There are several types of luminescence.

- **Photoluminescence** is when light energy, or photons, stimulate the emission of a photon.
- **Chemiluminescence**, is defined as when chemical energy stimulates the emission of a photon, and this includes bioluminescence, as seen in fire flies and many forms of sea life.
- **Electroluminescence**, is when electrical energy or a strong electric field, stimulates the emission of a photon, such as in some lighting applications.
- **Fluorescence**, specifically, is a type of photoluminescence where light raises an electron to an excited state.

  The excited state undergoes rapid thermal energy loss to the environment through vibrations, and then a photon is emitted from the lowest-lying singlet excited state. This process of photon emission competes for other non-radiative processes including energy transfer and heat loss.

When the term "fluorescence" is used, the same methods of measurement can typically be applied to any of the above categories of luminescence.

What is a Fluorescence Spectrum?

Steady state fluorescence spectra are when molecules, excited by a constant source of light, emit fluorescence, and the emitted photons, or intensity, are detected as a function of wavelength.

A fluorescence emission spectrum is when the excitation wavelength is fixed and the emission wavelength is scanned to get a plot of intensity vs. emission wavelength.

A fluorescence excitation spectrum is when the emission wavelength is fixed and the excitation monochromator wavelength is scanned.

In this way, the spectrum gives information about the wavelengths at which a sample will absorb so as to emit at the single emission wavelength chosen for observation.

It is analogous to absorbance spectrum, but is a much more sensitive technique in terms of limits of detection and molecular specificity.

Excitation spectra are specific to a single emitting wavelength/species as opposed to an absorbance spectrum, which measures all absorbing species in a solution or sample. The emission and excitation spectra for a given fluorophore are mirror images of each other. Typically, the emission spectrum occurs at higher wavelengths (lower energy) than the excitation or absorbance spectrum.

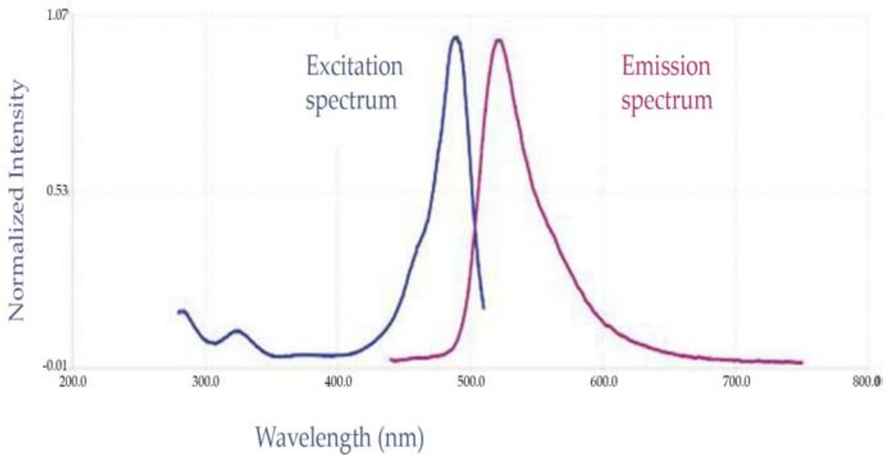


Figure 3. The emission and excitation spectra for a given fluorophore are mirror images of each other

These two spectral types (emission and excitation) are used to see how a sample is changing.

The spectral intensity and or peak wavelength may change with variants such as temperature, concentration, or interactions with other molecules around it.

This includes quencher molecules and molecules or materials that involve energy transfer. Some fluorophores are also sensitive to solvent environment properties such as pH, polarity, and certain ion concentrations.

What types of molecules and materials exhibit fluorescence?

Fluorescent molecules and materials come in all shapes and sizes. Some are intrinsically fluorescent, such as chlorophyll and the amino acid residue tryptophan (Trp), phenylalanine (Phe) and tyrosine (Tyr).

Others are molecules synthesized specifically as stable organic dyes or tags to be added to otherwise non-fluorescent systems. There are entire catalogs of these available.

Typically, organic fluorescent molecules have aromatic rings and pi-conjugated electrons in them. Depending on their size and structure, organic dyes can emit from the UV out into the near-IR.

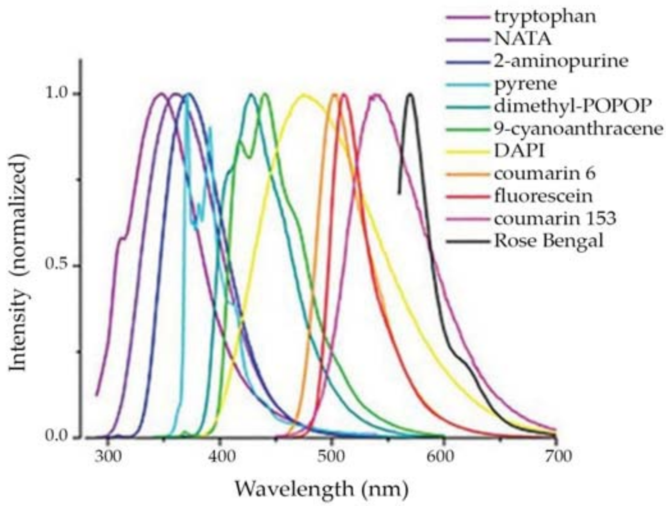


Figure 4. Fluorescence emission spectra of some common fluorophores across the UV and visible spectrum.

Here are a random sampling of a few common fluorophores that span the UV and Visible range. Some rare earth elements, or lanthanides, have higher electronic orbitals filled, where electrons transition due to metal ligand charge transfers happen between 4f-5d and even 4f-4f orbitals. (Bunzli, 1989) There are many molecules that are luminescent in nature such as a few of the amino acids, chlorophylls, and natural pigments. Others are highly engineered for very specific uses of fluorescence spectroscopy.

A few of the categories of fluorescent molecules and materials are:

- Amino acids (Trp, Phe, Tyr)
- Base pair derivatives (2-AP, 3-MI, 6-MI, 6-MAP, pyrrolo-C, tC)
- Chlorophylls
- Fluorescent Proteins (FPs)
- Organic dyes (fluorescein, rhodamine, N-aminocoumarins and derivatives of these)
- Rare earth elements (lanthanides)
- Semiconductors
- Quantum dots
- Single Walled Carbon Nanotubes (SWCNTs)
- Solar cells
- Pigments, brighteners
- Phosphors
- Many more…

Other molecules and materials such as fluorescent proteins, semiconductors, phosphors, and rare earth elements are among the commonly used fluorescent samples. Polymers with conjugated aromatics or dienes also commonly have fluorescent properties. Of course, new materials are being created all the time.

7.4.5 Raman Spectroscopy

Raman spectroscopy is a powerful analytical technique used to study the vibrational and rotational energy levels of molecules. It provides information about the molecular structure, chemical composition, and bonding characteristics of a sample by measuring the scattering of light. Raman spectroscopy is widely used in various scientific fields, including chemistry, biochemistry, materials science, and pharmaceutical analysis.

Raman spectroscopy is based on the Raman effect, discovered by C.V. Raman in 1928. When a sample is illuminated with a laser beam, a small fraction of the scattered light undergoes a change in energy due to interactions with the sample's molecular vibrations. This scattered light, known as the Raman scattering, contains information about the vibrational modes and energy levels of the sample.

Key features and applications of Raman spectroscopy include:

1. Molecular Identification: Raman spectroscopy is used for the identification and characterization of chemical compounds. Each compound exhibits a unique Raman spectrum, allowing researchers to identify unknown substances and differentiate between similar compounds.
2. Structural Analysis: Raman spectroscopy provides insights into the molecular structure and bonding of a sample. It can detect and analyze functional groups, identify the presence of specific chemical bonds, and determine the conformation of molecules.
3. Chemical Imaging: Raman spectroscopy can be coupled with microscopy to generate chemical images of samples. By raster scanning the laser beam across a sample, researchers can obtain spatially resolved Raman spectra, allowing for the visualization of chemical distributions and mapping of molecular components.
4. Biological and Medical Applications: Raman spectroscopy finds applications in biological and medical research. It can be used to analyze biomolecules, study cellular structures, and

detect diseases at the molecular level. Raman spectroscopy has been employed in cancer diagnosis, tissue imaging, and monitoring drug delivery systems.

5. Materials Science: Raman spectroscopy is widely used in materials science for the characterization of materials. It can provide information about crystallinity, phase transitions, and defects in materials. Raman spectroscopy is particularly useful for the analysis of carbon-based materials, polymers, catalysts, and semiconductors.
6. Quality Control and Forensic Analysis: Raman spectroscopy is employed in quality control processes to verify the composition and quality of products. It is also used in forensic analysis to identify trace evidence, analyze counterfeit substances, and investigate crime scenes.
7. Non-destructive Analysis: Raman spectroscopy is a non-destructive technique, allowing for the analysis of samples without altering their chemical or physical properties. This makes it suitable for the analysis of valuable or sensitive samples.

Raman spectroscopy offers several advantages, including high specificity, non-destructiveness, and the ability to analyze samples in various states (solid, liquid, or gas). It is a versatile technique that provides valuable insights into molecular structures, chemical compositions, and material properties, making it an important tool in scientific research and analysis.

REFERENCES

1. Aebersold, R., & Mann, M. (2016). Mass spectrometry-based proteomics. Nature, 537(7620), 347-355.
2. Ausubel, F. M., Brent, R., Kingston, R. E., Moore, D. D., Seidman, J. G., Smith, J. A., & Struhl, K. (Eds.). (2005). Current protocols in molecular biology. John Wiley & Sons.
3. Halket, J. M., & Zaikin, V. G. (2017). Derivatization in mass spectrometry: Principles and applications. John Wiley & Sons.
4. Hames, B. D., & Rickwood, D. (2005). Gel electrophoresis of proteins: A practical approach. Oxford University Press.
5. Marriott, P. J. (Ed.). (2012). Gas chromatography: Principles, techniques, and applications. CRC Press.

6. Snyder, L. R., Kirkland, J. J., & Glajch, J. L. (2010). Practical HPLC method development. John Wiley & Sons.
7. Watson, J. T., & Sparkman, O. D. (2007). Introduction to mass spectrometry: Instrumentation, applications, and strategies for data interpretation. John Wiley & Sons.
8. Westermeier, R., & Naven, T. (Eds.). (2006). Proteomics in practice: A laboratory manual of proteome analysis. Wiley-VCH.
9. Wu, J., & Liang, G. (Eds.). (2008). Chiral separations by liquid chromatography and related technologies. CRC Press.

CHAPTER

8

PRINCIPLES OF CLINICAL BIOCHEMISTRY

INTRODUCTION

Chemistry is a division of science which deals with the understanding of matter. The chemistry that deals with biological matter are known as Biochemistry. Biochemistry dealing with the human health and diseases is known as medical biochemistry. Clinical biochemistry is also referred to as clinical chemistry/chemical pathology or medical biochemistry. Basically, it is a branch of clinical pathology generally related to the understanding of body fluids diagnosis as well as therapeutic purpose. Clinical biochemistry is an application aspect of medical biochemistry. It deals methodology and interpretation of clinical (chemical) tests carried out for the diagnosis of disease.

Protein (amino acids), carbohydrate (sugars) and lipids (fatty acids and glycerol) are the chemical component of the human body. Besides DNA and RNA, the nucleic acids residing in the body are formed of nucleo- tides/nucleosides. For life, energy is the main requirement and is generated by the oxidization of food material we eat. Energy is produced in the fonn of ATP in the body. The component (plasma membrane, chromosomes, etc.) of the cells are comprised of different macromolecule (protein, carbohydrates, lipids, and nucleic acids). The food we take, is utilize for energy production and synthesis of cellular component (extracellular and intracellular) with the help of enzymatic anabolic and catabolic reactions. Body systems such as digestion, circulation, excretion, and nervous system utilize these reactions to perform various functions such as maintenance of the components of blood and plasma, enzyme synthesis and degradation, transportation of particles across the cell membrane, synthesis, and action of honnones, blood coagulation, signal transduction, etc. In a normal individual, the metabolism process (anabolism and catabolism) goes smoothly and the metabolites circulating through the biological fluids are in nonnal concentration. During a disease condition, the homeostasis between anabolism and catabolism gets altered and persists for a small/long time in the body.

8.1 PRINCIPLES OF CLINICAL BIOCHEMICAL ANALYSIS

Clinical biochemistry is a discipline which usually deals with the examination of body fluids such as blood, serum, and urine for therapeutic and diagnostic purposes. It is an applied form of biochemistry which involves the use and measurement of enzyme activities by spectrophotometry, electrophoresis, and immunoassay method.

This branch initiated in the late 19th century with the use of basic chemical reaction tests for numerous constituents of blood and urine. Nowadays, there are many blood and urine tests are available with extensive diagnostic abilities.

8.1.1 Basis of Analysis of Body Fluids for Diagnostic, Prognostic and Monitoring Purposes

Underlying most human diseases is a change in the amount or function of one or more proteins that in turn triggers changes in cellular, tissue or organ function. The dysfunction is commonly characterized by a significant change in the biochemical profile of body fluids.

The application of quantitative analytical biochemical tests to a large range of biological analyses in body fluids and tissues is a valuable aid to the diagnosis and management of the prevailing disease state. The general biological and analytical principles underlying these tests will be discussed and related to the general principles of quantitative chemical analysis. Body fluids such as blood, cerebrospinal fluid and urine in both healthy and diseased states contain a large number of inorganic ions and organic molecules. Whilst the normal biological function of some of these chemical species lies within that fluid, for the majority it does not. The presence of this latter group of chemical species within the fluid is due to the fact that normal cellular secretory mechanisms and the temporal synthesis and turnover of individual cells and their organelles within the major organs of the body, both result in the release of cell components, particularly those located in the cytoplasm, into the surrounding extracellular fluid and eventually into the blood circulatory system.

This in turn transports them to the main excretory organs, namely the liver, kidneys and lungs, so that these cell components and/or their degradation products are eventually excreted in faeces, urine, sweat and expired air. Examples of cell components in this category include enzymes, hormones, intermediary metabolites and small organic and inorganic ions.

The concentration, amount or activity of a given cell component that can be detected in these fluids of a healthy individual at any point in time depends on many factors that can be classified into one of three categories, namely chemical characteristics of the component, endogenous factors characteristic of the individual and exogenous factors that are imposed on the individual.

Chemical characteristics

Some molecules are inherently unstable outside their normal cellular environment. For example, some enzymes are reliant on the presence of their substrate and/or coenzyme for their stability and these may be absent or in too low concentrations in the extracellular fluid. Molecules that can act as substrates of catabolic enzymes found in extracellular fluids, in particular blood, will also be quickly metabolized. Cell components that fall into these two categories therefore have a short half-life outside the cell and are normally present in low concentrations in fluids such as blood.

Endogenous factors

These include age, gender, body mass and pregnancy. For example:

a. serum cholesterol concentrations are higher in men than premenopausal women but the differences decreases post-menopause;

b. serum alkaline phosphatase activity is higher in children than in adults and is raised in women during pregnancy;

c. serum insulin and triglyceride concentrations are higher in obese individuals than in the lean;
d. serum creatinine, a metabolic product of creatine important in muscle metabolism, is higher in individuals with a large muscle mass;
e. erSerum sex hormone concentrations differ between males and females and change with age.

Exogenous factors

These include time, exercise, food intake and stress. Several hormones are secreted in a time-related fashion. Thus cortisol and to a lesser extent thyroid stimulating hormone (TSH) and prolactin all show a diurnal rhythm in their secretion. In the case of cortisol, its secretion peaks around 9.00 am and declines during the day reaching a trough between 11.00 pm and 5.00 am. The secretion of female sex hormones varies during a menstrual cycle and that of 25-hydroxycholecalciferol (vitamin D3) varies with the seasons peaking during the late summer months. The concentrations of glucose, triglycerides and insulin in blood rise shortly after the intake of a meal. Stress, including that imposed by the process of taking a blood sample by puncturing a vein (venipuncture), can stimulate the secretion of a number of hormones and neurotransmitters including prolactin, cortisol, adrenocorticotropic hormone (ACTH) and adrenaline.

The influence of these various factors on the extent of release of cell components into extracellular fluids inevitably means that even in healthy individuals there is a considerable intraindividual variation (i.e. variation from one occasion to another) in the value of any chosen test analyte of diagnostic importance and an even larger interindividual variation (i.e. variation between individuals). More importantly, the superimposition of a disease state onto these causes of intra- and interindividual variation will result in an even greater variability between test occasions. Many clinical conditions compromise the integrity of cells located in the organs affected by the condition. This may result in the cells becoming more 'leaky' or, in more severe cases, actually dying (necrosis) and releasing their contents into the surrounding extracellular fluid. In the vast majority of cases the extent of release of specific cell components into the extracellular fluid, relative to the healthy reference range, will reflect the extent of organ damage and this relationship forms the basis of diagnostic clinical biochemistry. If the cause of the organ damage continues for a prolonged time and is essentially irreversible (i.e. the organ does not undergo self-repair), as is the case in cirrhosis of the liver for example, then the mass of cells remaining to undergo necrosis will progressively decline so that eventually the release of cell components into the surrounding extracellular fluid will decrease even though organ cells are continuing to be damaged. In such case the measured amounts will not reflect the extent of organ damage.

Clinical biochemical tests have been developed to complement in four main ways a provisional clinical diagnosis based on the patient's medical history and clinical examination:

- To support or reject a provisional diagnosis by detecting and quantifying abnormal amounts of test analytes consistent with the diagnosis. For example, serum myoglobin, troponin-I (part of the cardiac contractile muscle), creatine phosphokinase (specifically the CK-MB isoform) and aspartate transaminase all rise following a myocardial infarction (heart attack) that results in cell death in some heart tissue. The released cellular components also cause cell inflammation (leakiness) in surrounding cells causing an amplification of total cellular component release. Tests can also help a differential diagnosis, for example in distinguishing the various forms of jaundice (yellowing of the skin due to the presence of the yellow pigment bilirubin, a metabolite of haem) by the measurement of alanine transaminase (ALT) and aspartate transaminase (AST) activities and by determining whether or not the bilirubin is conjugated with b-glucuronic acid.
- To monitor recovery following treatment by repeating the tests on a regular basis and monitoring the return of the test values to those within the reference range. Following a myocardial infarction, for example, the raised serum enzyme activities referred to above usually return to reference range values within 10 days. Similarly, the measurement of serum tumour markers such as CA125 can be used to follow recovery or recurrence after treatment for ovarian cancer.
- To screen for latent disease in apparently healthy individuals by testing for raised levels of key analytes. For example, measuring serum glucose for diabetes mellitus and immunoreactive trypsin for cystic fibrosis. It is now common for serum cholesterol levels to be used as a measure of the risk of the individual developing heart disease. This is particularly important for individuals with a family history of the disease.

An action limit of serum cholesterol >5.2 mM has been set by the British Hyperlipidaemia Association for an individual to be counselled on the importance of a healthy (low fat) diet and regular exercise and a higher action limit of serum cholesterol >6.6 mM for cholesterol-lowering ‘statin’ drugs to be prescribed and clinical advice given.

- To detect toxic side effects of treatment, for example in patients receiving hepatotoxic drugs, by undertaking regular liver function tests. An extension of this is therapeutic drug monitoring in which patients receiving drugs such as phenytoin and carbamezepine (both of which are used in the treatment of epilepsy) that have a low therapeutic index (ratio of the dose required to produce a toxic effect relative to the dose required to produce a therapeutic effect) are regularly monitored for drug levels and liver function to ensure that they are receiving effective and safe therapy.

8.1.2 Reference Ranges

For a biochemical test for a specific analyte to be routinely used as an aid to clinical diagnosis, it is essential that the test has the required performance indicators, especially specificity and sensitivity. Sensitivity expresses the proportion of patients with the disease who are correctly identified by the test. Specificity expresses the proportion of patients without the disease who are correctly identified by the test. These two parameters may be expressed mathematically as follows:

$$\text{sensitivity} = \frac{\text{true positive tests} \times 100\%}{\text{total patients with the disease}}$$

$$\text{specificity} = \frac{\text{true negative tests} \times 100\%}{\text{total patients without the disease}}$$

Ideally both of these indicators for a particular test should be 100% but this is not always the case. This problem is most likely to occur in cases where the change in the amount of the test analyte in the clinical sample is small compared with the reference range values found in healthy individuals. Both of these indicators express the performance of the test but it is equally important to be able to quantify the probability that the patient with a positive test has the disease in question. This is best achieved by the predictive power of the test. This expresses the proportion of patients with a positive test who are correctly diagnosed as disease positive:

$$\text{positive predictive value} = \frac{\text{true positive patients}}{\text{total positive tests}}$$

$$\text{negative predictive value} = \frac{\text{true negative patients}}{\text{total negative tests}}$$

The concept of predictive power can be illustrated by reference to foetal screening for Down's syndrome and neural tube defects. Preliminary tests for these conditions in unborn children are based on the measurement of a-fetoprotein (AFP), human chorionic gonadotropin (hCG) and unconjugated oestriol (uE3) in the mother's blood. The presence of these conditions results in an increased hCG and decreased AFP and uE3 relative to the average in healthy pregnancies. The results of the tests are used in conjunction with the gestational and maternal ages to calculate the risk of the baby suffering from these conditions. If the risk is high, further tests are undertaken including the recovery of some foetal cells for genetic screening from the amniotic fluid surrounding the foetus in the womb by inserting a hollow needle into the womb (amniocentesis). The three tests detect two out of three cases (67%) of Down's syndrome and four out of five cases (80%) of neural tube defects. Thus the performance indicators of the tests are not 100% but they are sufficiently high to justify their routine use.

The correct interpretation of all biochemical test data is heavily dependent the use of the correct reference range against which the test data are to be judged. As previously pointed out, the majority of biological analytes of diagnostic importance are subject to considerable inter- and intraindividual variation in healthy adults, and the analytical method chosen for a particular analyte assay will have its own precision, accuracy and selectivity that will influence the analytical results. In view of these biological and analytical factors, individual laboratories must establish their own reference range for each test analyte using their chosen methodology and a large number (hundreds) of 'healthy' individuals. The recruitment of individuals to be included in reference range studies presents a considerable practical and ethical problem due to the difficulty of defining 'normal' and of using invasive procedures, such as venipuncture, to obtain the necessary biological samples. The establishment of reference ranges for children, especially neonates, is a particular problem.

Reference ranges are most commonly expressed as the range that covers the mean ±1.96 standard deviations of the mean of the experimental population. This range covers 95% of the population. The majority of reference ranges are based on a normal distribution of individual values but in some cases the experimental data are asymmetric often being skewed to the upper limits. In such cases it is normal to use logarithmic data to establish the reference range but even so, the range may overlap with values found in patients with the test disease state.

8.2 CLINICAL MEASUREMENTS AND QUALITY CONTROL

Quality measurement is a routine part of the clinical research laboratory to improve the reliability of diagnosis and prognosis. Inaccurate measurements might impede the handling of clinical studies such as randomized trials and epidemiological analyses. Progress in clinical measurement is proportional to new technology or techniques.

Measurements cover all the required parameters for the clinical and research purpose. Low-quality measurements affect a steady prognosis and diagnosis. Wrong measurements affect epidemiological studies and randomized trials. Statistical analysis deals with reference values, spreading of individual groups, outliers, and reference limits. Administration of entire testing such as test usage and practice guidelines, patient identification, turnaround time (TAT), laboratory logs, transcription errors, patient preparation, specimen collection, transport, and division and spreading of aliquots should be performed after and before analysis.

8.2.1 The Operation of Clinical Biochemistry Laboratories

The clinical biochemistry laboratory in a typical general hospital in the UK serves a population of about 400 000 containing approximately 60 General Practitioner (GP) groups depending upon the location in the UK. This population will generate approximately 1200 requests from GPs and hospital doctors each weekday for clinical biochemical tests on their patients. Each patient request will require the laboratory to undertake an average of seven specific analyte tests. The result is that a typical general hospital laboratory will carry out between 2.5 and 3 million tests each year. The majority of clinical biochemistry laboratories offers the local medical community as many as 200 different clinical biochemical tests that can be divided into eight categories as shown in Table 2

Most of the requests for biochemical tests will arise on a routine daily basis but some will arise from emergency medical situations at any time of the day. The large number of daily test samples coupled with the need for a 24-hour 7-day week service dictates that the laboratory must rely heavily on automated analysis to carry out the tests and on information technology to process the data.

Table 1. Typical reference ranges for biochemical analytes

Analyte	Reference range	Comment
Sodium	133–145 mM	
Potassium	3.5–5.0 mM	Values increased by haemolysis or prolonged contact with cells.
Urea	3.5–6.5 mM	Range varies with sex and age e.g. values up to 12.1 will be found in males over the age of 70.
Creatinine	75–115 mM (males) 58–93 mM (females)	Creatinine (a metabolite of creatine) production relates to muscle mass and is also a reflection of renal function. Values for both sexes increase by 5–20% in the elderly.
Aspartate transaminase (AST)	$<40\ \mathrm{IU\ dm^{-3}}$	Perinatal levels are $<80\ \mathrm{IU\ dm^{-3}}$ and fall to adult values by the age of 18. Some slightly increased values up to $60\ \mathrm{IU\ dm^{-3}}$ may be found in females over the age of 50. Results are increased by haemolysis.
Alanine transaminase (ALT)	$<40\ \mathrm{IU\ dm^{-3}}$	Higher values are found in males up to the age of 60.
Alkaline phosphatase (AP)	$<122\ \mathrm{IU\ dm^{-3}}$ (adults) $<455\ \mathrm{IU\ dm^{-3}}$ (children <12 y)	Significantly raised results of up to two- or three-fold would be experienced during growth spurts through teenage years. Slightly raised levels also seen in the elderly and in women during pregnancy.
Cholesterol	No reference range but recommended value of <5.2 mM	The measurement of cholesterol in an adult 'well' population does not show a Gaussian distribution but a very tailed distribution with relatively few low results. The majority are <10 mM but there is a long tail up to over 20 mM. There is a tendency for males to have higher cholesterol than females of the same age but after the menopause female values revert to those of males. Generally values increase with age.

To achieve an effective service, a clinical biochemistry laboratory has three main functions:

- to advise the requesting GP or hospital doctor on the appropriate tests for a particular medical condition and on the collection, storage and transport of the patient samples for analysis;
- to provide a quality analytical service for the measurement of biological analytes in an appropriate and timely way;
- To provide the requesting doctor with a data interpretation and advice service on the outcome of the biochemical tests and possible further tests.

Table 2. Examples of biochemical analytes used to support clinical diagnosis

Type of analyte	Examples
Foodstuffs entering the body	Cholesterol, glucose, fatty acids, triglycerides
Waste products	Bilirubin, creatinine, urea
Tissue-specific messengers	Adrenocorticotropic hormone (ACTH), follicle-stimulating hormone (FSH), luteinising hormone (LH), thyroid-stimulating hormone (TSH)
General messengers	Cortisol, insulin, thyroxine
Response to messengers	Glucose tolerance test assessing the appropriate secretion of insulin; tests of pituitary function
Organ function	Adrenal function – cortisol, ACTH; renal function – K^+, Na^+, urea, creatinine; thyroid function – free thyroxine (FT_4), free tri-iodothyronine (FT_3), TSH
Organ disease markers	Heart – troponin-I, creatine kinase (CKMB), AST, lactate dehydrogenase (LD); liver – ALT, AP, γ-glutamyl transferase (GGT), bilirubin, albumin
Disease-specific markers	Specific proteins ('tumour markers') secreted from specific organs – prostate-specific antigen (PSA), CA-125 (ovary), calcitonin (thyroid), $\alpha_1$fetoprotein (liver)

The advice given to the clinician is generally supported by a User Handbook, prepared by senior laboratory personnel, that includes a description of each test offered, instructions on sample collection and storage, normal laboratory working hours and the approximate time it will take the laboratory to undertake each test. This turnaround time will vary from less than one hour to several weeks depending upon the speciality of the test. The vast majority of biochemical tests are carried out on serum or plasma derived from a blood sample. Serum is the preferred matrix for biochemical tests but the concentrations of most test analytes are almost the same in the two fluids. Serum is obtained by allowing the blood to clot and removing the clot by centrifugation. To obtain plasma it is necessary to add an anticoagulant to the blood sample and remove red cells by centrifugation. The two most common anticoagulants are heparin and EDTA, the choice depending on the particular biochemical test required. For example, EDTA complexes calcium ions so that calcium in EDTA plasma would be undetectable. For the measurement of glucose, fluorideoxalate is added to the sample not as an anticoagulant but to inhibit glycolysis during the transport and storage of the sample. Special vacuum collection tubes containing specific anticoagulants or other additives are available for the storage of blood samples. Collection tubes are also available containing clot enhancers to speed the clotting process for serum preparation. Many containers incorporate a gel with a specific gravity designed to float the gel between cells and serum providing a barrier between the two for up to 4 days. During these 4 days, the cellular component will experience lysis, so any subsequent contamination of the serum will include intracellular components. Biochemical tests may also be carried out on whole blood, urine, cerebrospinal fluid (the fluid surrounding the spinal cord and brain), faeces, sweat, saliva and amniotic fluid.

It is essential that the samples are collected in the appropriate container at the correct time (particularly important if the test is for the measurement of hormones such as cortisol subject to diurnal release) and labelled with appropriate patient and biohazard details.

Samples submitted to the laboratory for biochemical tests are accompanied by a request form, signed by the requesting clinician, which gives details of the tests required and brief details of the reasons for the request to aid data interpretation and to help identify other appropriate tests.

Laboratory reception

On receipt in the laboratory both the sample and the request form will be assigned an acquisition number usually in an optically readable form but with a bar code. A check is made of the validity of the sample details on both the request form and sample container to ensure that the correct container for the tests required has been used.

Samples may be rejected at this stage if details are not in accordance with the set protocol. Correct samples are then split from the request form and prepared for analysis typically by centrifugation to prepare serum or plasma. The request form is processed into the computer system that identifies the patient against the sample acquisition number, and the tests requested by the clinician typed into the database. It is vital at this reception phase that the sample and patient data match and that the correct details are placed in the database.

These details must be adequate to uniquely identify the patient bearing in mind the number of potential patients in the catchment area, and will include name, address, date of birth, CHI number (a unique identifier for each individual in the UK for health purposes), hospital or Accident and Emergency number and acquisition number.

8.2.2 Analytical Organization

The analytical organization of the majority of clinical biochemical laboratories is based on three work areas:

- auto-analyser section,
- immunoassay section,
- Manual section.

Auto-analyser section

Auto-analysers dedicated to clinical biochemical analysis are available from many commercial manufacturers. The majority of analysers are fully automated and have carousels for holding the test samples in racks each carrying up to 15 samples, one or two carousels each for up to 60 different reagents that are identified by a unique bar code, carousels for sample washing/preparation and a reaction carousel containing up to 200 cuvettes for initiating and monitoring individual test reactions. The Abbott ARCHITECT c1600 (Fig. 1) utilizes three methodologies: spectrophotometry, immunoturbidometry and potentiometry. The spectrophotometry system is capable of measuring at 16 wavelengths simultaneously whilst the potentiometric system, based on the use of ion-selective electrodes (ISEs) that are combined into a single unit, is used to measure electrolyte concentrations of Na^+, K^+ and Cl^- simultaneously with an analytical time of less than 4 min on a sample size of only 25 mm^3.

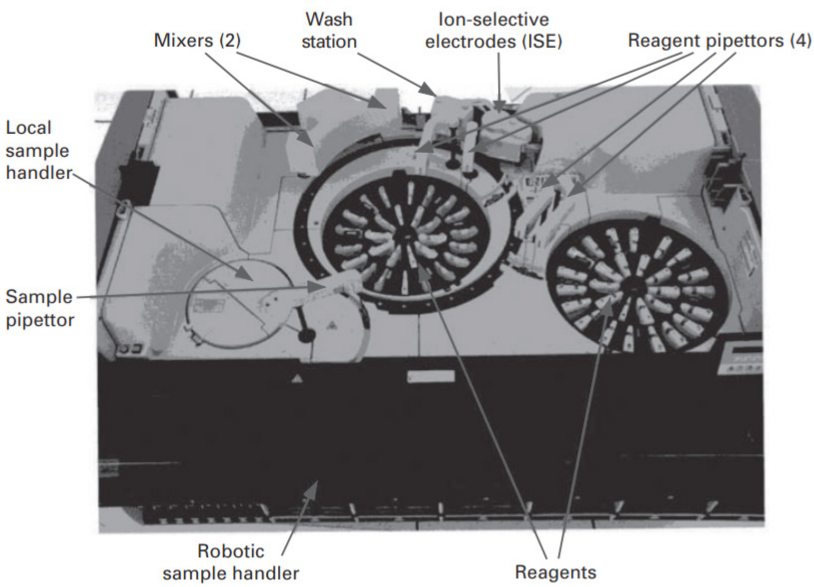


Figure 1. The Abbott ARCHITECT c1600 clinical chemical analyser.

The reaction carousel contains 330 cuvettes and before each assay each cuvette is automatically cleaned and dried using an eight-stage wash mechanism. Supplementary washes are available to reduce carryover when necessary. Reagents for up to 62 assays based on different analytical techniques may be loaded at the same time and to maximize throughput, testing is organized in two parallel lines each one of which is resource and analysis-controlled separately. The analyser uses a fixed cycle time. Every 4.5 seconds the reaction carousel moves approximately one-quarter turn and readings are taken each full turn. The total analysis time is dependent upon the assay being performed and is determined by the assay parameters. All tests are complete in less than 10 minutes and results are reported immediately whilst complete results for an individual patient sample are recorded as soon as the final assay is complete. The theoretical throughput for the analyser is 1800 tests per hour. As with the majority of analysers, the c1600 is an ‘open system’ in that in addition to the programmed tests, ‘in house’ tests may also be incorporated into the routine by laboratory personnel. These may include therapeutic drug monitoring, drugs of abuse tests and specific protein assays. Each laboratory will have at least two analysers each offering a similar analytical repertoire so that one can back up the other. The analyser reads the bar code acquisition number for each sample and on the basis of the reading interrogates the host computer database to identify the tests to be carried out on the sample. The identified tests are then automatically prioritorised into the most efficient order and the analyser programmed to take the appropriate volume of sample by means of a sampler that may also be capable of detecting microclots in the sample, add the appropriate volume of reagents in a specified order and to monitor the progress of the reaction. Internal quality control samples are also analysed on an identical basis at regular intervals. The analyser automatically monitors the use of all reagents so that it can identify when each will need replenishing. When the test results are calculated, the operator can validate them either on the analyser or on the main computer database. When appropriate, the results can also be checked against previous results on the same patient.

Micro Sensor Analysers

Recent advances in micro sensor technology have stimulated the development of miniaturised multi-analyte sensor blocks that are widely used in the clinical biochemistry laboratory for the routine measurement of pH, pCO_2, Na^+, K^+, Ca^{2+}, Cl^-, glucose and lactate. The basic component of the block is a ‘cartridge’ that contains the analyte sensors, a reference electrode, a flow system, wash solution, waste receptacle and a process controller. The block is thermostatically controlled at 37°C and its surface provides an interface to the analyte sensors (Fig. 2a). The sensors are embedded in three layers of plastic, the size and shape of a credit card. Each card may contain up to 24 sensors. A metallic contact under each sensor forms the electrical interface with the cartridge.

As the test sample passes over the sensors, a current is generated by mechanisms specific to the individual analyte and recorded. The size of the current is proportional to the concentration of analyte in the fluid in the sample path.

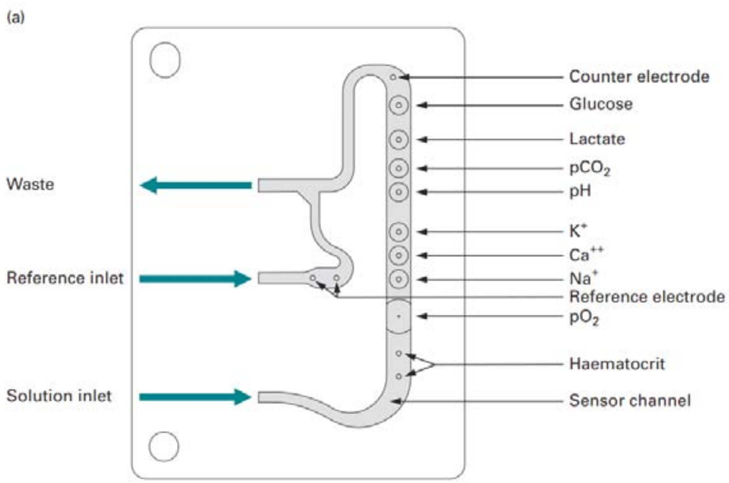
Calibration of the sensors with standard solutions of the analyte allows the concentration of the test sample to be evaluated.

The sensor card and the sample path are automatically washed after each test sample and can be used for the analysis of up to 750 whole blood samples before being discarded.

The glucose and lactate sensors have a platinum amperometric electrode with a positive potential relative to the reference electrode.

In the case of the measurement of glucose, the glucose oxidase reacts with the glucose and oxygen to generate hydrogen peroxide which then diffuses through a controlling layer (Fig. 2b) and is oxidized by the platinum electrode to release electrons and create a current flow, the size of which is proportional to the rate of hydrogen peroxide diffusion.

The GEM Premier 4000 system contains an active quality control process controller that monitors the operation of the system, validates the integrity of the cartridge and monitors the electrode response to detect microclots in the test sample that may invalidate the analytical results.



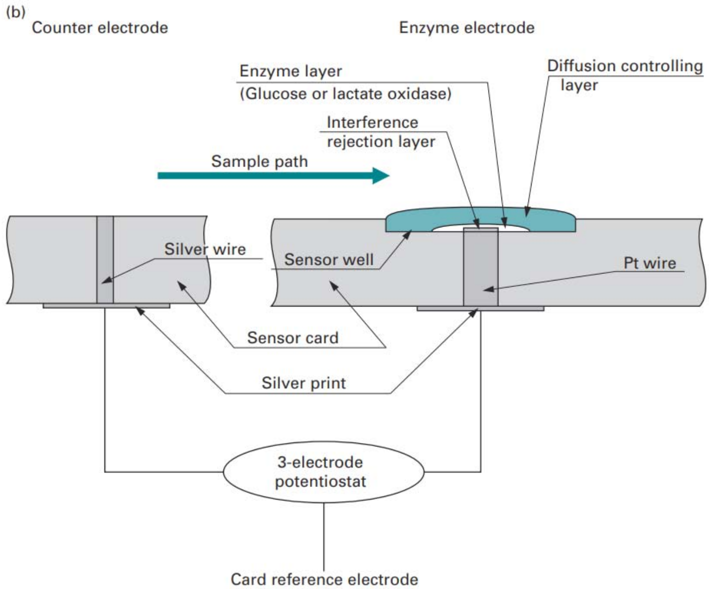


Figure 2. Micro sensor analyte detectors. (a) The GEM 3000 Sensor Card; (b) the Amperometric Glucose/Lactate Analyte Sensors.

Immunoassay section

Immunoassay procedures undertaken by modern auto-analysers are mostly based on fluorescence or polarised fluorescence techniques. The range of analytes varies from manufacturer to manufacturer but usually involves basic endocrinology (e.g. thyroid function tests), therapeutic drugs (theophylline, digoxin) and drugs of abuse (opiates, cannabis). The operation of auto-analysers in immunoassay mode is similar to that described above and the results are generally reported on the same day, and are generally compared with the previous set of results for the patient.

Manual assays section

This approach to biochemical tests is generally more labour intensive than the other two sections and covers a range of analytical techniques such as acetate or gel electrophoresis, immunoelectrophoresis and some more difficult basic spectrophotometric assays. Examples include the assays for catecholamines (for the diagnosis of phaeochromocytoma), 5-hydroxyindole acetic acid (for the diagnosis of carcinoid syndrome) or HbA_{1c} (for the monitoring of diabetes).

Result reporting

The instrument operator or the section leader initially validates analytical results. This validation process will, in part, be based on the use of internal quality control procedures for individual analytes. Quality control samples are analysed at least twice daily or are included in each batch of test analytes. The analytical results are then subject to an automatic process which identifies results that are either significantly abnormal or require clinical comment or interpretation against rules set by senior laboratory staff.

Neonatal screening

Neonatal or newborn screening is the process of testing newborn babies for certain potentially dangerous disorders. If these conditions are detected early, preventative measures can be adopted that help to protect the child from the disorders. However, such testing is not easy due to the difficulty of obtaining adequate samples of biological fluids for the tests. The development of tandem MS techniques has significantly alleviated this problem. It is possible to screen for a range of metabolic diseases using a single dried bloodspot 3 mm in diameter. There is no need for pre HPLC separation of the sample but the technique is used simply to deliver the sample to the MS. A large number of inherited metabolic diseases can be screened by the technique including aminoacidopathies such as phenylketonuria (PKU) caused by a deficiency of the enzyme phenylalanine hydroxylase, fatty acid oxidation defects such as medium chain acyl-CoA dehydrogenase deficiency (MCADD) and organic acidaemias such as propionic acidaemia caused by a deficiency of the enzyme propionyl-CoA carboxylase.

Quality assessment procedures

In order to validate the analytical precision and accuracy of the biochemical tests conducted by a clinical biochemistry department, the department will participate in external quality assessment schemes in addition to routinely carrying out internal quality control procedures that involve the repeated analysis of reference samples covering the full analytical range for the test analyte. In the UK there are two main national clinical biochemistry external quality assessment schemes: The majority of UK hospital clinical biochemistry departments subscribe to both schemes. UK NEQAS and WEQAS distribute test samples on a fortnightly basis, the samples being human serum based. In the case of

UK NEQAS the samples contain multiple analytes each at an undeclared concentration within the analytical range. The concentration of each analyte is varied from one distribution to the next. In contrast, WEQAS distributes four or five test samples containing the test analytes at a range of concentrations within the analytical range. Both UK NEQAS and WEQAS offer a number of quality assessment schemes in which the distributed test samples contain groups of related analytes such as general chemistry analytes, peptide hormones, steroid hormones and therapeutic drug monitoring analytes. Participating laboratories elect to subscribe to schemes relevant to their analytical services.

The participating laboratories are required to analyse the external quality assessment samples alongside routine clinical samples and to report the results to the organising centre.

Each centre undertakes a full statistical analysis of all the submitted results and reports them back to the individual laboratories on a confidential basis. The statistical data record the individual laboratory's data in comparison with all the submitted data and with the compiled data broken down into individual methods (e.g. the glucose oxidase and hexokinase methods for glucose) and for specific manufacturers' systems. Results are presented in tabular, histogram and graphical form and are compared with the results from recent previously submitted samples. This comparison with previous performance data allows longer-term trends in analytical performance for each analyte to be monitored.

Table 3. Selected UK NEQAS quality assessment data for serum glucose (mM).

Analytical method	n	Mean	SD	CV (%)
All methods	521	16.04	0.48	3.0
Dry slide	78	15.45	0.51	3.3
OCD (J & J) slides [1JJ]	78	15.45	0.51	3.3
Glucose oxidase electrode	59	15.74	0.32	2.0
Beckman reagents [11BK]	56	15.77	0.28	1.8
Hexokinase + G6PDH	347	16.03	0.44	2.8
Abbott reagents [15AB]	89	16.14	0.43	2.7
Olympus reagents [15OL]	113	16.06	0.40	2.5
Roche Modular reagents [15BO]	70	15.93	0.35	2.2
Glucose oxidase/dehydrogenase	112	16.20	0.47	2.9
Roche Modular reagents [4BO]	60	16.12	0.37	2.3

Notes: The Beckman Glucose Analyser uses the glucose oxidase method and measures oxygen consumption using an oxygen electrode. The Vitros method is a so-called 'dry slide' method that involves placing the sample on a slide, similar to a photographic slide that has the reagents of the glucose oxidase method impregnated in the emulsion.

A blue colour is produced and its intensity measured by reflected light. These data are for a laboratory that used the hexokinase method and reported a result of 16.0 mM. UK NEQAS calculates a Method Laboratory Trimmed Mean (MLTM) as a target value.

It is the mean value of all the results returned by all laboratories using the same method principle with results ±2 SD outside the mean omitted. Its value was 16.03. On the basis of the difference between the MLTM and the laboratory's result UK NEQAS also calculates a score of the specimen accuracy and bias together with a measurement of the laboratory's consistency of bias.

This involves aggregating the bias data from all specimens of that analyte submitted by the laboratory within the previous 6 months representing the 12 most recent distributions. This score is an assessment of the tendency of the laboratory to give an over-positive or under-negative estimate of the target MLTM values. The score indicated that the laboratory was performing consistently in agreement with the MLTM.

8.2.3 Audit and Accreditation

In addition to participating in external quality assessment schemes, clinical laboratories are also subject to clinical audit. This is a systematic and critical assessment of the general performance of the laboratory against its own declared standards and procedures and against nationally agreed standards. In the context of analytical procedures, the audit evaluates the laboratory performance in terms of the appropriateness of the use of the tests offered by the laboratory, the clinical interpretation of the results and the procedures that operate for the receipt, analysis and reporting of the test samples.

Thus whilst it includes the evaluation of analytical data, the audit is primarily concerned with processes leading to the test data with a view to implementing change and improvement. The ultimate objective of the audit is to ensure that the patient receives the best possible care and support in a cost-effective way. The audit is normally undertaken by junior doctors, laboratory staff or CPA assessors from the hospital, lasts for several days, and involves interaction with all laboratory personnel.

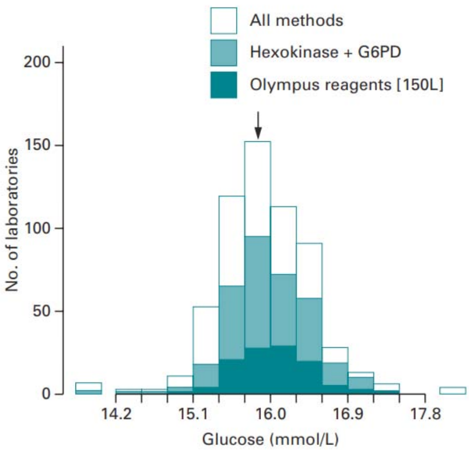


Figure 3. Histogram of UK NEQAS quality assessment data for serum glucose based on data in Table 2. The arrow indicates the location of the value submitted by the participating laboratory.

Closely allied to the process of clinical audit is that of accreditation. However, whereas clinical audit is carried out primarily for the local benefit of the laboratory and its staff and ultimately for the patient, accreditation is a public and national recognition of the professional quality and status of the laboratory and its personnel. The accreditation process and assessment is the responsibility of either a recognized public professional body or a government department or agency. Different models operate in different countries. In the UK, accreditation of clinical biochemistry laboratories is required by government bodies and is carried out by either Clinical Pathology Accreditation (UK) Ltd (CPA) or less commonly the United Kingdom Accreditation Service (UKAS). In the USA accreditation is mandatory and may be carried out by one of a number of 'deemed authorities' such as the College of American Pathologists. Accreditation organizations also exist for non-clinical analytical laboratories. Examples in the UK include the National Measurement Accreditation Service (NAMAS) and the British Standards Institution (BSI). The International Accreditation Co-operation (ILAC), the European Co-operation for Accreditation (EA) and the Asia-Pacific Laboratory Accreditation Co-operation (APLAC) are three of many international fora for the harmonization of national standards of accreditation for analytical laboratories.

Assessors appointed by the accreditation body assess the compliance by the laboratory with standards set by the accreditation body. The standards cover a wide range of issues such as those of accuracy and precision, timeliness of results, clinical relevance of the tests performed, competence to carry out the tests as judged by the training and qualifications of the laboratory staff, health and safety, the quality of administrative and technical support systems and the quality of the laboratory management systems and document control. The successful outcome of an assessment is the national recognition that the laboratory is in compliance with the standards and hence provides quality healthcare. The accreditation normally lasts for 4 years.

Table 4. WEQAS quality assessment data for serum glucose (mM).

		Sample number			
Analytical method		**1**	**2**	**3**	**4**
Reported result		7.2	3.7	17.2	8.7
Hexokinase	Mean	6.9	3.6	17.0	8.4
	SD	0.2	0.1	0.5	0.3
	Number	220	221	219	219
Aeroset	Mean	7.2	3.8	17.3	8.5
	SD	0.15	0.08	0.045	0.2
	Number	8	8	9	8
Overall	Mean	7.0	3.7	17.2	8.6
	SD	0.28	0.20	0.54	0.33
	Number	388	392	388	389
WEQAS SD		0.26	0.16	0.6	0.3
SDI		1.15	0.63	0.33	1.0

Notes: SD, standard deviation; SDI, standard deviation index. These data are for a laboratory that used the hexokinase method for glucose using an Aeroset instrument. Accordingly, the WEQAS report includes the results submitted by all laboratories using the hexokinase method and all results for the method using an Aeroset. The overall results refer to all methods irrespective of instrument. All the data are 'trimmed' in that results outside ±2 SD of the mean are rejected which explains why the total number for each test sample varies slightly. WEQAS SD is calculated from the precision profiles for each analyte and the SDI is equal to (the laboratory result – method mean result)/WEQAS SD at that level. SDI is a measure of total error and includes components of inaccuracy and imprecision. The four SDIs for the laboratory are used to calculate an overall analyte SDI, in this case 0.78. A value of less than 1 indicates that all estimates were within ±1 SD and is regarded as a good performance. An unacceptable performance would be indicated by a value greater than 2.

8.3 EXAMPLES OF BIOCHEMICAL AIDS TO CLINICAL DIAGNOSIS

The measurement of the activities or masses of selected enzymes in serum is a long-established aid to clinical diagnosis and prognosis.

8.3.1 Principles of Diagnostic Enzymology

Diagnostic enzymes are used to detect and quantify certain substances. As a marker in an enzyme immunoassay (EIA) system, clinical laboratories usually use many alternative techniques for diagnosis, including electrophoresis, chromatography, isoelectric focusing, etc.

The enzymes found in serum can be divided into three categories based on the location of their normal physiological function:

- ***Serum-specific enzymes***: The normal physiological function of these enzymes is based in serum. Examples include the enzymes associated with lipoprotein metabolism and with the coagulation of blood.
- ***Secreted enzymes***: These are closely related to the serum-specific enzymes. Examples include pancreatic lipase, prostatic acid phosphatase and salivary amylase.
- ***Non-serum-specific enzymes***: These enzymes have no physiological role in serum. They are released into the extracellular fluid and consequently appear in serum as a result of normal cell turnover or more abundantly as a result of cell membrane damage, cell death or morphological changes to cells such as those in cases of malignancy. Their normal substrates and/or cofactors may be absent or in low concentrations in serum.

Serum enzymes in this third category are of the greatest diagnostic value. When a cell is damaged the contents of the cell are released over a period of several hours with enzymes of the cytoplasm appearing first since their release is dependent only on the impairment of the integrity of the plasma membrane. The release of these enzymes following cell membrane damage is facilitated by their large concentration gradient, in excess of a thousand-fold, across the membrane. The integrity of the cell membrane is particularly sensitive to events that impair energy production, for example by the restriction of supply of oxygen. It is also sensitive to toxic chemicals including some drugs, microorganisms, certain immunological conditions and genetic defects.

Enzymes released from cells by such events may not necessarily be found in serum in the same relative amounts as were originally present in the cell. Such variations reflect differences in the rate of their metabolism and excretion from the body and hence of differences in their serum half-lives. This may be as short as a few hours (intestinal alkaline phosphatase, glutathione S-transferase, creatine kinase) or as long as several days (liver alkaline phosphatase, alanine aminotransferase, lactate dehydrogenase).

The clinical exploitation of non-serum-specific enzyme activities is influenced by several factors:

- ***Organ specificity***: Few enzymes are unique to one particular organ but fortunately some enzymes are present in much larger amounts in some tissues than in others. As a consequence, the relative proportions (pattern) of a number of enzymes found in serum are often characteristic of the organ of origin.
- ***Isoenzymes***: Some clinically important enzymes exist in isoenzyme forms and in many cases the relative proportion of the isoenzymes varies considerably between tissues so that measurement of the serum isoenzymes allows their organ of origin to be deduced.
- ***Reference ranges***: The activities of enzymes present in the serum of healthy individuals are invariably smaller than those in the serum of individuals with a diagnosed clinical condition such as liver disease. In many cases, the extent to which the activity of a particular enzyme is raised by the disease state is a direct indicator of the extent of cellular damage to the organ of origin.
- ***Variable rate of increase in serum activity***: The rate of increase in the activity of released enzymes in serum following cell damage in a particular organ is a characteristic of each enzyme. Moreover, the rate at which the activity of each enzyme decreases towards the reference range following the event that caused cell damage and the subsequent treatment of the patient is a valuable indicator of the patient's recovery from the condition.

The practical implication of these various points to the applications of diagnostic enzymology is illustrated by its use in the management of heart disease and liver disease.

8.3.2 Ischaemic Heart Disease and Myocardial Infarction

The healthy functioning of the heart is dependent upon the availability of oxygen. This oxygen availability may be compromised by the slow deposition of cholesterol-rich atheromatous plaques in the coronary arteries. As these deposits increase, a point is reached at which the oxygen supply cannot be met at times of peak demand, for example at times of strenuous exercise. As a consequence, the heart becomes temporarily ischaemic ('lacking in oxygen') and the individual experiences severe chest pain, a condition known as angina pectoris ('angina of effort'). Although the pain may be severe during such events, the cardiac cells temporarily deprived of oxygen are not damaged and do not release their cellular contents. However, if the arteries become completely blocked either by the plaque or by a small thrombus (clot) that is prevented from flowing through the artery by the plaque, the patient experiences a myocardial infarction ('heart attack') characterised by the same severe chest pain but in this case the pain is accompanied by the irreversible damage to the cardiac cells and the release of their cellular contents. This release is not immediate, but occurs over a period of many hours. From the point of view of the clinical management of the patient, it is important for the clinician to establish whether or not the chest pain was accompanied by a myocardial infarction. In about one-fifth of the cases of a myocardial event the patient does not experience the characteristic chest pain ('silent myocardial infarction') but again it is important for the clinician to be aware that the event has occurred. Electrocardiogram (ECG) patterns are a primary indicator of these events but in atypical presentations ECG changes may be ambiguous and additional evidence is sought in the form of changes in serum enzyme activities.

The activities of three enzymes are commonly measured:

- ***Creatine kinase (CK):*** This enzyme converts phosphocreatine (important in muscle metabolism) to creatine. CK is a dimeric protein composed of two monomers, one denoted as M (muscle), the other as B (brain), so that three isoforms exist: CK-MM, CK-MB and CK-BB. The tissue distribution of these isoenzymes is significantly different such that heart muscle consists of 80–85% MM and 15–20% MB, skeletal muscle 99% MM and 1% MB and brain, stomach, intestine and bladder predominantly BB. CK activity is raised in a number of clinical conditions but since the CK-MB form is almost unique to the heart, its raised activity in serum gives unambiguous support for a myocardial infarction even in cases in which the total CK activity remains within the reference range. A rise in total serum CK activity is detectable within 6 hours of the myocardial infarction and the serum activity reaches a peak after 24–36 hours. However, a rise in CK-MB is detectable within 3–4 hours, has 100% sensitivity within 8–12 hours and reaches a peak within 10–24 hours. It remains raised for 2–4 days.

- ***Aspartate aminotransferase (AST):*** This is one of a number of transaminases involved in intermediary metabolism. It is found in most tissues but is abundant in heart and liver. Its activity in serum is raised following a myocardial infarction and reaches a peak between 48 and 60 hours. It has little clinical value in the early diagnosis of heart muscle damage but is of use in the case of delayed presentation with chest pain.
- ***Lactate dehydrogenase (LD):*** This is a tetrameric protein made of two monomers denoted as H (heart) and M (muscle) so that five isoforms exist: LD-1 (H4), LD-2 (H3M), LD-3 (H2M2), LD-4 (HM3) and LD-5 (M4). LD-1 predominates in heart, brain and kidney and LD-5 in skeletal muscle and liver. Total LD activity and LD-1 activity in serum increases following a myocardial infarction and reaches a peak after 48–72 hours. The subsequent decline in activity is much slower than that of CK or AST. The diagnostic value of LD activity measurement is mainly confined to monitoring the patient's recovery from the myocardial infarction event.

Typical changes in the activities of these three enzymes following a myocardial infarction are shown in Fig. 4. All three enzymes are assayed by an automated method based on the following reactions.

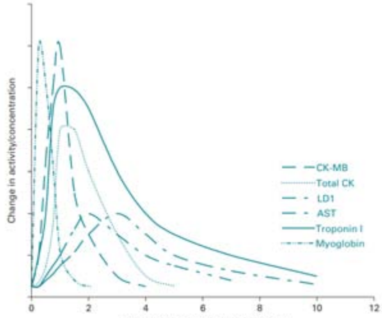


Figure 4. Serum enzyme activity and myoglobin and troponin-I concentration changes following a myocardial infarction. Changes are expressed as a multiple

of the upper limit of the reference range. Values vary according to the severity of the event, but the time course of each profile is characteristic of all events.

- ***Total CK activity:*** This is assessed by coupled reactions with hexokinase and glucose-6-phosphate dehydrogenase in the presence of N-acetylcysteine as activator, and the measurement of increase in absorbance at 340 nm or by fluorescence polarisation (primary wavelength 340 nm, reference wavelength 378 nm):

$$\text{creatine phosphate} + \text{ADP} \rightleftharpoons \text{creatine} + \text{ATP}$$
$$\text{ATP} + \text{D-glucose} \rightleftharpoons \text{ADP} + \text{D-glucose-6-phosphate}$$
$$\text{D-glucose-6-phosphate} + \text{NAD(P)}^+ \rightleftharpoons \text{D-6-phosphogluconate} + \text{NAD(P)H} + \text{H}^+$$

- ***CK-MB activity*:** This is assessed by the inhibition of the activity of the M monomer by the addition to the serum sample of an antibody to the M monomer. This inhibits CK-MM and the M unit of CK-MB. The activity of CK-BB is unaffected but is normally undetectable in serum hence the remaining activity in serum is due to the B unit of CK-MB. It is assayed by the above coupled assay procedure and the activity doubled to give an estimate of the CK-MB activity. An alternative assay uses a double antibody technique: CK-MB is bound to anti-CK-MB coated on microparticles, the resulting complex washed to remove non-bound forms of CK and anti-CK-MM conjugated to alkaline phosphatase added. It binds to the antibody–antigen complex, is washed to remove unbound materials and assayed using 4-methylumbelliferone phosphate as substrate, the released 4-methylumbelliferone being measured by its fluorescence and expressed as a concentration (ng cm^3) rather than as activity.
- ***Aspartate aminotransferase activity:*** This is assessed by a coupled assay with malate dehydrogenase and the measurement of the decrease in absorbance at 340 nm:

$$\text{L-aspartate} + \text{2-oxoglutarate} \rightleftharpoons \text{oxaloacetate} + \text{L-glutamate}$$
$$\text{oxaloacetate} + \text{NADH} + \text{H}^+ \rightleftharpoons \text{malate} + \text{NAD}^+$$

- ***Lactate dehydrogenase:*** The measurement of total activity is based on the measurement of the increase in absorbance at 340 nm using lactate as substrate. The measurement of LD-1 is based on the use of 2-hydroxybutyrate as substrate since only LD-1 and LD-2 can use it:

$$\text{Total LD:} \quad \text{lactate} + \text{NAD}^+ \rightleftharpoons \text{pyruvate} + \text{NADH} + \text{H}^+$$

$$\text{LD-1: 2-hydroxybutyrate} + \text{NAD}^+ \rightleftharpoons \text{2-ketobutyrate} + \text{NADH} + \text{H}^+$$

The clinical importance of obtaining early unambiguous evidence of a myocardial infarction has encouraged the development of markers other than enzyme activities and currently two tests are commonly run alongside enzyme activities. These are based on myoglobin and troponin-I:

- ***Myoglobin***: Concentrations in serum, assayed by HPLC or immunoassay, increase more rapidly than CK-MB after a myocardial infarction. An increase is detectable within 1–2 hours, has 100% sensitivity and reaches a peak within 4–8 hours and returns to normal within 12–24 hours. However, myoglobin changes are not specific for myocardial infarction since similar changes also occur in other syndromes such as muscle damage or crush injury such as that following a road accident.
- ***Troponin-I:*** This is one of three proteins (the others being troponin-T and troponin-C) of a complex which regulates the contractility of the myocardial cells. Its activity in serum increases at the same rate as CK-MB after a myocardial infarction, has a similar time for 100% sensitivity and for peak time, but it remains raised for up to 4 days after the onset of symptoms. Its reference range is less than 1ng cm^3 but its concentration in serum is raised to up to 30–50ng cm^3 within 24 hours of a myocardial infarction event. It is assayed by a 'sandwich' immunological assay in which the antibody is labelled with alkaline phosphatase. Using 4-methylumbelliferone phosphate as substrate, the release of 4-methylumbelliferone is measured by fluorescence. The measurement of serum troponin-I is widely used to exclude cardiac damage in patients with chest pain since it remains raised for several days following a myocardial infarction, but the timing of the test sample is important as a sample taken too early may give a false negative result. A limitation of its use is that its release into serum is not specific to myocardial infarction; an increase in serum mass may occur following a crush injury.

The measurement of enzyme activities and myoglobin and troponin-I concentrations, together with plasma potassium, glucose and arterial blood gases, is routinely used to monitor the recovery of patients following a myocardial infarction. A patient may experience a second myocardial infarction within a few days of the first. In such cases the pattern of serum enzymes shown in Fig. 4 is repeated, the pattern being superimposed on the remnants of the first profile. CK-MB is the best initial indicator of a second infarction since the levels of troponin-I may not reflect a secondary event. The sensitivity and specificity of ECG and diagnostic enzymology in the management of heart disease are complementary. Thus the specificity of ECG is 100% whilst that of enzyme measurements is 90%, and the sensitivity of ECG is 70% whilst that of enzyme measurements is 95%.

8.3.3 Liver Disease

Diagnostic enzymology is routinely used to discriminate between several forms of liver disease including:

- ***Hepatitis***: General inflammation of the liver most commonly caused by viral infection but which may also be a consequence of blood poisoning (septicaemia) or glandular fever. It results in only mild necrosis of the hepatic cells and hence of a modest release of cellular enzymes.
- ***Cirrhosis***: A general destruction of the liver cells and their replacement by fibrous tissue. It is most commonly caused by excess alcohol intake but is also a result of prolonged hepatitis, various autoimmune diseases and genetic conditions. They all result in extensive cell damage and release of hepatic cell enzymes.
- ***Malignancy***: Primary and secondary tumours.
- ***Cholestasis***: The prevention of bile from reaching the gut due either to blockage of the bile duct by gallstones or tumours or to liver cell destruction as a result of cirrhosis or prolonged hepatitis. This gives rise to obstructive jaundice (presence of bilirubin, a yellow metabolite of haem, in the skin).

Patients with these various liver diseases often present to their doctor with similar symptoms and a differential diagnosis needs to be made on the basis of a range of investigations including imaging techniques especially ultrasonography (ultrasound), magnetic resonance imaging (MRI), computerized tomography (CT) scanning, microscopic examination of biopsy samples and liver function tests.

Four enzymes are routinely assayed to aid differential diagnosis:

- ***Aspartate aminotransferase (AST) and alanine aminotransferase (ALT):*** As previously stated, these enzymes are widely distributed but their ratios in serum are characteristic of the specific cause of liver cell damage. For example, an AST/ALT ratio of less than 1 is found in acute viral hepatitis and fresh obstructive jaundice, a ratio of about 1 in obstructive jaundice caused by viral hepatitis and a ratio of greater than 1 in cases of cirrhosis.

- ***γ-Glutamyl transferase (GGT):*** This enzyme transfers a g-glutamyl group between substrates and may be assayed by the use of g-glutamyl-4-nitroaniline as substrate and monitoring the release of 4-nitroaniline at 400 nm. GGT is widely distributed and is abundant in liver, especially bile canaliculi, kidney, pancreas and prostate but these do not present themselves by contributing to serum levels. Raised activities are found in cirrhosis, secondary hepatic tumours and cholestasis and tend to parallel increases in the activity of alkaline phosphatase especially in cholestasis. Its synthesis is induced by alcohol and some drugs also cause its serum activity to rise.
- ***Alkaline phosphatase (AP):*** This enzyme is found in most tissues but is especially abundant in the bile canaliculi, kidney, bone and placenta. It may be assayed by using 4- nitrophenylphosphate as substrate and monitoring the release of 4-nitrophenol at 400 nm. Its activity is raised in obstructive jaundice and when measured in conjunction with ALT can be used to distinguish between obstructive jaundice and hepatitis since its activity is raised more than that of ALT in obstructive jaundice. Decreasing serum activity of AP is valuable in confirming an end of cholestasis. Raised serum AP levels can also be present in various bone diseases and during growth and pregnancy.

8.3.4 Kidney Disease

The kidneys, together with the liver, are the major organs responsible for the removal of waste material from the body. The kidneys also have other specific functions including the control of electrolyte and water homeostasis, and the synthesis of erythropoietin. Each of the two kidneys contains approximately 1 million nephrons that receive the blood flowing to the kidneys. Blood flowing to the kidneys is first presented to the glomerulus of each nephron which filters the plasma water to produce the ultra-filtrate or primary urine removing all the contents of the plasma except proteins. Each nephron produces approximately 100 mm^3 of primary urine per day giving a total production of primary urine by the two kidneys of approximately 100–140 cm^3 per minute or 200 dm^3 per day in a healthy adult person. This is referred to as the glomerular filtration rate (GFR). The primary urine then encounters the tubule of the nephron that is the site of the reabsorption of water and the active and passive reabsorption of lipophilic compounds and cellular nutrients such as sugars and amino acids and the active secretion of others. These two processes in combination result in the production of approximately 2 dm^3 of urine per day that is collected in the urinary bladder.

Glomerular filtration rate is the accepted best indicator of kidney function. Any pathology of the kidneys is reflected in a decreased GFR and this in turn has serious physiological consequences including anaemia and severe cardiovascular disease. Kidney disease is a progressive one, proceeding through subacute or intrinsic renal disease such as glomerular nephritis into chronic kidney disease (CKD). Complete kidney failure leads to the need for kidney dialysis and kidney transplantation. There is evidence that the incidence of CKD is increasing in developed countries and is associated with increasing risk of diabetes and an increasingly elderly population. There is thus a great clinical demand for accurate measurements of GFR in order to detect the onset of kidney disease, to assess its severity and to monitor its subsequent progression.

Measurement of glomerular filtration rate

The measurement of GFR is based on the concept of renal clearance which is defined as the volume of serum cleared of a given substance by glomerular filtration in unit time. It therefore has units of $cm^3\ min^{-1}$. In principle any endogenous or exogenous substance that is subject to glomerular filtration and is not reabsorbed could form the basis of the measurement. The polysaccharide inulin meets these criteria and is subject to few variables or interferences but because it is not naturally occurring in the body is inconvenient for routine clinical use but is commonly used as a standard for alternative methods. In practice serum creatinine is the most commonly used marker. It is the end product of creatine metabolism in skeletal muscle and meets the excretion criteria so that its serum concentration is inversely related to GFR. However, it is subject to a number of non-renal variables including:

- *Muscle mass:* Serum values are influenced by extremes of muscle mass as in athletes and in individuals with muscle-wasting disease or malnourished patients.

- *Gender*: Serum creatinine is higher in males than females for a given GFR.
- *Age*: Children under 18 years have a reduced serum creatinine and the elderly have an increased value.
- *Ethnicity*: African–Caribbeans have a higher serum creatinine for a given GFR than have Caucasians.
- *Drugs*: Some commonly used drugs such as cimetidine, trimethoprim and cephalosporins interfere with creatinine excretion and hence give elevated GFR values.
- *Diet*: Recent intake of red meats and oily fish can raise serum creatinine levels.

Routine laboratory estimations of GFR (referred to as eGFR) are based on the measurement of serum creatinine concentration and the calculation of eGFR from it using an equation that makes corrections for four of the above variables. Serum creatinine is routinely measured by one of two ways:

- *Spectrophotometric method based on the Jaffe reaction*: This involves the use of alkaline picric acid reagent which produces a red-coloured product that is measured at 510 nm. A limitation is that the reagent also reacts with some non-creatinine chromogens such as ketones, ascorbic acid and cephalosporins and as a result gives high values.
- Coupled enzyme assay: The method uses either creatinine kinase, pyruvate kinase or lactate dehydrogenase and measuring the change in absorption at 340 nm:

$$\text{creatinine} + H_2O \rightleftharpoons \text{creatine}$$

$$\text{creatine} + \text{ATP} \rightleftharpoons \text{phosphocreatine} + \text{ADP}$$

$$\text{ADP} + \text{phosphoenolpyruvate} \rightleftharpoons \text{ATP} + \text{pyruvate}$$

$$\text{pyruvate} + \text{NADH} + H^+ \rightleftharpoons \text{L-lactate} + \text{NAD}^+$$

or an alternative coupled assay using creatinine iminohydrolase and glutamate dehydrogenase.

- ***HPLC or GC/MS:*** HPLC uses a C18 column and water/acetonitrile (95:5 v/v) eluent containing 1-octanesulphonic acid as a cation-pairing agent. GC-MS is based on the formation of the t-butyldimethylsilyl derivative of creatinine.
- *Isotopic dilution*: This method is coupled with mass spectrometry (ID-MS). This involves the addition of 13C- or 15N-labelled creatinine to the serum sample, isolation

of creatinine by ion-exchange chromatography and quantification by mass spectrometry using selective ion monitoring. The lower limit of detection is about 0.5 ng

The lack of an internationally or even nationally agreed standard assay for creatinine leads to significant inter-laboratory differences in both bias and imprecision so that national external quality assurance schemes, such as UK NEQAS and WEQAS have important roles in alerting laboratories to assays that stray outside national control values. UK NEQAS provides clinical laboratories that participate in the eGFR scheme with an assay-specific adjustment factor (F) to correct for methodological variations in estimations of serum creatinine. The factor is obtained using calibration against a GC-MS creatinine assay. It is updated at 6-monthly intervals. A number of equations have been derived to calculate eGFR from serum creatinine values but the one currently used throughout the UK is the four-variable Modification of Diet in Renal Disease (MDRD) Study equation:

$$
\begin{aligned}
&eGFR \\
&= F \times 175 \times (\text{serum creatinine}/88.4)^{-1.154} \times \text{age}^{-0.203} \times 0.742(\text{if female}) \times 1.212(\text{if black})
\end{aligned}
$$

Serum creatinine concentrations are expressed in mM to the nearest whole number and are adjusted for variations in body size by normalising using a factor for body surface area (BSA) correcting to a BSA value of 1.73 m^2. The units of eGFR are therefore cm^3 min^{-1} 1.73 m^{-2} and values are reported to one decimal place. The equation has been validated in a large-scale study against the most accurate method based on the use of 125I-iothalamate GFR. Alternative equations exist for use with children.

Reference values for eGFR are 130 cm^3 min^{-1} 1.73 m^{-2} for males in the age range 20–30 and 125 cm^3 min^{-1} 1.73 m^{-2} for females of the same age. Values decline with increasing age becoming 95 and 85 cm^3 min^{-1} 1.73 m^{-2} for males and females respectively in the age range 50–60 and 70 and 65 cm^3 min^{-1} 1.73 m^{-2} for males and females in the age range 70–80.

Table 5. Stages of chronic kidney disease (CKD)

CKD stage	eGFR (cm^3 min^{-1} 1.73 m^{-2})	Clinical relevance
1	>90	Regard as normal unless other symptoms present[a]
2	60–89	Regard as normal unless other symptoms present[a]
3	30–59	Moderate renal impairment
4	15–29	Severe renal impairment
5	<15	Advanced renal failure

Note: [a]Symptoms include persistent proteinuria, haematuria, weight loss, hypertension.

Clinical Assessment of Renal Disease

Acute renal failure (ARF)

Acute renal failure is the failure of renal function over a period of hours or days and is defined by increasing serum creatinine and urea. It is a life-threatening disorder caused by the retention of nitrogenous waste products and salts such as sodium and potassium. The rise in potassium may cause ECG changes and a risk of cardiac arrest. Acute renal failure may be classified into pre-renal, renal and post-renal. Prompt identification of pre- or post-renal factors and appropriate treatment action may allow correction before damage to the kidneys occurs. Pre-renal failure occurs due to a lack of renal perfusion. This can occur in volume loss in haemorrhage, gastrointestinal fluid loss and burns or because of a decrease in cardiac output caused by cardiogenic shock, massive pulmonary embolus or cardiac tamponade (application of pressure) or other causes of hypertension such as sepsis. Post-renal causes include bilateral uretic obstruction because of calculi or tumours or by decreased bladder outflow/urethral obstruction e.g. urethral stricture or prostate enlargement through hypertrophy of carcinoma. Correction of the underlying problem can avoid any kidney damage. Renal causes of acute renal failure include glomerular nephritis, vascular disease, severe hypertension, hypercalcaemia, invasive disorders such as sarcoidosis or lymphoma and nephrotoxins including animal and plant toxins, heavy metals, aminoglycosides, antibiotics and non-steroidal anti-inflammatory drugs. Chronic kidney disease (CKD) CKD is a progressive condition characterised by a declining eGFR (Table 5). All CKD patients are subject to regular clinical and laboratory assessment and once Stage 3 has been reached to additional clinical management. This is aimed at attempting to reverse or arrest the disease by drug therapy saving the patient the inconvenience and the paying authority the cost of dialysis or transplantation.

Table 6. Examples of hormones of the hypothalamus–pituitary axis.

Secreted hormone	Pituitary effect	Gland effect
Thyrotropin-releasing hormone (TRH)	Release of thyroid-stimulating hormone (TSH)	Release of thyroxine (T4) and triiodothyronine (T3) by thyroid gland
Growth-hormone-releasing hormone (GHRH)	Release of growth hormone (GH)	Stimulates cell and bone growth
Corticotropin-releasing hormone (CRH)	Release of adrenocorticotropic hormone (ACTH)	Stimulates production and secretion of cortisol in adrenal cortex
Gonadotropin-releasing hormone (GnRH)	Release of follicle-stimulating hormone (FSH) and luteinising hormone (LH)	FSH – maturation of follicles/ spermatogenesis; LH – ovulation/ production of testosterone

8.3.5 Endocrine Disorders

Endocrine hormones are synthesized in the brain, adrenal, pancreas, testes and ovary, and most importantly in the hypothalamus and pituitary, but they act elsewhere in the body as a result of their release into the circulatory system. The result is a hormonal cascade that incorporates an amplification of the amount of successive hormone released into the circulatory system, increasing from micrograms to milligrams, as well as a negative feedback that operates to control the cascade when the level of the 'action' hormone has reached its optimum value. Most signals originate in the central nervous system as a result of an environmental (external) signal, such as trauma or temperature, or an internal signal. The response is a signal to the hypothalamus and the release of a hormone such as corticotropinreleasing hormone (CRH). This travels in the bloodstream to the anterior pituitary gland where it acts on its receptor and results in the release of a second hormone, adrenocorticotropic hormone (ACTH). It circulates in the blood to reach its target gland, the adrenal cortex, where it acts to release the 'action' hormone, cortisol, known as the stress hormone. The released cortisol raises blood pressure and blood glucose and is subject to a natural diurnal variation, peaking in early morning and being lowest around midnight. It has a negative feedback effect on the pituitary and adrenal cortex. Glands linked by the action of successive hormones are referred to as an axis, e.g. the hypothalamus–pituitary–adrenal axis. These coordinated cascades regulate the growth and function of many types of cell (Table 6). The hormones released act at specific receptors, which trigger the release of second messengers such as cAMP, cGMP, inositol triphosphate, Ca2+ and protein kinases. Diseases of the endocrine system result in dysregulated hormone release, inappropriate signalling response or, in extreme cases, the destruction of the gland. Examples include diabetes mellitus, Addison's disease, Cushing's syndrome, hyper- and hypothyroidism and obesity. Such medical conditions are characterized by their long-term nature. Laboratory tests are commonly employed to measure hormone levels in order to assist in the diagnosis of the condition and the subsequent care of the patient.

Thyroid function tests

Approximately 1% of the population suffer from some form of thyroid disease although in many cases the symptoms may be non-specific. Even so, over 1 million thyroid function tests are conducted annually in the UK.

As shown in Table 6, the hypothalamus releases thyrotropin-releasing hormone (TRH) which acts directly on the pituitary to produce thyroid-stimulating hormone (TSH) which in turn stimulates the thyroid gland to produce two thyroid hormones, thyroxine (T4) and triiodothyronine (T3). The gland produces approximately 10% of the circulating T3, the remainder being produced by the metabolism of T4 mainly in the liver and kidney. The majority of T4 and T3 are bound to thyroxine-binding globulin (TBG) but only the free unbound forms (fT4, fT3) are biologically active. Although the concentration of T3 is approximately one-tenth of that of T4, T3 is ten times more active. Both hormones act on nuclear receptors to increase cell metabolism and both have a negative feedback effect on the hypothalamus to switch off the secretion of TRH and on the pituitary to switch off TSH secretion. Hyperthyroidism is a consequence of the overproduction of the two hormones and common causes are thyroiditis, Grave's disease and TSH-producing pituitary tumours. Hypothyroidism, characterized by weakness, fatigue, weight gain and joint or muscle pain, may be primary due to the under-secretion of T4 and T3, possibly due to irradiation or drugs such as lithium, or secondary due to damage to the hypothalamus or pituitary. Normal laboratory tests for these conditions are based on the measurement of TSH and either total (bound and unbound) T4 and total T3 or fT4 and fT3 all by immunoassay.

8.3.6 Hypothalamus–Pituitary–Gonad Axis

In both sexes, the hypothalamus produces gonadotropin-releasing hormone (GRH) that stimulates the pituitary to release luteinising hormone (LH) and follicle-stimulating hormone (FSH). In males, the release of LH and FSH is fairly constant, whereas in females the release is cyclical. In males LH stimulates Leydig cells in the testes to produce testosterone which together with FSH causes the production of sperm. The testosterone has a negative feedback effect on both the hypothalamus and the pituitary thereby controlling the release of GRH. The testosterone acts on various body tissues to give male characteristics. In females, FSH acts on the ovaries to produce both oestradiol and the development of the follicle. The oestradiol and LH then act to stimulate ovulation. Oestradiol has a negative feedback effect on the hypothalamus and the pituitary and acts on body tissues to produce female characteristics.

8.3.7 Diabetes Mellitus

Diabetes is the most common metabolic disorder of carbohydrate, fat and protein metabolism, and is primarily due to either a deficiency or complete lack of the secretion of insulin by the β-cells of the islets of Langerhans in the pancreas. It affects 1–2% of Western populations and 5–10% of the population over the age of 40. It is characterized by hyperglycaemia (elevated blood glucose level) leading to long-term complications.

Diabetes can be classified into a number of types:

- *Insulin-dependent diabetes* (Type 1) (also called juvenile diabetes and brittle diabetes) is due to the autoimmune destruction of b-cells in the pancreas. Generally it has a rapid onset with a strong genetic link.
- *Non-insulin-dependent diabetes* (Type 2) (also called adult-onset diabetes and maturityonset diabetes), is a complex progressive metabolic disorder characterized by b-cell failure and variable insulin resistance. A subtype is maturity-onset diabetes of the young (MODY) which usually occurs before the age of 25 years. It is the first form of diabetes for which a genetic cause and molecular consequence have been established. Mutations of the genes for hepatocyte nuclear factor 4a (MODY1), glucokinase (MODY2), HNF1a (MODY3), insulin promotor factor 1 (MODY4), HNF1b (MODY5) and neurogenic differentiation factor 1 (MODY6) have all been characterized.
- *Impaired glucose tolerance* where there is an inability to metabolise glucose in the 'normal' way but not so impaired as to be defined as diabetes.
- *Gestational diabetes* that is any degree of glucose intolerance developed during pregnancy. It is characterized by a decrease in insulin sensitivity and an inability to compensate by increased insulin secretion. The condition is generally reversible after the termination of pregnancy, but up to 50% of women who develop it are prone to develop Type 2 diabetes later in life.
- *Other types* which include certain genetic syndromes, pancreatic disease, endocrine disease and drug or chemical induced diabetes.

Insulin-dependent diabetes (Type 1)

Between 5% and 10% of all diabetics have the insulin-dependent form of diabetes requiring regular treatment with insulin. Type 1 develops in young people with a peak incidence of around 12 years of age. In this type of diabetes the degree of insulin deficiency is so severe that only insulin replacement can avoid the complications of diabetes that are discussed later. Dietary control or oral drugs are not sufficient. The disease is caused by the autoimmune destruction of β-cells in the pancreas thus reducing the ability of the body to produce insulin. Islet cell antibodies (ICA), antibodies IA-2 and IA-2β to transmembrane protein tyrosine phosphatases in islet cells, autoantibodies to glutamic acid decarboxylase (GAD) found in β-cells and insulin autoantibodies (IAA) are all used as diagnostic markers of the disease.

Non-insulin-dependent diabetes (Type 2)

Type 2 accounts for 90% of all cases and develops later in life and can be exacerbated by obesity. MODY versions account for 1–5% of all cases and are not associated with obesity. From population screening studies it is thought that only half of those individuals with Type 2 have been diagnosed. Control of blood glucose levels in this group is normally by a combination of diet and oral drug therapy but occasionally it may require insulin injection. There is growing evidence that the increasing worldwide incidence of Type 2 diabetes may, in part, be linked to the increasing concentration of so-called persistent organic pollutants, such as bisphenol A, DDT and polychlorinated biphenyls (PCBs), in the environment. These compounds suppress adiponectin, a hormone that regulates fatty acid catabolism and glucose metabolism.

Diagnosis and monitoring of control of diabetes

Diabetes is frequently recognized by the symptoms it causes but can be confirmed by clinical biochemical measurements based on World Health Organization (WHO) recommendations:

- a fasting (12 hours) plasma glucose level greater than 7.0 mM;
- a random plasma glucose level greater than 11.1 mM;
- Application of an oral glucose tolerance test in which a 75 g dose of glucose is administered and the plasma level measured after 2 hours. Diabetes is characterised by a value greater than 11.1 mM.

The diagnostic cut-off values of 7.0 and 11.1 mM are based on the level at which retinopathy begins to appear in a population. The clinical aim in the treatment of Type 1 diabetes is to maintain plasma glucose levels in the healthy range of 4–6 mM. This is typically monitored by patients themselves by measuring their blood glucose at predetermined times that are interrelated to their meal times during the day. For example, the lowest blood glucose of the day is likely to be after the longest fast before breakfast and the highest blood glucose of the day is likely to be 1 hour after the main meal. By manipulating treatment around these highs and lows, good glycaemic control is generally maintained. The patients measure their blood glucose using hand-held, portable blood glucose meters based on glucose oxidase using dry stick technology to measure finger-prick blood samples.

Another measure of glycaemic control is by using haemoglobin A1c (Hb_{a1c}) measurements. This testing strategy works on the basis that most proteins (in this case haemoglobin A) will bind glucose dependent on the length of time they are in contact with glucose, the temperature and the concentration of glucose. Hence haemoglobin, having a typical half-life of 120 days and a standardized body temperature of 37°C, will bind the appropriate amount of glucose depending on the concentration of glucose. Hba1c is typically measured in the clinic using HPLC to separate the different haemoglobin pigments and is expressed as a percentage of total haemoglobin. In 2009 the International Federation of Clinical Chemistry and Laboratory Medicine (IFCC) published recommendations for the standardisation of Hb_{alc} using IFCC standards that allow traceability of the method back to the IFCC reference method. This caused a change in units from % to mM. This was introduced in June 2009 in the UK although most laboratories report results in both units for education purposes. The lower the result the better the control. This test is extremely useful in measuring long-term control of diabetes but is not without its pitfalls. For example, if the patient has very brittle diabetes having equal numbers of hypoglycaemic and hyperglycaemic periods, then the hypos will cancel out the hyper periods and the Hb_{a1c} will appear to show that the patient is in good glycaemic control.

Complications of Diabetes

The diabetic patient needs to have regular intake of carbohydrate to maintain their blood glucose level and appropriate levels of insulin treatment. If these are not in balance then hypoglycaemia or hyperglycaemia may take place. In hypoglycaemia the patient will become cold, clammy and sweaty and may become confused or even unconscious. Giving a sweet drink easily treats this complication. The major complication of hyperglycaemia is diabetic ketoacidosis. Almost one-third of insulin-dependent diabetic patients present for the first time with ketoacidosis that is often precipitated by infection. The biochemical features of presentation are a high or very high blood glucose level, glycosuria (glucose in the urine) and ketonuria (ketones in the urine). The patient's breath will often smell of acetone. Treatment consists of administration of fluids and an insulin infusion but this can often lead to precipitate falls in serum potassium and this must be monitored at all times.

All types of diabetes are also associated with several types of long-term complication. These can largely be split into macro-vascular disease, micro-vascular disease and others. Macro-vascular disease involves accelerated atherosclerosis in the large and medium-sized vessels. Macro-vascular disease accounts for most of the excessive mortality seen in diabetes. Micro-vascular disease leads to diabetic retinopathy and diabetic nephropathy and diabetic foot that may turn to gangrenous ulcers of the feet. The other major complications of diabetes are conditions such as gout, fatty changes in the liver, hypertension and diabetic dyslipidaemias (raised blood lipid levels).

8.3.8 Plasma Proteins

Plasma contains a very large number of proteins many of which are present only in trace amounts. The ones that have their main physiological role in plasma have three main functions:

- osmotic regulation;
- transport of ligands such as hormones, metal ions, bilirubin, fatty acids, vitamins and drugs;
- Response to infection or foreign bodies entering the body.

All plasma proteins are synthesized in the liver with the exception of the immunoglobulins which are synthesized in the bone marrow. Plasma proteins are readily separated by electrophoresis and this technique forms the basis of several clinical diagnostic tests. The tests are normally recorded subjectively, but a densitometer may be used to get a semi-quantitative result.

Albumin

Albumin is the commonest plasma protein making up some 50% of all plasma protein. Its half-life in plasma is about 20 days and in a good nutritional state the liver produces about 15 g a day to replace this loss. Albumin is the main regulator of the osmotic pressure of plasma but also acts as a transporter of haem, bilirubin (a metabolite of haem), biliverdin (a metabolite of bilirubin), free fatty acids, steroids and metal ions (e.g.

Cu^{2+}, Fe^{3+}). It also binds some drugs. Other specialist proteins found in plasma are also involved in transport, for example of steroids e.g. cortisol-binding globulin, sex-hormone-binding globulin (androgens and oestrogens) and metal ions, e.g. ceruloplasmin (Cu^{2+}) and transferrin (Fe^{3+}). Other transport plasma proteins include thyroid-binding globulin (thyroxine T4 and triiodothyronine T3) and haptoglobin (haemoglobin dimers).

Immunoglobulin's

Immunoglobulin's are synthesized in bone marrow in response to the exposure to a specific foreign body. Immunoglobulin's share a common Y-shaped structure of two heavy and two light chains, the light chains forming the upper arms of the Y. There are two types of light chains, these are either kappa (k) or lambda (l), and each are found in all classes of the immunoglobulin's. The class of immunoglobulin is determined by the heavy chain that gives rise to five types – IgG, IgA, IgM, IgD and IgE.

IgG accounts for approximately 75% of the immunoglobulin's present in the plasma of adults and has a half-life of approximately 22 days. It is present in extracellular fluids and appears to eliminate small proteins through aggregation and the reticuloendothelial system. IgA is the secretory immunoglobulin protecting the mucosal surfaces. IgA is synthesized by mucosal cells and represents approximately 10% of plasma immunoglobulin's and has a half-life of 6 days. It is found in bronchial and intestinal secretions and is a major component of colostrum (the form of milk produced by the mammary gland immediately after giving birth). IgA is the primary immunological barrier against pathogenic invasion of the mucosal membranes. IgM is found in the intravascular space and its role is to eliminate circulating microorganisms and antigens. IgM accounts for about 8% of plasma immunoglobulin's and has a half-life of 5 days. IgM is the first antibody to be synthesised after an antigenic challenge. IgD and IgE are minor immunoglobulin's whose roles are not clear since a deficiency of either seems to be associated with no obvious pathology. IgE plays a major part in allergy and may be significantly raised in situations of allergic response, for example in hay fever and atopic eczema.

Myeloma

Myeloma, also called multiple myeloma, is a malignant pathology of plasma cells in which there is a proliferation of a single β-cell clone in the bone marrow effectively behaving as a tumour. The replication of the cell is unregulated so it proliferates. The cells produce large quantities of a single identical antibody which runs as a single dense band in the gamma globulin region on electrophoresis of a serum sample. The protein is called the para protein and has been shown to be an immunoglobulin with two light chains and two heavy chains. Some myelomas produce an excess of light chains that appear in the serum and because they are small they also appear in the urine. Their detection is a bad prognosis as they indicate that the cell line may be more aggressive and replicating faster. In rare cases of myeloma, the marrow cells only produce light chains.

Acute phase response

Following a stimulus of tissue injury or infection, the body will respond by producing an acute phase response characterised by the release from the liver of a number of acute phase proteins which cause a change in the pattern of plasma protein electrophoresis. There will be an increased synthesis of some proteins such as α-1-antitrypsin, a proteinase inhibitor that down regulates inflammation, fibrinogen and prothroylian (coagulation) complement and C-reactive protein (CRP). These are referred to as positive acute phase proteins. There will also be a decrease in the production of other proteins such as albumin, transferrin and transcortin. These are known as negative acute phase proteins. The clinical measurement of acute phase proteins, particularly CRP, by immunoassay is widely used as a marker of inflammation in a variety of clinical conditions.

REFERENCES

1. Beckett, G. I., Walker, S. W., Rae, P. and Ashby, P. (2005). Lecture Notes on Clinical Biochemistry, 6th edn. Oxford: Blackwell Science. (An excellent reference text for all aspects of clinical biochemistry.)
2. Blau, N., Duran, M., Blaskovics, M. E. and Gibson, K. M. (eds.) (2003). Physician's Guide to the Laboratory Diagnosis of Metabolic Diseases, 2nd edn. Berlin: Springer-Verlag.
3. Bruns, D. E. and Ashwood, E. R. (2007). Tietz Fundementals of Clinical Chemistry, 6th edn. Philadelphia: W. B. Saunders. (A comprehensive coverage of the principles and practice of clinical biochemistry.)
4. Chace, D. H. and Kalas, T. A. (2005). A biochemical perspective on the use of tandem mass spectrometry for newborn screening and clinical testing. Clinical Biochemistry, 38, 296–309.

INDEX

Biochemistry and Molecular Biology: Principles and Applications

Biochemistry, with its wide-ranging applications, has had a profound impact on various industries and aspects of society. While its significance is most evident in the medical and pharmaceutical sectors, biochemistry permeates everyday life, influencing fields such as retail, food, cosmetics, fashion, and healthcare. One of the notable contributions of biochemistry lies in the development of medical products. Through biochemical research, scientists have discovered and improved upon various medications that have revolutionized healthcare. Furthermore, biochemistry has played a crucial role in the advancement of DNA recombinant technology, which enables the production of essential molecules like insulin and food additives. These breakthroughs have enhanced the treatment of diseases and improved the overall quality of life for countless individuals.

Beyond medicine, biochemistry has also had a positive impact on food production and safety. Agrochemicals developed through biochemistry have enhanced crop yield and protected plants from pests and diseases, leading to increased food production. Additionally, biochemistry has facilitated the creation of foods that promote human health, such as pre- and probiotics, which support gut health, and antioxidants, which help combat oxidative stress. These advancements in food science have contributed to a healthier and more sustainable food supply.

This book provides a comprehensive overview of the storage and flow of genetic information, serving as a valuable resource for students and researchers in the field. It emphasizes the integration of cutting-edge research into biochemistry and molecular biology education, highlighting the importance of a supportive academic environment with access to advanced facilities and mentoring.

The applications of molecular biology methods extend beyond fundamental scientific inquiries. These methods are instrumental in disease prevention and treatment, enabling the development of innovative therapies and diagnostic tools. They also facilitate the generation of novel protein products with various applications, from industrial processes to biotechnology. Moreover, molecular biology methods are employed in the manipulation of plants and animals to achieve desired phenotypic traits, enhancing agricultural practices and animal husbandry.

Fischer Dimitrion, PhD, is an accomplished Associate Professor in Biological Chemistry. He has refined his expertise through visiting scholar positions at multiple institutes. His research encompasses diverse areas such as neurochemistry, hormone and pheromone biochemistry, enzymes and metabolism, hormone action, and gene regulation. Likewise, he actively participates in gene characterization studies. His overarching goal is to unravel the complex mechanisms that underlie these fascinating fields of research.

ISBN 978-1-98469-331-0

00000

9 781984 693310

BIBLIOTEX
Digital Library